



Oak Stream Platform

Message of the Week (MoW)

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1.1 WW28_26

New:

- [Diamond Rapids BDG Release Notification](#) on page 15
- [Oak Stream Platforms System Boards Design Archive Release Notification](#) on page 16
- [Oak Stream 16-Channel Platform Socket Cover Availability Update](#) on page 18
- [Oak Stream 16-Channel Platform Socket DX Update](#) on page 18
- [WW28'26 BKC Release Notification](#) on page 19

Updated:

- [\(Updated WW28_26\) Oak Stream Key Collaterals Availability and Forecast](#) on page 16
- [\(Updated WW28_26\) Oak Stream Key Tools Availability and Forecast](#) on page 20

1.1.1 Documentation

1.1.1.1 Diamond Rapids BDG Release Notification

The *Diamond Rapids and Palm Ridge Processor BIOS Developer's Guide*, document number [839071](#), revision 1.0, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.0 changes:

- Updated title and content to comprehend Palm Ridge processors.
- Introduction
 - Updated Platform and Processor Publications.
 - Updated Related Publications.
- Overview
 - Added **Platform RAPL (PSYS)** section.
- Processor Initialization and Configuration
 - Added **Intel® Firmware Support Package** section.
- Memory Subsystem
 - Reorganized Supported Memory Configuration and Validation content under **Integrated Memory Controller (IMC)** section.
 - Added **Reclaim Lost Memory** section.

- Security Considerations for BIOS
 - Added Intel® Total Memory Encryption - Multi-Key (Intel® TME-MK), Intel® Vendor Authorized Boot, and Debug Tokens and Vendor Authorized Debug sections.
 - Removed Section UEFI VABD Tool.
- Appendix
 - Added **BIOS to Primecode Mailbox Usage and Message Definition** and **Unified IFWI Support** sections.

1.1.1.2 Oak Stream Platforms System Boards Design Archive Release Notification

The *Oak Stream Platform System Boards Design Archive*, document number [821025](#), Revision 1.03, is now available in the Resource and Documentation Center (RDC).

Overview of 1.03 Changes:

- Added front Panel schematic, layout, and BOM files. The front panel design includes circuitry for I/O button management such as Power, ID, and Reset buttons.
- There are no changes to the other boards included in design archive.

1.1.1.3 (Updated WW28_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 1. Table Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.15	1.16	July'26
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.15	1.2	WW32'26
<i>Diamond Rapids Processor Test Capability (Heat Map)</i>	873155	WW24'26	-	WW28'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	1.4	1.5	July'26
<i>continued...</i>				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
Diamond Rapids and Palm Ridge Processors BIOS Developer's Guide	839071	1.0	1.1	Q4'26
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	1.0	As needed	-
Oak Stream 16-Channel Platform CPLD Reference Code	860022	1.0	1.5	WW28'26
Oak Stream 16-Channel Platform Design Guide	788196	1.0	1.1	WW40'26
Oak Stream 16-Channel Platform Electrical Specification	787006	1.1	1.2	Q3'26
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	1.2	As needed	-
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258	1.2	As needed	-
Oak Stream 16-Channel Platform Pin List	636260	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	1.0	As needed	-
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	1.12	1.13	WW31'26
Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive	821026	0.55	0.7	WW50'26
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Oak Stream Platform Diamond Rapids 16-Channel Dashboard</i>	863599	WW27'26	-	July'26
Oak Stream Platforms System Boards Design Archive	821025	1.03	As needed	-
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	1.2	As needed	-

1.1.2 Enabling

1.1.2.1 Oak Stream 16-Channel Platform Socket Cover Availability Update

- Foxconn Interconnect Technology* (FIT*) socket cover hard-tooled prototypes are now approved by Intel for ordering.
- This update will be reflected in the Oak Stream 16-Channel Platform Thermal Mechanical Specification, document number 786521, revision 1.13, targeted to be released by the end of July 2026.
- Additional details on hard-tooled prototype status and use considerations are in the [Oak Stream 16-Channel Platform Hard-Tooled Prototype Availability Update](#) on page 67 article.

1.1.2.2 Oak Stream 16-Channel Platform Socket DX Update

Socket DX parts from all three enabled suppliers have been promoted to production candidate, DG0.7, status:

- "DG" refers to sockets based on Intel's socket design guide. This socket design guide is a detailed socket requirements document and is for the use between Intel and socket suppliers only.
- Criteria for DG0.7 status include time 0 shock, vibration, durability testing, and Cpk criteria.
- DG0.7 sockets may have design refinements compared to DG0.6 and will have improved supplier manufacturing capability and yields.
- DG0.7 sockets are the first versions recommended for beginning of life customer validation work.

With this milestone, CPUs and thermal solutions may now be shipped installed in a system.

The Part Numbers (PNs) for TE Connectivity* and LOTES* that are currently listed in the *Oak Stream 16-Channel Platform Thermal Mechanical Specification (TMS)*, revision 1.12, document number [786521](#), and will remain unchanged as parts are promoted to DG0.7.

Foxconn Interconnect Technology* (FIT*) is updating their PN. Contact them for their latest PN. This promotion will be reflected in revision 1.13 TMS, targeted for release by the end of July 2026.

Accelerated environmental stress testing of DG0.7 sockets is in progress and scheduled for completion in September 2026.

NOTE

DG0.7 sockets are not recommended to be used for environmental stress testing until they are approved at this milestone. Any environmental qualification testing is done at your own risk as the parts are subject to change until they pass all of Intel's validation testing.

Full validation of socket DX as production approved DG1.0 is targeted for completion in Q4 2026.

Contact your socket supplier for additional details on their DX product readiness.

1.1.3 Software and Firmware

1.1.3.1 WW28'26 BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platform.

The BKC consists of an update package and associated report, published on a regular basis, for the Johnson City reference platform.

The BKC package includes UEFI* firmware and onboard device drivers. These components have been screened through interoperability testing together over a period, with the aim of delivering customer confidence that not only have platform ingredients been individually tested, but also collectively tested to screen out interoperability issues.

Table 2. Table Johnson City WW28'26 BKC

Ingredient	Previous	Current
BKC	WW24	WW28
CRB BIOS	35.D61	36.D54
Unified Patch (MCU)	0x800009A6	0x800009AA
MMC FW	0.855.0	0.892.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	1.0.6422	1.0.6509
Download	921663	923224
Status	Released	Released

NOTE

For information on silicon availability, contact your Intel representative.

Memory Configuration Enforcement

This WW28'26 BKC release includes a memory configuration enforcement behavior that causes certain memory configurations to possibly observe memory initialization or system bring-up issues.

This memory configuration enforcement will be reverted in the next BKC release.

As temporary mitigation, customers that encounter memory initialization or system bring-up issues can apply a temporary BIOS patch included in the BKC GitHub*release to restore the previous memory configuration handling:

- 0001-Align-Memory-POR-verification.patch

Known Limitations

The following limitations maybe observed with WW28'26 BKC:

- (16030248887) On some systems, OS installation through BMC Virtual Media may fail, causing the system to hang in the BIOS menu.
 - Workaround: Disable EFI Network boot: **EDK II > Platform Configuration > Network Configuration > EFI Network > Disable**
- (15019338702) On 2S system you may observe PL_rtte error with SSC2 enabled.
 - Workaround: Disable SSC2 or Enabling Elastic Buffer Mode over Drift Buffer Mode in FBLP
- (16029885327) Warm reset fails with DXE_ASSERT on the system having with some NVMe* drives.
 - Status: Ensure usage of latest firmware version from vendor.
- (16030716083) DXE ASSERT is observed during reboot after Seamless OOB BIOS/ UCODE update.
 - Status: Under investigation.
- (14027896021) BIOS assert is observed when Pre-boot DMA Protection bios knob is enabled.
 - Status: Under investigation.
- 16030968586) On 2S systems, G3 cycling may intermittently cause the BMC root password to be lost and the system to hang at post code 0.5.
 - Workaround: do not update the HPM_PLD, AGG_PLD, or DCSCM_PLD firmware on 2S systems.

NOTE

Monitor the *Diamond Rapids Processor Test Capability Heatmap*, document number [873155](#), for updates on these exceptions.

1.1.4 Tools and Debug

1.1.4.1 (Updated WW28_26) Oak Stream Key Tools Availability and Forecast

NOTE

Updated information is highlighted in bold text.

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to Intel® ITP adapter.
- Close Chassis Adapter (CCA).
- Lauterbach* - Contact Lauterbach* for support. Two Streams Tracing supported:
 - LA-3907 preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports four trace ports).
 - LA-3892 converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60.
- ASSET* Arium* - Contact ASSET Arium for support.
- Compact Debug Port (CDP) + MIPI10 adapter for I3C*
- Type-C* connection: Voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- VTune™
- Intel® SoC Watch
- Intel® System Debugger
- Includes token generation for Intel® Vendor Authorize Boot / Vendor Authorize Debug (VAD).

NOTE

The Intel® System Bring-up Toolkit NDA requires registration. See *How to Obtain Intel® System Bring-up Toolkit NDA (Formerly Intel® System Studio NDA) User Guide*, document number [616675](#), and *Intel® System Bring-up Toolkit NDA (formerly Intel® System Studio NDA) Registration Form New Users*, document number [631579](#), for details.

Table 3. Table Oak Stream Tools Forecast

Type	Tool	RDC or DIT	Availability	Revision
Debug	ASSET Intertech Arium * Note: Contact Asset* for support.	-	Q3'26	-
Debug	ASSET* Close Chassis Controller (CCC) Note: Contact Asset* for support.	-	Q3'26	-
Debug	CScripts for Oak Stream Platforms Note: Refer to Release Notes for details.	821387	Available	2627.2000
Debug	Intel Remote FPGA Access Tool (RFAT) for Server Platforms	871932	Available	1.0
Debug	Intel Triage Tool (Crashdump Summarizer) Note: Available through Cscripts	N/A	Available	0.89
Debug	Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C*	99CJRJ	Available	N/A
Debug	Intel® I3C* USB Type-C Level Shift Adapter	99CN2C	Available	N/A
continued...				

Type	Tool	RDC or DIT	Availability	Revision
Debug	Intel® In-Target Probe (Intel® ITP) eXtended Debug Port (XDP)	940990	Available	N/A
Debug	Intel® Silicon View Technology (Intel® SVT) Closed Chassis Adapter	921521	Available	N/A
Debug	Intel® System Bring-up Toolkit NDA Note: Includes Intel® System Debugger NDA, Intel® VTune™ Profiler NDA and Intel® SoC Watch NDA.	634166	Available	N/A
Debug	Lauterbach*	-	Available	N/A
Debug	SelfTest 7 - Windows* x64	524152	Available	181
Debug	Server Security Toolkit Note: Server Security Toolkit includes Intel® Total Memory Encryption – Multi-Key (Intel® TME-MK), Intel® Trusted Execution Technology (Intel® TXT), BIOS TPM Provisioning with Intel® Platform Trust Technology (Intel® PTT) tools, and CBnT (Converged Intel® Boot Guard and Intel® TXT).	630398	Available	1.56
Manufacturing and Fleet Health	Autonomous Crashdump (ACD) Note: Autonomous Crash Dump (ACD) is available via BMC source code under license agreement	N/A	Available	0.8
Manufacturing and Fleet Health	Intel® At-Scale Debug (Intel® ASD)	N/A	Available	N/A
Manufacturing and Fleet Health	Intel® At-Scale Debug (Intel® ASD) BMC-JTAG* Software Guide	594948	Available	1.6.5
Manufacturing and Fleet Health	Intel® In Field Scan	-	Q2'27	5.x.x
Manufacturing and Fleet Health	Intel® System Stress and Memory Diagnostics (SSMD) for Linux	818881	Available	2026.05.04.fc55
Manufacturing and Fleet Health	Memory Latency Checker (MLC)	736633	Q3'26	3.12
Signal Integrity	Clock Jitter Tool (CJT)	803838	Available	6.0.4
Signal Integrity	Integrated Channel Analysis Tool (ICAT) Suite for Windows	782966	Available	6.200.50
Signal Integrity	Intel® Passive Component Checker Tool (Intel® PCC Tool)	600218	Available	1.8
Signal Integrity	Intel® Platform Clock Jitter Analysis Tool (CJAT) for Windows	619648	Available	6.0.4
Signal Integrity	PCIe* and Intel® UPI Load Board Solution Set for 16-Channel Platform	DMRUCCPLBULB	Available	N/A
Signal Integrity	PCIe* Load Board Solution Set for 16-Channel Platform	DMRUCCPLB	Available	N/A
Thermal / Power Delivery	Almondsville Thermal Mechanical Test Vehicle (TMTV) Application Note	852568	Available	1.0
Thermal / Power Delivery	Almondsville Thermal Test Vehicle (TTV) for 16-Channel Platform	-	Available	-
continued...				

Type	Tool	RDC or DIT	Availability	Revision
Thermal / Power Delivery	Blue Interposer for Gen5 VRTT for 16-Channel Platform	99D5W2	Available	N/A
Thermal / Power Delivery	Gen5 VR Test Tool (VRTT) for Windows	99CJ9X	Available	N/A
Thermal / Power Delivery	Red Interposer for Gen5 VRTT for 16-Channel Platform	99D5W1	Available	N/A
Thermal / Power Delivery	VRTT Mechanical Interposer Retention Mechanism Kit for 16-Channel Platform	99CN1R	Available	N/A
Validation	Diamond Rapids CPU Crash Log Content Decode Definition	917977	Available	1.0
Validation	Diamond Rapids CPU Crash Log Quick Start Guide	917972	Available	1.0
Validation	Diamond Rapids Processor Boundary-Scan Description Language (BSDL) Files	919398	Available	1.0
Validation	Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card	-	Available	-
Validation	Intel® Simics Note: SIMICs available under license agreement. Contact your Intel representative for details.	-	Available	N/A
Validation	Intel® Structural Test Plus (Intel® STP) Note: Contact a local representative to enable Intel® STP for customer designs	-	Available	N/A
Validation	MEI DDR5 Lite Card 8000 MT/s for Windows	99D5X2	Available	N/A
Validation	Memory Assistant for Intel® Xeon® Platforms	620497	Available	WW 23'26
Validation	Memory Error Injection (MEI) DDR5 Card for Windows	99CDV1	Available	N/A
Validation	Oak Stream Platform DDR Electrical Validation Tools and Methodology	914412	Available	1.0
Validation	Oak Stream Platform Intel® IO Margin Tool for Windows* User Guide	913382	Available	1.0
Validation	Oak Stream Platform Intel® IO Margin Tool for Windows Note: The WW30 release of the Intel® IO Margin Tool will deliver important usability and validation enhancements. Key updates include resolution of antivirus false-positive detections on main.dll, support for PCIe* 6.0/5.0/4.0/3.0 margin testing, On-target support for PCIe margin validation for Linux* and Windows*, and methodology improvements that replace FBER-based testing with margin testing to provide a more actionable and quantitative assessment of PCIe and Intel® UPI 3.0 channel health.	871016	Available	2.0.0
Validation	Oak Stream Platform Margin Testing	871015	Available	1.0
continued...				

Type	Tool	RDC or DIT	Availability	Revision
Validation	<i>Oak Stream Stitched Rank Margining Tool (RMT)</i> Note: Included in BKC.	-	Available	-
Validation	<i>Oak Stream Target Loader Rank Margining Tool (RMT)</i>	913381	Available	1.0
Validation	<i>PCI Express* (PCIe*) Lane Margin Tool (LMT) for PCIe* 4.0/5.0/6.0</i>	639433	Available	1.1
Validation	<i>PCIe* 5.0 Error Injection (PEI) Card</i>	99C3FA	Available	N/A
Validation	<i>PCIe* PEI 6.0 Card</i>	-	Q1'27	N/A
Validation	<i>PEI (PCIe Error Injector Card) Collaterals</i>	610667	Available	5.1
Validation	<i>PEI 5.0 (PCIe 5.0 Error Injection) + CXL 2.x Card</i>	99CN2D	Available	N/A
Validation	<i>RAS Automation ToolKit</i>	633201	Available	3.0.0

1.2 WW27_26

New:

- [Diamond Rapids Processor Sightings Report Release Notification](#)
- [Oak Stream Platform Dashboard WW27 2026 Release Notification](#)
- [CScripts for Oak Stream Platforms Release Notification WW27_26](#)
- [Diamond Rapids BSDL Files Release Notification](#)

Updated:

- [\(Updated WW27_26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.2.1 Documentation

1.2.1.1 Diamond Rapids Processor Sightings Report Release Notification

The *Diamond Rapids Processor Sightings Report*, document number [862869](#), Revision 1.4, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.4 changes are:

- Added sightings 106 to 125.
- Updated sightings 60, 62, 74, and 103.

Refer to the revision history for a complete list of changes.

1.2.1.2 Oak Stream Platform Dashboard WW27 2026 Release Notification

The *Oak Stream Platform Diamond Rapids 16-Channel Processor Dashboard*, document number [863599](#), Revision WW27 2026, is now available in the RDC.

Refer to document content for revision updates.

1.2.1.3 (Updated WW27_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 4. Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.15	1.16	July'26
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.15	1.2	WW32'26
<i>Diamond Rapids Processor Test Capability (Heat Map)</i>	873155	WW24'26	-	WW28'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	1.4	1.5	July'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i>	839071	0.9	1.0	WW30'26
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	1.0	1.1	Q3'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	1.1	1.2	Q3'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.2	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.2	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	1.0	As needed	-
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	1.0	As needed	-
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	1.11	-	Q3'26
Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive	821026	0.55	0.7	WW50'26
Oak Stream Platform Diamond Rapids 16-Channel Dashboard	863599	WW27'26	-	July'26
Oak Stream Platforms System Boards Design Archive	821025	1.02	As needed	-
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	1.2	As needed	-

1.2.2 Tools and Debug

1.2.2.1 CScripts for Oak Stream Platforms Release Notification WW27_26

The *CScripts for Oak Stream Platforms*, document number [821387](#), version 2627.2000, is now available in the RDC.

Overview of 2627.200 changes are:

- Introduced new remote diagnostics and PCI Express * (PCIe*) validation capabilities, including manual MCA decoding, on demand crashdump collection, PRBS generation, self-loopback testing, and enhanced command logging.
- Expanded in-band support for RAS, EI, MC, and PCIe modules across CentOS*, Windows*, and VMware* ESXi* applications.
- Completed Redfish* functionality.
- Added Mirror and ADDDC support in ras.mem.sa2da_table().
- Added a prerequisite checker for CScripts dependencies.
- Improved BMC communication stability, platform reset recovery, Intel® Ultra Path Interconnect (Intel® UPI) topology reporting, eMCA2 validation, PCIe 4.0 Leaky Bucket support, Intel® At-Scale Debug (Intel® ASD) synchronization, I3C* information display, memory error injection, command parity injection, and patrol scrub flows.

- Reduced unnecessary logging to improve usability and reliability.

Refer to the revision history for a complete list of changes.

1.2.2.2 Diamond Rapids BSDL Files Release Notification

The *Diamond Rapids Processor Boundary-Scan Description Language (BSDL) Files*, document number [919398](#), Revision 0.5, is now available in the RDC.

Overview of the Revision 0.5 change is:

- Initial release

1.3 WW26_26

New:

- [Oak Stream Platform RAS Technology Integration and Validation Guide Release Notification](#)
- [Enabling Intel® TXT on Oak Stream Platforms Requires a New TPM Provisioning Command Clarification](#)
- [Engineering Validation and Test \(EVT\) Enabling for Oak Stream 16-Channel Platform 1S/2S Designs Update](#)
- [Oak Stream IFWI Nomenclature, Recipe, and Compatibility](#)

1.3.1 Documentation

1.3.1.1 Oak Stream Platform RAS Technology Integration and Validation Guide Release Notification

The *Oak Stream Platform RAS Technology Integration and Validation Guide*, document number [846674](#), Revision 0.5, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 0.5 changes are:

- Removed all Intel® Simics® simulation-based test cases.
- Added content tested with Diamond Rapids UCC silicon.

1.3.2 Enabling

1.3.2.1 Enabling Intel® TXT on Oak Stream Platforms Requires a New TPM Provisioning Command Clarification

When enabling Intel® Trusted Execution Technology (Intel® TXT), incorrect TPM provisioning will cause Intel® TXT to fail. The provisioning command of the required TPM AUX index has changed from previous platforms. Now on the Oak Stream platform, there is an additional parameter that must be added: "server".

Intel® TXT enabling will fail if using the same TPM provisioning command as on previous platforms. The BIOS ACM will report the following error code which corresponds to ERR_TPM_NV_INDEX_INVALID_AUX_SIZE:

BIOSACM Error Code (0xFED0328[31:0]):0xC0069C40

The TPM AUX index size is not correct due to using the incorrect TPM provisioning command.

To avoid this error code, provision the TPM AUX index with the latest *TPM 2.0 Provisioning Tool EFI Shell binaries and samples*, document number [633967](#), RDC Version 1.6 or later, using the following command (use SHA256 or SHA384 as needed):

```
Tpm2_Cbnt_Prov.nsh SHA_____ server
```

For more details, refer to the TPM Provisioning chapter of the *Oak Stream Platform Converged Intel® Boot Guard and Intel® TXT Integration and Validation Guide*, document number [915484](#).

1.3.2.2 Engineering Validation and Test (EVT) Enabling for Oak Stream 16-Channel Platform 1S/2S Designs Update

This is an update to the MOW article published in the [WW49'24 MOW](#).

To enable customer 1S/2S Time To Market (TTM) launch readiness aligning to the Oak Stream 16-Channel platform schedule, Intel will enable 1S/2S customers' **EVT** activities by separating EVT into **EVT 1** and **EVT 2** phases.

EVT 1 activities will use UCC IMH1-based ES1 samples and **EVT 2** activities will use UCC-M IMH2-based EVS samples.

Refer to the latest Oak Stream Platform Diamond Rapids Dashboard for the detailed Oak Stream schedule and key sampling milestones.

A high-level mapping of expected customers EVT 1 and EVT 2 activities is detailed in Figure 1.

Figure 1. Customer Validation Phase Mapping

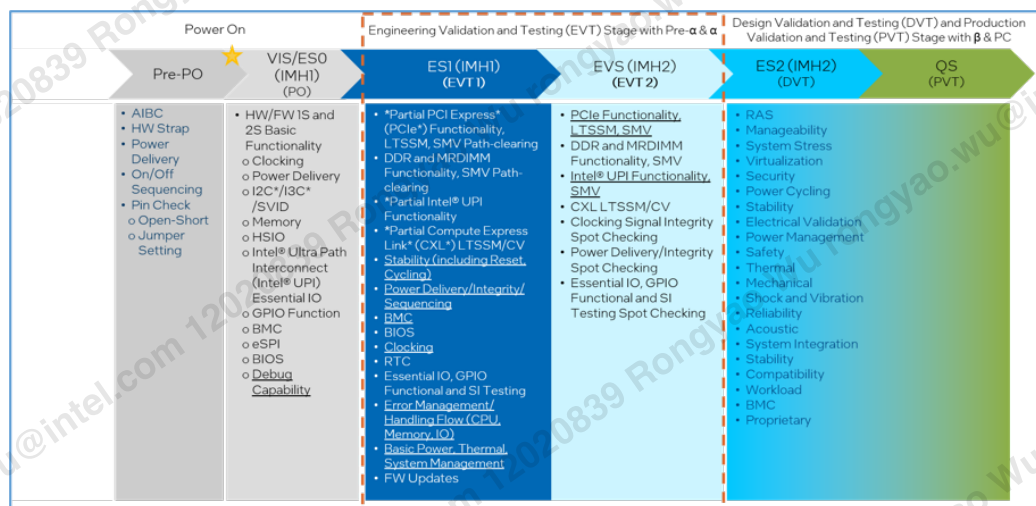


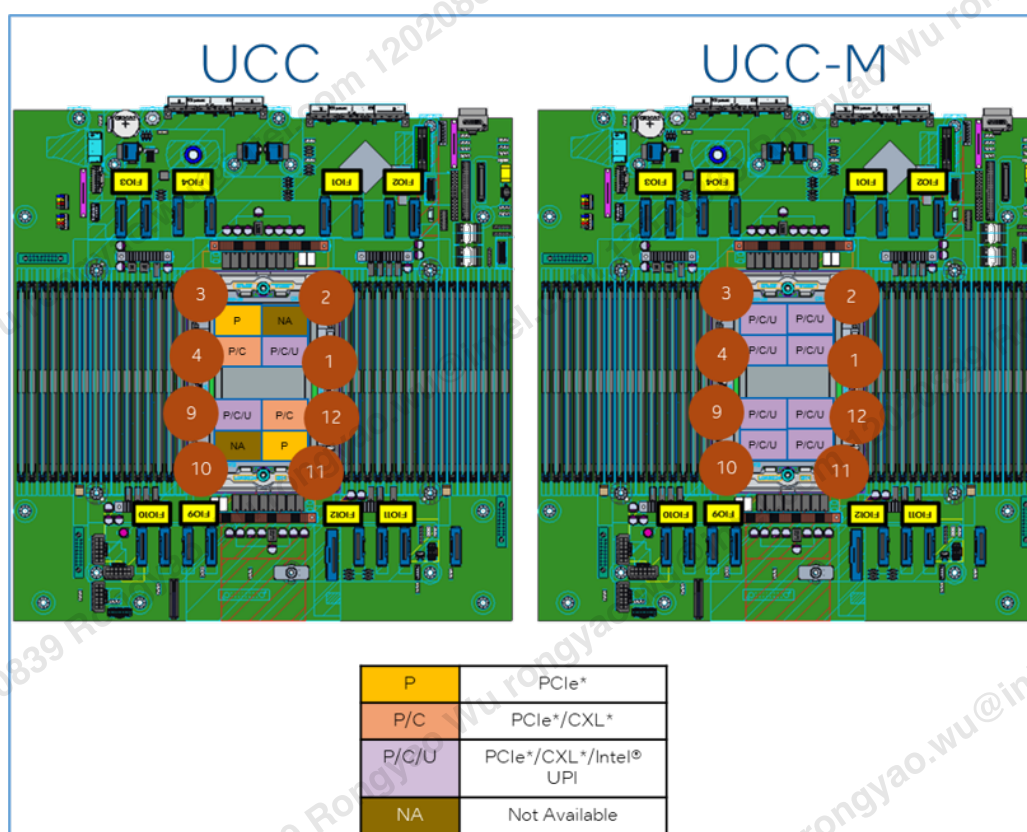
Figure 2 contains the high-level comparison of the UCC and UCC-MHSIO capabilities. Differences in the HSIO capabilities are comprehended in the mapping of customers' **EVT 1** and **EVT 2** activities.

IMH1-based ES1 samples have Intel® Ultra Path Interconnect (Intel® UPI) capabilities on ports 1 and 9. 2S designs must use ports 1 and 9 for Intel® UPI connectivity to be able to take advantage of ES1 samples for earlier 2S EVT start.

Intel® UPI connectivity for ES1 is:

- Socket 0 port 1 to Socket 1 port 1
- Socket 0 port 9 to Socket 1 port 9

Figure 2. UCC and UCC-M Port Mapping



NOTES:

PCIe represents PCI Express* (PCIe*).

CXL represents Compute Express Link* (CXL*).

1.3.3 Software and Firmware

1.3.3.1 Oak Stream IFWI Nomenclature, Recipe, and Compatibility

Oak Stream IFWI Nomenclature Changes

Inconsistent Birch Stream IFWI nomenclature caused confusion regarding which IFWI build to use and how to build IFWI releases. Starting with Oak Stream, Intel will align the unified patch, ACM, and IFWI nomenclature. The following table highlights the nomenclature changes in **bold text**.

Table 5. Oak Stream IFWI Ingredients Nomenclature

Sample Type	Unified Patch (UP)	ACMs (BIOS, SINIT, ACTM, SEAM_LDR)	Intel® Trust Domain Extensions (Intel® TDX) Module	IFWI
Engineering Samples (ES)	Debug	Debug	Debug	Non-Production -> Debug
Qualification Samples (QS)	SED-Signed -> non-production worthy	Non-production worthy	Debug	Production -> Non-production worthy
Revenue Samples (QSfD and PRQ)	Production	Production worthy	Production	PRQ -> Production Worthy

IFWI binaries use “Trace” versus “NoTrace” (instead of “debug”) to indicate whether the serial output trace messages are available.

Oak Stream IFWI Recipe

IFWI needs to be built with the appropriate firmware components for each CPU sample type. The following table summarizes the compatibility of the Oak Stream IFWI ingredients.

Table 6. Oak Stream IFWI Recipe

Sample Type	BKC IFWI	Unified Patch (UP)	Memory Microcontroller (MMC) FW	ACMs (BIOS, SINIT, ACTM, SEAM_LDR)	Intel® TDX Module
Engineering Samples (ES)	Debug	Debug, NPW-signed, or Production	Debug, NPW-signed, or Production	Debug	Debug
Qualification Samples (QS)	Non-production worthy	NPW-Signed or Production	NPW-Signed or Production	Non-production worthy ¹ or Production worthy	Debug for NPW SEAM_LDR Production for PW SEAM_LDR
Revenue Samples (QSfD and PRQ)	Production worthy	Production	Production worthy	Production worthy	Production

NOTE

¹Intel® Boot Guard and Intel® Trusted Execution Technology (Intel® TXT) PW BIOS ACMs and PW SINIT ACMs will increment their security versions, preventing rolling back to NPW BIOS ACMs and NPW SINIT ACMs.

CPU Samples and IFWI Compatibility

The following table summarizes the different CPU samples and IFWI type compatibility.

Table 7. Oak Stream CPU Samples and IFWI Mapping

Sample Type	Debug IFWI	Non-Production Worthy (NPW) IFWI	Production-worthy IFWI
Engineering Samples (ES)	Fully supported	Not validated or supported Expected to boot if Intel® Boot Guard enforcement is disabled. Intel® TXT, Intel® Boot Guard, Intel® Software Guard Extensions (Intel® SGX), Intel® TDX, and Intel® In-field Scan features will not be available.	Not validated or supported Expected to boot if Intel® Boot Guard enforcement is disabled. Intel® TXT, Intel® Boot Guard, Intel® SGX, Intel® TDX, and Intel® In-field Scan features will not be available.
Qualification Samples (QS)	Not validated or supported Will not boot with a debug unified patch.	Fully supported Pre-production Intel® TXT, Intel® SGX, and Intel® TDX attestation will be available.	Expected to boot Pre-production Intel® TXT, Intel® SGX, and Intel® TDX attestation will be available. There will be limited boot, security, and attestation validation.
Revenue Samples (QSfD and PRQ)	Not validated or supported Will not boot with a debug or NPW unified patch.	Not validated or supported Will not boot with an NPW unified patch.	Fully supported All features available. Production Intel® TXT, Intel® SGX, and Intel® TDX attestation will be available.

1.4 WW24_26

New:

- [Diamond Rapids Processor Test Capability Heat Map Release Notification](#)
- [Oak Stream 16-Channel Platform Electrical Specification Release Notification](#)
- [WW24'26 BKC Release Notification](#)

Updated:

- [\(Updated WW24_26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.4.1 Documentation

1.4.1.1 Diamond Rapids Processor Test Capability Heat Map Release Notification

The *Diamond Rapids Processor Test Capability Heat Map*, document number [873155](#), Revision WW24 2026, is now available in the Resource and Documentation Center (RDC).

Overview of revision WW24 2026 changes are:

- Added new page for processor identification.
- Updated multiple issues with BKC-intercept fixes information.

Changes from previous revisions are highlighted in bold purple text throughout the document.

1.4.1.2 Oak Stream 16-Channel Platform Electrical Specification Release Notification

The *Oak Stream 16-Channel Platform Electrical Specification*, document number [787006](#), Revision 1.1, is now available in the Resource and Documentation Center.

Overview of Revision 1.1 changes are:

- Updated Platform Clocking Signals, PCI Express* (PCIe*) Signals, Processor Essential I/O Signals, Debug Signals, and I3C* Signals chapters.

Refer to revision history for complete list of changes.

1.4.1.3 (Updated WW24_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 8. Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
Diamond Rapids Processor Registers Specification	793272	1.15	1.2	Q3'26
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.15	1.2	Q3'26
Diamond Rapids Processor Test Capability (Heat Map)	873155	WW24'26	-	WW28'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	1.3	1.4	WW25'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i>	839071	0.9	1.0	Late Q2'26
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	1.0	1.1	Late Q2'26
Oak Stream 16-Channel Platform Electrical Specification	787006	1.1	1.2	Q3'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.2	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.2	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As needed	-
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	1.11	-	Q3'26
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.55	0.7	WW50'26
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW22'26	-	Late June'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	1.02	As needed	-
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	1.2	As needed	-

1.4.2 Software and Firmware

1.4.2.1 WW24'26 BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platform.

The BKC consists of an update package and associated report, published on a regular basis, for the Johnson City reference platform.

The BKC package includes UEFI firmware and onboard device drivers. These components have been screened through interoperability testing together over a period, with the aim of delivering customer confidence that not only have platform ingredients been individually tested, but also collectively tested to screen out interoperability issues.

Table 9. Johnson City WW24'26 BKC

Ingredient	Previous	Current
BKC	WW22	WW24
CRB BIOS	34.D68	35.D61
Unified Patch (MCU)	0x800009A0	0x800009A6
MMC FW	0.827.0	0.855.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	1.0.6384_v2	1.0.6422
Download	921123	921663
Status	Released	Released

NOTE

For information on silicon availability, contact your Intel representative.

Known Limitations

The following limitations are observed with WW24'26 BKC:

- (16030244649) On a few systems, Intel is observing regression failure where the OS installation via the BMC Virtual Media fails, and the system hangs in the BIOS menu.
 - Workaround: Disable network boot (EDK II -> Platform Configuration -> Network Configuration -> EFI Network -> Disable)
- Observed PL_rtte errors with SSC2 enabled on 2-Socket configurations (Internal Sighting 15019338702).**
 - Workaround: Enable Elastic Buffer Mode over Drift Buffer Mode in FBLP
 - Workaround: Reduce the Intel® UPI 3.0 link speed: EDKII → Socket Configuration → Uncore Configuration → Uncore General Configuration → Link Frequency Select = **16.0 GT/s** or **32.0 GT/s** or **40.0 GT/s** or **48.0 GT/s**.
 - Workaround: Disable SSC2 in the BIOS: EDKII → Platform Configuration → Miscellaneous Configuration → ISCLK Configuration → SSC2 Enable = **Disabled**.
- (15018973360) 2 socket configuration observed instability when SSC is enabled.
 - Status: Fix included in WW20'26 BKC.**
- (16029885327) Warm reset fails with DXE_ASSERT on the system having some vendor's NVMe*.
 - Status: Update the third-party vendor's NVMe firmware to the latest version.

- (16030716083) DXE ASSERT is observed during reboot after a seamless Out of Band (OOB) BIOS or microcode update.
- (14027896021) A BIOS assert is observed when the Pre-boot DMA Protection BIOS knob is enabled.
 - Workaround: None identified.

NOTE

Monitor the *Diamond Rapids Processor Test Capability Heat Map*, document number [873155](#), for updates on these exceptions.

1.5 WW23_26

New:

- [Diamond Rapids Processor Registers Specification Release Notification](#)
- [Oak Stream Early Boot Error Debug Documentation Release Notification](#)
- [Oak Stream Thermal Mechanical Collateral Release Notifications](#)
- [Oak Stream Diamond Rapids Change Request Cut-off Date for Beta Firmware](#)
- [WW22'26 BKC Release Notification](#)

1.5.1 Documentation

1.5.1.1 Diamond Rapids Processor Registers Specification Release Notification

The *Diamond Rapids Processor Registers Specification*, document number [793272](#), Revision 1.15, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.15 changes are:

- Updated OOB access for all registers and System Management Service Accessible Registers information.
- Added IP instance to IMH1 and IMH2 mapping to all chapters.
- Updated register descriptions for registers on FIO, S3M, CXLCM, CXPSMB, MC, PUNIT, and MSR chapters.
- Added new sections:
 - TPMI Configuration Space Registers (CSR_ALL, CSR_COMPUTE_CSR_PKG_ROOT)
 - Intel® Ultra Path Interconnect (Intel® UPI) Link Agent (ULA) Core Building Block (CBB) Memory Space Registers
 - Universal Chiplet Interconnect Express Die-to-Die Adapter (UCIEDDA) Core Building Block (CBB) Memory Space Registers
- Updated Error Logging Registers Decoding section.
- Updated CCF chapter to split CCF SANTA register into Multicast and Unicast sections and added additional access details.
- Added CSRS and SCMS port IDs to the IMH table.

- Updated all IP instance mappings.
- Added new table for CBB IP Port IDs.

NOTE

Some MSRs have “OOB-Pending” access, meaning they are set to be available for OOB access with a future BKC release.

CXL* represents Compute Express Link*.

Refer to the revision history for a complete list of changes.

1.5.1.2 Oak Stream Early Boot Error Debug Documentation Release Notification

The *Oak Stream Early Boot Error Debug: Server Platform Implementation Guidelines and Recipes*, document number [918592](#), Revision 1.0, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.0 changes are:

- Initial Release

This document describes the early boot error debug technology and its evolution through server platform generations starting with the Eagle Stream. The document also includes boot error debug feature integration recipes for the Oak Stream platform.

1.5.1.3 Oak Stream Thermal Mechanical Collateral Release Notifications

The *Oak Stream 16-Channel Platform Thermal Mechanical Specification*, document number [786521](#), Revision 1.11, is now available in RDC.

Overview of Revision 1.11 changes are:

- Chapter 7: Thermal Design Guidelines
 - Updated table DDR5 1DPC 1U Tall DIMM power estimate.
 - Added table DDR5 2DPC 1U Tall DIMM power estimate.
 - Updated table 2U Tall DIMM power estimate.
- Chapter 10: Thermal Solution Designs
 - Updated 2U RMHS performance chart based on measured data.
 - Updated Cold Plate performance chart based on measured data.

1.5.2 Enabling

1.5.2.1 Oak Stream Diamond Rapids Change Request Cut-off Date for Beta Firmware

As communicated in the *Oak Stream Platform Quarterly Technical Review (QTR) Q1'26*, document number [916754](#), the next change request intercept is **ES2/Beta**, with August 2026 as the change request cut-off target. Intel has further narrowed this target to (week ending on **August 7**).

Ensure that any change requests targeting ES2/Beta intercept are submitted before August 7. The changes requested must be received in advance to intercept the indicated milestone for implementation and validation. Required receipt time varies based upon the feature change requested; contact your PAE partner for guidance.

Table 10. Feature Request Guidance for Oak Stream Development

Change Area	ES1/Alpha	ES2/Beta	QS/PC	Prod/PV	Post-Launch
Silicon Design	Not Considered	Not Considered	Not Considered	Not Considered	Not Considered
Platform Design	Not Considered	Not Considered	Not Considered	Not Considered	Not Considered
SoC FW (Microcode)	Not Considered	Limited	Not Considered	Not Considered	Limited
Platform FW (IFWI, BMC, CPLDs)	Not Considered	Limited	Not Considered	Not Considered	Limited
OS Kernel / Driver	Not Considered	Limited	Not Considered	Not Considered	Limited
Memory Configs (DIMM Types, Capacity)	Not Considered	Limited	Not Considered	Not Considered	Limited
HW Configs (Example: Endpoint Support)	Not Considered	Limited	Not Considered	Not Considered	Limited

Limited indicates limited opportunities for approvals to intercept the milestone due to validation, quality, and schedule.

NOTE

This table is for guidance and does not guarantee change request acceptance.

1.5.3 Software and Firmware

1.5.3.1 WW22'26 BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platform.

The BKC consists of an update package and associated report, published on a regular basis, for the Johnson City reference platform.

The BKC package includes UEFI firmware and onboard device drivers. These components have been screened through interoperability testing together over a period, with the aim of delivering customer confidence that not only have platform ingredients been individually tested, but also collectively tested to screen out interoperability issues.

Table 11. Johnson City BKC

Ingredient	Previous	Current
BKC	WW20	WW22
CRB BIOS	34.D68	34.D68
<i>continued...</i>		

Ingredient	Previous	Current
Unified Patch (MCU)	0x800009A0	0x800009A0
MMC FW	0.827.0	0.827.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	1.0.6384_v2	1.0.6384_v2
Download	919465	921123
Status	Released	Released

NOTE

For information on silicon availability, contact your Intel representative.

Known limitations

The same known limitations from WW20'26 BKC are applicable for this release:

- On few systems, Intel is observing regression failure where OS installation via the BMC Virtual Media fails, and the system hangs in the BIOS menu (Internal Sighting ID: 16030244649).
 - WA: Disable network boot (EDK II -> Platform Configuration -> Network Configuration -> EFI Network -> Disable).
- On the 2S system, you may observe a PL_rtte error with SSC2 enabled, the issue seems to be related to clocking or PLL (Internal Sighting 15019338702).
 - WA: Disable SSC2 or Enabling Elastic Buffer Mode over Drift Buffer Mode in FBLP.
- Warm reset fails with DXE_ASSERT on the system having some vendor's NVMe* (Internal Sighting 16029885327).
 - Status: Update the third-party vendor's NVMe firmware to the latest version.
- On some configurations, an OS unaware cold reset may cause the system to stop at the boot device / GRUB menu instead of booting to the OS. The issue was root-caused to a change introduced in **HPM_PLD v9.7** (Internal sighting ID: 16030463949).
 - WA: Revert **HPM_PLD** to the previous known-good version (**v9.5**).
 - Status: Fix to be included in a future BKC release.

NOTE

Monitor the *Diamond Rapids Processor Test Capability Heat Map*, document number [873155](#), for updates on these exceptions.

1.6

WW22_26

New:

- [Diamond Rapids Processor Test Capability Heat Map Release Notification](#)
- [Oak Stream Platform Dashboard Release Notification](#)
- [Oak Stream Platform Clocking Requirement Clarification](#)

Updated:

- (Updated WW22_26) Cadence* Version and Tool Usage Feedback Request
- (Updated WW22_26) Oak Stream Key Collaterals Availability and Forecast

1.6.1 Customer Feedback

1.6.1.1 (Updated WW22_26) Cadence* Version and Tool Usage Feedback Request

NOTE

Updated information is highlighted in bold text.

Intel is requesting feedback from customers on their current Cadence* Allegro* versions, schematic tools, PCB design tools, and migration plans, to align future Design Archive (DA) deliverables with customer environments.

Starting with Oak Stream 8-Channel platform and Coral Rapids, all Design Archive (DA) schematic files will be migrated to System Capture* (replacing DEHDL*).

Help answer the following questions:

Are you currently using Cadence Allegro version 23.1 or older versions?

- Yes
- No

Which Cadence Allegro version are you currently using?

- **22.1** or earlier
- 23.1
- 24.1
- Other (specify)

Which schematic design tool are you currently using?

- Concept* HDL (DEHDL*)
- System Capture*
- Other (specify)

What PCB design tool are you currently using?

- Cadence Allegro products (Enterprise PCB Design, OrCAD*)
- Other (Mentor*, Zuken*, ODB++*)

Do you have plans to migrate to newer Cadence versions (24.1, 25.1, or later)?

- Yes
- No
- Under evaluation

Do you plan to migrate to System Capture for schematic design?

- Yes
- No
- Under evaluation

Do you require conversion or downgrade support for your design files?

- Yes
- No
- Not sure

Provide your answers to your Intel representative by June 5, 2026.

NOTE

Supplier contact information is provided for convenience. Intel does not formally endorse any products or suppliers.

1.6.2 Documentation

1.6.2.1 Diamond Rapids Processor Test Capability Heat Map Release Notification

The *Diamond Rapids Processor Test Capability Heat Map*, document number [873155](#), Revision WW21 2026, is now available in the Resource and Documentation Center (RDC).

Overview of WW21 2026 changes are:

- Split UCC A0 readiness into 1-Socket (1S) and 2-Socket (2S) columns for better traceability of issues impacting 2S topologies.
- Broke down tables into Sub-Domain and Features.
- Updated seamless domain readiness.

Changes from previous revisions are highlighted in bold purple text throughout the document.

1.6.2.2 Oak Stream Platform Dashboard Release Notification

The *Oak Stream Platform Diamond Rapids 12/16-Channel Dashboard*, document number [863599](#), Revision WW22 2026, is now available in the Resource and Documentation Center (RDC).

1.6.2.3 (Updated WW22_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 12. Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.1	1.15	WW23'26
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.15	1.2	Q3'26
<i>Diamond Rapids Processor Test Capability (Heat Map)</i>	873155	WW 21 '26	WW2 4 '26	WW2 4 '26
<i>Diamond Rapids Processor Sightings Report</i>	862869	1.3	-	WW23'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i>	839071	0.9	1.0	Late Q2'26
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	1.0	1.1	Late Q2'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	1.0	1.1	Late Q2'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.2	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.2	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing</i>	827452	1.0	As needed	-

continued...

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
Development Platform 1 (MDP1) Design Archive				
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	1.0	As needed	-
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	1.0	1.1	Q2'26
Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive	821026	0.55	0.7	WW50'26
Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard	863599	WW 22 '26	-	Late June'26
Oak Stream Platforms System Boards Design Archive	821025	1.02	As needed	-
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	1.2	As needed	-

1.6.3 Enabling

1.6.3.1 Oak Stream Platform Clocking Requirement Clarification

Problem Statement

The *Oak Stream 16-Channel Platform Design Guide*, document number 788196, Revision 1.0, restricts the clock source and PCI Express* (PCIe*) lane assignment to originate from the same IMH die. While this implementation works for certain configurations, it limits routing flexibility, increases board design complexity.

Resolution

The restriction has been updated to allow PCIe lanes and reference clock to originate from different iMH dies within the same socket. This update improves routing flexibility and simplifies customer implementations. The updated behavior is defined as follows:

- The Diamond Rapids CPU provides two 100 MHz reference clocks, one from each iMH die. A PCIe/Compute Express Link* (CXL*) device connected to any PCIe/CXL port of a Diamond Rapids CPU can utilize either reference clock on the CPU when running in the supported clocking modes described as follows:
 - Common Clock mode with SSC ON and OFF
 - Separate Reference Clock with No Spread (SRNS)
 - Separate Reference Clock with Independent Spread (SRIS) mode

- The 100 MHz reference clocks must only be used for PCIe lanes that originate from the same socket. Using a 100 MHz reference clock from one socket to drive a device connected to data lanes from a different socket is not supported.

While the restriction requiring the clock source and PCIe lanes to originate from the same iMH die is being removed, the socket boundary restriction remains in place.

Cross-socket clocking configurations are still not supported and must continue to be avoided in customer designs.

Status

This update will be reflected in the next revision of the *Oak Stream 16-Channel Platform Design Guide*. Monitor the *Oak Stream Platform Message of the Week (MoW)* for documentation updates.

1.7 WW21_26

New:

- [Cadence* Version and Tool Usage Feedback Request](#)
- [Diamond Rapids I3C Enabling User Guide Release Notification](#)
- [Uncorrectable Errors Observed When Directed Refresh Management is Enabled](#)
- [Uncorrectable/Correctable Errors Observed When Directed Refresh Management is Enabled](#)
- [Post-Launch Firmware Update to Diamond Rapids for Post Quantum Cryptography](#)
- [WW20 BKC Release Notification](#)
- [CScripts for Oak Stream Platforms Release Notification](#)

1.7.1 Customer Feedback

1.7.1.1 Cadence* Version and Tool Usage Feedback Request

Intel is requesting feedback from customers on their current Cadence* Allegro* versions, schematic tools, PCB design tools, and migration plans, to align future Design Archive (DA) deliverables with customer environments. Help answer the following questions:

Are you currently using Cadence Allegro version 23.1 or older versions?

- Yes
- No

Which Cadence Allegro version are you currently using?

- 2.1 or earlier
- 23.1
- 24.1
- Other (specify)

Which schematic design tool are you currently using?

- Concept* HDL (DE-HDL*)

- System Capture*
- Other (specify)

What PCB design tool are you currently using?

- Cadence Allegro products (Enterprise PCB Design, OrCAD*)
- Other (Mentor*, Zuken*, ODB++*)

Do you have plans to migrate to newer Cadence versions (24.1, 25.1, or later)?

- Yes
- No
- Under evaluation

Do you plan to migrate to System Capture for schematic design?

- Yes
- No
- Under evaluation

Do you require conversion or downgrade support for your design files?

- Yes (specify downgrade version or required SW conversion)
- No
- Not sure

Provide your answers to your Intel representative by June 5, 2026.

NOTE

Supplier contact information is provided for convenience. Intel does not formally endorse any products or suppliers.

1.7.2 Documentation

1.7.2.1 Diamond Rapids I3C Enabling User Guide Release Notification

The *Diamond Rapids I3C* Enabling User Guide*, document number [919646](#), Revision 1.0, is now available in the Resource and Documentation Center (RDC).

Overview of the 1.0 change is:

- Initial release. This document covers instructions on how to set up I3C_DBG for close chassis debug using Closed Chassis Adapter (CCA) and I3C* Type-C interface.

1.7.3 Enabling

1.7.3.1 Uncorrectable Errors Observed When Directed Refresh Management is Enabled

Problem Statement

Intel has observed uncorrectable errors when Directed Refresh Management (DRFM) is enabled, localized to early Micron* H-die DIMMs that support DRFM. The WW20'26 BKC enables the DRFM feature.

Implication

Uncorrectable errors may be observed during boot or functional operation.

Root Cause

None identified.

Mitigation

Configure *DfxRowHammer ForceDisable = Enabled*.

- Disabling DRFM requires a BIOS image that has DFX options available. If using the pre-compiled IFWI from the BKC, use the "IPCleanDFXEnable" IFWI binaries.
- In setup: *Socket Configuration > Memory Configuration > Memory Dfx Configuration > Force RowHammer Disable = Enabled*

A BIOS source code workaround can also be implemented as follows:

- File:
 - ServerPlatformPkg/Library/PeiMemoryPolicyUpdateLib/PeiMemoryPolicyUpdateLib.c
- Change:
 - Setup->mem.RowHammerMode = SetupData->SocketConfig.MemoryConfig.RowHammerMode;
- To:
 - Setup->mem.RowHammerMode = ROWHAMMER_MODE_DISABLE;

Status

This issue is currently being investigated by Intel and Micron* to determine date codes for the affected DIMMs. Contact Micron* for additional details.

NOTE

Supplier contact information is provided for convenience. Intel does not formally endorse any products or suppliers.

1.7.3.2 Uncorrectable/Correctable Errors Observed When Directed Refresh Management is Enabled

Problem Statement

Intel has observed uncorrectable/correctable errors when Directed Refresh Management (DRFM) is enabled, and Samsung* DIMMs that support DRFM are populated. The WW20'26 BKC enables the DRFM feature.

Implication

Uncorrectable/correctable errors may be observed during boot or functional operation.

Root Cause

None identified.

Mitigation

Configure *DfxRowHammer ForceDisable* = *Enabled*.

- Disabling DRFM requires a BIOS image that has DFX options available. If using the pre-compiled IFWI from the BKC, use the "IPCleanDFXEnable" IFWI binaries.
- In Setup: *Socket Configuration > Memory Configuration > Memory Dfx Configuration > Force RowHammer Disable* = *Enabled*

A BIOS source code workaround can also be implemented as follows:

- File:
 - *ServerPlatformPkg/Library/PeiMemoryPolicyUpdateLib/PeiMemoryPolicyUpdateLib.c*
- Change:
 - *Setup->mem.RowHammerMode* = *SetupData->SocketConfig.MemoryConfig.RowHammerMode;*
- To:
 - *Setup->mem.RowHammerMode* = *ROWHAMMER_MODE_DISABLE;*

Status

This issue is currently being investigated by Intel and Samsung*.

NOTE

Supplier contact information is provided for convenience. Intel does not formally endorse any products or suppliers.

1.7.4 Software and Firmware

1.7.4.1 Post-Launch Firmware Update to Diamond Rapids for Post Quantum Cryptography

Quantum computing is an emerging computing paradigm that accelerates certain classes of algorithms using atomic scale quantum superposition, interference, and entanglement. Quantum computing is expected to eventually compromise current conventional cryptography.

Post Quantum Cryptography (PQC) is the set of encryption, signing, and hashing algorithms that are resistant to cryptographic attack by quantum computers.

Commercial National Security Algorithms 2.0 (CNSA 2.0) are the U.S. Government approved suite of standard PQC algorithms for encryption, signing, and hashing.

Table 13. CNSA 2.0 Algorithms

Usage	Conventional Encryption	Post Quantum Encryption (CNSA 2.0)
Firmware and software signing	RSA, ECDSA	XMSS, LMS, ML-DSA
Symmetric encryption and hashing	AES-128, SHA-256	AES-256, SHA-384
Asymmetric encryption (public/private keys)	RSA, ECDSA, Diffie-Hellman	ML-KEM, ML-DSA

Diamond Rapids was designed with several PQC features enabled at launch:

- SoC firmware signed with hybrid RSA3K + XMSS
- PQC authenticated boot with Intel® Boot Guard (single key LMS available now, ML-DSA available at Beta milestone)
- PQC memory encryption for Intel® Total Memory Encryption (Intel® TME), Intel® Trust Domain Extensions (Intel® TDX), and Intel® Software Guard Extensions (Intel® SGX) (AES-XTS-256)

In late 2024, the U.S. Government accelerated timelines for conversion to PQC in National Security Systems (NSS) from 2027 to 2030.

- Other governments and segments around the world also plan PQC conversions before 2035, but trending later than the U.S. Government requirements.

Meeting the Market Need for Accelerated Post-Quantum Cryptography on Diamond Rapids

To address the accelerated U.S. Government timetable, Intel will issue **a post-launch firmware update to enable PQC** on the following Diamond Rapids (UCC-M) features:

- Intel® TDX
- Intel® Trust Domain Extensions Connect (Intel® TDX Connect)
- DICE-based attestation for Intel® TDX and Intel® TDX Connect
- Hosting and attesting to Intel® TDX hosted in Compute Express Link* (CXL*) attached memory
- Vendor Authorized Debug
- CPU attestation with SPDM 1.4

PQC Firmware Update Timing and Flexibility

The firmware update will be delivered to system manufacturers approximately 4-6 months after Diamond Rapids-12/16 Channel (UCC-M) PRQ.

- 1-2 socket will be validated first.
- 4-socket validation expected to be complete approximately 1Q later.
- System manufacturers will integrate and distribute the update to their end customers.

U.S. Government officials have communicated that firmware upgradability to PQC for the functions described will make Diamond Rapids eligible for NSS considerations:

- “Cryptographic agility is necessary to accomplish this transition in a timely manner; even equipment purchased before support is mandated should have sufficient memory and processing power, when possible, to run new algorithms, as well as capacity for future algorithms and protocols so that **any future enhancements can be included via a software update.**” https://media.defense.gov/2022/Sep/07/2003071836/-1/-1/0/CSI_CNSA_2.0_FAQ_.PDF

After installing the firmware update, the system manufacturer or system owner can choose PQC algorithms or continue with the conventional encryption that was enabled at launch.

- Several features automatically negotiate appropriate cryptography based on service requests from software or other devices.
- Enabling PQC, Intel TDX with DICE-based attestation will require the owner to opt-in via a BIOS switch.
- Opting into PQC, Intel TDX will disable all Intel SGX based services on the platform due to a PQC coverage gap when both technologies are active.
- Owners can change the PQC Intel TDX BIOS switch setting as desired, but a reboot is required to activate the new state. Note that sealed data from previous Intel SGX or Intel TDX sessions will need to be migrated to prevent loss of data.

PQC Policy for Existing Intel® Xeon® Platforms

For Intel® Xeon® platforms currently shipping, if a feature originally shipped using PQC, future updates to that feature will continue to support PQC.

For Intel® Xeon® products prior to Diamond Rapids, Intel will not issue updates that convert features originally shipped with conventional cryptography to PQC.

Diamond Rapids is the only Intel® Xeon® platform planned to receive post-launch updates that enable new PQC features.

Intel expects that all Intel® Xeon® platforms launching in 2029 and later will be fully PQC. Intel recommends that customers plan their transition to PQC technologies using Intel® Xeon® 6 platforms and their successors.

1.7.4.2 WW20 BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platform.

The BKC consists of an update package and associated report, published on a regular basis, for the Johnson City reference platform.

The BKC package includes UEFI firmware and onboard device drivers. These components have been screened through interoperability testing together over a period, with the aim of delivering customer confidence that not only have platform ingredients been individually tested, but also collectively tested to screen out interoperability issues.

Table 14. Table Johnson City BKC

Ingredient	Previous	Current
BKC	WW18	WW20
CRB BIOS	34.D25	34.D68
Unified Patch (MCU)	0x8000099D	0x800009A0
MMC FW	0.794.0	0.827.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	1.0.6384_v2	1.0.6384_v2
Download	918556	919465
Status	Released	Released

NOTE

For information on silicon availability, contact your Intel representative.

2S Alpha Support Declaration

Intel is pleased to announce that Diamond Rapids 2-Socket (2S) Alpha-level support is now available with WW20'26 Best Known Configuration. 2-Socket support has the following exceptions:

- Observed failures while doing OS installation via BMC Virtual Media; system hangs in the BIOS menu (Internal Sighting 16030244649).
 - Workaround: Disable network boot (EDK II -> Platform Configuration -> Network Configuration -> EFI Network -> Disable).
- Observed PL_rtte errors with SSC2 enabled on 2-Socket configurations (Internal Sighting 15019338702).
 - Workaround: Enable Elastic Buffer Mode over Drift Buffer Mode in FBLP
 - Workaround: Reduce the Intel® UPI 3.0 link speed: EDKII → Socket Configuration → Uncore Configuration → Uncore General Configuration → Link Frequency Select = **16.0 GT/s** or **32.0 GT/s** or **40.0 GT/s** or **48.0 GT/s**.
 - Workaround: Disable SSC2 in the BIOS: EDKII → Platform Configuration → Miscellaneous Configuration → ISCLK Configuration → SSC2 Enable = **Disabled**.
- (15018973360) 2 socket configuration observed instability when SSC is enabled.
 - **Status: Fix included in WW20'26 BKC.**
- Warm reset fails with DXE_ASSERT on system with specific vendor NVMe* drives (Internal Sighting 16029885327).
 - Status: Update to vendor's latest NVMe firmware.
- OS-unaware cold reset may cause the system to stop at the boot device / GRUB menu instead of booting to the OS. This issue was root-caused to a change introduced in **HPM_PLD v9.7** (Internal Sighting 16030463949).
 - Workaround: Revert **HPM_PLD v9.5** (included in the *Oak Stream 16-Channel Platform CPLD Reference Code*, document number 860022).
 - Status: Fix will be included in an upcoming BKC release.

NOTE

Monitor the *Diamond Rapids Processor Test Capability Heat Map*, document number [873155](#), for updates on these exceptions.

1.7.5 Tools and Debug

1.7.5.1 CScripts for Oak Stream Platforms Release Notification

The *CScripts for Oak Stream Platforms*, document number [821387](#), Revision 2621, is now available in the Resource and Documentation Center (RDC).

Overview of 2621 revision changes are:

- **PCIe:** LTSSM logger integrated into link training flow, stricter bitrate checks in reset_port, and DLLLA status reporting per port.
- **Error Injection:** inject_three_strike now supports SMM exit for sequential injections and accepts an imh parameter. Automatic signaling verification added for mcerr/ierr/three_strike. Patrol UCE/CE fully supported in Intel® Simics® tool. AMEI supports OS-First and Firmware-First modes.
- **MSR:** Dump list expanded covering all Diamond Rapids registers. RLM support added for MSR input IP parameters.
- **API/Utilities:** funcs() now shows function descriptions. Module discoveries auto-reset on IPC restart to prevent register access loss.
- **S3M:** Early Boot Error Decode adds error type/severity verification and recommended actions.
- **Topology:** Bridge device abstraction, enhanced PCI Express* (PCIe*)/Compute Express Linke* (CXL*) discovery, per-BDF metadata, CXL address translation engine, and hierarchical port topology support.

These improvements enhance usability, reliability, and coverage of existing functions. The review expanded MSR support and error injection updates for your test workflows.

1.8 WW20_26

New:

- [SPDM GET_MEASUREMENT and CHUNK Feedback Request](#)
- [Intel® Boot Guard and Intel® TXT Documents Release Notification](#)
- [Oak Stream 16-Channel Platform Mechanical Hardware Availability Update](#)
- [Socket DX Update](#)

Updated:

- [\(Updated WW20_26\) Oak Stream Key Tools Availability and Forecast](#)

1.8.1 Customer Feedback

1.8.1.1 SPDM GET_MEASUREMENT and CHUNK Feedback Request

The number of SPDM indexes increased from 8 to 26 when compared to previous generations. This will result in a large SPDM message size. If all measurements are collected at once (bulk retrieval) and the number of indexes is large, this may result in the following error:

- SPDM Request: GET_MEASUREMENTS (0xFF) 26 indexes
- SPDM Response: ResponseTooLarge (0xD) error

Intel measurements can be requested for each index instead of bulk retrieval. SPDM supports CHUNK mode. This mode allows large messages to be broken into multiple frames.

Intel is requesting feedback from customers on:

- Use of the GET_MEASUREMENT (0xFF) command to get bulk measurements.
- Use of the SPDM CHUNK mode.

Provide your feedback to your Intel representative by May 22, 2026.

1.8.2 Documentation

1.8.2.1 Intel® Boot Guard and Intel® TXT Documents Release Notification

The *Bootable CPU Converged Intel® Boot Guard and Intel® TXT BIOS Specification*, document number [838171](#), Revision 1.1.2, is now available in the Resource and Documentation Center (RDC).

Overview:

- This document is for Birch Stream and Oak Stream platforms.
- It replaces the server usage of the *Intel® Converged Boot Guard and Intel® Trusted Execution Technology (Intel® TXT) BIOS Specification*, document number [575623](#).

The *Oak Stream Platform Converged Intel® Boot Guard and Intel® TXT Integration and Validation Guide*, document number [915484](#), Revision 1.0, is now available in the RDC.

Overview:

- This document provides guidance on tools, integration, debug, and testing required to enable and validate Converged Intel® Boot Guard and Intel® TXT on an Oak Stream platform.

1.8.3 Enabling

1.8.3.1 Oak Stream 16-Channel Platform Mechanical Hardware Availability Update

- Foxconn Interconnect Technology* (FIT*) backplate and bolster plate hard-tooled prototypes are now approved for ordering by Intel.

- This update will be reflected in the next release of the *Oak Stream 16-Channel Platform Thermal Mechanical Specification*, document number [786521](#).
- Additional details on hard-tooled prototype status and use considerations are in the Oak Stream 16-Channel Platform Hard-Tooled Prototype Availability Update article in the [WW17'26 Oak Stream Platform Message of the Week](#).

Intel is continuing to work with FIT on their socket cover hard-tooled prototypes. Similar approval of this part is targeted for June 2026.

As discussed in the [WW12'2026 Oak Stream Platform Message of the Week](#), there is no mixing of hard- and soft-tooled socket covers and bolster plates.

1.8.3.2 Socket DX Update

Currently available DX sockets are preliminary ones at the DG 0.6 level. These may continue to be used for early development work.

- "DG" refers to sockets based on Intel's socket design guide. This socket design guide is a detailed socket requirements document and is for the use between Intel and socket suppliers only.

Production candidate sockets are targeted for Intel approval as **DG 0.7** in **July 2026**.

- Criteria for DG 0.7 status includes time 0 shock, vibration, and durability testing.
- DG 0.7 sockets may have design refinements compared to DG 0.6 and will have improved supplier manufacturing capability and yields.
- Intel qualification for DG 0.7 is done with hard-tooled loading and heatsink hardware.
- DG 0.7 sockets are the first versions recommended for beginning of life customer validation work.

Accelerated environmental stress testing of DG 0.7 candidate sockets is in progress and scheduled for completion in September 2026.

- DG 0.7 sockets are not recommended to be used for environmental stress testing until they are approved at this milestone. Any environmental qualification testing is done at your own risk as the parts are subject to change until they pass all of Intel's validation testing.

Full validation as production approved DG 1.0 is targeted for completion in Q4 2026.

Although Intel's intent is for all sockets and related hardware to align with this schedule, hardware status may vary supplier to supplier. Updates will be provided in the [WW24'26 Oak Stream Platform Message of the Week](#).

1.8.4 Tools and Debug

1.8.4.1 (Updated WW20_26) Oak Stream Key Tools Availability and Forecast

NOTE

Updated information is highlighted in bold text.

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI-60 to Intel ITP adapter.
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support. Two Streams Tracing supported:
 - LA-3907 pre-processor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports four trace ports)
 - LA-3892 converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI-60
- ASSET* Arium* - Contact ASSET Arium for support.
- Compact Debug Port (CDP) + MIPI-10 adapter for I3C*
- Type-C connection: Voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- VTune™ tools
- Intel® SoC Watch
- Intel® System Debugger
- Includes token generation for Vendor Authorize Boot (VAB) / Vendor Authorize Debug (VAD).

NOTE

The Intel® System Bring-up Toolkit NDA requires registration. See *How to Obtain Intel® System Bring-up Toolkit NDA (Formerly Intel® System Studio NDA) User Guide*, document number [616675](#), and *Intel® System Bring-up Toolkit NDA (formerly Intel® System Studio NDA) Registration Form New Users*, document number [631579](#), for details.

Table 15. Oak Stream Key Tools

Type	Tool	RDC or DIT	Availability	Revision
Debug	ASSET Intertech* Arium* Note: Contact Asset* for support.	-	Q2'26	-
Debug	ASSET* Close Chassis Controller (CCC) Note: Contact Asset* for support.	-	Q2'26	-
Debug	CScripts for Oak Stream Platforms Note: Refer to Release Notes for details.	821387	Available	2615.2000
Debug	Intel Remote FPGA Access Tool (RFAT) for Server Platforms	871932	Available	1.0
Debug	Intel Triage Tool (Crashdump Summarizer) Note: Available through CScripts	N/A	Available	0.88
continued...				

Type	Tool	RDC or DIT	Availability	Revision
Debug	Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C*	99CJRJ	Available	N/A
Debug	Intel® I3C* USB Type-C Level Shift Adapter	99CN2C	Available	N/A
Debug	Intel® In-Target Probe (Intel® ITP) eXtended Debug Port (XDP)	940990	Available	N/A
Debug	Intel® Silicon View Technology (Intel® SVT) Closed Chassis Adapter	921521	Available	N/A
Debug	Intel® System Bring-up Toolkit NDA Note: Includes Intel® System Debugger NDA, Intel® VTune™ Profiler NDA and Intel® SoC Watch NDA.	634166	Available	Latest
Debug	Lauterbach*	-	Available	N/A
Debug	SelfTest 7 - Windows* x64	524152	Available	17 9
Debug	Server Security Toolkit Note: Server Security Toolkit includes Intel® Total Memory Encryption – Multi-Key (Intel® TME-MK), Intel® Trusted Execution Technology (Intel® TXT), BIOS TPM Provisioning with Intel® Platform Trust Technology (Intel® PTT) tools, and CBnT (Converged Intel® Boot Guard and Intel® TXT).	630398	Available	1.54
Manufacturing and Fleet Health	Autonomous Crashdump (ACD) Note: Autonomous Crash Dump (ACD) is available via BMC source code under license agreement.	N/A	Available	0.75
Manufacturing and Fleet Health	Intel® At-Scale Debug (Intel® ASD)	N/A	Available	N/A
Manufacturing and Fleet Health	Intel® At-Scale Debug (Intel® ASD) BMC-JTAG* Software Guide	594948	Available	1.6.5
Manufacturing and Fleet Health	Intel® In-field Scan	-	Q2'27	5.x.x
Manufacturing and Fleet Health	Intel® System Stress and Memory Diagnostics (SSMD) for Linux*	818881	Available	2026.05.04.fc55
continued...				

Type	Tool	RDC or DIT	Availability	Revision
Manufacturing and Fleet Health	Memory Latency Checker (MLC)	736633	Q2'26	3.12
Signal Integrity	Clock Jitter Tool (CJT)	803838	Available	6.0.4
Signal Integrity	Integrated Channel Analysis Tool (ICAT) Suite for Windows*	782966	Available	6.200.50
Signal Integrity	Intel® Passive Component Checker Tool (Intel® PCC Tool)	600218	Available	1.8
Signal Integrity	Intel® Platform Clock Jitter Analysis Tool (CJAT) for Windows	619648	Available	6.0.4
Signal Integrity	PCIe* and Intel® UPI Load Board Solution Set for 16-Channel Platform	DMRUCCPLBULB	Available	N/A
Signal Integrity	PCIe* Load Board Solution Set for 16-Channel Platform	DMRUCCPLB	Available	N/A
Thermal / Power Delivery	Almondsville Thermal Mechanical Test Vehicle (TMTV) Application Note	852568	Available	1.0
Thermal / Power Delivery	Almondsville Thermal Test Vehicle (TTV) for 16-Channel Platform	-	Available	-
Thermal / Power Delivery	Blue Interposer for Gen5 VRTT for 16-Channel Platform	99D5W2	Available	N/A
Thermal / Power Delivery	Gen5 VR Test Tool (VRTT) for Windows	99CJ9X	Available	N/A
Thermal / Power Delivery	Red Interposer for Gen5 VRTT for 16-Channel Platform	99D5W1	Available	N/A
Thermal / Power Delivery	VRTT Mechanical Interposer Retention Mechanism Kit for 16-Channel Platform	99CN1R	Available	N/A
Validation	Diamond Rapids CPU Crash Log Content Decode Definition	917977	Available	1.0
Validation	Diamond Rapids CPU Crash Log Quick Start Guide	917972	Available	1.0
Validation	Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card	-	Available	-
Validation	Intel® Simics Tools Note: Intel SIMICs tools available under license agreement.	-	Available	N/A

continued...

Type	Tool	RDC or DIT	Availability	Revision
	Contact your Intel representative for details.			
Validation	Intel® Structural Test Plus (Intel® STP) Note: Contact a local representative to enable Intel® STP for customer designs.	-	Available	N/A
Validation	MEI DDR5 Lite Card 8000 MT/s for Windows	99D5X2	Available	N/A
Validation	Memory Assistant for Intel® Xeon® Platforms	620497	Available	WW17
Validation	Memory Error Injection (MEI) DDR5 Card for Windows	99CDV1	Available	N/A
Validation	Oak Stream Platform DDR Electrical Validation Tools and Methodology	914412	Available	1.0
Validation	Oak Stream Platform Intel® IO Margin Tool for Windows* User Guide	913382	Available	1.0
Validation	Oak Stream Platform Intel® IO Margin Tool for Windows	871016	Available	2.0.0
Validation	Oak Stream Platform Margin Testing	871015	Available	1.0
Validation	Oak Stream Stitched Rank Margining Tool (RMT) Note: Included in the BKC.	-	Available	-
Validation	Oak Stream Target Loader Rank Margining Tool (RMT)	913381	Available	1.0
Validation	PCI Express* (PCIe*) Lane Margin Tool (LMT) for PCIe* 4.0/5.0/6.0	639433	Available	1.0
Validation	PCIe* 5.0 Error Injection (PEI) Card	99C3FA	Available	N/A
Validation	PCIe* PEI 6.0 Card	-	Q4'26	-
Validation	PEI (PCIe Error Injector Card) Collaterals	610667	Available	5.1
Validation	PEI 5.0 (PCIe 5.0 Error Injection) + CXL 2.x Card	99CN2D	Available	N/A
Validation	RAS Automation ToolKit	633201	TBD	2.5.0

1.9 WW19_26

New:

- [Diamond Rapids CPU Crash Log Quick Start Guide Release Notification](#)
- [Diamond Rapids Processor Sightings Report Release Notification](#)
- [Data Center Design Conference \(DCDC\) 2026 Asia Face-to-Face Events](#)
- [WW18 BKC Release Notification](#)

Updated:

- [\(Updated WW19_26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.9.1 Documentation

1.9.1.1 Diamond Rapids CPU Crash Log Quick Start Guide Release Notification

The *Diamond Rapids CPU Crash Log Quick Start Guide*, document number [917972](#), Revision 1.0, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.0 changes are:

- Initial release. This document contains the Oak Stream platform crash log feature integration recipes for critical use cases including three-strike timeout errors, S3M errors, MCERRs, and core triple fault scenarios with BIOS-based content offload methods.

Future revisions will cover more IERR scenarios originating in the PCU and memory sub-systems, multi-socket systems, BMC Out of Band (OOB) content offload methods and back-to-back trigger scenarios.

Refer to the *Diamond Rapids Processor Sightings Report*, document number [862869](#), for known limitations with Crash Log (SR104).

The *Diamond Rapids CPU Crash Log Content Decode Definition*, document number [917977](#), Revision 1.0, is also published alongside the CPU Quick Start Guide. This document is a ZIP file that contains CSVs representing the list of registers within each of the Diamond Rapids CPU Crash Log records.

1.9.1.2 Diamond Rapids Processor Sightings Report Release Notification

The *Diamond Rapids Processor Sightings Report*, document number [862869](#), Revision 1.3, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.3 changes are:

- Added sightings 87 to 105.
- Updated sightings 44, 52, 60, 68, 74, and 75.

Refer to the revision history for a complete list of changes.

1.9.1.3 (Updated WW19_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in them [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 16. Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.1	1.15	Mid Q2'26
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.15	1.2	Q3'26
<i>Diamond Rapids Processor Test Capability (Heat Map)</i>	873155	WW17'26	WW22'26	WW22'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	1.3	-	WW23'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i>	839071	0.9	1.0	Late Q2'26
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	1.0	1.1	Late Q2'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	1.0	1.1	Late Q2'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.2	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.2	As needed	-
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
Oak Stream 16-Channel Platform Pin List	636260	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	1.0	As needed	-
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	1.0	1.1	Q2'26
Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive	821026	0.55	0.7	WW50'26
Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard	863599	WW18'26	-	May'26
Oak Stream Platforms System Boards Design Archive	821025	1.02	As needed	-
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	1.2	As needed	-

1.9.2 Enabling

1.9.2.1 Data Center Design Conference (DCDC) 2026 Asia Face-to-Face Events

Intel is excited to welcome you to Intel's 2026 Data Center Design Conference (DCDC) Asia Events. Under the theme *Resilient Ecosystems Accelerating AI*, this year conferences highlights Intel® Xeon® processors and AI solutions, optimized compute for diverse workloads and segments, and ecosystem innovation at scale.

Table 17. DCDC 2026 Asia Information

Geography	Taiwan	India	Mainland China
Event Format	Face to Face	Face to Face	Face to Face
Event Date/Schedule	June 10, 2026	June 18, 2026	June 23, 2026
Language	Mandarin	English	Mandarin
Registration Period	Now to May 27, 2026	May 11 to June 10, 2026	May 19 to June 12, 2026
On-demand Videos	No recording at the event	No recording at the event	No recording at the event

NOTE

Seats for the face-to-face event are limited.

Contact your Intel representative for more information on the events agenda and registration details. Monitor the Message of the Week for updates on DCDC.

1.9.3 Software and Firmware

1.9.3.1 WW18 BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platform.

The BKC consists of an update package and associated report, published on a regular basis, for the Johnson City reference platform.

The BKC package includes UEFI firmware and onboard device drivers. These components have been screened through interoperability testing together over a period, with the aim of delivering customer confidence that not only have platform ingredients been individually tested, but also collectively tested to screen out interoperability issues.

Table 18. Johnson City BKC

Ingredient	Previous	Current
BKC	WW16	WW18
CRB BIOS	33.D87	34.D25
Unified Patch (MCU)	0x8000099B	0x8000099D
MMC FW	0.794.0	0.794.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	1.0.6384_v2	1.0.6384_v2
Download	917475	918556
Status	Released	Released

NOTE

For information on silicon availability, contact your Intel representative.

Known Limitations

Issue: On a few systems, Intel is observing regression failure where the OS installation via the BMC Virtual Media fails, and the system hangs in the BIOS menu (internal sighting ID 16030244649).

Workaround: Disable the EFI Network:

- EDKII -> Platform Configuration -> Network Configuration -> EFI Network = **Disabled**

Issue: On some 2S configurations with the SSC2 enabled (default), customers may observe Intel® Ultra Path Interconnect (Intel® UPI) 3.0 correctable errors (for example, pl_rtte / pl_error). In some configurations, system hang behavior has also been observed (internal sighting ID 15018973360).

Workaround: Customers may use either of the following BIOS mitigations to improve stability:

- Disable SSC2 in the BIOS:
 - EDKII → Platform Configuration → Miscellaneous Configuration → ISCLK Configuration → SSC2 Enable = **Disabled**
- Reduce the Intel® UPI 3.0 link speed:
 - EDKII → Socket Configuration → Uncore Configuration → Uncore General Configuration → Link Frequency Select = **16.0 GT/s** or **32.0 GT/s** or **40.0 GT/s** or **48.0 GT/s**

Status: Intel is actively investigating the issue. Until a permanent fix is validated and released, applying one of the above BIOS mitigations is recommended to improve platform stability.

Issue: Warm reset fails with DXE_ASSERT on the system having Misc Vendor NVMe* (internal sighting 16029885327).

Status: Update the NVMe firmware to the latest version.

1.10 WW18-26

New:

- [Oak Stream Platform Dashboard Release Notification](#)
- [PCIe 6 RxLEQ Phase3 Tx EQ Compliance Test Clarification](#)

Updated:

- [\(Updated WW17_26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.10.1 Documentation

1.10.1.1 Oak Stream Platform Dashboard Release Notification

The *Oak Stream Platform Diamond Rapids 12/16-Channel Dashboard*, document number [863599](#), Revision WW18 2026, is now available in the Resource and Documentation Center (RDC).

1.10.1.2 (Updated WW18_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in them [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 19. Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.1	1.15	Mid Q2'26
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.15	1.2	Q3'26
<i>Diamond Rapids Processor Test Capability (Heat Map)</i>	873155	WW17'26	WW21'26	WW21'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	1.2	-	WW19'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i>	839071	0.9	1.0	Late Q2'26
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	1.0	1.1	Late Q2'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	1.0	1.1	Late Q2'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.2	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.2	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	-
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	1.0	1.1	Q2'26
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.55	0.7	WW50'26
Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard	863599	WW18'26	-	May'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	1.02	As needed	-
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	1.2	As needed	-

1.10.2 Enabling

1.10.2.1 PCIe 6 RxLEQ Phase3 Tx EQ Compliance Test Clarification

Problem statement

During PCI Express* (PCIe*) 6 compliance testing, RxLEQ Phase 3 Tx EQ in the current recipe is fixed to the Q2 coefficient instead of selecting the optimal preset, which leads to RxLEQ failures on customer platforms.

The RxLEQ Phase 3 Tx EQ selection should allow selection of an appropriate EQ coefficient during compliance testing to achieve RxLEQ pass results.

Workaround

As a temporary workaround, Phase 3 Tx EQ coefficients may be manually overridden in the BERT to unblock RxLEQ testing on custom platforms. Based on Intel validation data across multiple vendor platforms, Q5 has shown the most consistent RxLEQ pass behavior and is recommended as the Phase 3 override target. This workaround is intended for temporary use only.

Status

Two fixes are under evaluation:

- Tx EQ override
- Recipe update

Monitor the Message of the Week for updates on this issue.

1.11 WW17_26

New:

- [MCTP EID Reservation on Diamond Rapids Feedback Request](#)
- [Diamond Rapids Processor Test Capability Heat Map Release Notification](#)
- [Oak Stream 16-Channel Platform Hard-Tooled Prototype Availability Update](#)

Updated:

- [\(Updated WW17_26\) Oak Stream Key Collaterals Availability and Forecast](#)
- [\(WW17_26 Update\) WW16 Silicon BKC Release Notification](#)

1.11.1 Customer Feedback

1.11.1.1 MCTP EID Reservation on Diamond Rapids Feedback Request

Intel is requesting feedback from customers on Management Component Transport Protocol (MCTP) Endpoint ID (EID) Reservation in Diamond Rapids. The Intel® Out-of-Band Management Services Module (Intel® OOBM Services Module) reserves two to three MCTP EIDs per processor socket. Current POR mapping for these is in the following table.

Table 20. Current MCTP EID Mapping

Instance	EID
Socket 0 Primary IMH	8-9
Socket 0 Secondary IMH	10 (0x0A)
Socket 1 Primary IMH	11 (0x0B)
Socket 1 Secondary IMH	12 (0x0C)
Socket 2 Primary IMH	13 (0x0D)
Socket 2 Secondary IMH	14 (0x0E)
Socket 3 Primary IMH	15 (0x0F)
Socket 3 Secondary IMH	16 (0x10)

Intel has received reports from some customers that they reserve EIDs in this range for their own devices. In response, Intel is proposing a different mapping of reserved EIDs per processor socket as shown in the following table.

Table 21. Proposed MCTP EID Mapping

Instance	EID
Socket 0 Primary IMH	17-18 (0x11-0x12)
Socket 0 Secondary IMH	19 (0x13)
Socket 1 Primary IMH	20 (0x14)
Socket 1 Secondary IMH	21 (0x15)
Socket 2 Primary IMH	22 (0x16)
continued...	

Instance	EID
Socket 2 Secondary IMH	23 (0x17)
Socket 3 Primary IMH	24 (0x18)
Socket 3 Secondary IMH	25 (0x19)

Is this proposed static EID assignment acceptable? (Yes/No)

Provide your feedback on the EID assignment by May 1, 2026 to your Intel representative.

1.11.2 Documentation

1.11.2.1 Diamond Rapids Processor Test Capability Heat Map Release Notification

The *Diamond Rapids Processor Test Capability Heat Map*, document number [873155](#), Revision WW17'16, is now available in the Resource and Documentation Center (RDC).

Overview of WW17'26 changes are:

- This is an initial roadmap release. The heat map includes one-page functionality tables for accelerators, I/O, memory, manageability, RAS, power management, security, virtualization, and stability. The heat map also includes individual feature status for RAS, Intel® QuickAssist Technology (Intel® QAT), seamless, EV, and memory.

1.11.2.2 (Updated WW17_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 22. Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.1	1.15	Mid Q2'26
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.15	1.2	Q3'26
<i>Diamond Rapids Processor Test Capability (Heat Map)</i>	873155	WW17'26	WW21'26	WW21'26
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
Diamond Rapids Processor Sightings Report	862869	1.2	WW18'26	WW18'26
Diamond Rapids Server Platform BIOS Developer's Guide	839071	0.9	1.0	Late Q2'26
Oak Stream 16-Channel 2S in 19-Inch Design Feasibility	843581	1.7	As needed	-
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	1.0	As needed	-
Oak Stream 16-Channel Platform CPLD Reference Code	860022	1.0	As needed	-
Oak Stream 16-Channel Platform Design Guide	788196	1.0	1.1	Late Q2'26
Oak Stream 16-Channel Platform Electrical Specification	787006	1.0	1.1	Late Q2'26
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	1.2	As needed	-
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258	1.2	As needed	-
Oak Stream 16-Channel Platform Pin List	636260	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	1.0	As needed	-
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	1.0	1.1	Q2'26
Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive	821026	0.55	0.7	WW50'26
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW12'26	-	April'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	1.02	As needed	-
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	1.2	As needed	-

1.11.3 Enabling

1.11.3.1 Oak Stream 16-Channel Platform Hard-Tooled Prototype Availability Update

Hard-tooled prototypes of the following parts may now be ordered from and shipped by the suppliers listed.

NOTE

Supplier contact information is provided for convenience. Intel does not formally endorse any products or suppliers.

Backplates and bolster plates from:

- TE Connectivity*
- LOTES* Co., Ltd.

Socket covers from:

- TE Connectivity
- LOTES Co., Ltd.
- KYZ*

Heatsink hardware (HSHW) from:

- Package carriers from Jentech Precision Industrial* Co.,Ltd.
- KYZ
- ITW*

Intel is continuing to work with Foxconn Interconnect Technology* on their hard-tooled prototypes. Similar approval of their parts is targeted for some time in May 2026.

These hard-tooled prototypes have undergone initial viability testing by Intel:

- Limited sample quantities of these hard-tooled prototypes have completed critical **time 0** assessments to support EVT which include CTFs, hardware functionality, cycle durability, and very limited dynamics.
- Initial qualification testing may be done with these parts at your own risk as the parts are subject to change until they pass all of Intel's validation testing.

- Intel validation testing is scheduled for completion in September 2026, including accelerated environmental stresses and the completion of testing on full validation sample sizes.
- **CPUs and thermal solutions are not to be shipped while installed in a system** until Intel shock and vibration testing of these prototypes is complete and passing validation requirements/criteria.
- Shock and vibration testing is scheduled for completion in July 2026 with the intent of enabling systems to ship with CPUs and thermal solutions installed.
- Previously available soft-tooled prototypes continue to not be approved for shipping with CPUs and thermal solutions installed in a system.

Hard-tooled part numbers are unchanged from those published in the WW12'26 *Oak Stream Platform Message of the Week (MoW)*. Between soft- and hard-tooled prototypes, some supplier part numbers are different while others remain the same.

The listed suppliers may be contacted for additional details on their products.

Part numbers, drawings, and CAD for the hard-tooled parts are targeted for Revision 1.1 releases for the following documents by early May 2026:

- *Oak Stream 16-Channel Platform Thermal Mechanical Specification*, document number [786521](#)
- *Oak Stream 16-Channel Socket Enabling Components Mechanical Drawings*, document number [786522](#)
- *Oak Stream 16-Channel Socket Enabling Components Mechanical Models*, document number [786523](#)

As discussed in the WW12 2026 *Oak Stream MoW*, there is no mixing of hard- and soft-tooled socket covers and bolster plates. See that MoW for additional details.

1.11.4 Software and Firmware

1.11.4.1 (WW17_26 Update) WW16 Silicon BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platforms.

The BKC consists of an updated package and associated report, published on a regular basis, for the Johnson City reference platforms.

The BKC package includes UEFI firmware and onboard device drivers. These components have been screened through interoperability testing together over a period, with the aim of delivering customer confidence that not only have platform ingredients been individually tested, but also collectively tested to screen out interoperability issues.

Table 23. Johnson City BKC

Ingredient	Previous	Current
BKC	WW14	WW16
CRB BIOS	33.D41	33.D87
<i>continued...</i>		

Ingredient	Previous	Current
Unified Patch (MCU)	0x80000999	0x8000099B
MMC FW	0.785.0	0.794.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	1.0.6384_v2	1.0.6384_v2
Download	915787	917475
Status	Released	Released

NOTE

For information on silicon availability, contact your Intel representative.

Diamond Rapids 1S Alpha Software Milestone Declaration

Intel is pleased to announce that the Diamond Rapids 1-Socket (1S) Alpha Software milestone has been officially declared with the WW16 Best Known Configuration (BKC). The 1S Alpha declaration has the following exceptions:

- **UC D2D errors observed during warm reset cycling.** During warm reset cycling, some systems may log D2D unrecoverable errors, including messages such as "PL TrainError" or "PL Cerror/nferror/Error," along with Machine Check errors reported in the BIOS log banks 7/4. The D2D health check may report a failure even though the system can continue running. This is not a functional issue, and errors can be ignored.
- **Intermittent BIOS hang during warm reset cycling in the IPMI/MMBI BIOS path.** An intermittent issue has been observed during internal stress testing where the BIOS-BMC communication via MMBI may fail under warm reset or G3 cycling. This behavior is not consistently reproducible and has not been observed in normal functional usage scenarios. Debug is ongoing.
- **RSRC Adapter Error Observed (CBB PUNIT MCA) During Warm Reset Cycling.** 1S and 2S systems may intermittently hang at PC10 during warm reset cycling and log RSRC Adapter-related CBB PUNIT MCA errors. The issue has been observed primarily under aggressive internal reset-stress scenarios.

NOTE

Monitor the *Diamond Rapids Processor Test Capability Heat Map*, document number [873155](#), for updates on these exceptions and updates on the 2S configuration.

1.12 WW16_26

New:

- [Oak Stream 16-Channel Platform 500W Cookbook Release Notification](#)
- [Oak Stream 16-Channel Platform Heatsink Screening Test Document Release Notification](#)
- **[WW16 BKC Silicon Release Notification](#)**

Updated:

- (WW16_26 Update) Diamond Rapids ES1 X1 and X4 Comparison

1.12.1 Documentation

1.12.1.1 Oak Stream 16-Channel Platform 500W Cookbook Release Notification

The *Oak Stream 16-Channel Platform 500W Cookbook*, document number [851059](#), Revision 1.17.07, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.17.07 changes are:

- Added "READ ME FIRST" file for additional VR validation recommendations and change history details.
- Updated cookbook itself to fix known issues and match the updated power specs.

1.12.1.2 Oak Stream 16-Channel Platform Heatsink Screening Test Document Release Notification

The *Oak Stream 16-Channel Platform Heatsink Screening Test*, document number [872723](#), Revision 1.0, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.0 changes are:

- Initial release

1.12.2 Enabling

1.12.2.1 (WW16_26 Update) Diamond Rapids ES1 X1 and X4 Comparison

NOTE

Updated information is highlighted in **bold text**.

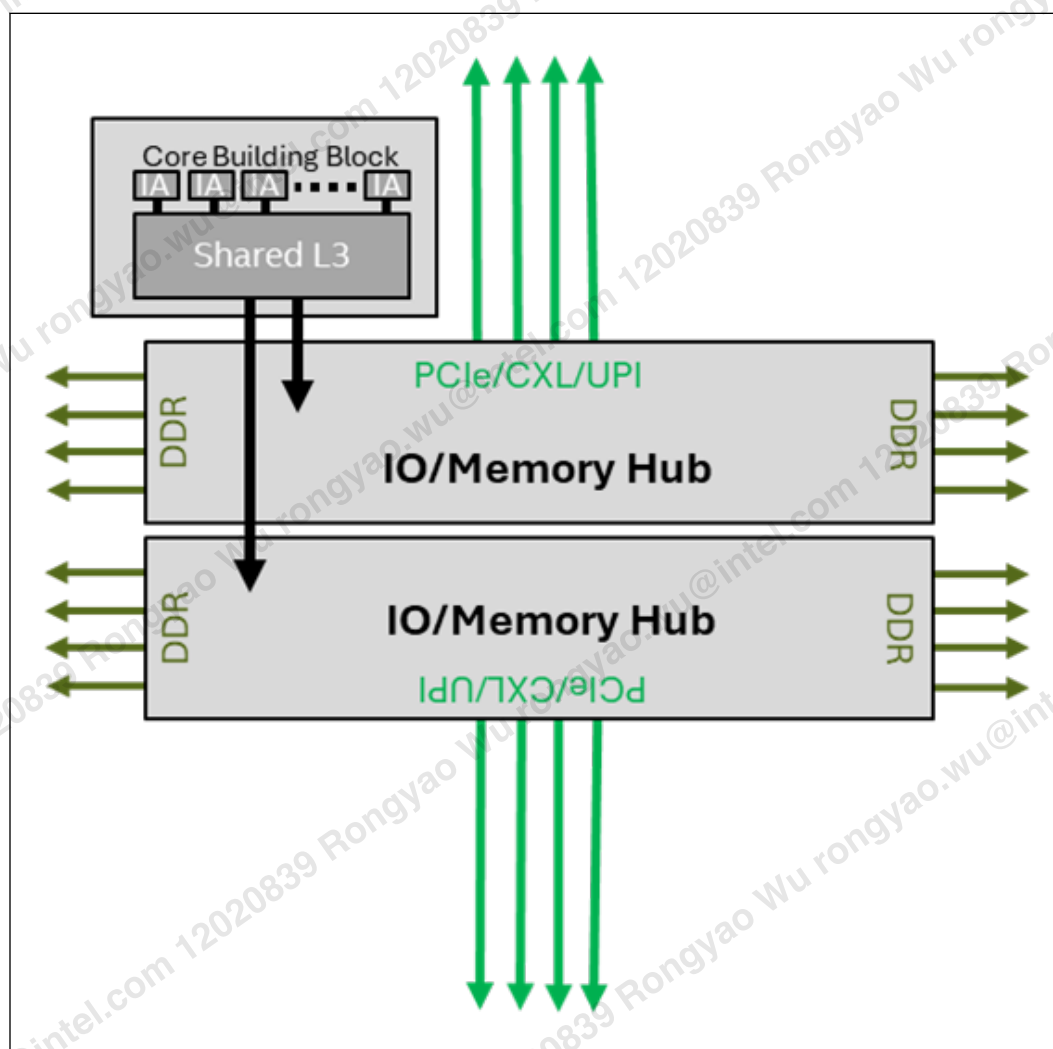
Diamond Rapids ES1 processors will have both X1 and X4 SKU offerings. The X1 configuration has one CBB while the X4 configuration has four CBBs. A comparison of X1 and X4 is provided in the following table to help with proper planning of the X1 ES1 and X4 ES1 usages.

Table 24. ES1 X1 and X4 Comparison

Area	X1 ES1	X4 ES1	Platform Testing Guidance
SoC Construction	Refer to X1 SoC Construction.	Refer to X4 SoC Construction.	While the internal Diamond Rapids SoC traffic and activities will be different on X1 ES1 and X4 ES1, at the board level all I/O and memory capabilities will be the same for EVT testing.
IMHs	2 x IMH1	2 x IMH1	The same I/O and memory capabilities can be tested at the board level as part of EVT with X1 ES1 or X4 ES1.
CBBs	1 x CBB	4 x CBB	The power specification across different TDPs is defined as an envelope that covers both X1 and X4 SKUs, following the power specs in the <i>Oak Stream 16-Channel Platform Design Guide</i>, document <i>continued...</i>

Area	X1 ES1	X4 ES1	Platform Testing Guidance
			number 788196, and the Oak Stream 16-Channel Platform 500W Cookbook, document number 851059, for the platform VR validation. Customers will need to revisit cycling testing with X4 ES1 samples. The package thermal profile will be different between one CBB versus four CBBs, hence thermal related testing with a dependency on granular CPU thermal information might need to be revisited with X4 ES1. Some PM related flow can also be affected by the number of CBBs/cores – for instance PkgC6 flow/timing, hence PM related testing might need to be revisited with X4 ES1. While no “stress testing” is expected to be done with ES1 as part of EVT, the additional cores with the four CBBs can drive higher I/O and memory utilization.

Figure 3. X1 SoC Construction



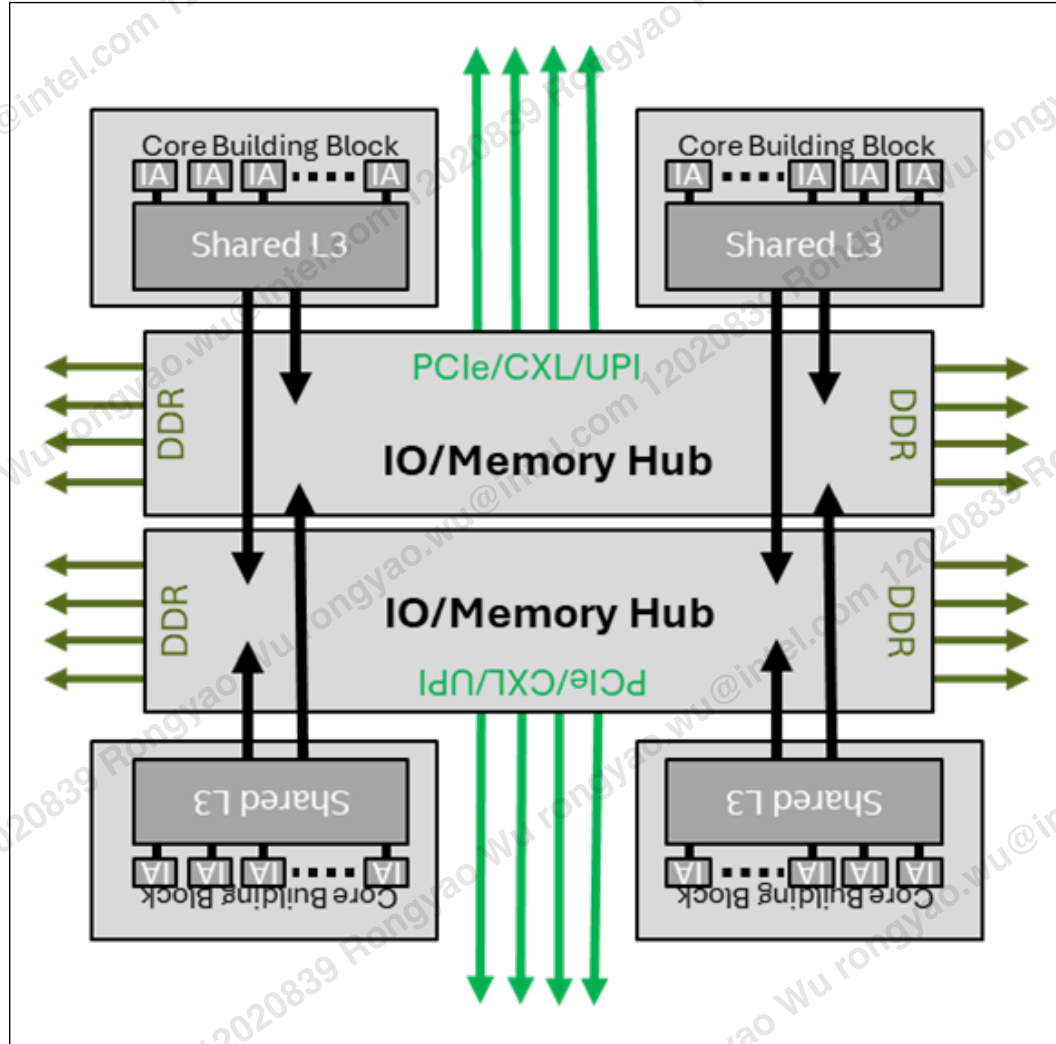
NOTES:

PCIe represents PCI Express* (PCIe*).

CXL represents Compute Express Link* (CXL*).

UPI represents Intel® Ultra Path Interconnect (Intel® UPI).

Figure 4. X4 SoC Construction



1.12.3 Software and Firmware

1.12.3.1 WW16 Silicon BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platforms.

The BKC consists of an update package and associated report, published on a regular basis, for the Johnson City reference platforms.

The BKC package includes UEFI firmware and onboard device drivers. These components have been screened through interoperability testing together over a period, with the aim of delivering customer confidence that not only have platform ingredients been individually tested, but also collectively tested to screen out interoperability issues.

Table 25. Johnson City BKC

Ingredient	Previous	Current
BKC	WW14	WW16
CRB BIOS	33.D41	33.D87
Unified Patch (MCU)	0x80000999	0x8000099B
MMC FW	0.785.0	0.794.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	1.0.6384_v2	1.0.6384_v2
Download	915787	917475
Status	Released	Released

NOTE

For information on silicon availability, contact your Intel representative.

Diamond Rapids 1S Alpha Software Milestone Declaration

Intel is pleased to announce that the Diamond Rapids 1-Socket (1S) Alpha Software milestone has been officially declared with WW16 Best Known Configuration (BKC). The 1S Alpha declaration has the following exceptions:

- **UC D2D errors observed during warm reset cycling.** During warm reset cycling, some systems may log D2D unrecoverable errors, including messages such as "PL TrainError" or "PL Cerror/nferror/Error," along with Machine Check errors reported in BIOS log banks 7/4. The D2D health check may report a failure even though the system can continue running. This is not a functional issue, and errors can be ignored.
- **Intermittent BIOS hang during warm reset cycling in the IPMI/MMBI BIOS path.** An intermittent issue has been observed during internal stress testing where BIOS-BMC communication via MMBI may fail under warm reset or G3 cycling. This behavior is not consistently reproducible and has not been observed in normal functional usage scenarios. Debug is ongoing.
- **RSRC Adapter Error Observed (CBB PUNIT MCA) During Warm Reset Cycling.** 1S and 2S systems may intermittently hang at PC10 during warm reset cycling and log RSRC Adapter-related CBB PUNIT MCA errors. The issue has been observed primarily under aggressive internal reset-stress scenarios.

NOTE

Monitor the Diamond Rapids Processor Test Capability Heatmap, document number [873155](#), for updates on these exceptions.

1.13 WW15_26

New:

- [Diamond Rapids Processor Sightings Report Release Notification](#)
- [Oak Stream 16-Channel Platform Johnson City Design Archive Release Notification](#)
- [Diamond Rapids ES1 X1 and X4 Comparison](#)
- [WW14 BKC Release Notification](#)
- [Oak Stream Intel® IO Margin Tool Release Notification](#)

Updated:

- [\(Updated WW15_26\) Oak Stream Key Tools Availability and Forecast](#)

1.13.1 Documentation

1.13.1.1 Diamond Rapids Processor Sightings Report Release Notification

The *Diamond Rapids Processor Sightings Report*, document number [862869](#), Revision 1.2, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.2 changes are:

- Initial roadmap release. Documented sightings SR1 to SR86.

Starting with Diamond Rapids, the Sightings Report will be published as a PDF and browsable through a web version.

1.13.1.2 Oak Stream 16-Channel Platform Johnson City Design Archive Release Notification

The following documents are Revision 1.2 and are now available in the Resource and Documentation Center (RDC):

- *Oak Stream 16-Channel Platform Johnson City 1 Design Archive*, document number [823065](#)
- *Oak Stream 16-Channel Platform Johnson City 2 Design Archive*, document number [636258](#)

Overview of Revision 1.2 changes are:

- Added missing GND vias in the DIMM region.

1.13.2 Enabling

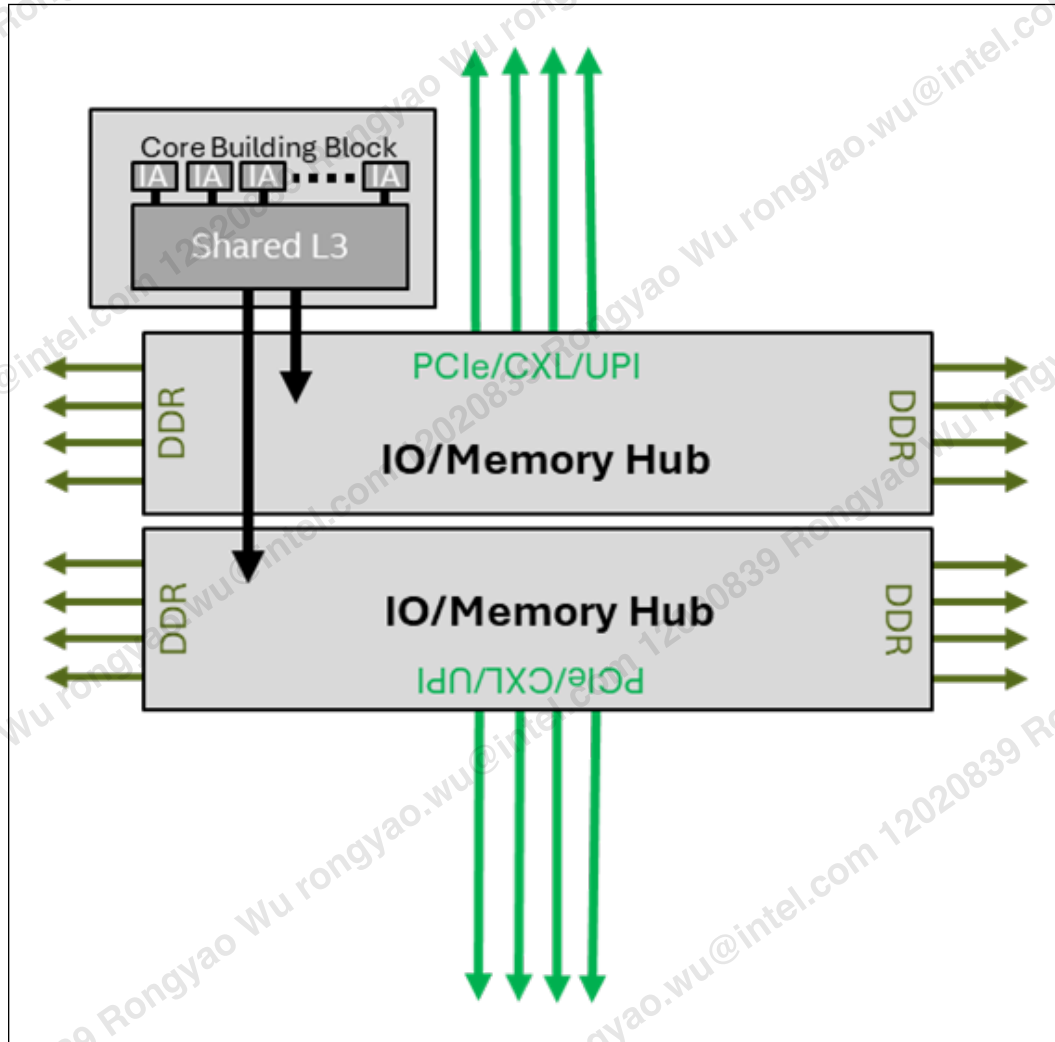
1.13.2.1 Diamond Rapids ES1 X1 and X4 Comparison

Diamond Rapids ES1 processors will have both X1 and X4 SKU offerings. The X1 configuration has one CBB while the X4 configuration has four CBBs. A comparison of X1 and X4 is provided in the following table to help with proper planning of the X1 ES1 and X4 ES1 usages.

Table 26. ES1 X1 and X4 Comparison

Area	X1 ES1	X4 ES1	Platform Testing Guidance
SoC Construction	Refer to X1 SoC Construction.	Refer to X4 SoC Construction.	While the internal Diamond Rapids SoC traffic and activities will be different on X1 ES1 and X4 ES1, at the board level all I/O and memory capabilities will be the same for EVT testing.
IMHs	2 x IMH1	2 x IMH1	The same I/O and memory capabilities can be tested at the board level as part of EVT with X1 ES1 or X4 ES1.
CBBs	1 x CBB	4 x CBB	Power being delivered to four CBBs on X4 ES1, hence customers will need to retest the board's CPU VR and power delivery. As VR is an element of cycling testing, customers will need to redo cycling testing with X4 ES1. The package thermal profile will be different between one CBB versus four CBBs, hence thermal related testing with a dependency on granular CPU thermal information might need to be revisited with X4 ES1. Some PM related flow can also be affected by the number of CBBs/cores – for instance PkgC6 flow/timing, hence PM related testing might need to be revisited with X4 ES1. While no "stress testing" is expected to be done with ES1 as part of EVT, the additional cores with the four CBBs can drive higher I/O and memory utilization.

Figure 5. X1 SoC Construction



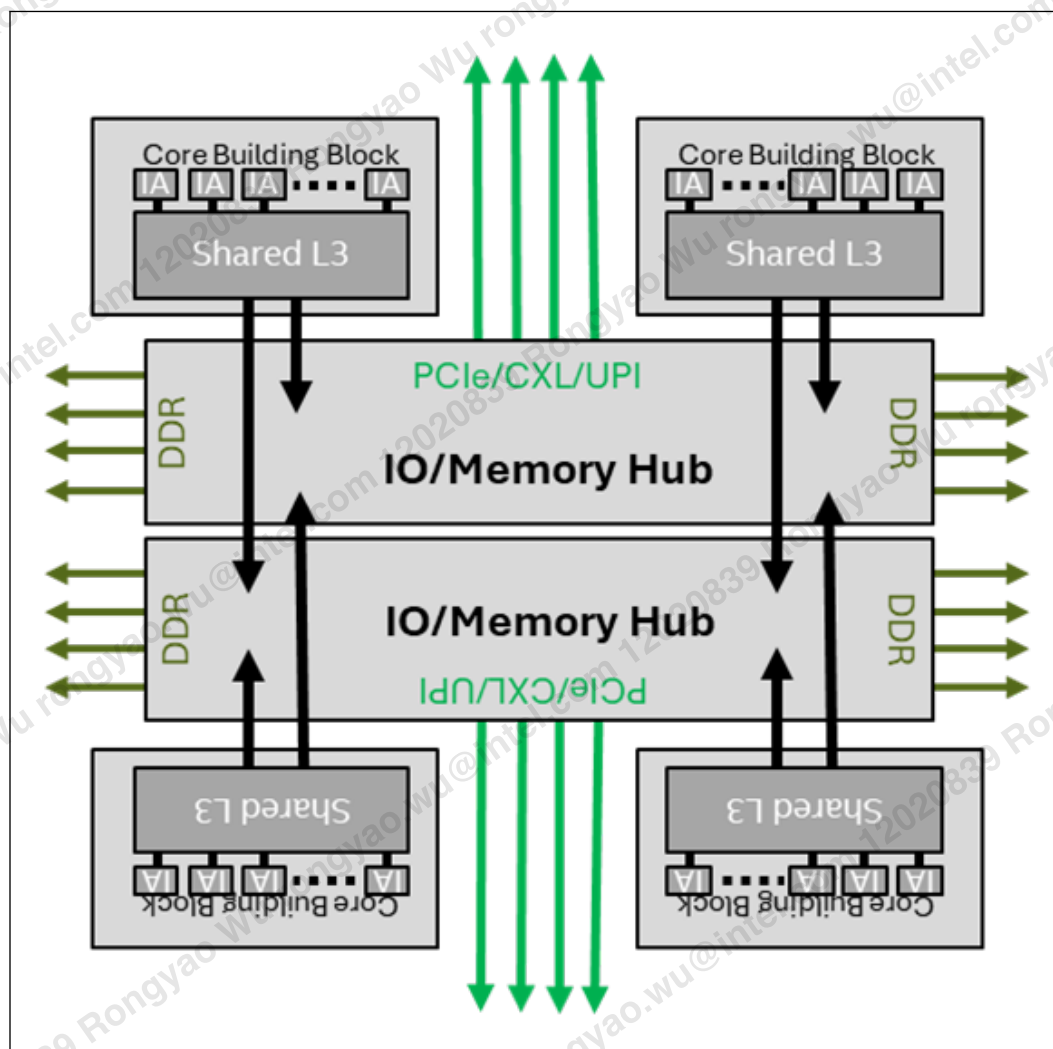
NOTES:

PCIe represents PCI Express* (PCIe*).

CXL represents Compute Express Link* (CXL*).

UPI represents Intel® Ultra Path Interconnect (Intel® UPI).

Figure 6. X4 SoC Construction



1.13.3 Software and Firmware

1.13.3.1 WW14 BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platform.

The BKC consists of an update package and associated report, published on a regular basis, for the Johnson City reference platform.

The BKC package includes UEFI firmware and onboard device drivers. These components have been screened through interoperability testing together over a period, with the aim of delivering customer confidence that not only have platform ingredients been individually tested, but also collectively tested to screen out interoperability issues.

Table 27. Johnson City BKC

Ingredient	Previous	Current
BKC	WW12 Silicon	WW14
CRB BIOS	32.D91	33.D41
Unified Patch (MCU)	0x80000994	0x80000999
MMC FW	0.770.0	0.785.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	1.0.6366	1.0.6384_v2
Download	915787	916482
Status	Released	Released

NOTE

For information on silicon availability, contact your Intel representative.

1.13.4 Tools and Debug

1.13.4.1 Oak Stream Intel® IO Margin Tool Release Notification

The *Oak Stream Intel® IO Margin Tool*, document number [871016](#), Revision 2.0, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 2.0 changes are:

- Initial release. This revision supports Host-Mode for PCI Express* (PCIe*) 5 and PCIe 6 margin testing. Support for On-Target Mode, PCIe 4 and Intel® Ultra Path Interconnect (Intel® UPI) will be added in a future revision.

Refer to the *Oak Stream Platform IO Margin Tool for Windows User Guide*, document number [913382](#), for details on how to use the Intel® IO Margin Tool.

Refer to the *Oak Stream Platform Margin Testing*, document number [871015](#), for details on the HSIO receiver port margin testing.

Limitations

Due to a silicon bug on the Diamond Rapids UCC silicon, the Intel® I/O Margin Tool does not work even after enabling Delayed Authentication Mode (DAM). This bug is expected to be resolved with Diamond Rapids UCC-M silicon.

Refer to [SR77](#) in the *Diamond Rapids Processor Sightings Report*, document number [862869](#), for details.

Workaround

To use the Intel® I/O Margin Tool with the Diamond Rapids UCC silicon, a higher privileged state must be entered known as **OEM Unlock**.

The OEM Unlock state can be entered by setting 2 bits in the IFWI's FIT 4 Boot Policy Record using the Intel® Modular Flash Image Tool (Intel® Modular FIT).

To enter the OEM Unlock:

1. Download, extract, and install the Intel® Modular FIT from the latest Oak Stream Best Known Configuration (BKC).
2. Extract the configuration .xml file from the IFWI:
 - `>python fit_cmd.py -i bkc_ifwi.bin -s bkc_ifwi.xml`
3. Once the .xml is extracted, set the bits and rebuild the image.

Rebuild with a modified .xml:

1. Locate the "btg" container within the .xml, set the following settings to value 0x1, and save the .xml:
 - `<setting name="Dam" value="0x1"/>`
 - `<setting name="oem_unlock" value="0x1"/>`
2. Build a new IFWI with the modified .xml:
 - `>python fit_cmd.py -b -i bkc_ifwi_modified.xml`

Rebuild with additional parameters:

- `>python fit_cmd.py -b -i bkc_ifwi.xml --params btg:Dam=1
btg:oem_unlock=1`

1.13.4.2 (Updated WW15_26) Oak Stream Key Tools Availability and Forecast

NOTE

Updated information is highlighted in bold text.

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI-60 to Intel ITP adapter.
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support. Two Streams Tracing supported:
 - LA-3907 pre-processor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports four trace ports)
 - LA-3892 converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI-60
- ASSET* Arium* - Contact ASSET Arium for support.
- Compact Debug Port (CDP) + MIPI-10 adapter for I3C*
- Type-C connection: Voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- VTune™ tools
- Intel® SoC Watch
- Intel® System Debugger
- Includes token generation for Vendor Authorize Boot (VAB) / Vendor Authorize Debug (VAD).

NOTE

The Intel® System Bring-up Toolkit NDA requires registration. See *How to Obtain Intel® System Bring-up Toolkit NDA (Formerly Intel® System Studio NDA) User Guide*, document number [616675](#), and *Intel® System Bring-up Toolkit NDA (formerly Intel® System Studio NDA) Registration Form New Users*, document number [631579](#), for details.

Table 28. Oak Stream Key Tools

Type	Tool	RDC or DIT	Diamond Rapids Support	Revision
Debug	ASSET Intertech* Arium* Note: Contact ASSET* for support.	-	Q2'26	-
Debug	ASSET* Close Chassis Controller (CCC) Note: Contact ASSET* for support.	-	Q2'26	-
Debug	CScripts for Oak Stream Platforms Note: Refer to Release Notes for details.	821387	Available	2615.2000
Debug	Intel Remote FPGA Access Tool (RFAT) for Server Platforms	871932	Available	1.0
Debug	Intel Triage Tool (Crashdump Summarizer) Note: Available through CScripts	N/A	Available	0.86.1.2
Debug	Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C*	99CJRJ	Available	N/A
Debug	Intel® I3C* USB Type-C Level Shift Adapter	99CN2C	Available	N/A
Debug	Intel® In-Target Probe (Intel® ITP) eXtended Debug Port (XDP)	940990	Available	N/A
Debug	Intel® Silicon View Technology (Intel® SVT) Closed Chassis Adapter	921521	Available	N/A
Debug	Intel® System Bring-up Toolkit NDA Note: Includes Intel® System Debugger NDA, Intel® VTune™ Profiler NDA and Intel® SoC Watch NDA.	634166	Available	Latest
Debug	Lauterbach*	-	Available	N/A
Debug	SelfTest 7 - Windows* x64	524152	Available	178
continued...				

Type	Tool	RDC or DIT	Diamond Rapids Support	Revision
Debug	<i>Server Security Toolkit</i> Note: Server Security Toolkit includes Intel® Total Memory Encryption – Multi-Key (Intel® TME-MK), Intel® Trusted Execution Technology (Intel® TXT), BIOS TPM Provisioning with Intel® Platform Trust Technology (Intel® PTT) tools, and CbNT (Converged Intel® Boot Guard and Intel® TXT).	630398	Available	1.54
Manufacturing and Fleet Health	<i>Autonomous Crashdump (ACD)</i> Note: Autonomous Crash Dump (ACD) is available via the BMC source code under a license agreement.	N/A	Available	0.75
Manufacturing and Fleet Health	<i>Intel® At-Scale Debug (Intel® ASD)</i>	Intel-BMC/asd	Available	N/A
Manufacturing and Fleet Health	<i>Intel® At-Scale Debug (Intel® ASD) BMC-JTAG* Software Guide</i>	594948	Available	1.6.5
Manufacturing and Fleet Health	<i>Intel® In Field Scan</i>	-	Q2'27	5.x.x
Manufacturing and Fleet Health	<i>Intel® System Stress and Memory Diagnostics (SSMD) for Linux</i>	818881	Available	2025.11.10.4ed3
Manufacturing and Fleet Health	<i>Memory Latency Checker (MLC)</i>	736633	Q2'26	3.12
Signal Integrity	<i>Clock Jitter Tool (CJT)</i>	803838	Available	6.0.4
Signal Integrity	<i>Integrated Channel Analysis Tool (ICAT) Suite for Windows</i>	782966	Available	6.200.50
Signal Integrity	<i>Intel® Passive Component Checker Tool (Intel® PCC Tool)</i>	600218	Available	1.8
Signal Integrity	<i>Intel® Platform Clock Jitter Analysis Tool (CJAT) for Windows</i>	619648	Available	6.0.4
Signal Integrity	<i>PCIe* and Intel® UPI Load Board Solution Set for 16-Channel Platform</i>	DMRUCCPLBULB	Available	N/A
Signal Integrity	<i>PCIe* Load Board Solution Set for 16-Channel Platform</i>	DMRUCCPLB	Available	N/A

continued...

Type	Tool	RDC or DIT	Diamond Rapids Support	Revision
Thermal / Power Delivery	Almondsville Thermal Mechanical Test Vehicle (TMTV) Application Note	852568	Available	1.0
Thermal / Power Delivery	Almondsville Thermal Test Vehicle (TTV) for 16-Channel Platform	-	Available	-
Thermal / Power Delivery	Blue Interposer for Gen5 VRTT for 16-Channel Platform	99D5W2	Available	N/A
Thermal / Power Delivery	Gen5 VR Test Tool (VRTT) for Windows	99CJ9X	Available	N/A
Thermal / Power Delivery	Red Interposer for Gen5 VRTT for 16-Channel Platform	99D5W1	Available	N/A
Thermal / Power Delivery	VRTT Mechanical Interposer Retention Mechanism Kit for 16-Channel Platform	99CN1R	Available	N/A
Validation	Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card	-	Available	-
Validation	Intel® Simics Tools Note: Intel Simics tools are available under a license agreement. Contact your Intel representative for details.	-	Available	N/A
Validation	Intel® Structural Test Plus (Intel® STP) Note: Contact a local representative to enable Intel® STP for customer designs.	-	Available	N/A
Validation	MEI DDR5 Lite Card 8000 MT/s for Windows	99D5X2	Available	N/A
Validation	Memory Assistant for Intel® Xeon® Platforms	620497	WW16'26	WW16-26
Validation	Memory Error Injection (MEI) DDR5 Card for Windows	99CDV1	Available	N/A
Validation	Oak Stream DDR EV Tools & Methodology	914412	WW16'26	1.0
Validation	Oak Stream Platform Intel® IO Margin Tool for Windows* User Guide	913382	Available	1.0
Validation	Oak Stream Platform Intel® IO Margin Tool for Windows	871016	Available	2.0.0
continued...				

Type	Tool	RDC or DIT	Diamond Rapids Support	Revision
Validation	Oak Stream Platform Margin Testing	871015	Available	1.0
Validation	<i>Oak Stream Stitched Rank Margining Tool (RMT)</i> Note: Included in the BKC.	-	Available	-
Validation	Oak Stream Target Loader Rank Margining Tool (RMT)	913381	WW16'26	1.0
Validation	<i>PCI Express* (PCIe*) Lane Margin Tool (LMT) for PCIe* 4.0/5.0/6.0</i>	639433	Available	1.0
Validation	<i>PCIe* 5.0 Error Injection (PEI) Card</i>	99C3FA	Available	N/A
Validation	<i>PCIe* PEI 6.0 Card</i>	-	Q4'26	-
Validation	<i>PEI (PCIe Error Injector Card) Collaterals</i>	610667	Available	5.1
Validation	<i>PEI 5.0 (PCIe 5.0 Error Injection) + CXL 2.x Card</i>	99CN2D	Available	N/A
Validation	<i>RAS Automation ToolKit</i>	633201	Available	3.0.0

1.14 WW14-26

New:

- [Diamond Rapids Processor Architecture Specification Release Notification](#)
- [Oak Stream 16-Channel Platform Design Guide Release Notification](#)
- [PTLP Egress Blocking Sighting Update](#)

Updated:

- [\(Updated WW14_26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.14.1 Documentation

1.14.1.1 Diamond Rapids Processor Architecture Specification Release Notification

The *Diamond Rapids Processor Architecture Specification*, document number [792359](#), Revision 1.15, is now available on the Resource and Documentation Center (RDC).

Overview of Revision 1.15 changes are:

- Overview, Features, and Topologies
 - Updated Intel® Ultra Path Interconnect (Intel® UPI) maximum speed (2S) in the Features Overview table.

- Flexible I/O (FIO)
 - Updated UCC/UCC-M naming in IO Port Numbering and various PCI Express* (PCIe*)/Compute Express Link* (CXL*) figures.
 - Updated Supported CXL Device Populations per the Host Bridge table.
- Memory Subsystem
 - Added Memory Access Flow subsection, covering perspectives of both cache hit and miss.
- Reliability, Availability, and Serviceability (RAS)
 - Updated Leaky Bucket Example Flow diagram.

1.14.1.2 Oak Stream 16-Channel Platform Design Guide Release Notification

The *Oak Stream 16-Channel Platform Design Guide (PDG)*, document number [788196](#), Revision 1.0, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.0 changes are:

- **Chapter 1: Introduction and Overview**
 - Removed references to "A-Step" and "B-Step". Replaced with UCC or UCC-M.
- **Chapter 3: General High-Speed Guidelines**
 - Added the #3 note for Dog-Bone Guideline.
- **Chapter 4: Platform Clocking**
 - Updated the clocking architecture description to add support for cross-IMH die REFCLK in SRNS and SRIS clock modes.
- **Chapter 6: DDR5 Memory Routing Guidelines**
 - From DIMM Connector Layout Design Considerations section:
 - Added a third bullet on DIMM pitch support.
 - Added an additional DIMM pitch comment that DIMM pitches smaller than 0.262 in. will not be validated.
 - From the Raw Card Support section:
 - Changed the 1-DIMM support from 1Rx4 32 GB raw card to 2Rx8 32 GB.
 - Added support for 4Rx4 128 GB and 4Rx4 192 GB 2U Tall MRDIMMs, post-TTM intercept.
 - Added support for 2Rx4 64 GB at 2DPC in 4S Configuration, post-TTM support.
 - From the Memory Copper Max Routing Requirements section:
 - Updated the range value for the "S_track" parameter from the Copper Max Parameters table.
- **Chapter 7: I3C* Design Guidelines**
 - Updated Note in Timing/Frequency calculations sections clarifying the PDG is compliant with *MIPI Alliance I3C Specification*, v1.1.1.
 - I3C for MNG:
 - Updated HUB weak pull-up resistance: Changed from 2 kohm to 500 ohm for all I3C MNG topologies.
 - I3C for DBG:

Updated Trace Lengths for *I3C DBG HUB BMC Solution Single Socket Routing* and *I3C DBG HUB BMC Direct Attached Solution Modular Routing* tables.

Removed 0-ohm resistors from some topologies.

- **Chapter 11: System Management**

- Added PERST_N Method 1 and Method 2 clarification.
- Added TPM over eSPI support.

- **Chapter 12: Platform Debug Guidelines**

- Updated trace lengths for *JTAG TDO Single Socket Routing*, *JTAG PRDY_N Single Socket Routing*, and *Hook[9:8] MBP[1:0] Single Socket Routing* tables.

- **Chapter 13: Platform Power Sequencing Flows**

- Updated T46 start from S0_PWR_OK in Global Reset, Aysc Global reset, and PSU AC Loss diagram in Section 13.2.1.
- Updated T23 from 20 us to 65 us.
- Updated T41 from 0.5/1.5 us min/max to 1.0 ms minimum.
- Added a note in T21 comments as multi-socket use only (for EXT_PFT_IN mode).
- Removed T10 to avoid confusion.
- Changed T11 as T11a.
- Added T11b timing with description from GLOBAL_RESET_N de-asserted to GLB_RST_WARN_N asserted minimum 15 ms.
- Updated G3-S5 diagram with T11a and T11b.
- Updated T14 description as from GLOBAL_RESET_N de-asserted to eSPI_RESET_N de-asserted.
- Updated notes to T24, applies to isCLK PFT_IN mode only, removed default value.
- Updated timing T32, T36, and T37 as minimum only.
- Updated T25, T27, T29, T30, T31, T46, T61, T63b, T66.
- Updated T47 as minimum, added T47b for maximum.
- Added MON_FAIL_N signal in S5 entry flow diagram.
- Added a pin strap table in Section 13.1 and a few clarification notes, removed the notes at Section 13.2.1.
- Added MON_FAIL_N in S5 Exit flow.

- **Chapter 14: Power Distribution Guidelines**

- Updated table *Other Processor VR Power Parameters Table 2*.
- Updated tables *RP Psys Power Limit*, *RP Psys Power Limit High Line*, and *RP Psys Power Limit Low Line* due to PLATFORM_RAPL_POWER_UNIT granularity update from 1/8W to 1W.
- Added clarification to the *CPU Voltage Rail Turning Off Timing Requirements* section.
- Added new *VR Stability Tuning Recommendations For CPU Rails* and *Isys Error Issue on PDB with Cold Redundancy Mode Enabled on PSUs* sections.
- Added recommendations to *Power Planes and Ground Planes* section.

Refer to the document revision history for full list of changes.

1.14.1.3 (Updated WW14_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 29. Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.1	1.15	Mid Q2'26
Diamond Rapids Processor Architecture Specification	792359	1.1 5	1.2	Q3'26
<i>Diamond Rapids Processor Test Capability (Heat Map)</i>	873155	-	-	April'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	-	1.2	WW15'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i>	839071	0.9	1.0	WW9'26
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
Oak Stream 16-Channel Platform Design Guide	788196	1.0	1.1	Late Q2'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	1.0	1.1	Late Q2'26
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	1. 2	As needed	-
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258	1. 2	As needed	-
Oak Stream 16-Channel Platform Pin List	636260	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	1.0	As needed	-
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	1.0	1.1	Q2'26
Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive	821026	0.55	0.7	WW50'26
Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard	863599	WW12'26	-	April'26
Oak Stream Platforms System Boards Design Archive	821025	1.02	As needed	-
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	1.2	As needed	-

1.14.2 Enabling

1.14.2.1 PTLP Egress Blocking Sighting Update

Intel has collected customer feedback on [root port poison TLP \(PTLP\) egress blocking enablement](#).

Problem

In FLIT mode, when PTLP egress blocking (outbound poisoned data) is enabled, a logic bug is preventing AER logging/signaling of the fault in the AER error logs. The poisoned transaction is blocked but the subsequent transactions are allowed to move forward. eDPC will not trigger even if it was configured to do so. Non-FLIT mode logging and signaling works as expected.

Implication

PTLP egress blocking events are not recoverable and the system is exposed to unpredictable behavior.

Workaround

The Oak Stream BKC will keep root port PTLP blocking disabled for the Diamond Rapids UCC processor. This will allow the outbound poisoned packet to be delivered to the end device.

This issue will be fixed on Diamond Rapids UCC-M. When the PTLP egress blocking is enabled on Diamond Rapids UCC-M, the outbound poisoned data will be blocked, and the error event will be logged/signaled per the eDPC flow. The AER HDR log will not be valid for this error event.

1.15 WW13_26

New:

- [Diamond Rapids Processor Registers Specification Release Notification](#)
- [TDX Module Removal from Oak Stream IFWI](#)
- [WW12 Silicon BKC Release Notification](#)

Updated:

- [\(Updated WW13_26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.15.1 Documentation

1.15.1.1 Diamond Rapids Processor Registers Specification Release Notification

The *Diamond Rapids Processor Registers Specification*, document number [793272](#), Revision 1.1, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.1 changes are:

- Added new subsections for merged banks, MSRs, and TPMI Out of Band (OOB) access in the **System Management Service Accessible Registers** section.
- Added the **Intel® Volume Management Device (Intel® VMD)** section.
- Added the FBLP PCI Configuration Space and Memory Space registers to the **PCI Express* (PCIe*) 4 Registers** and **Flexible I/O Registers** chapters.
- Added additional registers to the **Root RASIP Registers** section.
- Updated all chapter introductions to include die mapping for each IP instance.
- Updated register definitions in multiple chapters: Accelerator, HIOP, Intel® Out-of-Band Management Services Module (Intel® OOBM Services Module), S3M, FIO, CXLCM, I3C*, HAMVF, MC, SCA, UBOX, PUNIT, and MSRs.
- Updated the registers OOB access information in all chapters.

Refer to the revision history for a complete list of changes.

1.15.1.2 (Updated WW13_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 30. Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
Diamond Rapids Processor Registers Specification	793272	1 . 1	1. 15	Mid Q2'26
Diamond Rapids Processor Architecture Specification	792359	1.1	1.15	Mid Q2'26
Diamond Rapids Processor Test Capability (Heat Map)	873155	-	-	April'26
Diamond Rapids Processor Sightings Report	862869	-	-	April'26
Diamond Rapids Server Platform BIOS Developer's Guide	839071	0.9	1.0	WW9'26
Oak Stream 16-Channel 2S in 19-Inch Design Feasibility	843581	1.7	As needed	-
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	1.0	As needed	-
Oak Stream 16-Channel Platform CPLD Reference Code	860022	1.0	As needed	-
Oak Stream 16-Channel Platform Design Guide	788196	0.9	1.0	WW13'26
Oak Stream 16-Channel Platform Electrical Specification	787006	1.0	1.1	Late Q2'26
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	1.15	As needed	-
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258	1.15	As needed	-
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
Oak Stream 16-Channel Platform Pin List	636260	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	1.0	As needed	-
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	1.0	As needed	-
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	1.0	1.1	Q2'26
Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive	821026	0.55	0.7	WW50'26
Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard	863599	WW12'26	-	April'26
Oak Stream Platforms System Boards Design Archive	821025	1.02	As needed	-
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	1.2	As needed	-

1.15.2 Enabling

1.15.2.1 TDX Module Removal from Oak Stream IFWI

Intel® Trust Domain Extensions (Intel® TDX) firmware is delivered in a TDX module binary. This module has been previously integrated into the Oak Stream IFWI binary.

As new features are added, such as, Trust Domain migration, Intel® Trust Domain Extensions Connect (Intel® TDX Connect), and Compute Express Link* (CXL*) memory support, the TDX module is expected to grow to about 3 MB compressed, eventually not fitting in the FV_BINARY IFWI region.

Starting with the WW12 BKC, **the TDX module will not be included in the Oak Stream IFWI image.**

The TDX module binary will be distributed as part of the Oak Stream BKC release. The TDX module can be installed in the EFI System Partition /EFI/TDX, which is mapped to Linux* as /boot/efi/EFI/TDX. The Intel reference BIOS will load the TDX module from this EFI System Partition directory.

Intel is investigating other options for installing the TDX module.

1.15.3 Software and Firmware

1.15.3.1 WW12 Silicon BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platform.

The BKC consists of an update package and associated report, published on a regular basis, for the Johnson City reference platform.

The BKC package includes UEFI firmware and onboard device drivers. These components have been screened through interoperability testing together over a period, with the aim of delivering customer confidence that not only have platform ingredients been individually tested, but also collectively tested to screen out interoperability issues.

Table 31. Johnson City BKC

Ingredient	Previous	Current
BKC	WW10	WW12 Silicon
CRB BIOS	32.D57	32.D91
Unified Patch (MCU)	0x80000991	0x80000994
MMC FW	0.758.0	0.770.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	ModularFIT_cmd_1.0.6357	ModularFIT_cmd_1.0.6366
Download	914432	915787
Status	Released	Released

NOTE

For information on silicon availability, contact your Intel representative.

Important Note for this Release:

For this release, customers must ensure the UBR VN Credit Loss mitigation is configured to Full Mitigation.

If Full Mitigation is not enabled, systems may experience known stability issues, including:

- Intermittent loss of IP address after OS boot with E810 adapters
- System hangs due to fatal errors
- BMC virtual-media-based OS installation failures

NOTE

This mitigation is applied through the **S3M softstrap** and will **not** take effect if **DfxS3mSoftStrap** is disabled. The reference code build flows using the non-default -y 1P0 option may enable DMR_PO_FLAG, which forces DfxS3mSoftStrap = 0 (disabled). In this case, the workaround will **not** take effect even if the mitigation selection is set to *Full Mitigation*, unless this override is removed.

Refer to the BIOS release notes for more information about the issue and this mitigation.

1.16 WW12_26

New:

- [Oak Stream Platform Dashboard Release Notification](#)
- [Intermittent Write Training Failures on Gen2 MRDIMM with Renesas* MRCD B0 and MDB B0](#)
- [Oak Stream 16-Channel Platform Narrow Hardware Update](#)
- [CScripts Release Cadence Clarification](#)

Updated:

- [\(Updated WW12_26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.16.1 Documentation

1.16.1.1 (Updated WW12_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 32. Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.01	1.1	WW13'26
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.1	1.15	Mid Q2'26
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Test Capability (Heat Map)</i>	873155	-	-	April'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	-	-	April'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i>	839071	0.9	1.0	WW9'26
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	WW13'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	1.0	1.1	Late Q2'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.15	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.15	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	1.0	1.1	Q2'26
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.55	0.7	WW50'26
Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard	863599	WW 12 '26	-	April'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	1.02	As needed	-
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	1.2	As needed	-

1.16.1.2 Oak Stream Platform Dashboard Release Notification

The *Oak Stream Platform Diamond Rapids 12/16-Channel Dashboard*, document number [863599](#), Revision WW12 2026, is now available in the Resource and Documentation Center (RDC).

1.16.2 Enabling

1.16.2.1 Intermittent Write Training Failures on Gen2 MRDIMM with Renesas* MRCD B0 and MDB B0

Issue Observation

Intel is seeing intermittent write training failures on Diamond Rapids UCC processors when using Gen2 MRDIMMs with Renesas* MRCD B0 and MDBB0 components at 12800 MT/s POR speed. The issue is observable as training failures/warnings:

- Write Pre-DFE 2D Centering. Checkpoint Code: 0xB7, 0x84
- Write Post-DFE 2D Centering. Checkpoint Code: 0xB7, 0x85

Workaround

Temporary workaround: Reboot system; if failure persist after reboot, replace the failing MRDIMM.

Status

Renesas is working on a permanent work around for B0 components. Contact Renesas for details on the workaround.

NOTE

Supplier contact information is provided for convenience. Intel does not formally endorse any products or suppliers.

1.16.2.2 Oak Stream 16-Channel Platform Narrow Hardware Update

Enabled supplier and Intel part numbers for the Oak Stream 16-Channel **hard-tooled** backplates, bolster plate, socket cover, and heatsink hardware are in the following tables. These parts continue to be targeted for initial ordering approval in April 2026.

These parts will be hard-tooled prototypes that have undergone initial viability testing by Intel. With these prototypes, CPUs, and thermal solutions are not to be shipped installed in a system. Initial reliability testing may be done with these parts; however, the parts are subject to change until full validation is complete. Full validation of these ingredients is targeted for September 2026.

After the initial ordering approval checkpoint in April 2026, updated drawings and CAD files will be included in the relevant mechanical collaterals.

NOTE

Supplier contact information is provided for convenience. Intel does not formally endorse any products or suppliers.

Table 33. Hard-Tooled Part Numbers: Backplates, Bolster Plate, Socket Cover

Component	Intel	Foxconn Interconnect Technology* (FIT)	OTES Co., Ltd. TE Connectivity*	TE Connectivity*	KYZ*
Socket DX Backplate (93-106 mil Board Thickness) (POR, Narrow)	N84950-001	PMGP100-01NK1-4 H	AHSA143-P002C01SX	2536351-1	---
Socket DX Backplate (107-120 mil Board Thickness) (POR, Narrow)	N84910-001	PMGP100-02NK1-4 H	AHSA143-P001C01SX	2536351-2	---
Socket DX Backplate (121-130 mil Board Thickness) (POR, Narrow)	N84947-001	PMGP100-03NK1-4 H	AHSA143-P003C01SX	2536351-3	---
Socket DX Backplate (131-141 mil Board Thickness) (POR, Narrow)	N84948-001	PMGP100-04NK1-4 H	AHSA143-P004C01SX	2536351-4	---
Socket DX Bolster Plate	N74522-002	PMGL100-01NK1-4 H	AZIFA438-P002C01SX	2536350-1	---
Socket DX Cover	N74818-002	PMGD000-01NK1-EH	AZIFA444-P002C01SX	2536352-1	A30528H

Table 34. Hard-Tooled Part Numbers: Heatsink Hardware

Component	Intel	KYZ*	ITW*
DX-HSHW, without Anti-Tilt Nuts	N70218-002	A30529H	FT2501L
DX-HSHW, with Anti-Tilt Nuts	N70219-002	A30530H	FT2501R

POR Narrow Backplate

The hard-tooled backplate is the POR narrow version that is compatible with the Oak Stream 16-Channel Bottom KOZ (Narrow) as described in the *Oak Stream 16-Channel Socket Enabling Components Mechanical Drawings*, document number [786522](#).

Hardware Instructions

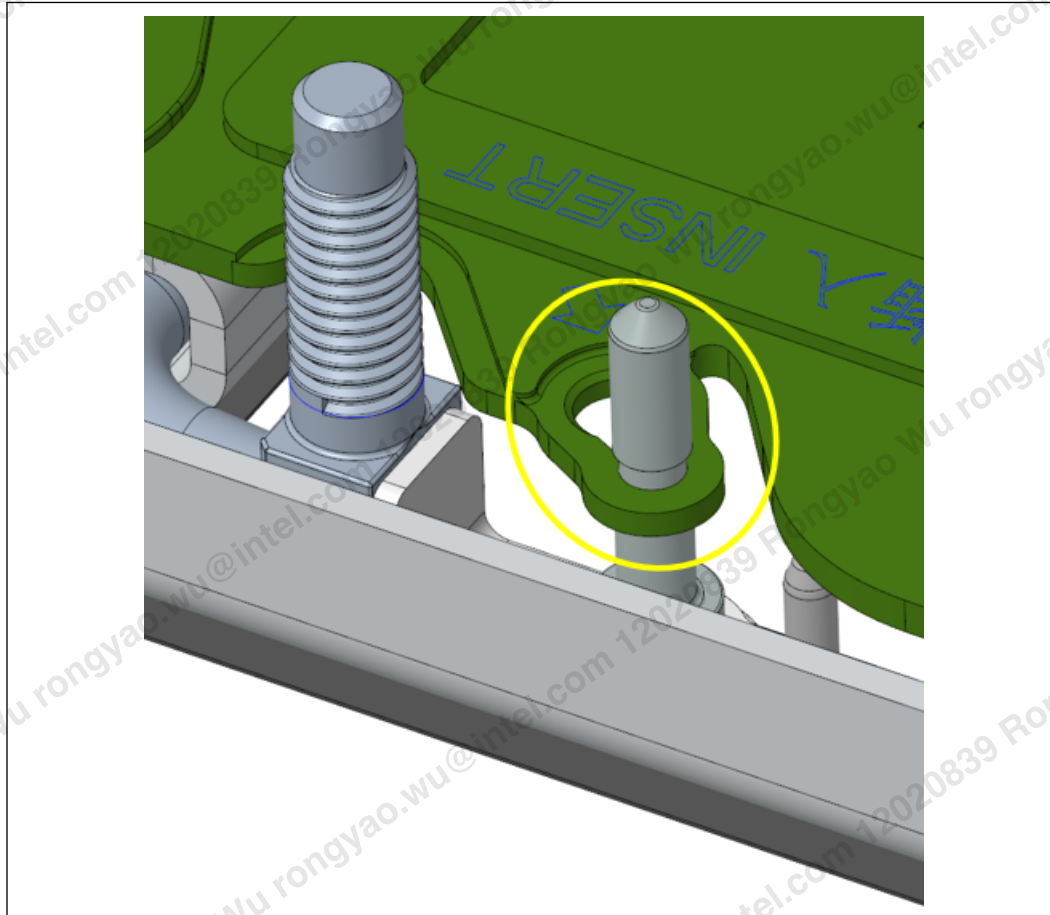
Use of all of this hardware is discussed in the *Oak Stream 16-Channel Preliminary Loading Mechanism Hardware Assembly Guidance*, document number [864097](#). This document will soon be replaced by *Manufacturing Advantage Service (MAS) document: Manufacturing with the Intel® Xeon® Platform: Oak Stream 16-Channel (LGA9324-1 / Socket-DX) Including:: Intel® Xeon® Processor (Diamond Rapids UCC-M)*, document number 872889.

Hard-tooled versions of both the Oak Stream 16-Channel socket cover and bolster plate have undergone minor refinements compared to the soft-tooled versions. These refinements improve socket cover retention during shock and overall ergonomics.

There must not be any mixing of hard- and soft-tooled socket covers and bolster plates. There are design differences between the soft- and hard-tooled versions of both parts that make them incompatible with each other:

- Soft-tooled socket covers must be used with soft-tooled bolster plates.
- Hard-tooled socket covers must be used with hard-tooled bolster plates.

The primary refinements are related to how the socket cover engages with the two bolster plate pins as shown in the following figure. Other minor refinements have also been made. There are no changes to the general socket cover assembly to socket process discussed in the previously listed Hardware Instructions.

Figure 7. Primary Areas Refined on Hard-Tooled Socket Cover and Bolster Plate

1.16.3 Tools and Debug

1.16.3.1 CScripts Release Cadence Clarification

Currently, CScripts releases for the Oak Stream platform follow a 12-week cadence. While this schedule has provided structured updates, it has limited the ability to respond quickly to customer needs and deliver timely improvements.

To enhance agility and provide faster support, the CScripts release cadence is being transitioned to a **six-week cycle**. This new approach enables more frequent updates and a more effective response to customer feedback. The next release is targeted for WW15'26, with a subsequent release planned for WW21'26 and continuing every six weeks thereafter.

Customers are expected to align with the new release schedule starting in WW15 and plan their activities accordingly.

1.17 WW11_26

New:

- [Boot Hang Caused by FBRK# Signal Overlap in Multi-Socket Systems](#)
- [WW10 BKC Release Notification](#)

Updated:

- [\(Updated WW11_26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.17.1 Documentation

1.17.1.1 (Updated WW11_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 35. Table 1 Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.01	1.1	WW1 3 '26
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.1	1.15	Mid Q2'26
<i>Diamond Rapids Processor Test Capability (Heat Map)</i>	873155	-	-	April'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	-	-	April'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i>	839071	0.9	1.0	WW9'26
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	WW13'26
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	1.0	1.1	Late Q2'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.15	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.15	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	1.0	1.1	Q2'26
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.55	0.7	WW50'26
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW9'26	-	March'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	1.02	As needed	-
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	1.2	As needed	-

1.17.2 Enabling

1.17.2.1 Boot Hang Caused by FBRK# Signal Overlap in Multi-Socket Systems

Problem Statement

In a multi-socket system, each socket asserts a fuse break signal (FBRK#) to the platform for approximately 30 microseconds after reset de-assertion (Global_Reset# for primary IMHs, S0_PWR_OK for secondary IMHs).

FBRK# is used for early boot halt in Diamond Rapids; Diamond Rapids latches the input value of FBRK# at reset de-assertion. If the platform “wired-ORs” the FBRK# signals from all sockets to support debug, there is a high likelihood that one socket will enter a false “boot halt” hang.

Implication

If one socket exits reset slightly before another—due to differences in reset signal propagation through the board logic/interconnects—within the ~30 us assertion window, the later socket(s) may incorrectly latch FBRK# to 0 and enter a permanent “Boot_Halt” state, causing a boot hang even though neither the platform nor a debug tool intentionally asserted FBRK#.

Mitigation

- Workaround #1: If the FBRK tool control is not required, remove the series resistor on the FBRK# net. This will improve 2-socket (2S) boot and cycling robustness.
- Workaround #2: If the FBRK tool control is required, insert a 40–45 microsecond (us) delay between the S0_PWR_OK signals of Socket 0 and Socket 1 (either socket can be sequenced first).

Status

The incorrectly latching issue is fixed in the Diamond Rapids UCC-M A0 silicon.

Workaround #2 will be intercepted in a future revision of the *Oak Stream 16-Channel Platform CPLD Reference Code*, document number 860022.

NOTE

In the Diamond Rapids UCC-M A0 silicon, it is not necessary to remove the workarounds if implemented.

1.17.3 Software and Firmware

1.17.3.1 WW10 BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platform.

The BKC consists of an update package and associated report, published on a regular basis, for the Johnson City reference platform. The BKC package includes UEFI firmware and onboard device drivers. These platform components have been screened through interoperability testing individually and collectively over a period of time with the aim of delivering customer confidence.

Table 36. Johnson City BKC

Ingredient	Previous	Current
BKC	WW06 Silicon	WW10
CRB BIOS	31.D94	32.D57
Unified Patch (MCU)	0x8000098C	0x80000991
continued...		

Ingredient	Previous	Current
MMC FW	0.735.0	0.758.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	ModularFIT_cmd_1.0.6269	ModularFIT_cmd_1.0.6357
Download	874180	914432
Status	Released	Released

NOTE

WW10 BKC includes support for silicon and Intel® Simics® software. For information on silicon availability or Intel SIMICs software support, contact your Intel representative.

NOTE

This release provides support for 1S only. The 2S BKC validation is blocked due to stability issues.

NOTE

From this BKC release onwards, there is a dependency on the BMC (from oks-2026.07.1) and BIOS (from 32.D44) due to the eSPI MMBI base address change. Both ingredients should be upgraded once together; otherwise, the BMC-BIOS IPMI-over-MMBI communication could break and increase the BIOS boot time. Refer to article [Support for 64-MB BIOS-BMC MMBI MMIO Windows](#) for more information.

1.18**WW10_26**

New:

- [Oak Stream 16-Channel Socket Cover Retention Improvement Recommendations](#)
- [Oak Stream Platform EINJ Memory Fatal Error Change](#)
- [Q1'26 Oak Stream Quarterly Technical Review \(QTR\) Online Session](#)

Updated:

- [\(Updated WW10_26\) Oak Stream Key Tools Availability and Forecast](#)

1.18.1**Enabling****1.18.1.1****Oak Stream 16-Channel Socket Cover Retention Improvement Recommendations**

Intel has observed isolated cases where an Oak Stream 16-Channel socket cover can detach during dynamic events. To mitigate this risk happening on a motherboard that has a socket with no pick-and-place capacitor over the contact field, it is recommended to implement one of the following socket cover retention improvement options:

1. Flange Nuts

2. Elastic Bands
3. Foam (only for use in a chassis)

One of these options should be done whether the motherboard is an uninstalled part or installed in a system.

The addition of any of these options is done after the socket cover is assembled and inspected as specified in the *Manufacturing Advantage Service (MAS) document: Manufacturing with the Intel Xeon Platform: Oak Stream 16-Channel (LGA9324-1 / Socket-DX) Including: Intel Xeon Processor (Diamond Rapids UCC-M)*, document number 872889.

NOTE

This document replaces the *Oak Stream 16-Channel Preliminary Loading Mechanism Hardware Assembly Guidance*, document number 864097.

These socket cover retention options are temporary steps for the currently available soft-tooled socket covers and bolster plates. Refinements to both of these parts have been made to further improve socket cover retention. See the Hard-Tooled Socket Cover Retention Improvements section at the end of this article for further detail.

Option # 1 – Flange Nuts

Overview: Add two flange nuts onto the bolster plate studs for additional socket cover retention.

Details:

1. Install and inspect the socket cover.
2. Drive one steel serrated flange nut with M5 x 0.8 mm thread onto each of the two bolster plate load studs diagonally opposite each other, that are nearest the socket cover grip holes as shown in Figure 1.
 - Use a deep 8 mm hexdriver.
 - Driver torque can be anywhere in the 4 – 12 in-lb range.
3. Remove the socket cover in the reverse order.

Space considerations:

- Due to size and placement tolerances of the components within the flange nut installation area, the edges of the flange nut could extend slightly outside of the loading mechanism KOZ.
- The outer surface of the 8 mm hex driver could potentially contact the DIMM(s) immediately adjacent to the loading mechanism during the installation.
- DIMM(s) immediately adjacent to the loading mechanism may need to be removed when installing the flange nut. This need is highly dependent upon the system design, including DIMM spacing to the loading mechanism KOZ and the DIMM thermal solution.
- Example flange nuts: 18-8 stainless steel serrated flange nut, M5 x 0.8 mm thread, McMaster-Carr* part number [97400A858](#).

NOTE

Supplier contact information is provided for convenience. Intel does not formally endorse any products or suppliers.

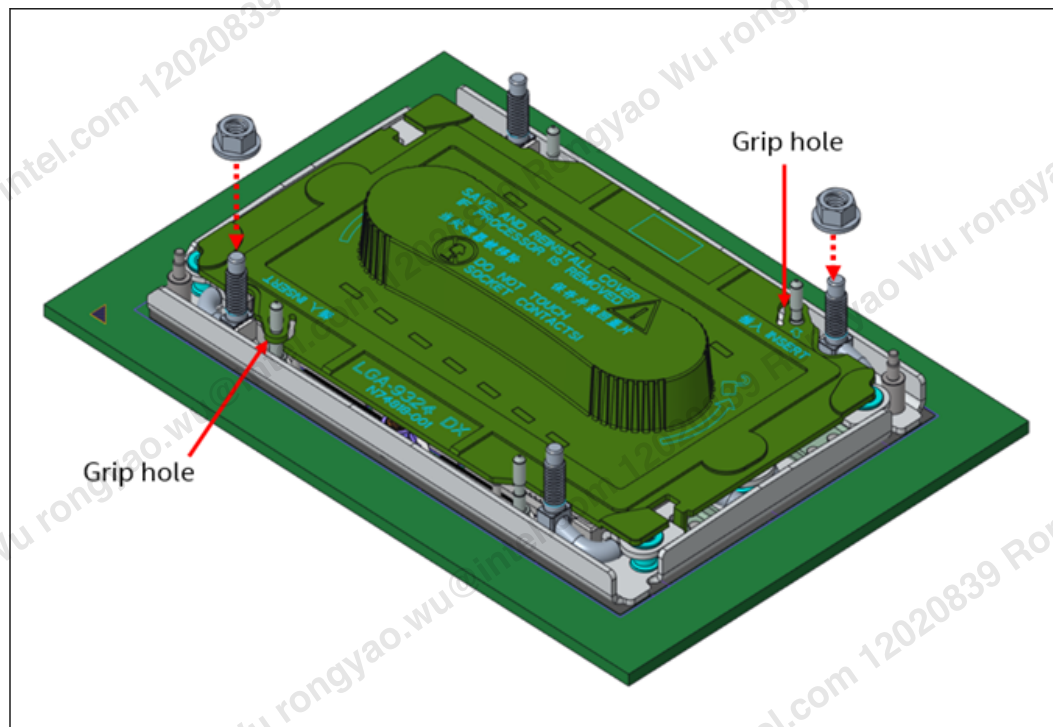
Figure 8. Flange Nut Installation Locations

Figure 9. Example of Socket Cover with Flange Nuts Installed



Option # 2 – Elastic Bands

Overview: Add two elastic bands to the bolster plate studs in an X-pattern over the socket cover for additional retention.

Details:

1. Install and inspect the socket cover.
2. Loop two elastic bands between diagonally opposite load posts to form an X-pattern as shown in Figure 3.
3. Elastic bands must be placed on load posts such that the loop end is touching the load post base as shown in Figures 4 and 5.

Additional considerations:

- Elastic bands can degrade over time and are not recommended to be stored on the loading mechanism for an extended time.
- Once boards are received and no longer being shipped, it is recommended to discard the elastic bands.
- It is recommended to use new elastic bands each time a motherboard is to be shipped.

Example elastic bands:

- Size Number: #62 bands
- Nominal dimension: 2 ½ X ¼ X 0.040"
- Composition: 55% latex rubber, 45% clay mixture
- Performance: 700% elongation

- Example: rubber bands - 2 1/2 x 1/4", #62, Uline* part number [S-13706BL](#)

Figure 10. Example of Elastic Bands X-Pattern

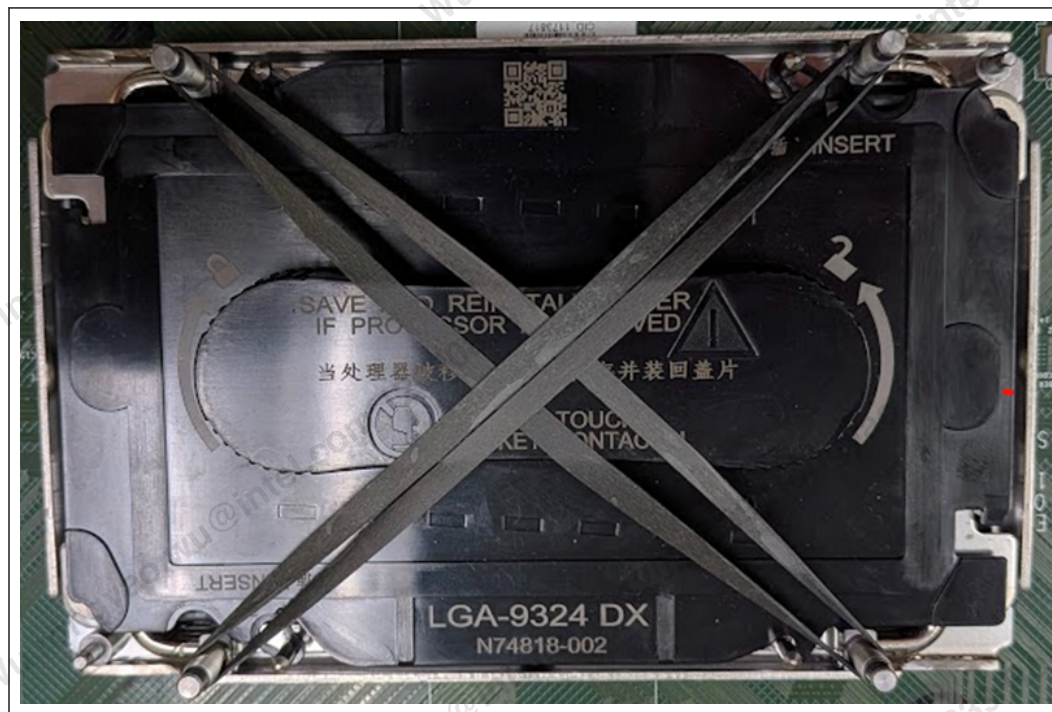


Figure 11. Elastic Bands Properly Placed on a Bolster Plate Load Stud

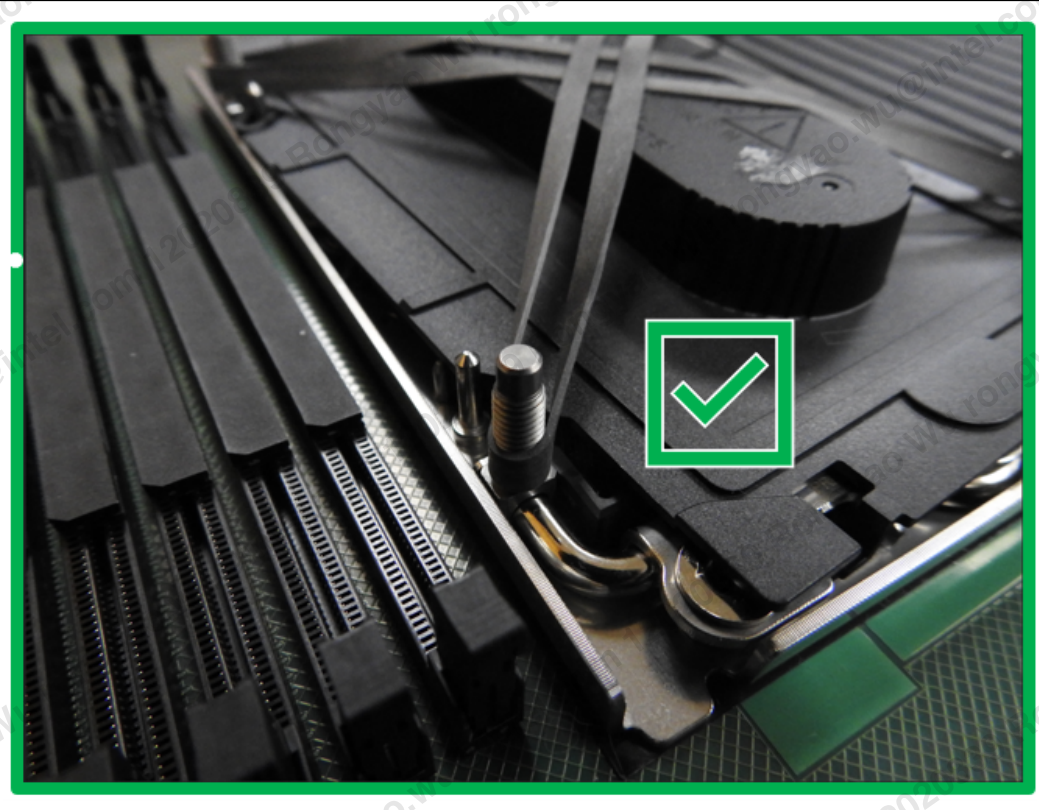
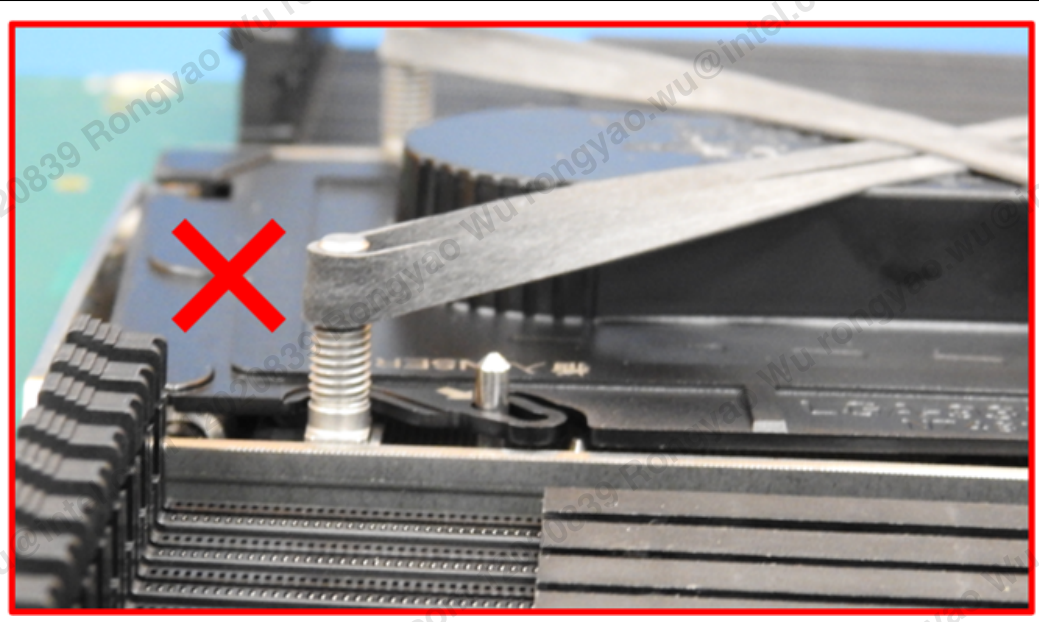


Figure 12. Elastic Bands Improperly Placed on a Bolster Plate Load Stud



Option # 3 - Foam

Overview: In a chassis, add a block of ESD-safe foam between the duct and socket cover for additional socket cover retention.

Details:

1. Install and inspect the socket cover.
2. Integrate foam into the system over the socket cover:
 - The specific size of the foam is dependent on the system. It should be tall enough that it is contacted by the duct and slightly compressed. A rule of thumb is for the total foam thickness to be approximately equal to the distance between the top of the motherboard and the bottom of the duct over the socket.
 - The foam should be installed in a manner such that it will not move during shipment, such as the use of double-sided tape between the duct and the foam.
 - A version of this could be feasible in a closed chassis that does not have a duct.
3. Remove the socket cover in the reverse order. Depending on the system, the could remain installed in the duct.

Example: The following is an example used on a Johnson City 3U reference platform (RP):

- Two 4.25 x 3" blocks are cut from antistatic super-cushioning foam, 1.5", McMaster-Carr part number [87035K47](#).
- First foam block is installed into the duct, bonded by double-sided tape between the primary foam surface and the duct.
- Second foam block is installed on to the first foam block, with double-sided tape between them.
- The duct is assembled into the system.
- A slight downward force is used in some cases to compress the foam when tightening the duct retention nuts.

Figure 13. Example Length and Width of Foam for Johnson City 3U RP



Figure 14. Example Length and Width of Stacked Foam in a Johnson City 3U RP Duct

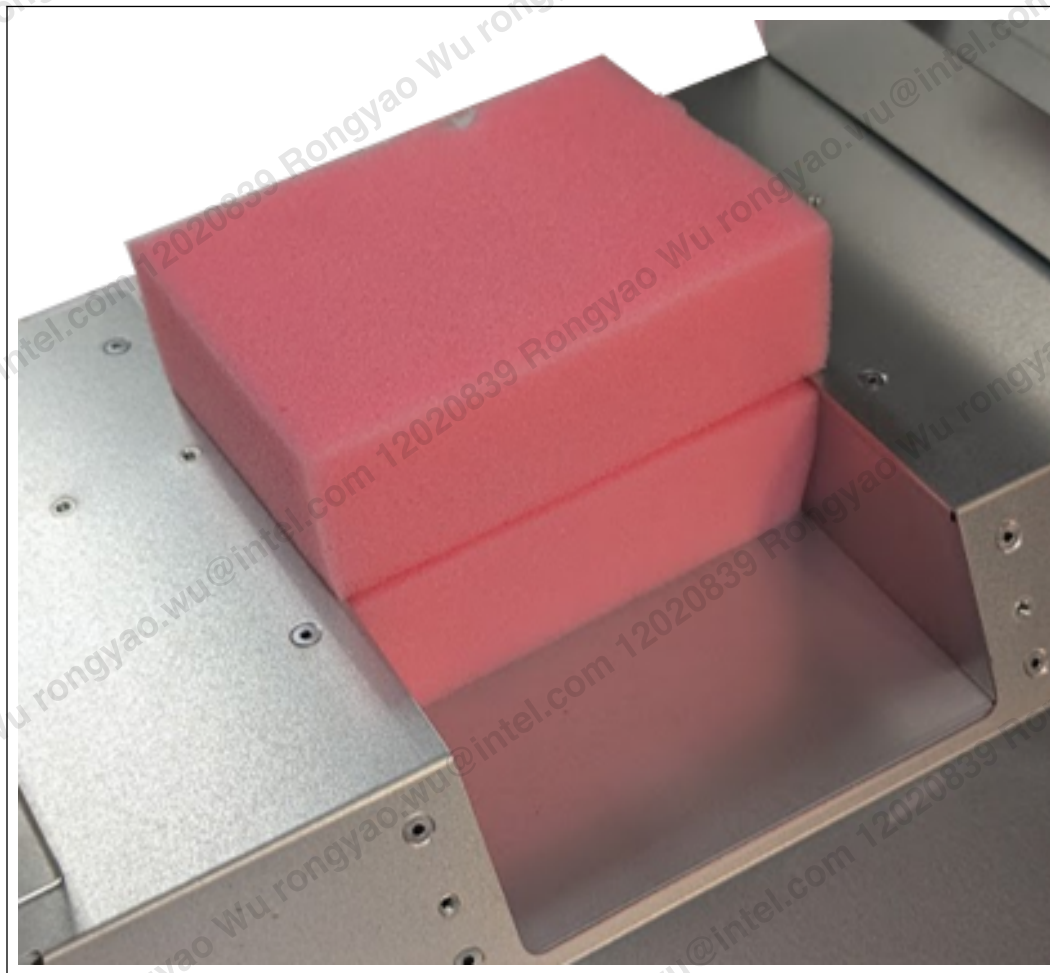
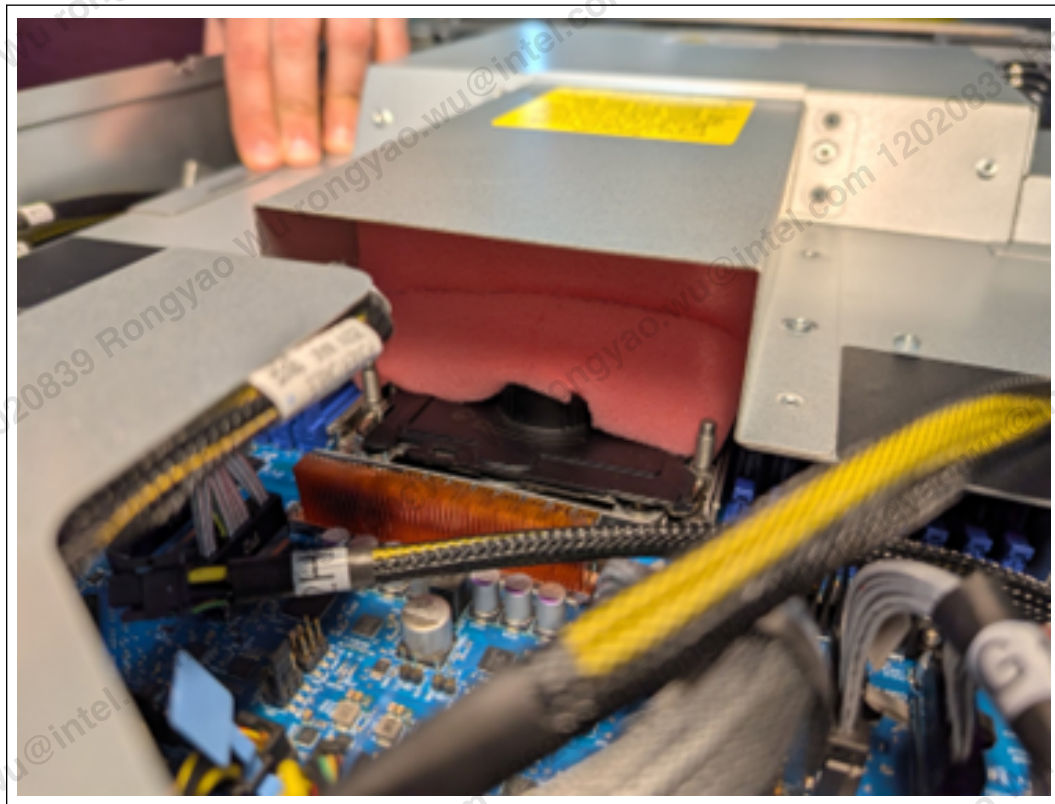


Figure 15. Example of Properly Installed Foam in a Johnson City 3U RP Duct (1 of 2)



Figure 16. Example of Properly Installed Foam in a Johnson City 3U RP Duct (2 of 2)



Hard-Tooled Socket Cover Retention Improvements

Hard-tooled versions of both the Oak Stream 16-Channel socket cover and bolster plate have undergone minor refinements compared to the soft-tooled versions. These refinements improve socket cover retention during shock and overall ergonomics.

Design differences between the soft-tooled and hard-tooled versions of both parts are not compatible with the other. No mixing.

- Soft-tooled socket covers must be used with soft-tooled bolster plates.
- Hard-tooled socket covers must be used with hard-tooled bolster plates.

Hard-tooled narrow backplates (POR), bolster plates, socket cover, and heatsink hardware continue to be targeted for initial ordering approval in April 2026, with updated part numbers provided by the end of March 2026.

These will be hard-tooled prototypes that have undergone initial viability testing by Intel. With these prototypes, CPUs and thermal solutions are not to be shipped installed in a system. Initial reliability testing may be done with these parts; however, the parts are subject to change until full validation is complete.

Full validation of these ingredients is targeted for mid-2H 2026.

1.18.1.2 Oak Stream Platform EINJ Memory Fatal Error Change

Starting with the Oak Stream platform, the Intel BKC SMM handler's EINJ memory fatal error injection flow will be updated to more closely emulate real fault conditions. In this new approach, the integrated Memory Controller (iMC) will log a "link fail" error. This type of error is likely to trigger nested machine checks, which would be an escalation of the MCERR to an IERR.

Refer to the Diamond Rapids Best Known Configuration (BKC) for notifications on new error injection implementation.

This new injection method replaces the legacy approach, where the PCC=1 condition was artificially created by placing the iMC in non-poison mode and then consuming an uncorrectable ECC error.

NOTE

The Oak Stream platform does not support dynamic modification of the "MCG_CONTAIN" MSR.

The new error injection method will be documented in a future revision of the *Diamond Rapids Server Platform BIOS Developer's Guide*, document number [839071](#).

1.18.1.3 Q1'26 Oak Stream Quarterly Technical Review (QTR) Online Session

Intel is excited to invite you to the Q1'26 Oak Stream Quarterly Technical Review (QTR) online session. This session will allow customers to find the latest enabling updates for the next generation Oak Stream platform.

The online session will be hosted on April 6, 2026 at 4:00-5:30 PM Pacific Standard Time (PST).

To join the online session, go to:

[Join OKS Q1'26 QTR Teams Live Event](#).

1.18.2 Software and Firmware

1.18.2.1 (Updated WW10_26) Oak Stream Key Tools Availability and Forecast

NOTE

Updated information is highlighted in bold text.

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes the MIPI 60 to Intel ITP adapter.
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach for support. Two streams tracing supported:
 - LA-3907 preprocessor for Intel® Processor Trace (Intel® PT) autofocus MIPI 60-P (supports four trace ports)
 - LA-3892 converged MIPI 60-Q/-P or MIPI 60-Cv2/-P to MIPI 60
- ASSET* Arium* - Contact ASSET Arium for support.
- Compact Debug Port (CDP) + MIPI 10 adapter for I3C*
 - Type-C connection: Voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

NOTE

Supplier contact information is provided for convenience. Intel does not formally endorse any products or suppliers.

Intel® System Bring-up Toolkit NDA includes the following tools:

- VTune™ tools
- Intel® SoC Watch
- Intel® System Debugger
 - Includes token generation for Vendor Authorize Boot (VAB) / Vendor Authorize Debug (VAD).

NOTE

The Intel® System Bring-up Toolkit NDA requires registration. See *How to Obtain Intel® System Bring-up Toolkit NDA (Formerly Intel® System Studio NDA) User Guide*, document number [616675](#), and *Intel® System Bring-up Toolkit NDA (formerly Intel® System Studio NDA) Registration Form New Users*, document number [631579](#), for details.

Table 37. Oak Stream Key Tools

Tool Type	Title	RDC or DIT	Availability	Revision
Debug	ASSET* InterTech Arium* Note: Contact ASSET for support.	-	Q2'26	-
Debug	Lauterbach*	-	Available	N/A
Debug	Intel® System Bring-up Toolkit NDA Note: Includes Intel® System Debugger NDA, Intel VTune™ Profiler NDA and Intel® SoC Watch NDA.	634166	Available	Latest
Debug	Intel® In-Target Probe (Intel® ITP) eXtended Debug Port (XDP)	940990	Available	N/A
Debug	Intel Triage Tool (Crashdump Summarizer) Note: Available through CScripts.	N/A	Available	0.86.1.2
Debug	Server Security Toolkit Note: Server Security Toolkit includes Intel® Total Memory Encryption – Multi-Key (Intel® TME-MK), Intel® Trusted Execution Technology (Intel® TXT), BIOS TPM Provisioning with Intel® Platform Trust Technology (Intel® PTT) tools, and CBnT (Converged Intel® Boot Guard and Intel® TXT).	630398	Available	1.54
Debug	Intel Remote FPGA Access Tool (RFAT) for Server Platforms	871932	Available	1.0
Debug	Intel® Silicon View Technology (Intel® SVT) Closed Chassis Adapter	921521	Available	N/A
Debug	Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C*	99CJRJ	Available	N/A
Debug	CScripts for Oak Stream Platforms Note: Refer to the release notes for details.	821387	Available	2608.2000
Debug	ASSET Close Chassis Controller (CCC) Note: Contact ASSET for support.	-	Q2'26	-
continued...				

Tool Type	Title	RDC or DIT	Availability	Revision
Debug	Intel® I3C* USB Type-C Level Shift Adapter	99CN2C	Available	N/A
Debug	SelfTest 7 - Windows* x64	524152	Available	177
Manufacturing and Fleet Health	System Health Check (SHC) kit Note: Includes Test Seed Loader (TSL), Sandstone, and iMunch.	-	Q1'26	-
Manufacturing and Fleet Health	Intel® In-field Scan	-	Q1'26	-
Manufacturing and Fleet Health	Intel® At-Scale Debug (Intel® ASD) BMC-JTAG* Software Guide	594948	Available	1.6.5
Manufacturing and Fleet Health	Intel® At-Scale Debug (Intel® ASD)	-	Alpha	-
Manufacturing and Fleet Health	Autonomous Crashdump (ACD) Note: Autonomous Crash Dump (ACD) is available by license agreement.	-	Q1'26	-
Manufacturing and Fleet Health	Intel® System Stress and Memory Diagnostics (Intel® SSMD) for Linux*	818881	Available	2025.11.10.4ed3
Manufacturing and Fleet Health	Memory Latency Checker (MLC)	736633	Q1'26	3.12
Signal Integrity	Clock Jitter Tool (CJT)	803838	Available	6.0.4
Signal Integrity	Integrated Channel Analysis Tool (ICAT) Suite for Windows	782966	Available	6.200.50
Signal Integrity	Intel® Platform Clock Jitter Analysis Tool (CJAT) for Windows	619648	Available	6.0.4
Signal Integrity	PCI Express* (PCIe*) and I Intel® Ultra Path Interconnect (Intel® UPI) Load Board Solution Set for 16-Channel Platform	DMRUCCPLBULB	Available	N/A
Signal Integrity	PCIe Load Board Solution Set for 16-Channel Platform	DMRUCCPLB	Available	N/A
Signal Integrity	Intel® Passive Component Checker Tool (Intel® PCC Tool)	600218	Available	1.8
Thermal / Power Delivery	Blue Interposer for Gen5 VRTT for 16-Channel Platform	99D5W2	Available	N/A
continued...				

Tool Type	Title	RDC or DIT	Availability	Revision
Thermal / Power Delivery	Almondville Thermal Mechanical Test Vehicle (TMTV) Application Note	852568	Available	1.0
Thermal / Power Delivery	Almondville Thermal Test Vehicle (TTV) for 16-Channel Platform	-	Available	-
Thermal / Power Delivery	Gen5 VR Test Tool (VRTT) for Windows	99CJ9X	Available	N/A
Thermal / Power Delivery	Red Interposer for Gen5 VRTT for 16-Channel Platform	99D5W1	Available	N/A
Thermal / Power Delivery	VRTT Mechanical Interposer Retention Mechanism Kit for 16-Channel Platform	99CN1R	Available	N/A
Validation	Intel® IO Margin Tool (IOMT)	871016	Q2'26	-
Validation	RAS Automation ToolKit	633201	Q2'26	2.5.0
Validation	PEI (PCIe Error Injector Card) Collaterals	610667	Available	5.1
Validation	PEI 5.0 (PCIe 5.0 Error Injection) + CXL 2.x Card	99CN2D	Available	N/A
Validation	Intel® Simics tool Note: Contact your Intel representative for availability.	-	Available	N/A
Validation	PCIe* PEI 6.0 Card	-	Q4'26	-
Validation	PCIe* 5.0 Error Injection (PEI) Card	99C3FA	Available	N/A
Validation	PCI Express* (PCIe*) Lane Margin Tool (LMT) for PCIe* 4.0/5.0/6.0	639433	Available	0.90
Validation	Memory Error Injection (MEI) DDR5 Card for Windows	99CDV1	Available	N/A
Validation	Memory Assistant for Intel® Xeon® Platforms	620497	Q1'26	2.8.27
Validation	MEI DDR5 Lite Card 8000 MT/s for Windows	99D5X2	Available	N/A
Validation	Intel® Structural Test Plus (Intel® STP)	-	Available	N/A
continued...				

Tool Type	Title	RDC or DIT	Availability	Revision
	Note: Contact a local representative to enable Intel® STP for customer designs.			
Validation	Stitched Rank Margining Tool (RMT)	-	Q2'26	-
Validation	Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card	-	Available	-

1.19 WW09_26

New:

- [Diamond Rapids Processor Architecture Specification Release Notification](#)
- [Oak Stream Platform Dashboard Release Notification](#)
- [Oak Stream Platform RAS Deep Dive Technical Training Release Notification](#)
- [Support for 64-MB BIOS-BMC MMBI MMIO Windows](#)

Updated:

- [\(Updated WW9_26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.19.1 Documentation

1.19.1.1 (Updated WW9_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

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Diamond Rapids Processor Architecture Specification	792359	1. 1	1. 15	Mid Q2'26
<i>continued...</i>				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
Diamond Rapids Processor Test Capability (Heat Map)	873155	-	-	April'26
Diamond Rapids Processor Sightings Report	862869	-	-	April '26
Diamond Rapids Server Platform BIOS Developer's Guide	839071	0.9	1.0	WW9'26
Oak Stream 16-Channel 2S in 19-Inch Design Feasibility	843581	1.7	As needed	-
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Oak Stream 16-Channel Platform CPLD Reference Code	860022	1.0	As needed	-
Oak Stream 16-Channel Platform Design Guide	788196	0.9	1.0	WW13'26
Oak Stream 16-Channel Platform Electrical Specification	787006	1.0	1.1	Late Q2'26
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	1.15	As needed	-
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258	1.15	As needed	-
Oak Stream 16-Channel Platform Pin List	636260	1.0	As needed	-
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Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	1.0	1.1	Q2'26
continued...				

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Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard	863599	WW9'26	-	March' 26
Oak Stream Platforms System Boards Design Archive	821025	1.02	As needed	-
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	1.2	As needed	-

1.19.1.2 Diamond Rapids Processor Architecture Specification Release Notification

The *Diamond Rapids Processor Architecture Specification*, document number [792359](#), Revision 1.1, is now available in the Resource and Documentation Center (RDC).

Summary of the Revision 1.1 changes are:

- **Overview, Features, and Topologies:** Added LLC details to the Core Building Block (CBB) subsection; added CBB to FIO and FIO to memory behavioral description in the Hub and Spoke subsection.
- **Flexible I/O (FIO):**
 - *PCI Express* (PCIe*)*: Added "Intel® Volume Management Device (Intel® VMD)" sub-chapter, "Interrupts" sub-chapter, "Supported Degraded Modes and Lane Reversal" table, and Configuration Request Retry Status (RRS) Handling.
 - *Compute Express Link* (CXL*)*: Added "Memory Pooling" and "Intel Validation Scope" sections.
- **Virtualization:** Added feature support and IOMMU details.
- **Accelerators:**
 - *Intel® QuickAssist Technology (Intel® QAT)*: Updated Intel® QAT Upstream Traffic subsection.
 - Added Intel Single Root I/O Virtualization (SR-IOV) support note.
- **Intel® Out-of-Band Management Services Module (Intel® OOBM Services Module):** Added Crashdump return data subsection; updated platform manageability overview and removed MCTP over PCIe support for system management APIs.
- **Power Controller Unit (PCU):** Added Adaptive Thermal Monitor and On-Demand Mode to Processor Temperature subsections.
- **Reliability, Availability, and Serviceability (RAS):** Updated the Supported RAS Features and ERR[2:0] subsections.

Refer to the revision history for the complete list of changes.

1.19.1.3 Oak Stream Platform Dashboard Release Notification

The *Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard*, document number [863599](#), Revision WW09 2026, is now available in the RDC.

1.19.1.4 Oak Stream Platform RAS Deep Dive Technical Training Release Notification

The *Oak Stream Platform RAS Deep Dive Technical Training*, document number [824347](#), Revision 0.9, is now available in the RDC.

Overview of the Revision 0.9 changes are:

- Added "Memory Mirroring" section.
- Added "DDR5 Link Errors" section.
- Added "MRDIMM RAS" section.
- Added "Runtime S3M RAS" section.
- Added "Platform Runtime Mechanism" section.
- Updated "Diamond Rapids Processor RAS Features" section.
- Updated "Compute Express Link* (CXL*) 3.1 Module" section.
- Updated "RAS Offload" section.
- Updated "Intel BKC Implementation for Handling Storm of Errors" section.
- Updated "PCI Express* (PCIe*) Legacy" section.
- Updated "Shutdown Suppression" section.
- Updated "Oak Stream RAS Collaterals" section.

1.19.2 Enabling

1.19.2.1 Support for 64-MB BIOS-BMC MMBI MMIO Windows

To support customers who want to implement ASPEED* AST2700 in their designs, Intel is adding support for a larger BIOS-BMC MMBI window.

NOTE

Intel has added support for larger BIOS-BMC MMBI windows in Oak Stream platforms to aid customers who want to implement ASPEED's AST2700/AST2750 series BMCs. While AST2600 remains the Plan of Record (POR) for the Oak Stream platform, this enhancement enables customers seeking to migrate to ASPEED's newer BMC solutions.

Supplier contact information is provided for convenience. Intel does not formally endorse any products or suppliers.

Problem Statement

ASPEED* introduced a new series of BMC chip solutions that require a significantly larger 64 MB MMBI MMIO window; this series does not have a reconfiguration option present in the previous generation. To adopt the new BMC chip solution in the Oak Stream platform generation, Intel had to increase the currently supported 8 MB

window to 64 MB. This also required system memory map adjustments on the CPU side and corresponding BMC FW stack modification, for the BIOS-BMC communication channel to function.

Root Cause

The originally planned 8 MB MMBI MMIO window compatible with AST2600 is not applicable to the new AST2700 series. A CPU and BMC re-alignment to the larger fixed MMIO window is required.

Resolution

To have a coherent solution for both AST2600 and AST2700 in the Oak Stream platform, Intel has:

- Expanded MMBI window support from 8 MB to 64 MB.
Adjusted the system memory map on the CPU side; the MMBI window static base address is mapped to ESPI Chip Select 0 (CS#0).

NOTE

The second 64 MB MMBI window via eSPI CS#1 is not enabled to minimize excessive memory consumption. Intel has not identified any use cases that require the second 64 MB MMBI window. Support for the second MMBI can be considered on an as needed basis.

- Modified the BMC firmware stack to maintain proper BIOS-BMC communication functionality.

These enhancements ensure seamless BIOS-BMC communication while supporting customers' transition to ASPEED's latest BMC technology.

Implementation Details

The platform solution involves the following system components and settings:

- S3M FW handles flexible window size: 64 KB (legacy on the Birch Stream platform), 8 MB (Oak Stream platform), and 64 MB (a new requirement in the Oak Stream platform) are set through platform soft straps and further configured by the S3M ESPI ESPI_LGMR register.
- The BMC follows the new paradigm by changing the eSPI Source Target Address.
The BMC FW update needs to be coordinated with the BIOS update on the platform.
- Intel® Modular Flash Image Tool (Intel® Modular FIT) is required to configure the MMBI window size. The default MMBI window size is 8 MB.
- The window setting flexibility allows for maintaining backward compatibility with the previous generation and with the current Diamond Rapids plan, which means it supports both ASPEED* generations (AST2600 and AST27xx), and all other BMC vendors.
- The BIOS supports the adjusted MMBI window at the static new location in MMIO_L by programming the ESPI_LGMR register: 0xEC000000 - 0xFFFFFFFF (MMBI via eSPI CS#0).

Contact your Intel representative if you have concerns with the new system address map.

Status

The new 64 MB MMBI window solution will be integrated into the BKC WW10 release. Refer to the BKC release notes for details on updated ingredients.

NOTE

Ensure the update of the BIOS and BMC are coordinated appropriately for both BMC versions, AST2600 and AST2700, preferably in a single cycle for the system to work as expected. In some scenarios, a hang may occur if the BIOS and BMC are not synchronized.

1.20 WW08_26

New:

- JTAG Chain Communication Enhancement for Non-Legacy Socket Implementation
- Oak Stream Collaterals Appendix

Updated:

- (Updated WW8_26) Oak Stream Key Collaterals Availability and Forecast

1.20.1 Documentation

1.20.1.1 (Updated WW8_26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

A list of Oak Stream collaterals is also available in the [Appendix](#) section.

Table 39. Oak Stream Key Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.01	1.1	WW11'26
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.0	1.1	WW9'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	-	1.0	WW9'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i>	839071	0.9	1.0	WW9'26
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	WW13'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	1.0	1.1	Q2'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.15	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.15	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	1.0	1.1	Q2'26
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.55	0.7	WW9'26
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Revision ETA
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW51'2025	WW9'26	WW9'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	1.02	As needed	-
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	1.2	As needed	-

1.20.2 Enabling

1.20.2.1 JTAG Chain Communication Enhancement for Non-Legacy Socket Implementation

To ensure proper JTAG chain communication with the Non-Legacy (NL) socket, the JTAG chain should only be active when the socket is powered.

Problem Statement

In 2-Socket (2S) topology configurations, the NL socket fails to be detected, resulting in no available device for unlocking. Although the system reaches the Fuse Break (FBRK) halt state (boots to S5), the JTAG chain is compromised. This issue is specific to 2S platforms and has not been reproduced on 1S platform designs.

Root Cause

The JTAG chain attempts to access all devices (both legacy and NL sockets) while the system is in an S5 state. However, the NL socket is not powered in S5 and only receives power when the system reaches an S0 state, which is expected behavior. This power state mismatch causes the JTAG chain to break when trying to communicate with the unpowered NL socket.

Resolution

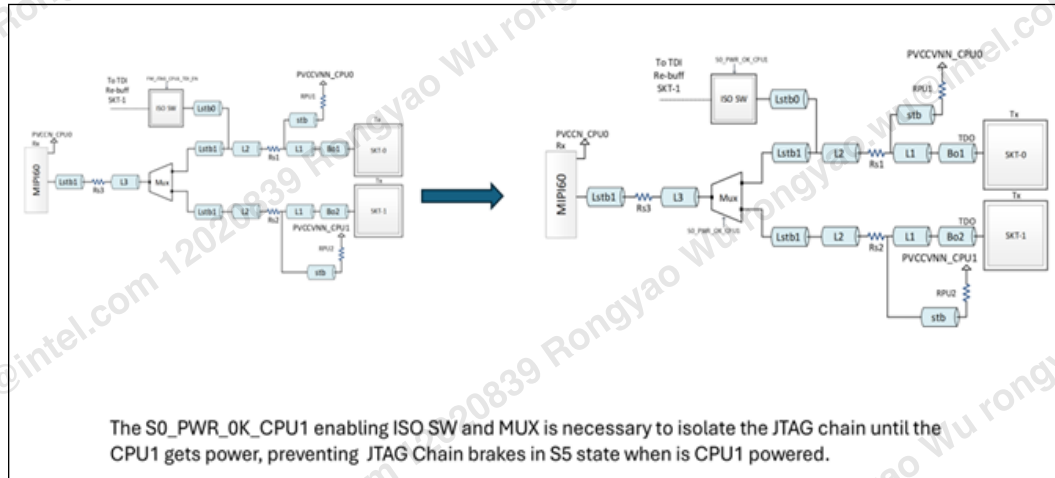
The solution utilizes the S0_PWROK signal from CPU 1, translated to 3.3V, to control the switches and buffers. This ensures that JTAG chain operations occur exclusively when the NL socket is powered. Depending on 2:1 Mux and buffers control signal logic (active high or active low), an inversion of the signal may be required as well.

Implementation Details

- **MDP2 systems:** Due to the absence of available 3.3V spare pins in the PLD, the S0_PWR_OK_CPU1 signal is translated to 3.3V to directly control the Mux/Isolator buffer.
- **2S systems with spare 3.3V pins:** These systems can route the S0_PWR_OK_CPU1 logic to a dedicated 3.3V pin for Mux/Isolator control, eliminating the need for voltage translation.

This approach ensures reliable JTAG chain operation across different system configurations while maintaining compliance with power state requirements.

Figure 17. JTAG Chain Communication Enhancement for Non-Legacy Socket



Status

The solution has been integrated into the latest MDP2 board revision. This solution will be reflected in upcoming revisions of:

- Oak Stream 16-Channel Platform Design Guide, document number [788196](#)
- Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive, document number [828483](#)

1.21 WW07_26

New:

- Oak Stream Platform Security Collaterals Release Notification
- Update to Intel® Xeon® Processors Frequency Specifications
- WW06 Silicon BKC Release Notification

1.21.1 Documentation

1.21.1.1 Oak Stream Platform Security Collaterals Release Notification

The *Diamond Rapids Vendor Authorized Boot (VAB) and Debug (VAD) Design Specification*, document number [851899](#), revision 0.95, is now available in the Resource and Documentation Center (RDC).

Overview of the 0.95 revision:

- Initial Release. The design specification provides details on the architecture of these features and their structure details.

The *Oak Stream Platform Tokens, Vendor Authorized Debug (VAD), and Intel® Vendor Authorized Boot Integration and Validation Guide*, document number [863386](#), revision 0.8, is now available in the RDC.

Overview of the 0.8 revision:

- Initial Release. The integration and validation guide provides a general overview of features and feature enabling information for customers.

1.21.2 Enabling

1.21.2.1 Update to Intel® Xeon® Processors Frequency Specifications

Intel is simplifying frequency specifications for its Intel® Xeon® processors. The simplification applies to:

- Turbo frequency
 - **All Core Turbo Frequency:** It is an opportunistic maximum turbo frequency the CPU can attain while all cores are active (not in C6/disable).
 - **Max Turbo Frequency:** It is an opportunistic maximum turbo frequency the CPU can attain when only a certain number of cores are active while remaining cores are in the C6/disabled state.
- Base frequency
 - **Base frequency:** It is the deterministic frequency that a defined workload should be able to sustain at TDP.

NOTE

Turbo frequency is considered as opportunistic, meaning the higher turbo frequency is only attainable if relevant conditions like thermal, electrical, and headroom are met.

Turbo Frequency Specification Update

- Starting with the 4th Gen Intel® Xeon® processor Scalable family, the turbo (opportunistic) operating frequency for any given instruction is determined dynamically at runtime.
- Turbo frequencies are no longer statically associated with specific instruction classes. Consequently, an instruction such as Intel® Advanced Vector Extensions 2 (Intel® AVX2) may operate at a turbo frequency higher than the published Intel® AVX2 all core or maximum turbo specifications.
- Therefore, starting with Diamond Rapids, Intel will publish only the all core and maximum turbo frequencies associated with the lowest CDYN value, corresponding to the highest turbo frequency achievable by the processor.

Base Frequency Specification Update

- CPU performance is tied to the turbo frequency, and base frequency is not relevant for performance KPIs.
- Starting with Diamond Rapids, Intel will simplify base frequency corresponding to SIR heavy workloads for "Server" mode.
 - **Important:** There is no change to the 'Compute' or 'Network' mode.
- Intel will not publish other instruction sets [Intel® AVX2, Intel® Advanced Vector Extensions 512 (Intel® AVX-512), Intel® Advanced Matrix Extensions (Intel® AMX)] specific base frequency.

The following figures illustrate an example of the previous frequency specifications from an Intel® Xeon® 6 processor Dear Customer Letter (DCL) example and examples of the new format of the frequency specifications from future Diamond Rapids DCLs.

Figure 18. Intel® Xeon® 6 Processor DCL Turbo Frequencies per Number of Active Cores

Table 13. Intel® SST-PP Config 0 Turbo Frequencies per Number of Active Cores					
Number of Active Cores	Turbo Frequency	Intel® AVX2	Intel® AVX-512	Intel® AVX-512 [heavy]	Intel® AMX
0 to 60	3.9	3.9	3.7	3.6	3
61 to 72	3.8	3.6	3.4	3.3	2.7
73 to 84	3.7	3.4	3.2	3.1	2.6
85 to 90	3.6	3.3	3.1	3	2.5
91 to 102	3.5	3.1	2.8	2.6	2.2
103 to 108	3.4	3.1	2.8	2.6	2.1
109 to 114	3.3	3	2.7	2.5	2.1
115 to 120	3.3	3	2.7	2.5	2.1

Figure 19. Diamond Rapids DCL Turbo Frequencies per Number of Active Cores

Number of Active Cores	Turbo Frequency
0 to 60	3.9
61 to 72	3.8
73 to 84	3.7
85 to 90	3.6
91 to 102	3.5
103 to 108	3.4
109 to 114	3.3
115 to 120	3.3

Figure 20. Intel® Xeon® 6 Processor DCL Frequencies

Table 4. Intel® Advanced Vector Extensions (Intel® AVX) Frequencies			
Intel® SST-PP Config	Base Frequency	All Core Turbo Frequency	Max Turbo Frequency
0	2.2 GHz	3.3 GHz	3.9 GHz

Table 5. Intel® Advanced Vector Extensions 2 (Intel® AVX2) Frequencies			
Intel® SST-PP Config	Base Frequency	All Core Turbo Frequency	Max Turbo Frequency
0	2.1 GHz	3 GHz	3.9 GHz

Table 6. Intel® Advanced Vector Extensions 512 (Intel® AVX-512) Frequencies			
Intel® SST-PP Config	Base Frequency	All Core Turbo Frequency	Max Turbo Frequency
0	1.8 GHz	2.7 GHz	3.7 GHz

Table 7. Intel® Advanced Vector Extensions 512 Heavy Usages (Intel® AVX-512 [heavy]) Frequencies		
Intel® SST-PP Config	All Core Turbo Frequency	Max Turbo Frequency
0	2.5 GHz	3.6 GHz

Table 8. Intel® Advanced Matrix Extensions (Intel® AMX) Frequencies			
Intel® SST-PP Config	Base Frequency	All Core Turbo Frequency	Max Turbo Frequency
0	1.6 GHz	2.1 GHz	3 GHz

Figure 21. Diamond Rapids DCL Frequencies

Intel® SST-PP Config	Base Frequency	All Core Turbo Frequency	Max Turbo Frequency
0	2.2 GHz	3.3 GHz	3.9 GHz

1.21.3 Software and Firmware

1.21.3.1 WW06 Silicon BKC Release Notification

Intel has published a Diamond Rapids Best Known Configuration (BKC) package for the Johnson City reference platform.

The BKC consists of an update package and associated report, published on a regular basis, for the Johnson City reference platform.

The BKC package includes UEFI firmware and onboard device drivers. These components have been screened through interoperability testing together over a period, with the aim of delivering customer confidence that not only have platform ingredients been individually tested, but also collectively tested to screen out interoperability issues.

Table 40. Johnson City BKC

Ingredient	Previous	Current
BKC	-	2026 WW06 Silicon
CRB BIOS	-	31.D94
Unified Patch (MCU)	-	0x8000098C
MMC FW	-	0.735.0
Intel® Modular Flash Image Tool (Intel® Modular FIT)	-	ModularFIT_cmd_1.0.6269
Download	-	874180
Status	-	Released

NOTE

The WW06 BKC is the first BKC release with silicon support. For information on silicon availability, contact your Intel representative.

1.22 WW06_26

New:

- [Oak Stream 16-Channel Johnson City 1 and 2 Design Archives Release Notification](#)
- [MON_FAIL_N Signal Clarification](#)

Updated:

- [\(Updated WW6'26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.22.1 Documentation

1.22.1.1 Oak Stream 16-Channel Johnson City 1 and 2 Design Archives Release Notification

The *Oak Stream 16-Channel Johnson City 1 Design Archive*, document number [823065](#), and *Oak Stream 16-Channel Johnson City 2 Design Archive*, document number [636258](#), revisions 1.15, are now available in the Resource and Documentation Center (RDC).

Overview of revision 1.15 changes:

- Added Narrow TFLM KOZ TOP and BOTTOM to the board file.

1.22.1.2 (Updated WW6'26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 41. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.0	1.1	WW9'26
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	WW9'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	WW9'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces <i>Oak Stream Platforms BIOS Writer's Guide</i> , document number 820107.	839071	0.9	1.0	WW9'26
<i>Oak Stream 16-Channel 2-Socket in 19-Inch Rack Design Guidance</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	WW9'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	1.0	1.1	Q2'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.15	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.15	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	1.0	1.1	Q2'26
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.55	0.7	WW9'26
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW51'2025	WW9'26	WW9'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	1.02	As needed	-
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	1.2	As needed	-

1.22.2 Enabling

1.22.2.1 MON_FAIL_N Signal Clarification

MON_FAIL_N is one of the error monitor signals. When it is asserted in the valid period, it is an indication to the platform that hard faults within the CPU have occurred (FIVR or PLLs initialization failure).

The MON_FAIL_N signal is optional for platform usage. If it is used, it is required that the platform logic qualifies the MON_FAIL_N signal only during its guaranteed valid period for power up/down flows.

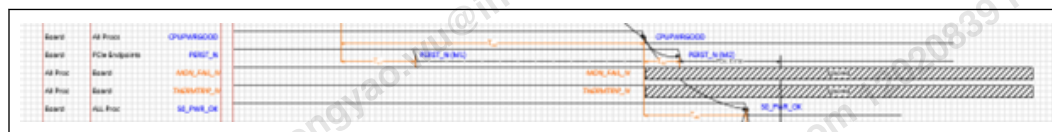
In the power up flow, MON_FAIL_N is valid only after S0_PWR_OK is asserted to CPUPWRGOOD assertion; while in power down flow, the signal stops being valid at the CPUPWRGOOD de-assertion.

The *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#), Revision 0.9, missed including the MON_FAIL_N update described in the revision history:

- [S5 Entry]: Added MON_FAIL_N timing and specified it as invalid on CPUPWRGOOD de-assertion instead of S0_PWROK de-assertion.

The following figure illustrates the update to the **S5 Entry Processor Power Sequence Timing Diagram**.

Figure 22. Snippet of Updated S5 Entry Processor Power Sequence Timing Diagram



This update will be intercepted in Revision 1.0.

1.23 WW05_26

New:

- [Oak Stream 16-Channel Platform Electrical Specification Release Notification](#)
- [Oak Stream Platform DC-SCM Design Archive Release Notification](#)
- [Oak Stream Thermal Mechanical Collaterals Release Notifications](#)
- [Oak Stream 16-Channel Socket Cover Installation and Inspection](#)
- [PM_FAST_WAKE_N Maximum Propagation Delay from Legacy Socket to Non-Legacy Sockets](#)
- [PTLP Egress Blocking Sighting Feedback Request](#)

Updated:

- (Updated WW5'26) [Oak Stream Key Collaterals Availability and Forecast](#)

1.23.1 Documentation

1.23.1.1 Oak Stream 16-Channel Platform Electrical Specification Release Notification

The *Oak Stream 16-Channel Platform Electrical Specification*, document number [787006](#), Revision 1.0, is now available in the Resource and Documentation Center (RDC).

Overview of the Revision 1.0 changes are:

- General Guidelines
 - Updated Signal Group Summary PMODE signal to Legacy Only.
 - Updated the THERMTRIP_N signal direction.
 - Added the ESPI_EDAF_EN signal to Table 11: Signals Terminations with On-Die Weak Pull Up (PU)/Pull Down (PD).
 - Removed the "Unused IO" column from Table 11: Signals Terminations with On-Die Weak Pull Up (PU)/Pull Down (PD).
 - Updated the Slew Rate Limits for the I2C/SMBus signal type.
 - Updated the Package Power C-state value.
 - Updated chapter names to align to the PDG's naming convention.
- Platform Clocking Signals
 - Added note to the SSC_SYNC AC parameter.
 - Added Frequency to the PFT_OUT signal.
 - Included timing diagrams for the PFT and SSC_SYNC clock signals.
 - Added notes to the PFT and SSC_SYNC clock signals.
- DDR5 Memory Signals
 - Removed all information from SINGLE_DIMM_CFG, as it is now fully described in the Essential IO chapter.
 - Included additional required DC specifications details for the DDR_DRAM_PWR_OK signal.
- I3C* Signals
 - Updated the I3C SPD signal "Skew between Clock and Data" ACIO numbers.
 - Updated the I3C debug signal's ACIO frequency values.
 - Updated the notes section for the I3C SPD and I3C Management signals.
 - Updated the I3C Overshoot and Undershoot specifications.
- Processor Essential IO Signals
 - Updated the PM_FAST_WAKE_N signal description.
 - Removed the Board No_Connect Termination column in the Processor Essential IO Unused Signal Terminations table.
 - Added the TEST, TEST_PU/PD signal details.
 - Updated I2C_RTC, SMBUS_HOST, CXPSMBUS, and SPI_Flash RX Delay.
 - Updated the SPI_TPM and SPI_TPM RX Delay ACIO data.
 - Changed tclk to Tcycle in the Overshoot/Undershoot table.

- Added notes to the ACIO SBLINK and SVID signals.

1.23.1.2 Oak Stream Platform DC-SCM Design Archive Release Notification

The *Oak Stream Platform DC-SCM 2.1 Design Archive*, document number [636262](#), Revision 1.2, is now available in the Resource and Documentation Center (RDC).

Overview of the Revision 1.2 changes are:

- A detailed list of the changes is available inside the design archive package.

1.23.1.3 Oak Stream Thermal Mechanical Collaterals Release Notifications

The *Oak Stream 16-Channel Platform Thermal Mechanical Specification*, document number [786521](#), Revision 1.0, is now available in the Resource and Documentation Center (RDC).

Overview of the Revision 1.0 changes are:

- Chapter 6: Processor Thermal Management Guidelines
 - Added Diamond Rapids UCC-M Processor Thermal Reporting.
 - Added Processor T_{CONTROL} Based Fan Speed Control Guidance.
 - Added Thermal Monitoring Status Reporting Implementation.
- Chapter 7: Thermal Design Guideline
 - Updated the Memory Cooling Capability Range Based on Thermal Tier for 1U Tall DIMM Liquid Cool table.
 - Updated the 2U Tall DIMM Power Estimate table.
 - Added Memory Cooling Mode Selection and Closed Loop Thermal Throttling (CLTT).
 - Updated Single Refresh (1x Refresh) versus Double Refresh (2x Refresh).
- Chapter 8: Processor Heatsink Module and Loading Mechanism
 - Added a reference to the *Oak Stream 16-Channel Platform Heatsink Screening Test*, document number 872723, to Heatsink Mechanical Requirements. (Targeted document availability at the end of February 2026.)
- Chapter 10: Thermal Solution Designs
 - Added 1U Remote Heatsink.
 - Added 2U Remote Heatsink.
 - Added Cold Plate.
- Chapter 11: Supplier Listing
 - Added references to *Manufacturing with the Intel® Xeon® Platform: Oak Stream 16-Channel (LGA9324-1 / Socket-DX) Including: Intel® Xeon® Processor*, document number 872889. (Targeted document availability in Q1 2026.)
 - Added Socket DX-1A Narrow Carrier, Socket DX Thermal Solutions.
 - Various updates to part time, notes, supplier listing, and so on.

The *Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings*, document number [786522](#), Revision 1.0, is now available in RDC.

Overview of the revision 1.0 changes are:

- Added drawings for the Oak Stream 16-channel cold plate concept.
- Added a high-level Oak Stream 16-Channel 2U RMHS drawing and a related assembly instructions label.

The *Oak Stream 16-Channel Mechanical Models*, document number [786523](#), Revision 1.0, is now available in RDC.

Overview of the Revision 1.0 change is:

- Added Oak Stream 16-channel cold plate concept.

1.23.1.4 (Updated WW5'26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 42. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.0	1.1	WW9'26
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	WW9'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	WW9'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces <i>Oak Stream Platforms BIOS Writer's Guide</i> , document number 820107.	839071	0.9	1.0	WW9'26
<i>Oak Stream 16-Channel 2-Socket in 19-Inch Rack Design Guidance</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	WW9'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	1.0	1.1	Q2'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.1	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.1	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	As needed	-
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	1.0	1.1	Q2'26
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.55	0.7	WW9'26
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW51'2025	WW9'26	WW9'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	1.01	As needed	-
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	1.2	As needed	-

1.23.2 Enabling

1.23.2.1 Oak Stream 16-Channel Socket Cover Installation and Inspection

To ensure proper retention of the Oak Stream 16-Channel socket cover, be sure to follow the assembly and inspection steps in the following figures and as outlined in the *Oak Stream 16-Channel Preliminary Loading Mechanism Hardware Assembly Guidance*, document number 864097. Additional details on the use of the socket cover procedure can be found in this document.

Figure 23. Socket Cover Assembly to Socket (1 of 2)

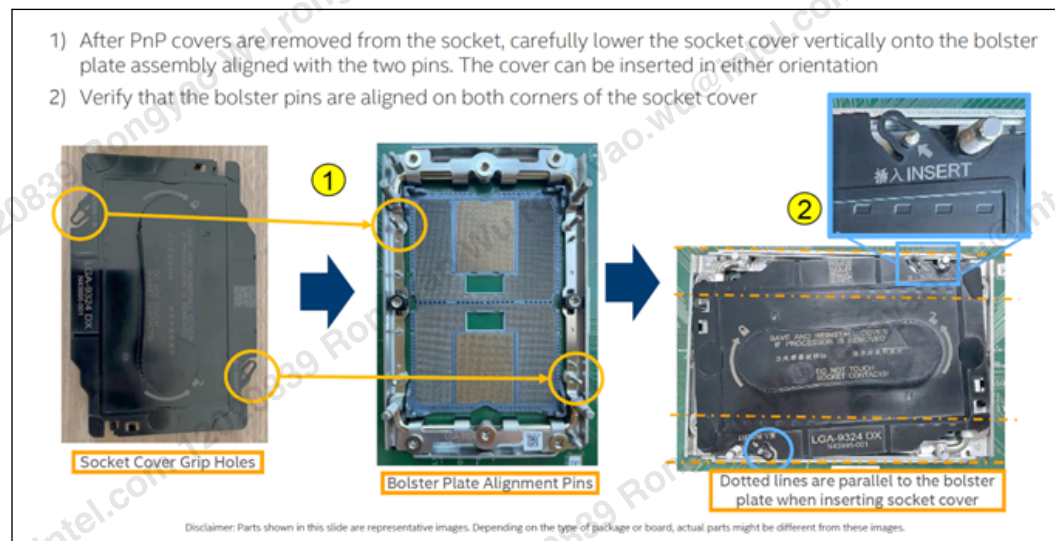


Figure 24. Socket Cover Assembly to Socket (2 of 2)

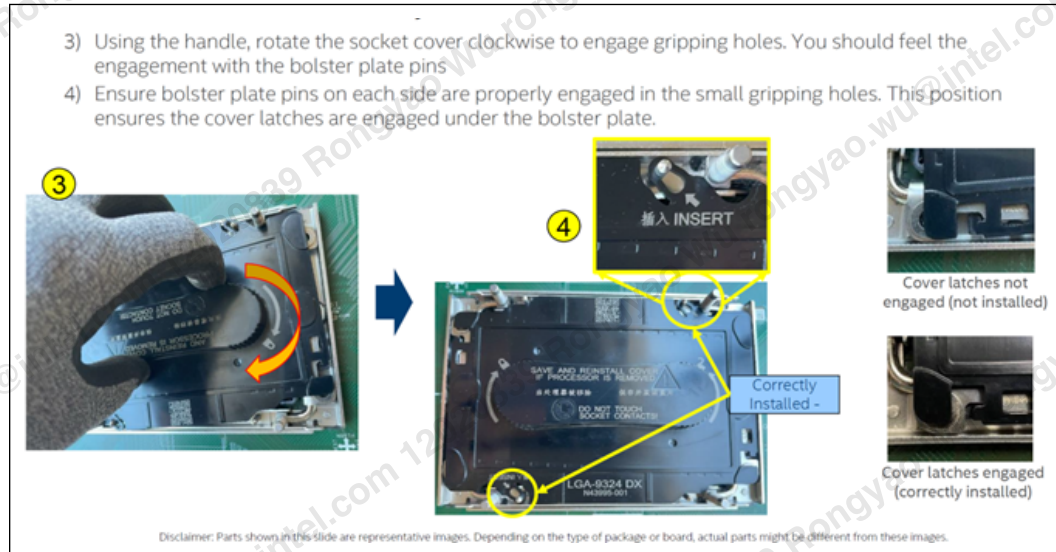
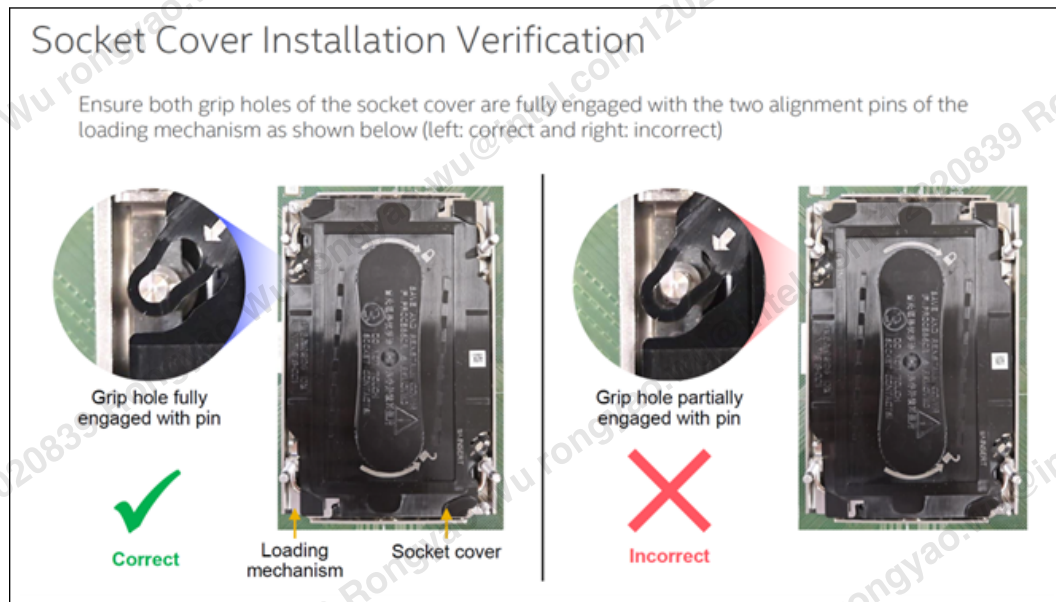


Figure 25. Socket Cover Installation Verification



1.23.2.2 PM_FAST_WAKE_N Maximum Propagation Delay from Legacy Socket to Non-Legacy Sockets

The PM_FAST_WAKE_N is a bidirectional signal and is connected to all sockets. PM_FAST_WAKE_N intent is to reduce entry and exit latency from the package C6 state. The legacy socket generates the assertion of low events, which must be propagated to all other processors within 100 ns. There is no requirement for a de-assertion event.

The latest revision of the *Oak Stream 16-Channel Platform Electrical Specification*, document number 787006, adds the maximum propagation delay requirement to the signal description.

Figure 26. PM_FAST_WAKE_N Definition

PM_FAST_WAKE_N	Package C-state Exit	The signal is bidirectional and connected to all sockets. Its main intent is to reduce the entry into and exit latency from the Package C6 state. The legacy socket generates the assertion in low event, which must be propagated to all other processors within 100 ns. There is no a requirement for de-assertion event. In Multi-Chip Package (MCP) configurations, the CPU die can forward this signal to each Companion Die (CD), if applicable. A dedicated signal between the CPU and each CD is required to wake the CD from the Package C state and to prevent it from transitioning into a low-power state when necessary.
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1.23.3 Customer Feedback

1.23.3.1 PTLP Egress Blocking Sighting Feedback Request

Intel is requesting customer feedback on root port poison TLP (PTLP) egress blocking enablement on Diamond Rapids processors.

Problem: In FLIT mode, when PTLP egress blocking (outbound poisoned data) is enabled, a logic bug is preventing AER logging/signaling of the fault in the AER error logs. The poisoned transaction is blocked but the subsequent transactions are allowed to move forward. eDPC will not trigger even if it was configured to do so. Non-FLIT mode logging and signaling works as expected.

Implications: PTLP egress blocking events are not recoverable and the system is exposed to unpredictable behavior.

Suggested Workaround: Keep root port PTLP egress blocking disabled. Allow outbound poisoned data to be delivered to the PCI Express* (PCIe*) end-device which will manage the fault as advisory non-fatal. This workaround also reduces the blast radius of PTLP egress blocking. For legacy devices that cannot handle poison, PTLP egress blocking in the PCIe switch can be enabled (legacy devices cannot be connected to the PCIe 6 root port directly and need to use PCIe switch hierarchy).

Status: Fix under investigation, trending toward the above suggested workaround as a permanent workaround.

Provide your feedback to your Intel representative **by February 4** to confirm if the suggested workaround aligns with your Oak Stream design needs.

1.24 WW04_26

New:

- Deprecation of Legacy Boot Support for All Server Programs
- Update to Intel® UPI 3.0 Internal Layer Topology Support

Updated:

- (Updated WW4'26) Oak Stream Key Collaterals Availability and Forecast

1.24.1 Enabling

1.24.1.1 Deprecation of Legacy Boot Support for All Server Programs

Starting with the Birch Stream platform, Intel has officially dropped Legacy Boot (Compatibility Support Module [CSM]) support. This decision aligns all other server programs, which are all now POR UEFI Class 3 Programs (UEFI without CSM).

HW IP blocks to support CSM are removed in the Oak Stream platform, and there is no mechanism to enable them.

1.24.1.2 Update to Intel® UPI 3.0 Internal Layer Topology Support

For Intel® Ultra Path Interconnect (Intel® UPI) topologies that do not use connectors, Intel® UPI at 64 GT/s speed can be supported if:

1. All PDG Intel® UPI design recommendations are met for 56 GT/s.
2. The maximum Intel® UPI trace length is 17000 mil or less.
3. The Intel® UPI topology is the same or a subset of the one used in Marketing Development Platform (MDP2) topology: "[Configuration #1: 2 Socket 2 or 4 Link Monoplanar Spread Core System Topology](#)".

This support update will be added to the next revision of the *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#). The *Diamond Rapids Processor Test Capability (Heat Map)*, document number 873155, targeted for release in Q2'26, will include Intel® UPI at 64 GT/s validation updates.

Customers with designs that deviate from MDP2's Intel® UPI topology but meet requirements 1 and 2, can contact their Intel representative for potential support of 64 GT/s through a co-engineering engagement with Intel. If you have questions on 64 GT/s support, contact your Intel representative.

There is no PDG support for 64 GT/s for topologies that use cables or connectors. For customers with designs with such topologies that are interested in 64 GT/s operation, Intel plans to provide simulation collaterals and a technical paper to enable innovations in this area through co-engineering engagement with Intel. Monitor the *Oak Stream Platform Message of the Week* for updates.

1.24.2 Documentation

1.24.2.1 (Updated WW4'26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 43. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.0	1.1	WW9'26
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	WW9'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	WW9'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces <i>Oak Stream Platforms BIOS Writer's Guide</i> , document number 820107.	839071	0.9	1.0	WW9'26
<i>Oak Stream 16-Channel 2-Socket in 19-Inch Rack Design Guidance</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	WW9'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	WW4'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.1	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.1	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.96	1.0	WW9'26
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.55	0.7	WW9'26
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW51'2025	WW9'26	WW9'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	1.01	As needed	-
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	-

1.25 WW03_26

New:

- [MC Firmware Feature Support Changes in Oak Stream BKC and Sample Code](#)
- [Oak Stream MCTP and PLDM BMC Firmware Stacks Transition to OpenBMC* Code](#)

- [Oak Stream 16-Channel Johnson City 1 Design Archive Release Notification](#)
- [Oak Stream 16-Channel Johnson City 2 Design Archive Release Notification](#)
- [Oak Stream System Boards Design Archive Release Notification](#)

Updated:

- [\(Updated WW3'26\) Oak Stream Key Collaterals Availability and Forecast](#)

1.25.1 Enabling

1.25.1.1 MC Firmware Feature Support Changes in Oak Stream BKC and Sample Code

Intel will retire the following BMC firmware features in the Oak Stream BKC release and NDA sample code. These features have low to no known or anticipated usage. Some of these will be removed, while others will remain in the sample code but will not be validated or supported by Intel. Refer to the NDA sample code for details as the changes are rolled out.

- FFUJ – Force Firmware Update Jumper
- PSU and fan redundancy
- NVMe* Management Interface (NVMe-MI) over MCTP
- Embedded Web Service – WebUI
- Mutual TLS Authentication
- Front panel buttons
- Serial Over LAN (SOL)
- Keyboard/Video/Mouse (KVM)
- Keyboard Controller Style (KCS) Trust policy
- LED status
- IPMI manufacturing mode commands
- Most of the FRU/SDR, IPMI and IPMI OEM commands
- IPMI power control commands
- Out of Band (OOB) BIOS configuration
- Intel® Platform Monitoring Technology (Intel® PMT) Redfish* sensors
- OpenBMC* Telemetry Service
- Virtual Media Write Support
- Host-Error-Monitor (CPU population fault)
- Intel® Node Manager (Intel® NM):
 - PCI Express* (PCIe*) domain (PCIe add-in cards via PLDM)
 - HW protection domain
 - PSU Error Monitoring (Smart and CLST software)
 - Intel NM related IPMI commands
 - Redfish power schema

- Domain limit bias
- Utilization triggered policies
- GPIO triggered policies
- Per-policy statistics
- Performance domain and policies:
 - Turbo ratio limits policy
 - HWP performance preference policy
 - HWP performance bias policy
 - Prochot response power policy

1.25.1.2 Oak Stream MCTP and PLDM BMC Firmware Stacks Transition to OpenBMC* Code

With the Diamond Rapids UCC-M SW/FW Alpha milestone, Intel will transition both the Management Component Transport Protocol (MCTP) and Platform Level Data Model (PLDM) BMC firmware stacks in the Oak Stream Best Known Configuration (BKC) release and NDA sample code to the upstreamed OpenBMC* community-supported code. This has been requested by customers and is intended to support easier code integration and collaboration as these are updated over time.

The prior Intel specific code will still be limitedly present in the NDA sample code and selectable via a build switch. The availability of this temporary code will be removed sometime after the Alpha milestone.

1.25.2 Documentation

1.25.2.1 Oak Stream 16-Channel Johnson City 1 Design Archive Release Notification

The *Oak Stream 16-Channel Platform Johnson City 1 Design Archive*, document number [823065](#), Revision 1.1, is now available in the Resource and Documentation Center (RDC).

Overview of the Revision 1.1 changes are:

- A detailed list of the changes is available inside the design archive package.

1.25.2.2 Oak Stream 16-Channel Johnson City 2 Design Archive Release Notification

The *Oak Stream 16-Channel Johnson City 2 Design Archive*, document number [636258](#), Revision 1.1, is now available in the Resource and Documentation Center (RDC).

Overview of the Revision 1.1 change is:

- Added back missing EV layers from Revision 1.0.

1.25.2.3 Oak Stream System Boards Design Archive Release Notification

The *Oak Stream Platforms System Boards Design Archive*, document number [821025](#), Revision 1.0, is now available in the Resource and Documentation Center (RDC).

Overview of the System Boards, Revision 1.0 changes are:

- Added new system boards:
 - 1U HSBP EDSFF
 - CEM cable riser
 - Fan board
 - EDSFF

NOTE

There is a known issue with the CEM cable riser board file that prevents the files from opening correctly in the Allegro* application. The issue will be fixed in Revision 1.01 targeted for release in WW4 2026.

1.25.2.4 (Updated WW3'26) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 44. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.0	1.1	WW9'26
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	WW9'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	WW9'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces <i>Oak Stream Platforms BIOS Writer's Guide</i> , document number 820107.	839071	0.9	1.0	WW9'26
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.7	As needed	-
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	-
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9.2	1.0	WW3'26
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	WW9'26
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	WW4'26
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.1	As needed	-
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.1	As needed	-
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As needed	-
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.96	1.0	WW9'26
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.55	0.7	WW9'26
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW51'2025	WW9'26	WW9'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	1.0	1.01	WW4'26
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	-

1.26 WW02-2026

New:

- [Oak Stream 16-Channel 2-Socket in 19-Inch Racks Design Guidance Release Notification](#)
- [Oak Stream 16-Channel Platform Mackinaw City Design Archive](#)
- [Oak Stream 16-Channel Platform Shadow Core Design Considerations Release Notification](#)

1.26.1 Documentation

1.26.1.1 Oak Stream 16-Channel 2-Socket in 19-Inch Racks Design Guidance Release Notification

The *Oak Stream 16-Channel 2-Socket in 19-Inch Racks Design Guidance*, document number [843581](#), Revision 1.7, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 1.7 changes are:

- Updated Mechanical section to align to latest narrow hardware guidelines and information.
- Added signal integrity section.
- Added information for NVIDIA MGX* based designs, including narrower 257-mil DIMM pitch guidance.

1.26.1.2 Oak Stream 16-Channel Platform Mackinaw City Design Archive

The *Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive*, document number [821026](#), Revision 0.55, is now available in the Resource and Documentation Center (RDC).

Overview of the Revision 0.55 change is:

- Added Bill of Materials (BOM) information.

1.26.1.3 Oak Stream 16-Channel Platform Shadow Core Design Considerations Release Notification

The *Oak Stream 16-Channel Platform Shadow Core Design Considerations Technical Article*, document number [845670](#), Revision 0.7, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 0.7 changes are:

- Updated HSIO design considerations.
- Added detailed alternative PCB layout strategies for shadow core designs with their estimated solution space for different design scenarios.

Note: Intel has no plans to build a shadow core reference platform. This technical article acts as guidance only. It is the responsibility of the system designer to verify their specific implementation to assure adherence to all applicable component and interface specifications.

2.0 2025

2.1 WW51_25

New:

- Oak Stream Platform Message of the Week (MoW) Release WW52-WW1 Notification
- Address Range Mirroring Scope Reduction Feedback Request
- Oak Stream RAS Machine Check Latched Corrected Mode Clarification
- Oak Stream RAS Features Scope Reduction Update
- Oak Stream 16-Channel Platform Johnson City 2 Design Archive Release Notification
- Oak Stream Platform Dashboard Release Notification

Updated:

- (Updated WW51'25) Oak Stream Key Collaterals Availability and Forecast
- (Updated WW51'25) Oak Stream Key Tools Availability and Forecast

2.1.1 Enabling

2.1.1.1 Oak Stream Platform Message of the Week (MoW) Release WW52-WW1 Notification

The Oak Stream Platform MOW will not be released during WW52-WW1.

2.1.1.2 Address Range Mirroring Scope Reduction Feedback Request

Intel is considering reducing the Best Known Configuration (BKC) supported and validated Address Range Mirroring (ARM) percentages to only 5%. Full memory mirroring is still supported.

Provide your feedback to your Intel representative by January 9th, 2026, if you have opens or concerns with this RAS feature scope reduction change.

2.1.1.3 Oak Stream RAS Machine Check Latched Corrected Mode Clarification

Machine Check Banks will always be in the "latched corrected" mode. Diamond Rapids SoC memory controllers will continue to support first and last corrected error logging through the RETRY_RD_ERR_LOG registers.

2.1.1.4 Oak Stream RAS Features Scope Reduction Update

As a follow up to [WW36'25](#), Intel has decided to move forward with the following Diamond Rapids RAS feature updates:

- Intel will validate and the BKC will support up to two virtual lockstep (VLS) regions per DDR5 subchannel. Each region can be used as bank or rank sparing regions.
- Intel will validate LMCE and Permanent Fault Detection (PFD) as always enabled. There will not be validation with these features disabled.

The Oak Stream program continues to evaluate opportunities to streamline RAS offerings and simplify the validation scope of platform RAS features.

2.1.2 Documentation

2.1.2.1 Oak Stream 16-Channel Platform Johnson City 2 Design Archive Release Notification

The *Oak Stream 16-Channel Platform Johnson City 2 Design Archive*, document number [636258](#), Revision 1.0, is now available in the Resource and Documentation Center (RDC).

A detailed list of this revision's list of changes is available inside the design archive package.

2.1.2.2 Oak Stream Platform Dashboard Release Notification

The *Oak Stream Platform Diamond Rapids 12/16-Channel Dashboard*, document number [863599](#), Revision WW51-25, is now available in the Resource and Documentation Center (RDC).

2.1.2.3 (Updated WW51'25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 45. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.0	1.2	Q1'26
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	Q1'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	Q1'26
<i>continued...</i>				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> <i>Note: This collateral replaces Oak Stream Platforms BIOS Writer's Guide, document number 820107.</i>	839071	0.9	1.0	WW52'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9.2	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	Q1'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	WW52'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.01	1.1	WW52'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.96	1.0	Q1'26
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.5	0.7	Q1'26
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW51' 2025	TBD	Q1'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.6	1.0	WW52'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

2.1.2.4 (Updated WW51'25) Oak Stream Key Tools Availability and Forecast

Updated information is highlighted in **bold text**.

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to Intel ITP adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support.
 - Two Streams Tracing:
 - LA-3907 preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports four trace ports)
 - LA-3892 converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- ASSET* Arium* - Contact ASSET Arium for support.
- Compact Debug Port (CDP) + MIPI10 adapter for I3C*

- Type-C connection: Voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- VTune™
- Intel® SoC Watch
- Intel® System Debugger
 - Includes token generation for Vendor Authorize Boot (VAB) / Vendor Authorize Debug (VAD).

NOTE

The Intel® System Bring-up Toolkit NDA requires registration. See *How to Obtain Intel® System Bring-up Toolkit NDA (Formerly Intel® System Studio NDA) User Guide*, document number [616675](#), and *Intel® System Bring-up Toolkit NDA (formerly Intel® System Studio NDA) Registration Form New Users*, document number [631579](#), for details.

Table 46. Oak Stream Tools

Tool Type	Tool	RDC / DIT	Availability
Debug	ASSET Intertech Arium*	N/A	Q1'26
Debug	ASSET* Close Chassis Controller (CCC)	N/A	Q1'26
Debug	CScripts for Oak Stream Platforms	821387	Available
Debug	Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C*	MM 99CJRJ	Available
Debug	Intel® I3C* USB Type-C Level Shift Adapter	MM 99CN2C	Available
Debug	Intel® In-Target Probe (Intel® ITP) eXtended Debug Port (XDP)	MM 940990	Available
Debug	Intel® SVT (Silicon View Technology) Closed Chassis Adapter	MM 921521	Available
Debug	Intel® System Bring-up Toolkit NDA Note: Includes Intel® System Debugger NDA, Intel® VTune™ Profiler NDA and Intel® SoC Watch NDA.	634166	Available
Debug	Lauterbach*	N/A	Available
Debug	SelfTest 7 - Windows* x64	524152	Available
Debug	Server Security Toolkit	819397	Available
Manufacturing and Fleet	Autonomous Crashdump (ACD)	TBD	Q1'26
Manufacturing and Fleet	IDAS Agent & Analyzer	TBD	Q1'26
Manufacturing and Fleet	Intel® At-Scale Debug (Intel® ASD)	TBD	Alpha
Manufacturing and Fleet	Intel® At-Scale Debug (Intel® ASD) BMC-JTAG* Software Guide	594948	Alpha
Manufacturing and Fleet	Intel® In-field Scan	TBD	Q1'26
continued...			

Tool Type	Tool	RDC / DIT	Availability
Manufacturing and Fleet	Intel® System Stress & Memory Diagnostics - Linux	818881	Available
Manufacturing and Fleet	Memory Latency Checker (MLC)	TBD	Q1'26
Manufacturing and Fleet	Intel® System Health Check (SHC) Kit <i>Note:</i> Includes Test Seed Loader (TSL), Sandstone, and iMunch.	TBD	Q1'26
Signal Integrity	Clock Jitter Tool (CJT)	803838	Available
Signal Integrity	Integrated Channel Analysis Tool (ICAT)	782966	Available
Signal Integrity	Intel® Platform Clock Jitter Analysis Tool (CJAT)	619648	Available
Signal Integrity	PCIe* and Intel® UPI Load Board Solution Set for 16-Channel Platform	DMRUCCPLBULB	Available
Signal Integrity	PCIe* Load Board Solution Set for 16-Channel Platform	SKU DMRUCCPLB	Available
Thermal / Power Delivery	Almondsville Thermal Mechanical Test Vehicle (TMTV) <i>Application Note</i>	852568	Available
Thermal / Power Delivery	Almondsville Thermal Test Vehicle (TTV) for 16-Channel Platform	N/A	Available
Thermal / Power Delivery	Blue Interposer for Gen5 VRTT for 16-Channel Platform	MM 99D5W2	Available
Thermal / Power Delivery	Gen5 VR Test Tool (VRTT) for Windows	MM 960764	Available
Thermal / Power Delivery	Red Interposer for Gen5 VRTT for 16-Channel Platform	MM 99D5W1	Available
Thermal / Power Delivery	VRTT Mechanical Interposer Retention Mechanism Kit for 16-Channel Platform	MM 99CN1R	Available
Validation	Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card. <i>Note:</i> Contact your Intel representative for availability.	N/A	Available
Validation	Intel® IO Margin Tool (IOMT)	TBD	Q2'26
Validation	Intel® Structural Test Plus (Intel® STP)	TBD	Available
Validation	MEI DDR5 Lite Card 8000 MT/s for Windows*	MM 99D5X2	Available
Validation	Memory Assistant for Intel® Xeon® Platforms	620497	Q1'26
Validation	Memory Error Injection (MEI) DDR5 Card for Windows	MM 99CDV1	Available
Validation	PCI Express* (PCIe*) Lane Margin Tool (LMT) for PCIe* 4.0/5.0/6.0	639433	Available
Validation	PCIe* 5.0 Error Injection (PEI) Card	MM 99C3FA	Available
Validation	PCIe* PEI 6.0 Card	TBD	Q3'26
Validation	Stitched Rank Margining Tool (RMT)	TBD	Q2'26
Validation	Target Loader RMT (EFI)	TBD	Q2'26
Validation	Intel® Simics Tool <i>Note:</i> Contact your Intel Representative for Intel® Simics support.	N/A	Available

NOTE

See the *Signal Integrity Collateral List for the Oak Stream Platform*, document number [822391](#), for a complete list of Signal Integrity tools.

2.2

WW50_2025

New:

- [Intel® Boot Guard Post-Quantum Cryptography Feedback Request](#)
- [Update to Platform RAPL Power Unit From 1/8W to 1W](#)
- [VCCVNN Ripple Guidance and Continuous DC Vmax and DC Vmin Specifications Update](#)
- [PCI Express* \(PCIe*\) 6.0 Supplemental Validation Guide Release Notification](#)

2.2.1

Enabling

2.2.1.1

Intel® Boot Guard Post-Quantum Cryptography Feedback Request

Intel is requesting feedback from customers on their Intel® Boot Guard post-quantum cryptography algorithm requirements for the Diamond Rapids processor.

Intel® Boot Guard for Diamond Rapids supports the Leighton-Micali Signature (LMS) or hybrid RSA 3K + LMS signing of the key manifest and boot policy manifest with a single OEM root key. LMS is a stateful signing algorithm, and NIST* SP-800-208/CNSA 2.0 does not allow backup of Hardware Security Module (HSM) LMS keys.

Intel is evaluating adding Module Lattice-based Digital Signing Algorithm (ML-DSA) support to Intel® Boot Guard. Due to the cryptography library and Intel HSM signing dependencies, if feasible, ML-DSA support would not be available in the Intel® Boot Guard Authenticated Code Module (ACM) until mid-2026.

Intel would like to know your Intel® Boot Guard PQC algorithm requirements:

- LMS/48 or hybrid RSA 3K + LMS/48 with a single OEM root key
- ML-DSA-87 or hybrid RSA 3K + ML-DSA-87 with a single OEM root key with mid-2026 availability if feasible

See the *Oak Stream Platform Security Update*, RDC document number 869452, for more information about Intel's plans for post-quantum cryptography algorithm support.

Provide your feedback by December 19, 2025 to your Intel representative.

2.2.1.2

Update to Platform RAPL Power Unit From 1/8W to 1W

Platform RAPL power unit is calculated with the following formula:

- $1W / \text{Power}(2, \text{PWR_UNIT})$

The default value of PLATFORM_RAPL_POWER_UNIT.PWR_UNIT is 3h, resulting in a power unit of 0.125W. This power unit does not satisfy the power ranges of heavy-load platforms, such as high-power AI systems. As a result, the default value will be **updated from 3h (0.125W) to 0h (1W)** to support and enable the Psys feature for heavy power-request platforms.

Table 47. Platform RAPL Power Unit Register (PLATFORM_RAPL_POWER_UNIT) TPMI Register Update

Bit Range	Default and Access	Field Name (ID): Description
3:0	3h 0h RO/V	PWR_ UNIT: Power units used for power control registers The actual unit value is calculated by 1 W / Power(2, PWR_UNIT).

Table 48. PLATFORM RAPL LIMIT (PLATFORM_RAPL_LIMIT_CFG) PUNIT Register Update

Bit Range	Default and Access	Field Name (ID): Description
48:32	00000h RW/L	PPL 1 (PPL 2): This field indicates the platform power limitation #2. Define the value of the platform power limits in Watts. Format is 14.3 and resolution is 0.125W 1W . Locked by: PLATFORM_RAPL_LIMIT_CFG
...	...	...
16:0	00000h RW/L	PPL 1 (PPL1): This field indicates the platform power limitation #1. Define the value of the platform power limits in Watts. Format is 14.3 and resolution is 0.125W 1W . Locked by: PLATFORM_RAPL_LIMIT_CFG

These updates will be reflected in the next revision of the *Diamond Rapids Processor Registers Specification*, document number [793272](#).

The power unit update impacts Psys settings including Platform Power Info Max/Min PPL2/PPL1 and Platform Power PPL2/PPL1 registers settings. The Johnson City reference platforms Psys values are updated to align with the new power unit. The updated values are highlighted in **bold text**.

Table 49. RP Psys Power Limit

CRPS PSU Config	CRPS PSU Power Rating (W)	Platform Power Information Max TW (Hex)	Platform Power Information Max PPL2 (Hex)	Platform Power Information Min PPL1 (Hex)	Platform Power Information Max PPL1 (Hex)
1+0	1600	2C	03A30 00 746	0	03080 00 610
2+0	1600	2C	07468	0	06100
continued...					

CRPS PSU Config	CRPS PSU Power Rating (W)	Platform Power Information Max TW (Hex)	Platform Power Information Max PPL2 (Hex)	Platform Power Information Min PPL1 (Hex)	Platform Power Information Max PPL1 (Hex)
			00E8D		00C20
1+0	2130	2C	04748 008E9	0	04090 00 812
2+0	2130	2C	08E98 011D3	0	08120 0 1024

Table 50. RP Psys Power Limit High Line

CRPS PSU Config	CRPS PSU Power Rating (W)	PPL1 (W)	PLATFORM_M_RAPL_LIMIT_CFG.PPL1 (hex)	TW1 (s)	PLATFORM_M_RAPL_LIMIT_CFG.PTW1 (Hex)	PPL2 (W)	PLATFORM_M_RAPL_LIMIT_CFG.PPL2 (Hex)	TW2 (ms)	PLATFORM_M_RAPL_LIMIT_CFG.PWT2 (Hex)
1+0	1600	1552	03080 00 610	1	0A	1862	03A30 00 746	12	43
2+0	1600	3104	06100 00C20	1	0A	3725	07468 00E8D	12	43
1+0	2130	2066	04090 00 812	1	0A	2281	04748 008E9	12	43
2+0	2130	4132	08120 0 1024	1	0A	4563	08E98 011D3	12	43

Table 51. RP Psys Power Limit Low Line

CRPS PSU Config	CRPS PSU Power Rating (W)	PPL1 (W)	PLATFORM_M_RAPL_LIMIT_CFG.PPL1 (hex)	TW1 (s)	PLATFORM_M_RAPL_LIMIT_CFG.PTW1 (Hex)	PP L2 (W)	PLATFORM_M_RAPL_LIMIT_CFG.PPL2 (Hex)	TW2 (ms)	PLATFORM_M_RAPL_LIMIT_CFG.PWT2 (Hex)
1+0	1600	986	01ED0 003DA	1	0A	1041	02088 00 411	12	43
2+0	1600	1973	03DA8 007B5	1	0A	2083	04118 00 823	12	43
1+0	2130	978	01E90 003D2	1	0A	1032	02040 00 408	12	43
2+0	2130	1956	03D20 007A4	1	0A	2064	04080 00 810	12	43

These updates will be reflected in the next revision of the *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#).

2.2.1.3 VCCVNN Ripple Guidance and Continuous DC Vmax and DC Vmin Specifications Update

The VCCVNN Ripple guidance and Continuous DC Vmax and DC Vmin are updated as indicated in the following table.

Table 52. Other Processor VR Power Parameters Table 2 Snippet

Power Rails	VI Do Max Required (V)	The Offset Used in RP Design (mV)	The Load Line Used in RP Design (mohm)	VID DAC Accuracy for V R14 (mV)	DC Voltage Regulation Tolerance Guidance (+/-mV)	Ripple Tolerance Guidance (+/-mV)	Continuous DC Vmax Required at the Measurement Location (V)	Continuous DC Vmin Required at the Measurement Location (V)
PVCCVNN	N/A	0	0	5	5	8 10	1.013 1.015	0.987 0.985

This update will be reflected in the next revisions of the *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#), and the *Oak Stream 16-Channel Platform 500W Cookbook*, document number [851059](#).

2.2.2 Documentation

2.2.2.1 PCI Express* (PCIe*) 6.0 Supplemental Validation Guide Release Notification

The *PCI Express* (PCIe*) 6.0 Supplemental Validation Guide – Oak Stream*, document number 867608, Revision 0.3, is now available in the Resource and Documentation Center.

Overview of Revision 0.3 changes:

- Initial release

This guide serves as an initial reference to perform a first check-up of PCIe on an Oak Stream-based design, becoming familiar with commonly used tools and BIOS knobs.

2.3 WW49-2025

New:

- [Diamond Rapids S3M UART Logging Capability Update](#) on page 154
- [Oak Stream 16-Channel Platform Alberta City Design Archive Release Notification](#) on page 155
- [Oak Stream CXL Supplemental Validation Guide Release Notification](#) on page 156
- [Oak Stream Platform Security Update Release Notification](#) on page 156

Updated:

- [\(Updated WW49'25\) Oak Stream Key Collaterals Availability and Forecast](#) on page 154

2.3.1 Enabling

2.3.1.1 Diamond Rapids S3M UART Logging Capability Update

As follow up to [WW36'25 MOW](#) "Oak Stream Platform Debug Probe Interfaces for S3M Streaming Tracing" and based on customer feedback, Intel has decided to **keep Intel® Secured Startup Services (S3M) UART logging capability in the Diamond Rapids generation** to ensure a smooth transition in debug capabilities from previous generation.

S3M UART logging is a fallback mechanism to the S3M tracing functionality included in Diamond Rapids advanced firmware streaming traces capability. S3M UART logging capability is planned to be deprecated starting with Coral Rapids.

Intel strongly **recommends supporting I3C* over MIPI10 or Type-C in Oak Stream based designs** as the debug port for enabling Diamond Rapids advanced firmware streaming traces capabilities. This mechanism offers the following advantages:

- **Coverage:** Supports capture synchronized and timestamped traces from multiple Intel IPs firmware like Intel® Secured Startup Services (S3M), Intel® OOBM Services Module, Power Management FW, and Essential Security Engine (ESE).
- **Complex Flow:** Supports multi-socket and multi-IMH trace capture to facilitate the debug of complex IP interdependency flows.
- **Debug Trace Infrastructure:** Incorporates I3C* and Diamond Rapids enhanced debug trace infrastructure to deliver improved trace performance and overall debugging efficiency.

Advanced firmware streaming traces capabilities overall help speed up customer enabling and debug support.

S3M UART logging remains a non-Plan of Record (POR) feature in Birch Stream with limited Intel support. The known limitations with S3M UART logging are:

- **Validation Coverage:** Lacks internal validation coverage. This feature is for debugging purposes only.
- **Limited Scope:** Supports only legacy socket and primary IMH S3M tracing. It cannot capture multi-socket, multi-IMH, or different IPs traces required for complex debug flow scenarios.
- **Performance:** Impacts boot time, timeouts during SoC firmware handshakes, and platform flows, potentially leading to unexpected behavior. When UART logging is enabled, time sensitive scenarios may be impossible to reproduce.

2.3.2 Documentation

2.3.2.1 (Updated WW49'25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 53. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.0	1.2	Q1'26
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	Q1'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	Q1'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces <i>Oak Stream Platforms BIOS Writer's Guide</i> , document number 820107.	839071	0.9	1.0	WW52'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9.2	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	Q1'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	WW52'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.01	1.1	WW52'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	WW50'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As Needed	As Needed
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.96	1.0	Q1'26
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.5	0.7	Q1'26
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW45'2025	WW51'2025	WW51'2025
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.6	1.0	WW52'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

2.3.2.2 Oak Stream 16-Channel Platform Alberta City Design Archive Release Notification

The *Oak Stream 16-Channel Alberta City Design Archive*, document number [870365](#), revision 1.0, is now available in the Resource and Documentation Center.

Overview of 1.0 changes:

- Initial release

Alberta City is a Cloud Reference Design (CRD) optimized for Hyperscalers requirements, and OCP* DC-SCM Type2 compliant. The design archive covers Host Processor Module (HPM), Aggregator Board, Power Distribution Board (PDB), and DC-SCM reference designs.

2.3.2.3 Oak Stream CXL Supplemental Validation Guide Release Notification

The *Compute Express Link* (CXL*) Supplemental Validation Guide – Oak Stream*, document number [869657](#), revision 0.5, is now available in Resource and Documentation Center.

Overview of 0.5 changes:

- Initial release

This document provides initial debug methodologies and step-by-step procedures for qualifying the CXL interface on an Oak Stream platform-based systems. These are some of the main topics and tools covered:

- BIOS debug
- CScripts
- Intel® Memory Latency Checker (Intel® MLC)
- cxi-cli
- cv_app

2.3.2.4 Oak Stream Platform Security Update Release Notification

The *Oak Stream Platform Security Update*, document number [869452](#), revision 1.0, is now available in the Resource and Documentation Center.

The Security Update includes the following topics:

- Debug security improvements and support of advanced debug using Vendor Authorized Debug and cryptographic tokens
- Intel® Boot Guard BIOS authentication extension with Intel® Vendor Authorized Boot to authorize all platform firmware to continue to boot the OS kernel.
- Intel® Platform Firmware Resilience deprecation
- Intel® Software Guard Extensions and Intel® Trust Domain Extensions DICE-based attestation updates.
- Intel® Software Guard Extensions plan of record (POR) features focusing on the most requested system configurations.
- Diamond Rapids Post-Quantum Cryptography compliance status

2.4 WW48-2025

New:

- [Diamond Rapids PCI Express* \(PCIe*\) 6.0 IBIS-AMI Models Release Notification](#) on page 158
- [Diamond Rapids PCI Express* \(PCIe*\) and Intel® Ultra Path Interconnect \(Intel® UPI\) Estimated Package Losses Release Notification](#) on page 158
- [Oak Stream 16-Channel Platform MDP2 Design Archive Release Notification](#) on page 159

Updated:

- [\(Updated WW48'25\) Oak Stream Key Collaterals Availability and Forecast](#) on page 157

- (Updated WW48'25) Oak Stream Key Tools Availability and Forecast on page 159

2.4.1 Documentation

2.4.1.1 (Updated WW48'25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 54. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.0	1.2	Q1'26
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	Q1'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	Q1'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces Oak Stream Platforms BIOS Writer's Guide, document number 820107.	839071	0.9	1.0	WW49'25
<i>Oak Stream 16-Channel 25 in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9.2	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	Q1'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.01	1.1	WW52'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	WW50'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	1.0	As Needed	As Needed
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.95	0.96	WW49'25
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.5	0.7	Q1'26
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW45'2025	WW51'2025	WW51'2025
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.6	0.7	Q1'26
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

2.4.1.2 Diamond Rapids PCI Express* (PCIe*) 6.0 IBIS-AMI Models Release Notification

The *Diamond Rapids PCI Express* (PCIe*) 6.0 IBIS-AMI Models*, document number [868826](#), revision 1.0, is now available in the Resource and Documentation Center.

This archive contains the IBIS-AMI signal integrity models for both Transmitter (TX) and Receiver (RX) PCI Express* 6 PAM4 buffers on the Diamond Rapids processor used on the Oak Stream platform. These models are bitstream (time-domain) only. Both Microsoft* Windows and Linux operating systems are supported.

NOTE

There is a known issue where the signs of individual transmitter equalization settings ("taps") used in simulation may not be correctly reported by the model. This does not affect presets used in simulation. The actual transmitter equalization values applied in any simulation will be correct. The next model release will include a correction to this behavior, as well as demonstration topologies supporting multiple signal integrity simulation tools.

2.4.1.3 Diamond Rapids PCI Express* (PCIe*) and Intel® Ultra Path Interconnect (Intel® UPI) Estimated Package Losses Release Notification

The *Diamond Rapids PCI Express* (PCIe*) and Intel® Ultra Path Interconnect (Intel® UPI) Estimated Package Losses*, document number [861617](#), revision 0.75, is now available in the Resource and Documentation Center.

This document contains estimated minimum and maximum losses for all ports, in both transmit and receive directions, on the UCC-M A-Step and UCC-M B-Step SKUs of the Diamond Rapids processor for the PCI Express* (PCIe*) and Intel® Ultra Path Interconnect (Intel® UPI) interfaces. Details on how the losses are calculated, as well as limitations and assumptions, are included. This update revises the PCIe* losses to cover package routing and buffer corners and therefore includes the effects of impedance mismatches.

NOTE

The Intel® UPI losses are unchanged in this document from previous releases; the next update will include Intel® UPI corner variations.

2.4.1.4 Oak Stream 16-Channel Platform MDP2 Design Archive Release Notification

The *Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive*, document number [828483](#), revision 1.0, is now available in the Resource and Documentation Center.

Overview of 1.0 changes:

- Implemented design updates to adopt narrow DIMM pitch
- Updated I3C_MNG, RMCA, CATERR, and WAKE_N topologies

See the *Change List* document included in the Design Archive compressed file for a detailed list of changes.

2.4.2 Tools and Debug

2.4.2.1 (Updated WW48'25) Oak Stream Key Tools Availability and Forecast

Updated information is highlighted in **bold text**.

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to ITP Adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support
 - Two Streams Tracing:
 - LA-3907 preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports four trace ports)
 - LA-3892 converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- ASSET* Arium - contact ASSET Arium for support
- Compact Debug Port (CDP) + MIPI10 Adapter for I3C*
 - Type-C connection: voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- Vtune™
- Intel® SoC Watch
- Intel® System Debugger
 - Includes token generation for Vendor Authorize Boot (VAB) / Vendor Authorize Debug (VAD)

NOTE

The Intel® System Bring-up Toolkit NDA requires registration. See *How to Obtain Intel® System Bring-up Toolkit NDA (Formerly Intel® System Studio NDA) User Guide*, document number [616675](#), and *Intel® System Bring-up Toolkit NDA (formerly Intel® System Studio NDA) Registration Form New Users*, document number [631579](#) for details.

Table 55. Oak Stream Tools

Tool Type	Tool	RDC / DIT	Availability
Debug	ASSET IntertechArium*	N/A	Q1'26
Debug	ASSET* Close Chassis Controller (CCC)	N/A	Q1'26
Debug	CScripts for Oak Stream Platforms	821387	Available
Debug	Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C*	MM 99CJRJ	Available
Debug	Intel® I3C* USB Type-C Level Shift Adapter	MM 99CN2C	Available
Debug	Intel® In-Target Probe (Intel® ITP) eXtended Debug Port (XDP)	MM 940990	Available
Debug	Intel® SVT (Silicon View Technology) Closed Chassis Adapter	MM 921521	Available
Debug	Intel® System Bring-up Toolkit NDA <i>Note:</i> Includes: Intel® System Debugger NDA, Intel® VTune™ Profiler NDA and Intel® SoC Watch NDA	634166	Q4'25
Debug	Lauterbach*	N/A	Available
Debug	SelfTest 7 - Windows* x64	524152	Available
Debug	Server Security Toolkit	819397	Available
Manufacturing and Fleet	Autonomous Crashdump (ACD)	TBD	Q1'26
Manufacturing and Fleet	IDAS Agent & Analyzer	TBD	Q1'26
Manufacturing and Fleet	Intel® At-Scale Debug (Intel® ASD)	TBD	Alpha
Manufacturing and Fleet	Intel® At-Scale Debug (Intel® ASD) BMC-JTAG* Software Guide	594948	Alpha
Manufacturing and Fleet	Intel® In Field Scan	TBD	Q1'26
Manufacturing and Fleet	Intel® System Stress and Memory Diagnostics (SSMD) for Linux	818881	Available
Manufacturing and Fleet	Memory Latency Checker (MLC)	TBD	Q1'26
Manufacturing and Fleet	System Health Check (SHC) kit. <i>Note: Includes: Test Seed Loader (TSL), Sandstone, and iMunch</i>	TBD	Q1'26
Signal Integrity	Clock Jitter Tool (CJT)	803838	Available
Signal Integrity	Integrated Channel Analysis Tool (ICAT) Suite for Windows	782966	Available
Signal Integrity	Intel® Platform Clock Jitter Analysis Tool (CJAT) for Windows	619648	Available
Signal Integrity	PCIe* and Intel® UPI Load Board Solution Set for 16-Channel Platform	DMRUCCPLBULB	Available
Signal Integrity	PCIe* Load Board Solution Set for 16-Channel Platform	SKU DMRUCCPLB	Available
Thermal / Power Delivery	Almondsville Thermal Mechanical Test Vehicle (TMTV) Application Note	852568	Available
Thermal / Power Delivery	Almondsville Thermal Test Vehicle (TTV) for 16-Channel Platform	N/A	Available
continued...			

Tool Type	Tool	RDC / DIT	Availability
Thermal / Power Delivery	Blue Interposer for Gen5 VRTT for 16-Channel Platform	MM 99D5W2	Available
Thermal / Power Delivery	Gen5 VR Test Tool (VRTT) for Windows	MM 960764	Available
Thermal / Power Delivery	Red Interposer for Gen5 VRTT for 16-Channel Platform	MM 99D5W1	Available
Thermal / Power Delivery	VRTT Mechanical Interposer Retention Mechanism Kit for 16-Channel Platform	MM 99CN1R	Available
Validation	Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card. Note: Contact your Intel representative for availability.	N/A	Available
Validation	Intel® IO Margin Tool (IOMT)	TBD	Q2'26
Validation	Intel® Structural Test Plus (Intel® STP)	TBD	Available
Validation	MEI DDR5 Lite Card 8000 MT/s for Windows	MM 99D5X2	Available
Validation	Memory Assistant for Intel® Xeon® Platforms	620497	Q1'26
Validation	Memory Error Injection (MEI) DDR5 Card for Windows	MM 99CDV1	Available
Validation	PCI Express* (PCIe*) Lane Margin Tool (LMT) for PCIe* 4.0/5.0/6.0	639433	Available
Validation	PCIe* 5.0 Error Injection (PEI) Card	MM 99C3FA	Available
Validation	PCIe* PEI 6.0 Card	TBD	Q3'26
Validation	Stitched Rank Margining Tool (RMT)	TBD	Q2'26
Validation	Target Loader RMT (EFI)	TBD	Q2'26
Validation	Intel® Simics. Note: Contact your Intel Representative for Intel® Simics support.	N/A	Available

NOTE

See the *Signal Integrity Collateral List for the Oak Stream Platform*, document number [822391](#), for a complete list of Signal Integrity tools.

2.5 WW47-2025

New:

- [Oak Stream 16-Channel Platform Mechanical Hardware and Collateral Update](#) on page 162
- [Oak Stream BIOS Reference Code Migration to Clang 20.1 Compiler](#) on page 164

2.5.1 Documentation

2.5.1.1 Oak Stream 16-Channel Platform Mechanical Hardware and Collateral Update

Revision 0.96 of the *Oak Stream 16-Channel Platform Thermal Mechanical Specification*, document number [786521](#) will be available in the Resource and Documentation Center (RDC) by the end of WW49 2025.

Revision 0.96 of the following two documents are now available in RDC:

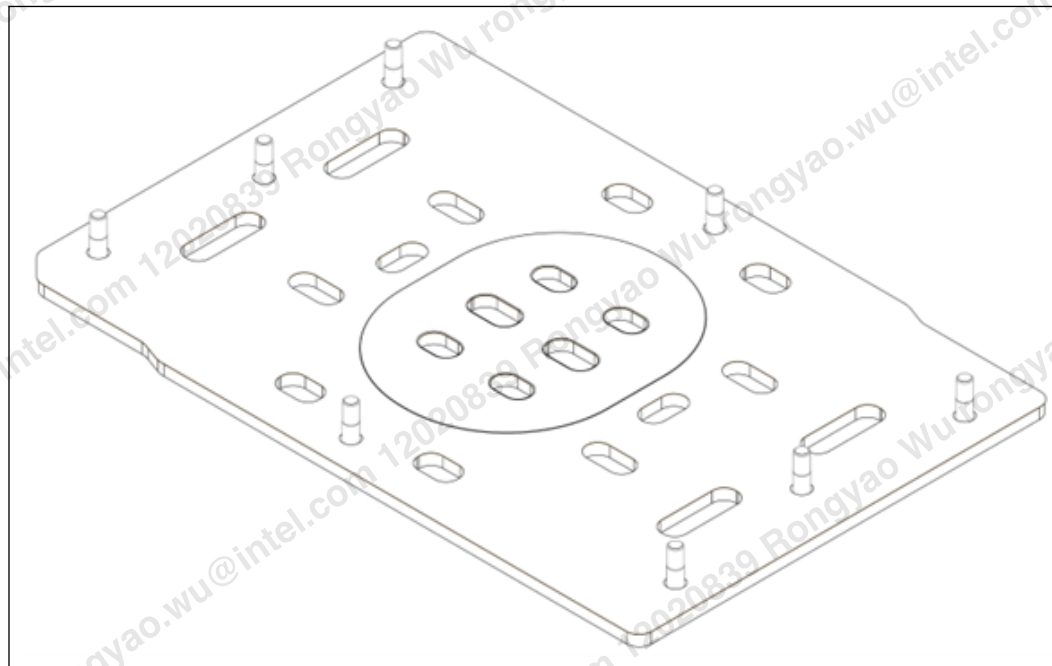
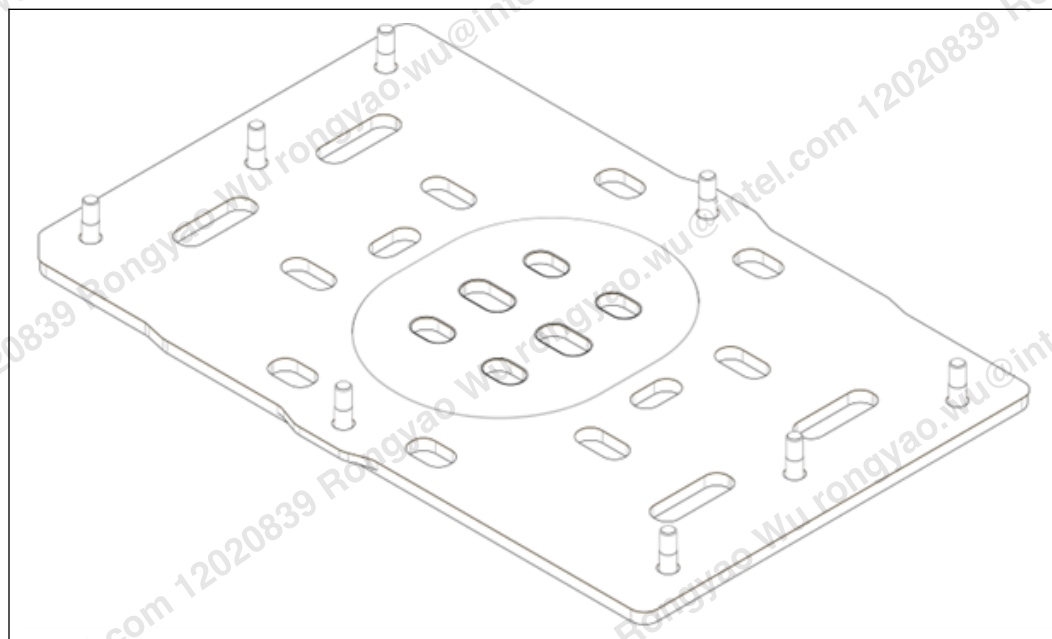
- *Oak Stream 16-Channel Socket Enabling Components Mechanical Drawings*, document number [786522](#)
- *Oak Stream 16-Channel Socket Enabling Components Mechanical Models*, document number [786523](#)

Information on the new Oak Stream 16-Channel Narrow Bottom KOZ is included in these new revisions. The narrow bottom KOZ replaces the wide Oak Stream 16-Channel Bottom KOZ.

The Oak Stream 16-Channel Platform Narrow Topside KOZ has had refinements in the way board component zones heights are specified. MAX Component Heights have been added, which do not include manufacturing contributions such as component placement tilt and solder joint thickness. The previous heights listed are now called MAX Zone Heights and continue to account for manufacturing contributions.

Drawings and CAD of the Plan of Record (POR) Narrow Backplates, compatible with the Oak Stream 16-Channel Narrow Bottom KOZ are included in revision 0.96 of the above documents. POR Narrow Backplates are undergoing hard-tooling development with suppliers and have not completed initial verification testing by Intel. Hard-tooled prototypes are targeted to be approved for initial ordering by April 2026.

Soft-tooled prototypes of the Interim Narrow Backplates continue to be available. These may be used for initial power-on development work with special-use considerations as discussed in [WW38'25 MOW](#). This part is not compatible with Oak Stream 16-Channel Narrow Bottom KOZ and will no longer be supported once the POR Narrow Backplate is available.

Figure 27. Interim Narrow Backplate**Figure 28. POR Narrow Backplate**

Hard-tooled prototypes of Oak Stream 16-Channel Narrow Backplates (POR), Bolster Plates, Socket Cover, and Heatsink Hardware are targeted to be approved for initial ordering by April 2026. Full validation of these ingredients and DG1.0 Socket DX is targeted for Q3 2026.

Additionally, revision 1.0 of *LGA9324 Thermal Solution Stiffness Guideline Application Note*, document number [795863](#), is now available in RDC. The test methodology for this thermal solution now supports narrow hardware.

2.5.2 Enabling

2.5.2.1 Oak Stream BIOS Reference Code Migration to Clang 20.1 Compiler

To standardize releases between Client and Server Products, Firmware Support Package (FSP) and BIOS reference code will be developed, validated, and compiled with Clang 20.1.x. Migration to Clang will start with Diamond Rapids UCC-M Alpha SW milestone.

Implication

- Oak Stream Best Known Configuration (BKC) will only include clang compiled images.
- Intel will do limited validation with compilers Visual Studio* 2022 (VS2022), GCC 11, GCC 14, and Clang 12.x.
- Intel will provide support to customers for VS2022 and GCC 14 compiler related issues.

NOTE

Starting with Oak Stream, Visual Studio 2019 (VS2019) will not be supported.

Contact your Intel representative if you have questions about this migration.

2.6 WW46-2025

New:

- [Direct BMC Ownership of SPD Bus Remains POR in the Oak Stream Platform](#) on page 165
- [RAS Full Memory Mirroring Support Remains POR in the Oak Stream Platform](#) on page 166

Updated:

- [\(Updated WW46'25\) Oak Stream Key Collaterals Availability and Forecast](#) on page 164

2.6.1 Documentation

2.6.1.1 (Updated WW46'25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 56. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.0	1.2	Q1'26
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	Q1'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	Q1'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces <i>Oak Stream Platforms BIOS Writer's Guide</i> , document number 820107.	839071	0.9	1.0	WW49'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9.2	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	Q1'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.01	1.1	WW51'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	WW49'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	1.0	WW47'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.95	0.96	WW47'25
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.5	0.7	Q1'26
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW45'2025	WW51'2025	WW51'2025
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.6	0.7	Q1'26
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

2.6.2 Enabling

2.6.2.1 Direct BMC Ownership of SPD Bus Remains POR in the Oak Stream Platform

As follow up to the [WW40'25 MOW](#) article "Removing Support for Direct BMC Ownership of SPD Bus Feedback Request", Intel has decided to keep direct BMC ownership of the SPD Bus (Option 2) as a Plan Of Record (POR) feature for the Oak Stream platform.

2.6.2.2 RAS Full Memory Mirroring Support Remains POR in the Oak Stream Platform

As follow up to the [WW36'25 MOW](#) article "Oak Stream RAS Features Scope Reduction Feedback Request", Intel has decided to keep full memory mirroring support as a Plan of Record (POR) feature for the Oak Stream platform.

2.7 WW45-2025

New:

- [Oak Stream Platforms Dashboard Release Notification](#) on page 166
- [Oak Stream 16-Channel Platform Pin List Release Notification](#) on page 166

Updated:

- [\(Updated WW45'25\) Oak Stream Key Collaterals Availability and Forecast](#) on page 166

2.7.1 Documentation

2.7.1.1 Oak Stream Platforms Dashboard Release Notification

The *Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard*, document number [863599](#), revision WW45 2025, is now available in the Resource and Documentation Center (RDC).

This collateral includes the most up-to-date schedules for Oak Stream platforms and Diamond Rapids processors.

2.7.1.2 Oak Stream 16-Channel Platform Pin List Release Notification

The *Oak Stream 16-Channel Platform Pin List*, document number [636260](#), Revision 1.0, is now available in the Resource and Documentation Center (RDC).

Overview of 1.0 changes:

- Added missing _PU suffix to Pin CC132 TEST_PU.
- Added missing _PD suffix to Pin CE133 TEST_PD.

2.7.1.3 (Updated WW45'25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 57. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	1.0	1.2	Q1'26
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	Q1'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	Q1'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces <i>Oak Stream Platforms BIOS Writer's Guide</i> , document number 820107.	839071	0.9	1.0	WW49'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9.2	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	Q1'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.01	1.1	WW49'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	WW47'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	1.0	WW46'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.95	0.96	WW47'25
<i>Oak Stream 8-Channel Platform Pin List</i>	821023	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Late Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	865794	0.9	As needed	As needed
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	WW49'25
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.5	0.7	Q1'26
<i>Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard</i>	863599	WW45' 2025	WW51' 2025	WW51'2025
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.6	0.7	Q1'26
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.8 WW44-2025

New:

- [Diamond Rapids DDR Memory Validation Guide Release Notification](#) on page 168
- [Oak Stream 16-Channel Reference Designs TEST_PD Clarification](#) on page 168

2.8.1 Documentation

2.8.1.1 Diamond Rapids DDR Memory Validation Guide Release Notification

The *Diamond Rapids Double Data Rate Memory Validation Guide*, document number [867684](#), revision 0.5, is now available in RDC.

Overview of Revision 0.5 changes:

- Initial release

2.8.2 Enabling

2.8.2.1 Oak Stream 16-Channel Reference Designs TEST_PD Clarification

The Oak Stream 16-Channel reference designs needs 1K pull down on TEST_PDKK165 pin.

All Oak Stream 16-Channel based designs must follow latest PDG guidelines for [Processor Essential IO Unused Signal Terminations](#):

- TEST – pin CC132: Pull to VCCVNN via 10 kohm.
- TEST – pin CE133: Pull down to GND via 10 kohm.
- TEST_PU – pins KM164, KP161, KP163: Pull to 3.3V AUX via 10 kohm.
- TEST_PD – pins KK165: Pull to GND via 1 kohm.

Refer to the *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#), for details.

The following collaterals will be updated to reflect the guidance:

- *Oak Stream 16-Channel Johnson City 1 Design Archive*, document number [823065](#)
- *Oak Stream 16-Channel Johnson City 2 Design Archive*, document number [636258](#)
- *Oak Stream 16-Channel Marketing Development Platform 1 (MDP1) Design Archive*, document number [827452](#)
- *Oak Stream 16-Channel Marketing Development Platform 2 (MDP2) Design Archive*, document number [828483](#)

Monitor the *Oak Stream Platform Message of the Week* for updates on collaterals availability.

2.9 WW43-2025

New:

- [Diamond Rapids Processor Architecture Specification Release Notification](#) on page 170
- [Oak Stream 16-Channel Mackinaw City Design Archive Release Notification](#) on page 171
- [Oak Stream 16-Channel Platform Narrow Hardware Update and Parts Availability](#) on page 172
- [Oak Stream Design Archive Collaterals Release Notification](#) on page 171

Updated:

- (Updated WW43'25) [Oak Stream Key Collaterals Availability and Forecast](#)

2.9.1 Documentation

2.9.1.1 (Updated WW43'25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 58. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
Diamond Rapids Processor Architecture Specification	792359	1.0	1.2	Q1'26
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	Q1'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	Q1'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces <i>Oak Stream Platforms BIOS Writer's Guide</i> , document number 820107.	839071	0.9	1.0	WW49'25
<i>Oak Stream 16-Channel 25 in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	1.0	As needed	As needed
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9.2	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	Q1'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.01	1.1	WW49'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	WW47'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	1.0	WW45'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.95	0.96	WW47'25
<i>Oak Stream 8-Channel Platform Pin List</i>	821023	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Late Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	865794	0.9	As needed	As needed
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	WW49'25
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.5	0.7	Q1'26
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.6	0.7	Q1'26
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.9.1.2 Diamond Rapids Processor Architecture Specification Release Notification

The *Diamond Rapids Processor Architecture Specification*, document number [792359](#), revision 1.0, is now available on the Resource and Documentation Center (RDC).

Chapters updated in Revision 1.0:

- Overview, Features, and Topologies
- Flexible I/O (FIO)
- S3M

- Intel® Out-of-Band Management Services Module (Intel® OOBM Services Module)
- Memory Subsystem
- Power Controller Unit (PCU)
- Reliability, Availability, and Serviceability (RAS)
- Processor Information ROM (PIROM)
- Appendix

See the document revision history for the complete list of changes.

2.9.1.3 Oak Stream 16-Channel Mackinaw City Design Archive Release Notification

The *Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive*, document number [821026](#), revision 0.5, is now available in the Resource and Documentation Center (RDC).

Overview of 0.5 changes:

- Added preliminary board, schematic and stack-up files.

See the *Oak Stream 16-Channel Platform 2-Socket in 19-Inch Rack NVIDIA MGX* Design Guidance*, document number [860677](#), for HW, SI, thermal, mechanical, and power design considerations for NVIDIA MGX* reference designs. Oak Stream 16-Channel Platform Design Guidelines (PDG) rules are applicable to this reference design.

NOTE

Customers are responsible for final signal and power integrity simulations and validation.

2.9.1.4 Oak Stream Design Archive Collaterals Release Notification

Three Oak Stream Design Archive Collaterals are now available in the Resource and Documentation Center (RDC):

- *Oak Stream 16-Channel Platform Aggregator Board Design Archive*, document number [820599](#), revision 1.0
- *Oak Stream 8-Channel Platform Aggregator Board Design Archive*, document number [865794](#), revision 0.9
- *Oak Stream Platforms System Boards Design Archive*, document number [821025](#), revision 0.6

Overview of Oak Stream 16-Channel Platform Aggregator Board Design Archive, revision 1.0, changes:

- Resistor value updates across I3C, JTAG, USB, and clock subsystems
- Connector replacements and depopulations
- Pull-up/pull-down adjustments for signal tuning
- Capacitor upgrades for power stability
- Pin reassignments and schematic corrections

Overview of *Oak Stream 8-Channel Platform Aggregator Board Design Archive*, document revision 0.9, changes

- Initial release

NOTE

This board is developed using Cadence Allegro System Capture. Files must be open with this software.

Overview of the *Oak Stream Platforms System Boards Design Archive*, revision 0.6, changes:

- Added Isys note added to the schematic pdf. See the [WW42'25 MOW](#) for additional information.

2.9.2 Enabling

2.9.2.1 Oak Stream 16-Channel Platform Narrow Hardware Update and Parts Availability

The Oak Stream 16-Channel narrow hardware from Foxconn Interconnect Technology (FIT)* have passed initial verification by Intel. These parts will be included in a 0.96 revision release of the *Oak Stream 16-Channel Platform Thermal Mechanical Specification (TMS)*, document number [786521](#), in November 2025.

The parts listed below from FIT*, as well as those in the *Enabling Component Part Numbers - Torsion Fastener Loading Mechanism (TFLM) Stack* and *Component Part Numbers - Heatsink Hardware (DX-HSHW)* [TMS Supplier Listing Tables](#) may now be both ordered from and shipped by FIT*, LOTES Co Ltd*, Tyco Electronics Connectivity (TE)*, KYZ*, and ITW*.

All of these parts are **soft-tooled prototypes** that may be used for initial power-on development work with special-use considerations as discussed in [WW38'25 MOW](#).

The back plates listed in this article are still preliminary; These are interim designs that are narrower than the original backplates. Intel will provide an update on backplate design by early November 2025.

Table 59. FIT Part Numbers: Backplates, Bolster Plate, Socket Cover

Component	Intel	Foxconn Interconnect Technology (FIT)*
Socket DX Backplate (93-106 mil Board Thickness) (Interim, Narrow)	N80140-001	PMGP100-01NK1-4H
Socket DX Backplate (107-120 mil Board Thickness) (Interim, Narrow)	N80146-001	PMGP100-02NK1-4H
Socket DX Backplate (121-130 mil Board Thickness) (Interim, Narrow)	N80147-001	PMGP100-03NK1-4H
Socket DX Backplate (131-141 mil Board Thickness)	N80148-001	PMGP100-04NK1-4H
continued...		

Component	Intel	Foxconn Interconnect Technology (FIT)*
(Interim, Narrow)		
Socket DX Bolster Plate	N74522-001	PMGL100-01NK1-4H
Socket DX Cover	N74818-001	PMGD000-01NK1-EH

See the [WW38'25 MOW](#) for additional details on these ingredients.

2.10 WW42-2025

New:

- [Isys Error Issue on PDB when Cold Redundancy Mode Enabled on PSUs](#) on page 173

2.10.1 Enabling

2.10.1.1 Isys Error Issue on PDB when Cold Redundancy Mode Enabled on PSUs

Problem Statement

The Oak Stream platform Power Distribution Board (PDB) supports up to two Power Supply Units (PSU). When the Cold Redundancy mode is enabled and only one PSU is active while the other PSU is in Cold Standby, the Isys signal error on the PDBs is double than expected. When the loading condition is increased and both PSUs are active, the Isys signal error is reduced to comply with Psys accuracy specification. The following figure showcases the Isys error measurement results based on Johnson City PDB design.

Figure 29. Isys Measurement Results on Oak Stream PDB

Output Current [A]	Isys expected [uA]	Isys measured [uA]	Error %	Isys measured [uA]	Error %
		CR off		CR active	
4	40	46.7	16.75%	85.6	114.00%
8	80	85.9	7.38%	164.3	105.38%
12	120	126.3	5.25%	243	102.50%
16	160	165.5	3.44%	322	101.25%
20	200	204	2.00%	400.2	100.10%
24	240	243.7	1.54%	479.5	99.79%
28	280	283.8	1.36%	558.9	99.61%
32	320	323.4	1.06%	638.6	99.56%
36	360	362.5	0.69%	719.3	99.81%
40	400	401.8	0.45%	799.1	99.78%
44	440	444.6	1.05%	879	99.77%
48	480	484.2	0.87%	959	99.79%
52	520	523.4	0.65%	1040.6	100.12%
56	560	563.3	0.59%	1120.4	100.07%
60	600	603.4	0.57%	1201.1	100.18%
64	640	643.5	0.55%	1281	100.16%
68	680	683.6	0.53%	1363	100.44%
70	700	704.5	0.64%	706.9	0.99%

NOTE

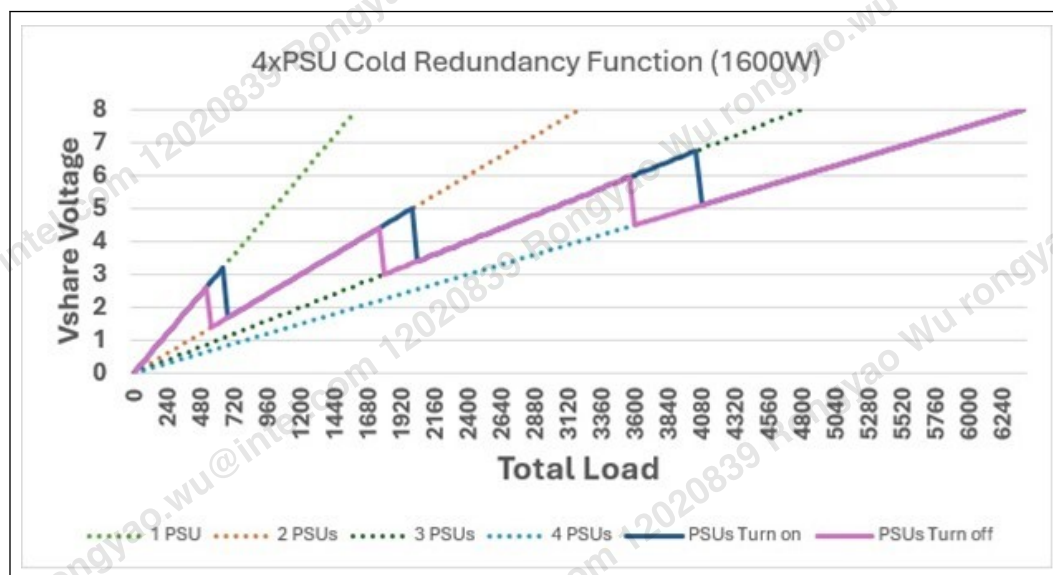
There is no issue with the Vsys circuitry

Root Cause

Voltage on Ishare bus (called Vshare) is set depending on the number of power supplies connected. When only one PSU is connected, the Ishare bus is programmed to set 8V when PSU is delivering maximum DC power (indicated in Green dot line in the following figure); When two PSUs are connected, the Ishare bus is programmed to set 8V when the two PSUs are delivering maximum DC power connected in parallel (indicated in orange dot line in the following figure); In the same way it is

programmed when three or four PSUs are connected. The envelop graph is changed based on the number of PSUs connected to the system. The following figure shows the envelop graph for different scenarios for 1600W PSU.

Figure 30. Voltage on Ishare Bus (Vshare)



When there are two or more PSUs connected to the system, it is possible to program them in cold redundant mode. In this mode, the PSUs can be:

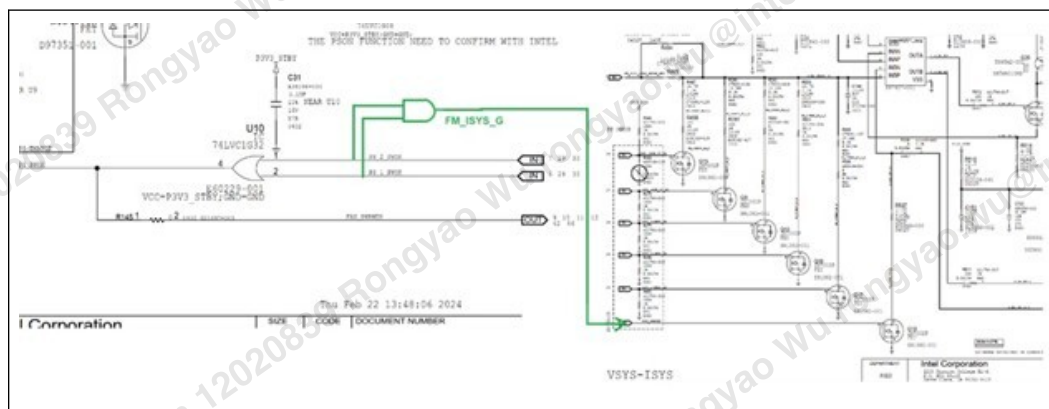
- Cold redundant active: PSU providing power.
- Cold redundant standby: PSU waiting to load condition exceed a threshold to provide power.

During the cold redundant mode, the Ishare bus voltage changes from different envelopes according to the number of PSUs active. The default status for the PSUs is to have the cold redundant mode disabled and it needs an interaction with an I2C controller to program the PSUs as cold redundant active or cold redundant standby.

Figure 31. Cold Redundancy Config (D0h) Value Decoding

Value	State	Description
00h	Standard Redundancy (default power on state)	Turns the power supply ON into standard redundant load sharing mode. The power supply's CR_BUS signal shall be in Tri-state but still pull the bus low if a fault occurs to activate any power supplies still in Cold Standby state.
01h	Cold Redundant Active ¹	Defines this power supply to be the one that is always ON in a cold redundancy configuration.
02h	Cold Standby 1 ¹	Defines the power supply that is first to turn on in a cold redundant configuration as the load increases.
03h	Cold Standby 2 ¹	Defines the power supply that is second to turn on in a cold redundant configuration as the load increases.
04h	Cold Standby 3 ¹	Defines the power supply that is third to turn on in a cold redundant configuration as the load increases.
05h	Always Standby ¹	Defines this power supply to be always in cold redundant configuration no matter what the load condition.
06h	Cold Standby 4 ¹	Defines the power supply that is 4 th to turn on in a cold redundant configuration as the load increases.
07h	Cold Standby 5 ¹	Defines the power supply that is 5 th to turn on in a cold redundant configuration as the load increases.
08h – FFh	Reserved	

When using non-M-CRPS compliant Intel CRPS 1600W PSUs, the Isys circuitry on the PDB utilizes the Vshare voltage to report the total current supplied by the PSUs and a circuit that measures the PWOK signals from each PSU to set the gain of a transconductance amplifier. However, when the PSUs are programmed in cold redundant mode, the PWOK signal is asserted to "high" even when the PSU is in cold redundant standby, which introduces an error in the gain selection for the transconductance amplifier. This error only occurs when the loading condition is low to have one PSU active and the second PSU in Cold redundant standby, once the loading is increased and the second PSU change from Cold redundant standby to active, the error in the gain selector is not present.

Figure 32. Gain Selector Using PWOK signals


Mitigation

The solution for Johnson City Reference Platform with 2 of Intel CRPS 1600W PSUs that are not compliant with M-CRPS specs is:

- PDB design will remain unchanged.
- Cold Redundancy mode for all PSUs will be disabled (default configuration).

NOTE

Psys validation does not get impacted as PPL1/2 levels are higher levels. Even in cold redundant mode (if enabled), the erroneous power telemetry does not reach those levels, which will not cause any CPU throttling

The recommendation for customers that use Oak Stream PDB is:

- Use all M-CRPS models and toggle Isys selection switch. The M-CRPS specification includes a pin for Imon which directly reports the output current information based on 12V main output loading condition, with 10 uA/A sensitivity.

Figure 33. M-CRPS Imon Pin Definition

PMBus SDA	S1	A19	B19	S8	A0 (SMBus address) ²
PMBus SCL	S2	A20	B20	S9	A1 (SMBus address) ⁴
PSON#	S3	A21	B21	S10	+12VSB
SMBAlert#	S4	A22	B22	S11	Cold Redundancy Bus
Return Sense / PS_KILL'	S5	A23	B23	S12	12V load share bus
+12V/+54V Remote sense	S6	A24	B24	S13	Imon ¹
PWOK	S7	A25	B25	S14	VINOK

Figure 34. Isys Selection Switch on PDB

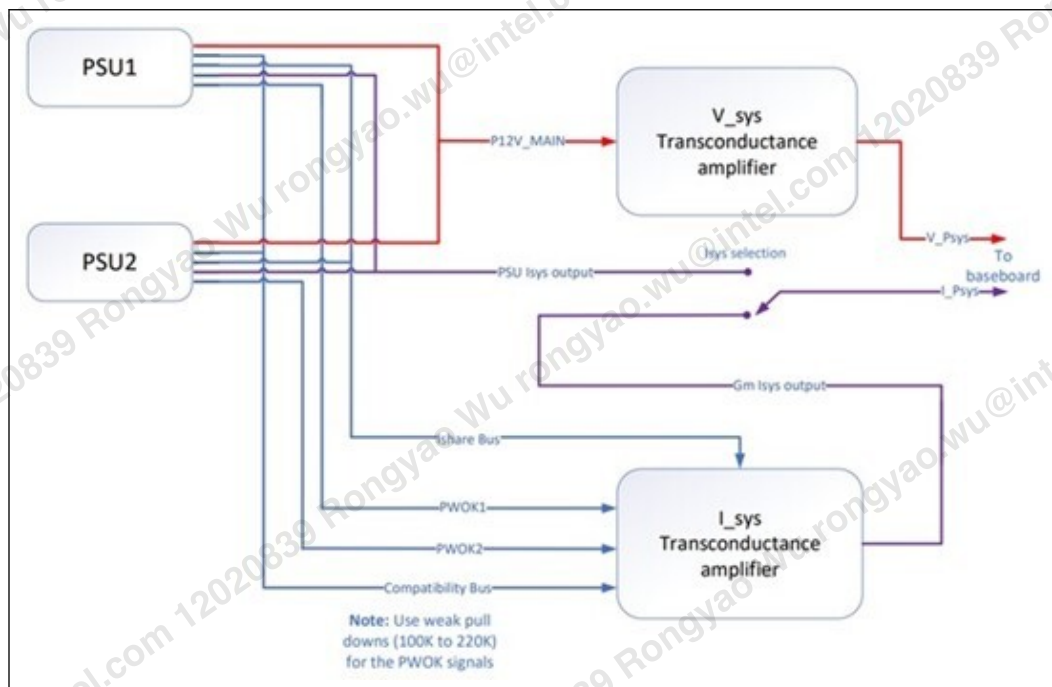


Figure 35. M-CRPS PSU Imon Measurement Results

Output Current [A]	Isys expected [uA]	Isys measured [uA]	Error %
		CR active	
3	30	33.2	10.67%
6	60	64	6.67%
12	120	124	3.33%
18	180	183	1.67%
24	240	243	1.25%
30	300	303.5	1.17%
36	360	364	1.11%
42	420	424	0.95%
48	480	484.5	0.94%
54	540	545	0.93%
60	600	608	1.33%

Cold Redundancy mode for all non-M-CRPS compliant PSUs is disabled (default configuration) as Intel RP design.

Status

The information on this article will be included in the next revision of *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#).

The following note **"To meet Psys error design target, the PDB design only supports using M-CRPS PSUs or using non-M-CRPS PSUs with cold redundancy mode disabled"** will be included in the *Oak Stream Platforms System Boards Design Archive*, document number [821025](#).

2.11 WW41-2025

New:

- Reminder - Oak Stream Features Dropped or Scope Reduced - Feedback Request on page 182

- [Diamond Rapids Out-of-Band \(OOB\) Accessible Registers Release Notification and Call to Action](#) on page 179
- [Oak Stream 16-Channel Mechanical Collateral Release Notification](#) on page 179
- [Intel® Platform Monitoring Technology \(Intel® PMT\) Update](#) on page 181

Updated:

- [\(Updated WW41'25\) Oak Stream Key Collaterals Availability and Forecast](#) on page 180

2.11.1 Documentation

2.11.1.1 Diamond Rapids Out-of-Band (OOB) Accessible Registers Release Notification and Call to Action

The *Diamond Rapids Processor Out-Of-Band (OOB) Accessible Registers*, document number [864991](#), revision 1.01, is now available in the Resource and Documentation Center.

This spreadsheet provides a simplified list of all OOB accessible registers in the Diamond Rapids processor.

Overview of 1.01 changes:

- Initial release. Sourced from revision 1.01 of the *Diamond Rapids Processor Registers Specification*, document number [793272](#).

Call To Action

Customers are invited to review the list of OOB accessible registers to assess if the current set meets their BMC* software requirements. Submit an Intel Premier Support (IPS) case to request additional registers to be added to the list. Make sure to select the **Documentation/Release Package** Category, **Registers Specification (EDS-R)** Subcategory.

2.11.1.2 Oak Stream 16-Channel Mechanical Collateral Release Notification

The revision 0.95 of the following three documents is now available in the Resource and Documentation Center (RDC).

- *Oak Stream 16-Channel Platform Thermal Mechanical Specification*, document number [786521](#)
- *Oak Stream 16-Channel Socket Enabling Components Mechanical Drawings*, document number [786522](#)
- *Oak Stream 16-Channel Socket Enabling Components Mechanical Models*, document number [786523](#)

These updates are primarily driven to transition to the narrow hardware communicated in [WW38'25 MOW](#). With the 0.95 releases of the above documents, these preliminary narrow hardware documents will be retired from RDC:

- *Preliminary Narrow Keep Out Zone (KOZ) and Interface Control Drawings (ICD) for Oak Stream 16-Channel*, document number [851010](#)
- *Oak Stream 16 Channel Narrow Mechanical Models*, document number [852564](#)

Additionally, the *Oak Stream 16-Channel Preliminary Loading Mechanism Hardware Assembly Guidance*, document number [864097](#), revision 1.0, is now available in RDC. This document provides preliminary loading mechanism hardware assembly guidance for Oak Stream 16-Channel Platforms.

2.11.1.3 (Updated WW41'25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 60. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	WW43'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	Q1'26
<i>Diamond Rapids Processor Sightings Report</i>	862869	NA	1.0	WW42'25
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces <i>Oak Stream Platforms BIOS Writer's Guide</i> , document number 820107.	839071	0.9	1.0	WW49'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	1.0	WW44'25
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9.2	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.9	1.0	Q1'26
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.01	As Needed	As needed
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Late Q4'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	1.0	WW46'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.95	0.96	WW47'25
<i>Oak Stream 8-Channel Platform Pin List</i>	821023	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.5	0.7	Late Q4'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Late Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	865794	N/A	1.0	WW44'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	WW49'25
<i>Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive</i>	821026	0.3	0.5	WW43'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Late Q4'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.11.2 Enabling

2.11.2.1 Intel® Platform Monitoring Technology (Intel® PMT) Update

XML Release Notification

The Pre-Alpha revision of *Intel(R) PMT XML Files for Diamond Rapids on Oak Stream*, document number [865036](#), is now available in the Resource and Documentation Center (RDC). This is the initial release of the document. These XML files can be used to interpret bulk telemetry via BMC or off-platform software.

Intel® PMT Windows* OS Driver Support Drop

Intel no longer supports the Intel® PMT driver for Windows* OS for datacenter platforms. This driver is no longer included in datacenter platforms Best Known Configuration (BKC) releases. Intel® PMT drivers for Linux* OS are still supported.

Telemetry Standardization on Oak Stream and Beyond DCDC Course Notification

The *Telemetry Standardization on Oak Stream and Beyond DCDC* session is available on-demand in English as part of the Data Center Design Conference (DCDC) Americas 2025. The DCDC course presentation material, *Telemetry Standardization on Oak Stream and Beyond*, document number [862898](#), is available in RDC.

Intel welcomes feedback on specific telemetry that customers want to see prioritized to be exposed via DMTF* Platform Level Data Model (PLDM). Submit an Intel Premier Support (IPS) case to share your PLDM requests.

2.11.3 Customer Feedback

2.11.3.1 Reminder - Oak Stream Features Dropped or Scope Reduced - Feedback Request

There are planned changes to Plan of Record (POR) features in the Oak Stream Platform. This is a reminder to **contact your Intel representative by October 17th** if you have concerns with these planned changes.

- [Intel® SGX + Intel® CET Cross Feature Support Drop](#)
- [One-Touch Activation Support Drop](#)
- [Removing Support for Direct BMC Ownership of SPD Bus Feedback Request](#)
- [Oak Stream RAS Features Scope Reduction](#)
- [FISH/FISHer Tool will be End of Life \(EOL\)](#)
- [Intel® Platform Firmware Resilience \(Intel® PFR\) Plan of Record \(POR\) Update](#)

2.12 WW40-2025

New:

- [Intel® SGX + Intel® CET Cross Feature Support Drop Feedback Request](#) on page 182
- [One-Touch Activation Support Drop Feedback Request](#) on page 183
- [Removing Support for Direct BMC Ownership of SPD Bus Feedback Request](#) on page 183
- [Oak Stream 16-Channel Platform 500W Cookbook Release Notification](#) on page 185
- [Oak Stream 16-Channel Platform Design Guide Release Notification](#) on page 186
- [Oak Stream 8-Channel Platform Electrical Specification Release Notification](#) on page 187
- [Data Center Design Conference \(DCDC\) 2025 India Event Registration Now Open](#) on page 189

Updated:

- [\(Updated WW40'25\) Oak Stream Key Collaterals Availability and Forecast](#) on page 184
- [\(Updated WW40'25\) Data Center Design Conference \(DCDC\) Americas 2025 Registration Open](#) on page 189

2.12.1 Customer Feedback

2.12.1.1 Intel® SGX + Intel® CET Cross Feature Support Drop Feedback Request

Intel is not planning to support the cross-feature functionality of Intel® Software Guard Extensions (Intel® SGX) and Intel® Control-Flow Enforcement Technology (Intel® CET) on Diamond Rapids generation.

Intel® SGX and Intel® CET will still be supported as standalone features.

Contact your Intel representative by October 17 th if you have concerns about this plan.

2.12.1.2 One-Touch Activation Support Drop Feedback Request

Intel previously provided One-Touch Activation (OTA) BMC and BIOS reference code to remotely enable and disable security features in the BIOS setup. Intel is planning to discontinue the One-Touch Activation reference code on the Oak Stream platform.

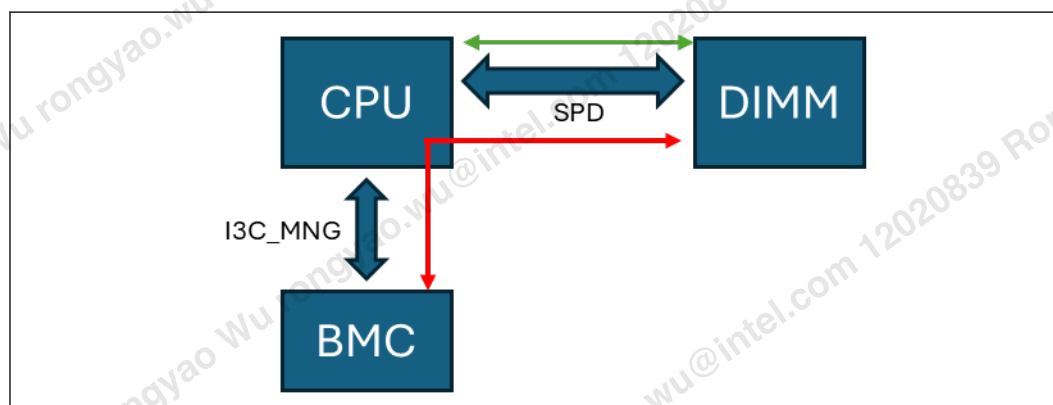
Contact your Intel representative by October 17th, 2025 if you have concerns about this change.

2.12.1.3 Removing Support for Direct BMC Ownership of SPD Bus Feedback Request

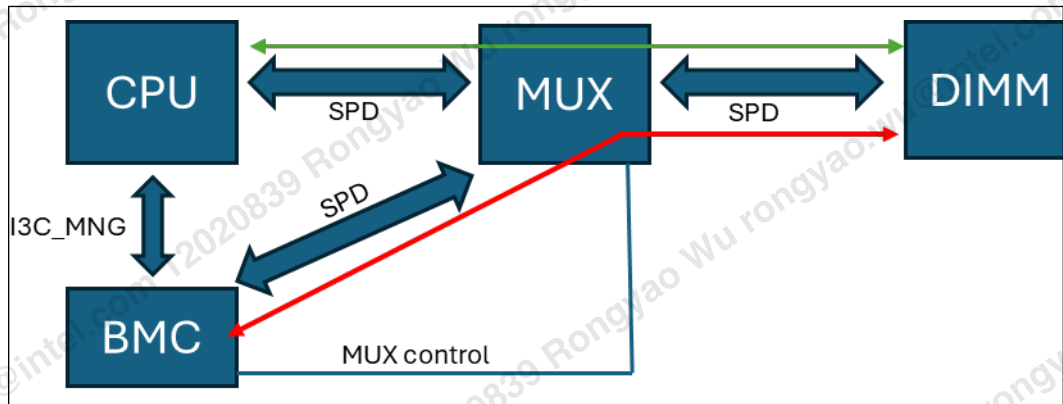
Currently, there are two options for BMC access to the SPD bus to DIMMs.

Option 1: The BMC uses the processor as a proxy to access SPD bus.

Figure 36. BMC Access Through Processor



Option 2: The Processor owns the SPD bus during initial power-on and MRC, and subsequently the BMC can control a mux to take ownership of the SPD bus. The SPD bus ownership will go back to processor after a system reset.

Figure 37. BMC Access Through MUX


Intel is planning to remove support for Option 2 (access through MUX) on the Oak Stream Platform.

Contact your Intel representative by October 17th, 2025 if you have concerns about this proposed change.

2.12.2 Documentation

2.12.2.1 (Updated WW40'25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 61. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	WW43'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	Q1'26
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<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
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continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
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<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Late Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	865794	N/A	1.0	WW44'25
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<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Late Q4'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.12.2.2 Oak Stream 16-Channel Platform 500W Cookbook Release Notification

The *Oak Stream 16-Channel Platform 500W Cookbook*, document number [851059](#), revision 1.14.06, is now available in Resource and Documentation Center. This revision must be run under Gen5 Unified Power Delivery Tester (UPDT) 1.8.

Overview of Revision 1.14.06 changes:

- Updated Imon, Thermaldrift, FastVmode scripts.
- Updated test parameters to match the latest power spec changes in OKS-16CH PDG Rev 0.9.
- Removed PVCCINF_NE and PVCCINF_NW Vdroop and Vovershoot transient test.

- For PVCCD_HV0/1 rails, added the second 3D sweep for Vovershoot and added the worst-case Vmax measurement.

2.12.2.3 Oak Stream 16-Channel Platform Design Guide Release Notification

The *Oak Stream 16 -Channel Platform Design Guide*, document number [788196](#), revision 0.9, is now available in the Resource and Documentation Center (RDC).

Overview of key revision 0.9 changes:

- Introduction and Overview
 - Updated the high-level block diagrams in section 1.1
- Processor Essential IO Signals Guidelines
 - Updated SKTOCCx_N description and added boot truth table.
 - Updated recommendations on the "Error and Thermal Protection Signals" table
 - Updated strap sampling note in "Processor Strap Signals" table
- DDR5 Memory Routing Guidelines
 - Updated the loopback signal naming from the sub-channel signals description.
 - Included a note about the updated DIMM 2 DIMM pitch
 - Removing MRDIMM 24Gb 48GB from POR
 - Removing 2S support for 24Gb 2Rx8 48GB and adding 2S support for 16Gb 2Rx8 32GB
 - Changing/Swapping 64GB MRDIMM from 32Gb 2Rx8 to 16Gb 4Rx8
 - Updated the values from the 3H4 (104 mils) Stackup Spacing Guidelines
 - Updated diagrams (Added Single DIMM CFG)
 - Added and updated FPGA to DIMM DDRxxx_RESET_N topologies for 2SPC – 1S and 2S monoplanar designs
- I3C* Design Guidelines
 - Updated I3C_DBG-BMC Down Hubless Topology, update to show effectively the solution for Diamond Rapids where there are two I3C channels (one per IMH).
 - Updated topology and the passive tables in the "OKS_DBG_I3C_DBG_BMC_HUB_solution" and "OKS_DBG_I3C_MIPI10_Hub_solution" due to confusion caused by inconsistencies with the series resistor Rs1.
- Intel® UPI 3.0 Guidelines
 - Added 6th configuration to 2S Intel® UPI 3.0 port map for stacked topology based on customer feedback
 - Modified Intel® UPI 3.0 port map configuration for 4S based on customer feedback
- System Management
 - Updated Intel® Platform Firmware Resilience (Intel® PFR) Section - added a subsection on Oak Stream Intel® PFR rescope
 - Removed the reset suppression subsection. In Oak Stream, there is only validation reset suppression, no need for crash log suppression

- Removed Intel® Virtual RAID on CPU (Intel® VROC) subsection
- Platform Debug Guidelines
 - Updated Figure "CDP 1 Socket Connectivity" - removed I3C_EXTCLK signal
 - Added two missing figures for the Platform Signal Tracing (PST) section
- Platform Power Sequence Flows
 - Added MON_FAIL_N Timing and specify it as invalid on CPUPWRGOOD De-assertion instead of S0_PWROK de-assertion
 - Changed the REFCLK invalid signal from CPUPWRGD to S0_PWROK
 - Added T75 to specify the minimum time from Vaux RDY deassertion in S5 to G3 flow to Seq Control Logic in deterministic or VCCFA_EHV/VNN Out of Spec.
 - Changed T40 to T44 to match Timing Table
 - Updated T32a from 1us min to 100us max to leave sufficient time to switch from BIST_EN mode to NMI Mode
 - Separated T63 to T63a and T63b with T63a defined as Min time that platform shall wait from GLB_RST_WARN# to 1st UART Error Code and T63b as minimum time that platform shall wait from 1st UART Error Code received until Timeout.
 - Added T53 to specify the minimum invalid window from RESET_N to CATERR_N/ERROR[]/RMCA_N valid.
- Power Distribution Guidelines
 - Updated PVCCIN_EHV1 SMBus address from 0x61 to 0x64 per RP update in Table DC-DC Converter List Table.
 - Updated "VCCIN_EHV_2nd_droop_Vmin (V)" and "VCCIN_EHV_3rd_droop_Vmin (V)" specs and removed "VCCIN_EHV_2nd_droop_Vmin_Error_Factors" (V) and "VCCIN_EHV_3rd_droop_Vmin_Error_Factors (V)" in Table VCCIN Power Parameters Table 2.
 - Added Section "PVCCIN_EHV Power Plane Cutout Filling Recommendation"
 - Updated Table "Power Rail SVID requirements" Table 2 by changing Capability Reg 06h_Bit 5 for PVCCDDR_HV1 from 1 to x.
 - Removed VCCSA_3P3_AUX rail in multiple tables to prevent confusion since on Intel RP, it is one VCC3P3_AUX rail powers both VCC3P3_AUX pins and VCCSX_3P3_AUX_0/1 pins.

See the document revision history for a complete list of changes.

2.12.2.4 Oak Stream 8-Channel Platform Electrical Specification Release Notification

The *Oak Stream 8-Channel Platform Electrical Specification*, document number [794719](#), revision 0.51, is now available in Resource and Documentation Center.

Overview of 0.5 and 0.51 changes:

- General Guidelines
 - Added Signal Buffer Behavior During Power Ramp and Signal Input Slew Rate Requirements topics.

- Added package C-state power specification.
- Added PROC_ID and PKG_ID decodes.
- Essential IO
 - Fixed PWRGOOD to CPUPWRGOOD.
 - Updated SPD I3CBUS dynamic pull up values.
 - Updated SPD signals to SPD_I3CBUS_SCL[3:0] and SPD_I3CBUS_SDA[3:0].
 - Updated SBLINK signals to SBLINK_RX_0 and SBLINK_TX_0.
 - Updated PROC_ID to PROC_ID_[2:0].
 - Updated PMAX_TRIGGER to PMAX_TRIGGER_IO_0.
 - Updated EIO signals with DC specification numbers.
 - Removed DRAM_PWR_OK.
 - Added SPI_CS[1:0] to signal list.
 - Added RESET_N_[1:0], SINGLE_DIMM_CFG, SKTOCC[2:1] *N to signal descriptions.*
 - Updated REFCLK_READY, ERROR*N, MEMHOT_OUT_N, MEMTRIP_N and THERMTRIP_N Ron DC value.
 - Updated SPI_FLASH, SPI_TPM, I2C_RTC, SMBUS_HOST, CXPSMBUS AC values and notes descriptions.
 - Removed test pins termination.
- Clocking
 - Updated DC specification values.
- Intel® Ultra Path Interconnect (Intel® UPI) (New Chapter)
- PCI Express* (PCIe*)
 - Removed PE0_RX.
 - Removed PE0_TX.
- DDR5
 - Removed SINGLE_DIMM_CFG and consolidated into Essential IO chapter.
 - Included additional DC Specifications required details for the DDR_DRAM_PWR_OK signal.
- I3C
 - Updated I3C* SPD signals AC values.
 - Updated I3C Management Signals notes.

2.12.3 Enabling

2.12.3.1 (Updated WW40'25) Data Center Design Conference (DCDC) Americas 2025 Registration Open

NOTE

Updated information in **bold text**

Intel is excited to welcome customers to Intel's 2025 Data Center Design Conference (DCDC). This year's conference focuses on Diamond Rapids, AI, PCI Express* (PCIe *) 6.0, and more. We offer over 45 courses, categorized into seven distinct technical tracks.

[View Course List and Register for Sessions](#)

The live virtual event will allow customers and Intel employees to view all sessions as they are happening on a secure streaming platform. The virtual event begins on Monday, October 6th, and concludes on Friday, October 10th, 2025. One to two sessions per track will be presented each day. All sessions will be recorded and available on-demand after the event.

Reminder - DCDC Americas begins next week! If you have not already registered, [sign up here](#). For those already registered, [log in](#) to add courses to your agenda.

2.12.3.2 Data Center Design Conference (DCDC) 2025 India Event Registration Now Open

Intel is excited to welcome customers to the first ever **Intel's 2025 Data Center Design Conference (DCDC) – India**.

Table 62. DCDC India Event Information

Geography	India
Event Format	Face-to-Face
Event Date/Schedule	October 30th, 2025, 8 AM (local time)
Place	Intel India Bengaluru campus (SRR4)
Language	English
Registration Period	Now until October 23rd, 2025
On-demand Videos	No recording at the event

Registration

Seats for the face to face (F2F) event are limited. Contact your local Intel representative if you are interested in attending.

2.13 WW39-2025

Updated:

- [\(Updated WW39'25\) Oak Stream Key Collaterals Availability and Forecast](#) on page 190

New:

- [CScripts for Oak Stream Platforms Release Notification](#) on page 195
- [Intel® Volume Management Device \(VMD\) Plan of Record \(POR\) Update](#) on page 195
- [Johnson City Theory of Operations \(TOPS\) Release Notification](#) on page 191
- [Oak Stream 8-Channel Platform Design Guide Release Notification](#) on page 191
- [Oak Stream Key Tools Availability and Forecast](#) on page 193

2.13.1 Documentation

2.13.1.1 (Updated WW39'25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 63. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
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<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9.2	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW39'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.01	As Needed	As needed
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	As needed
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	0.9	Q4'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	0.95	WW40'25
<i>Oak Stream 8-Channel Platform Pin List</i>	821023	0.5	0.7	Q4'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.5	0.7	Mid Q4'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q4'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.5	0.7	Late Q4'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	WW41'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q4'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.13.1.2 Johnson City Theory of Operations (TOPS) Release Notification

The *Johnson City Theory of Operations (TOPs)*, document number 864699, revision 0.5, first release, is now available in the Resource and Documentation Center.

Overview of Revision 0.5:

- Reference Platform (RP) Overview, board features and main differences between JC1 and JC2.
- Bring-up recipe for 1S and 2S: How to bring up the baseboard and how to set up jumpers and cable connections.
- DC-SCM, Aggregator and High-Speed I/O Overview
- Straps
- FPGA, BMC and IFWI Programming

2.13.1.3 Oak Stream 8-Channel Platform Design Guide Release Notification

Oak Stream 8-Channel Platform Design Guide, document number 813596, revision 0.5 is now available in RDC.

Overview of key revision 0.5 changes:

- All chapters:
 - Added detailed design guidelines to all chapters

- Restructured chapter hierarchy for consistency and alignment with the Oak Stream 16-Channel Platform Design Guide
- Chapter 1: Introduction and Overview
 - Updated Introduction to reflect latest platform definition, including 2-socket support
 - Updated the high-level block diagrams
- Chapter 2: Printed Circuit Board (PCB) Design Guidelines
 - Updated to reflect the learnings from Oak Stream 16-channel design
 - Added stack up
- Chapter 3: General High-Speed Guidelines **(New chapter)**
- Chapter 4: Platform Clocking
 - Included PCIe common clock description to Clocking Architectures chapter
 - Fixed Differential NEXT at probing spot figure under chapter Routing Spacing and Crosstalk
- Chapter 5: Processor Essential IO Signals Guidelines
 - Added all routing topologies and routing lengths
- Chapter 6: DDR5 Memory Routing Guidelines
 - Updated the Memory DIMM configurations.
 - Updated the CXL configurations.
- Chapter 7: I3C* Design Guidelines
 - Updated all the routing topologies and routing lengths
- Chapter 8: Flexible I/O Guidelines **(New chapter)**
- Chapter 9: PCIe and CXL Guidelines
 - Updated the PCIe Interface features section
- Chapter 10: Intel Ultra Path Interconnect Version 3.0 **(New chapter)**
- Chapter 11: System Management
 - Updated PFR Section.
 - Updated Platform logic implementation guideline section
- Chapter 12: Platform Debug Guidelines **(New Chapter)**
- Chapter 13: Platform Power Sequencing Flows
 - Corrected multiple timing parameters
 - Updated signals references
 - Updated structure and flows diagrams
- Chapter 14: Power Distribution Guidelines
-
- Updated power specifications
- Refreshed Hardware layout diagrams

Refer to the document revision history for full list of changes.

NOTE

All table formats will be updated in the next PDG revision.

2.13.2 Tools and Debug

2.13.2.1 Oak Stream Key Tools Availability and Forecast

NOTE

This article substitutes *Diamond Rapids Tools Readiness* article.

Updated information highlighted in **bold text**

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to ITP Adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support
 - Two Streams Tracing
 - LA-3907 preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports four trace ports)
 - LA-3892 converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- ASSET* Arium - contact ASSET Arium for support
- Compact Debug Port (CDP) + MIPI10 Adapter for I3C*
 - Type-C connection: voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- Vtune™
- Intel® SoC Watch
- Intel® System Debugger
 - Includes token generation for Vendor Authorize Boot (VAB) / Vendor Authorize Debug (VAD)

NOTE

Intel® System Bring-up Toolkit NDA requires registration. See *How to Obtain Intel® System Bring-up Toolkit NDA (Formerly Intel® System Studio NDA) User Guide*, document number [616675](#), and *Intel® System Bring-up Toolkit NDA (formerly Intel® System Studio NDA) Registration Form New Users*, document number [631579](#) for details.

Table 64. Oak Stream Tools

Tool	Availability and Forecast
ASSET Intertech Arium*	Q1'26
ASSET* Close Chassis Controller (CCC)	Q1'26
Autonomous Crashdump (ACD)	Late Q4'26
CScripts for Oak Stream Platforms (document number 821387)	Available Note: For stub mode and Intel Simics
Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card Note: Contact your Intel representative for availability.	Available
Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C* (MM 99CJRJ)	Available
Intel® I3C* USB Type-C Level Shift Adapter (MM 99CN2C)	Available
Intel® At-Scale Debug (Intel® ASD) Intel® At-Scale Debug (Intel® ASD) BMC-JTAG* Software Guide (document number 594948)	Alpha
W: Integrated Channel Analysis Tool (ICAT) Suite (document number 782966)	Available
W: Intel® Platform Clock Jitter Analysis Tool (document number 619648)	Available
16-Ch: PCIe* and Intel® UPI Load Board Solution Set	Available
16-Ch: PCIe* Load Board Solution Set	Available
8-Ch: Load Board Solution Sets	Q4'25
W: SelfTest 7 (document number 524152)	Available
Intel® SVT (Silicon View Technology) Closed Chassis Adapter (MM 921521)	Available
Intel® System Bring-up Toolkit NDA	Q4'25
Intel® In Field Scan	Q4'25
W: Intel® In-Target Probe (Intel® ITP) eXtended Debug Port (XDP) (MM 940990)	Available
Intel® IO Margin Tool and User Guide	Q4'25
Lauterbach*	Available
Memory Assistant	Q4'25
W: Memory Error Injection (MEI) DDR5 Card (MM 99CDV1)	Available
W: MEI DDR5 Lite Card (for 8000 MT/s) (MM 99D5X2)	Available
PCIe* 5.0 Error Injection (PEI) Card (MM 99C3FA)	Available
PCIe* PEI 6.0 Card	TBD
Intel® System Stress and Memory Diagnostics - Linux (document number 818881) Note: Replaces Intel® PTAT	Available
Stitched Rank Margining Tool (RMT), Target Loader RMT (EFI)	Q4'25
Intel® Structural Test Plus (Intel® STP)	Available
16-ch: Almondsville Thermal Test Vehicle (TTV)	WW42'25
8-ch: Turkey Canyon TTV	Mid Q2'26
Server Security Toolkit, document number 819397	Available
W: Gen5 VR Test Tool (VRTT)	Available
continued...	

Tool	Availability and Forecast
16-Ch: Red Interposer for Gen5 VRTT (MM 99D5W1)	Available
16-Ch: Blue Interposer for Gen5 VRTT (MM 99D5W2)	Available
16-Ch: VRTT Mechanical Interposer Retention Mechanism Kit (MM 99CN1R)	Available
8-Ch: VRTT Interposers and Retention Mechanism Kit	TBD

NOTE

W: Windows*. L: Linux*. If not specified, the tool is OS agnostic.

NOTE

Contact your Intel Representative for Intel® Simics support.

2.13.2.2 CScripts for Oak Stream Platforms Release Notification

CScripts for Oak Stream Platforms, document number [821387](#), version 2538.1000, is now available in the Resource and Documentation Center (RDC).

This version is a pre-silicon release intended to be run in stub mode (no hardware required).

CScripts for Oak Stream include enhancements and new capabilities compared to previous generation; One new capability is the introduction of **API Mode**. This mode allows customers to access CScripts output programmatically using Python. This enables automation, integration, and enhanced post-processing of results.

This version migrates the General User Guide, Release Notes, and per-module User Guides documentation to a consolidated PDF, included in the release ZIP. A [web version](#) of the user guide and release information is also available in RDC.

2.13.3 Enabling

2.13.3.1 Intel® Volume Management Device (VMD) Plan of Record (POR) Update

Following the recent evaluation of portfolio investments and capabilities, Intel has decided to maintain support for Intel® VMD on Diamond Rapids processors.

Diamond Rapids will support Intel® VMD version 5.0:

- Software and drivers will not be backwards compatible with previous generation of Intel® Xeon® processors
- Intel® VMD 5.0 will be enabled starting with Diamond Rapids UCC-M B-step (no support on UCC-M A-step).
- Supported features include Direct Assign, Error isolation from the Host OS, Hot-Plug and surprise removal, and Standardized LED Management.

Upcoming revision of Oak Stream and Diamond Rapids collaterals will reflect Intel® VMD POR update.

2.14 WW38-2025

New:

- [Diamond Rapids Processor Registers Specification Release Notification](#) on page 197
- [Oak Stream 16-Channel Platform Marketing Development Platform 1 \(MDP1\) Design Archive Release Notification](#) on page 198
- [Oak Stream 16-Channel Platform Johnson City 1 Design Archive Release Notification](#) on page 198
- [Oak Stream 8-Channel Platform Basin City Design Archive Release Notification](#) on page 198
- [Data Center Design Conference \(DCDC\) Americas 2025 Registration Open](#) on page 200

Updated:

- [\(Updated WW38'25\) Oak Stream Key Collaterals Availability and Forecast](#) on page 196
- [\(Updated WW38'25\) Oak Stream 16-Channel Platform Narrow Hardware Update and Parts Availability](#) on page 199

2.14.1 Documentation

2.14.1.1 (Updated WW38'25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 65. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	WW42'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	1.0	1.1	Q1'26
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> <i>Note: This collateral replaces Oak Stream Platforms BIOS Writer's Guide, document number 820107.</i>	839071	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9.2	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW39'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	1.01	As Needed	As needed
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	0.9	Q4'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	0.95	WW40'25
<i>Oak Stream 8-Channel Platform Pin List</i>	821023	0.5	0.7	Q4'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.5	0.7	Mid Q4'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q4'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	WW39'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	WW41'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q4'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.14.1.2 Diamond Rapids Processor Registers Specification Release Notification

The *Diamond Rapids Processor Registers Specification*, document number [793272](#), revision 1.0, is now available in the Resource and Documentation Center.

Overview of revision 1.0 changes:

- New and updated chapters and sections:
 - Added Out of Band (OOB) access table to all registers
 - Updated Introduction Chapter
 - Registers Overview Chapter: System Management Service Accessible Registers section

- Secure Standard Module (S3M) Registers Chapter: S3M IOAPIC Indirect Memory Space Registers
- Converged eXpress SMBUS (CXPSMB) Controller Registers Chapter
- Cache IP Registers Chapter
- Reliability, Availability, and Serviceability IP (RASIP) Registers Chapter:
 - Error Logging Registers Decoding section
 - Root-1 RASIP PCI Configuration Space Registers section
 - Non-Coherent Unit (NCU) Registers: NCU Multicast Memory Space Registers section
- Model Specific Registers (MSRs) Chapter:
 - Machine Check Architecture (MCA) Decoding section
 - DOE Mailbox Interface Virtual MSRs section

See the [revision history](#) for complete list of changes.

System Management Service Accessible Registers

Diamond Rapids Processor Out-Of-Band (OOB) Accessible Registers Datasheet, document number 864991, will document the list of OOB accessible registers in the *Diamond Rapids Processor Registers Specification*, revision 1.0. The ETA for this release is WW41'25.

2.14.1.3 Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive Release Notification

The *Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive*, document number [827452](#), revision 1.0, is now available in the Resource and Documentation Center.

Overview of revision 1.0 changes:

- Initial Release

2.14.1.4 Oak Stream 16-Channel Platform Johnson City 1 Design Archive Release Notification

The *Oak Stream 16-Channel Platform Johnson City 1 Design Archive*, document number [823065](#), revision 1.01, is now available in the Resource and Documentation Center.

Revision 1.0 included files not applicable to Johnson City. Revision 1.01 corrects the zip package release.

2.14.1.5 Oak Stream 8-Channel Platform Basin City Design Archive Release Notification

The *Oak Stream 8-Channel Platform Basin City Design Archive*, document number [636257](#), revision 0.5, is now available in the Resource and Documentation Center.

Overview of revision 0.5 changes:

- Updated design and CPU symbols aligning to latest *Oak Stream 8-Channel Platform Pin List*.

2.14.2 Enabling

2.14.2.1 (Updated WW38'25) Oak Stream 16-Channel Platform Narrow Hardware Update and Parts Availability

NOTE

Updated information is highlighted in **bold text**.

Intel has completed and approved initial verification testing of the Oak Stream 16-Channel narrow hardware components listed in the following tables.

- These parts may now be ordered from the listed suppliers.
- Intel approval for suppliers to start shipping parts is targeted for early October 2025.
- Additionally, Foxconn Interconnect Technology* (FIT*) is working with Intel to offer the components listed in the table *Part Numbers: Backplates, Bolster Plate, Socket Cover*. Updates on parts availability will be communicated in Q4'25.

These parts are currently soft-tooled; These prototypes that may be used for initial power-on development work with the following considerations:

- CPUs and thermal solutions are not to be shipped installed in a system.
- Do not use the parts for reliability testing.

These considerations are primarily due to the parts being soft-tooled prototypes, the limited sample quantities tested, and the limited reliability testing completed including dynamics. Further details on these, additional considerations, and hardware usage instructions will be in an initial release of *Oak Stream 16-Channel Preliminary Loading Mechanism Hardware Assembly Guidance*, document number 864097, to be available on RDC by early October 2025.

Hard-tooling of these parts is intended to begin later in Q4'25. Approval for suppliers to begin receiving orders for their hard-tooled parts is targeted for late Q1'26.

The backplates listed are preliminary, interim designs that are narrower than the original backplates listed in previous collaterals. Intel is continuing to refine the backplate design as discussed in the MOW and TA listed next. The POR narrow backplate design is targeted to be aligned on in October 2025 and will be communicated in a related MOW article.

- *Oak Stream Platform Message of the Week WW34'25 article "Oak Stream 16-Channel Narrow Backplate Constraints for Pre-ES1 Narrow TFLM HW"*.
- *Oak Stream 16-Channel 2S in 19-Inch Design Feasibility Technical Article*, document number [843581](#), section "Backplate Interference Considerations".

Additional details covered in this article will be available in revision 0.95 of the following collaterals by early October 2025.

- *Oak Stream 16-Channel Platform Thermal Mechanical Specification*, document number [786521](#)
- *Oak Stream 16-Channel Socket Enabling Components Mechanical Drawings*, document number [786522](#)

- Oak Stream 16-Channel Socket Enabling Components Mechanical Models, document number 786523

Table 66. Part Numbers: Backplates, Bolster Plate, Socket Cover

Component	Intel	LOTES Co Ltd*	Tyco Electronics Connectivity (TE)*	KYZ*
Socket DX Backplate (93-106 mil Board Thickness) (Interim, Narrow)	N80140-001	AHska140-P002C01SC	2509581-1	---
Socket DX Backplate (107-120 mil Board Thickness) (Interim, Narrow)	N80146-001	AHska140-P001C01SC	2509581-2	---
Socket DX Backplate (121-130 mil Board Thickness) (Interim, Narrow)	N80147-001	AHska140-P003C01SC	2509581-3	---
Socket DX Backplate (131-141 mil Board Thickness) (Interim, Narrow)	N80148-001	AHska140-P004C01SC	2509581-4	---
Socket DX Bolster Plate	N74522-001	AZIFA438-P001C01SC	2509580-1	---
Socket DX Cover	N74818-001	AZIFA444-P001C01SC	2509582-1	A30501H

Table 67. Part Numbers: Socket DX HSHW

Component	Intel	KYZ*	ITW*
DX-HSHW, without AT (Anti-Tilt)	N70218-001	A30487H	FT2501L
DX-HSHW, with AT	N70219-001	A30488H	FT2501R

2.14.2.2 Data Center Design Conference (DCDC) Americas 2025 Registration Open

Intel is excited to welcome customers to Intel's 2025 Data Center Design Conference (DCDC). This year's conference focuses on Diamond Rapids, AI, PCI Express* (PCIe *) 6.0, and more. We offer over 45 courses, categorized into seven distinct technical tracks.

[View Course List and Register for Sessions](#)

The live virtual event will allow customers and Intel employees to view all sessions as they are happening on a secure streaming platform. The **virtual event begins on Monday, October 6th, and concludes on Friday, October 10th, 2025.** One to two sessions per track will be presented each day. All sessions will be recorded and available on-demand after the event.

2.15 WW37-2025

New:

- [Oak Stream 16-Channel Platform Narrow Hardware Update and Parts Availability](#) on page 202

Updated:

- (Updated WW37) Oak Stream Key Collaterals Availability and Forecast on page 201

2.15.1 Documentation

2.15.1.1 (Updated WW37) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 68. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	WW42'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW38'25
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces Oak Stream Platforms BIOS Writer's Guide, document number 820107.	839071	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW39'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	TBD	N/A	0.5	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	0.9	Q4'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	0.95	WW40'25
<i>Oak Stream 8-Channel Platform Pin List</i>	821023	0.5	0.7	Q4'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.31	0.5	WW38'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q4'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	WW39'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	WW41'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q4'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.15.2 Enabling

2.15.2.1 Oak Stream 16-Channel Platform Narrow Hardware Update and Parts Availability

Intel has completed and approved initial verification testing of the Oak Stream 16-Channel narrow hardware components listed in the following tables.

- These parts may now be ordered from the listed suppliers.
- Intel approval for suppliers to start shipping parts is targeted for early October 2025.
- Additionally, Foxconn Interconnect Technology* (FIT*) is working with Intel to offer the components listed in the table *Part Numbers: Backplates, Bolster Plate, Socket Cover*. Updates on parts availability will be communicated in Q4'25.

These parts are currently soft-tooled; These prototypes that may be used for initial power-on development work with the following considerations:

- CPUs and thermal solutions are not to be shipped installed in a system.
- Do not use the parts for reliability testing.

These considerations are primarily due to the parts being soft-tooled prototypes, the limited sample quantities tested, and the limited reliability testing completed including dynamics. Further details on these, additional considerations, and hardware usage instructions will be an initial release of *Oak Stream 16-Channel Preliminary Loading Mechanism Hardware Assembly Guidance*, document number 864097, to be available on RDC by early October 2025.

Hard-tooling of these parts is intended to begin later in Q4'25. Approval for suppliers to begin receiving orders for their hard-tooled parts is targeted for late Q1'26.

The backplates listed are preliminary, interim designs that are narrower than the original backplates listed in previous collaterals. Intel is continuing to refine the backplate design as discussed in the MOW and TA listed next. The POR narrow backplate design is targeted to be aligned on in October 2025 and will be communicated in a related MOW article.

- *Oak Stream Platform Message of the Week* WW34'25 article "[Oak Stream 16-Channel Narrow Backplate Constraints for Pre-ES1 Narrow TFLM HW](#)".
- *Oak Stream 16-Channel 2S in 19-Inch Design Feasibility Technical Article*, document number [843581](#), section "Backplate Interference Considerations".

Additional details covered in this article will be available in revision 0.95 of the following collaterals by early October 2025.

- *Oak Stream 16-Channel Platform Thermal Mechanical Specification*, document number [786521](#)
- *Oak Stream 16-Channel Socket Enabling Components Mechanical Drawings*, document number [786522](#)
- *Oak Stream 16-Channel Socket Enabling Components Mechanical Models*, document number [786523](#)

Table 69. Part Numbers: Backplates, Bolster Plate, Socket Cover

Component	Intel	LOTES Co Ltd*	Tyco Electronics Connectivity (TE)*	KYZ*
Socket DX Backplate (93-106 mil Board Thickness) (Interim, Narrow)	N80140-001	AHska140-P002C01SC	2509581-1	---
Socket DX Backplate (107-120 mil Board Thickness) (Interim, Narrow)	N80146-001	AHska140-P001C01SC	2509581-2	---
Socket DX Backplate (121-130 mil Board Thickness) (Interim, Narrow)	N80147-001	AHska140-P003C01SC	2509581-3	---
Socket DX Backplate (131-141 mil Board Thickness) (Interim, Narrow)	N80148-001	AHska140-P004C01SC	2509581-4	---
Socket DX Bolster Plate	N74522-001	AZIFA438-P001C01SC	2509580-1	---
Socket DX Cover	N74818-001	AZIFA444-P001C01SC	2509582-1	A30501H

Table 70. Part Numbers: Socket DX HSHW

Component	Intel	KYZ*	ITW*
DX-HSHW, without AT (Anti-Tilt)	N70218-001	A30487H	FT2501L
DX-HSHW, with AT	N70219-001	A30488H	FT2501R

2.16 WW36-2025

New:

- [Oak Stream RAS Features Scope Reduction Feedback Request](#) on page 204
- [Oak Stream 16-Channel Platform New DDR_RESET Topologies for 2-Socket Designs](#) on page 206
- [Oak Stream Platform Debug Probe Interfaces for S3M Streaming Tracing](#) on page 207
- [FISH/FISHer Tool will be End of Life \(EOL\) in Oak Stream Platform](#) on page 208

Updated:

- [\(Updated WW36\) Oak Stream Key Collaterals Availability and Forecast](#) on page 204

2.16.1 Customer Feedback

2.16.1.1 Oak Stream RAS Features Scope Reduction Feedback Request

Intel is considering the following RAS features changes in the Oak Stream generation:

- **Reduce the number of supported Virtual Lockstep (VLS) regions per DDR5 subchannel to 2** (currently supporting 4 per subchannel). Each region will continue to support Bank or Rank VLS.
- **Drop full memory mirroring support.** The most efficient form of memory mirroring is offered via Address Range Mirroring (ARM), where the critical OS kernel code (memory regions that the OS will not be able to recover from) gets to operate in mirrored mode. ARM to target support ~1%, ~10% of the total memory. Intel is considering dropping full memory mirroring support, which is equivalent to 100% ARM.
- **Drop support for LMCE disabled and Permanent Fault Detection (PFD) disabled.** These two features are always enabled by default.

Intel continues to evaluate opportunities to streamline RAS offerings and simplify the validation scope of platform RAS features.

Contact your Intel representative by September 19th if you have opens or concerns with these RAS features scope reduction changes.

2.16.2 Documentation

2.16.2.1 (Updated WW36) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 71. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	WW42'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW37'25
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces Oak Stream Platforms BIOS Writer's Guide, document number 820107.	839071	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.5	1.0	WW37'25
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW39'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	TBD	N/A	0.5	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	0.9	Q4'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	0.95	WW40'25
<i>Oak Stream 8-Channel Platform Pin List</i>	821023	0.5	0.7	Q4'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.31	0.5	WW38'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	WW39'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q3'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.16.3 Enabling

Background

Implication

Status

The following tables and figures showcase the new 2-Socket DDR_RESET topologies. These guidelines are designed for 2S Mono Planar systems but can also be used for 1S systems that need to support longer max routing lengths than the 1S topologies.

Figure 38. DDR5 Miscellaneous Memory Signal Guidelines, RESET. FPGA to DIMM, 1SPC - 2S Monoplanar

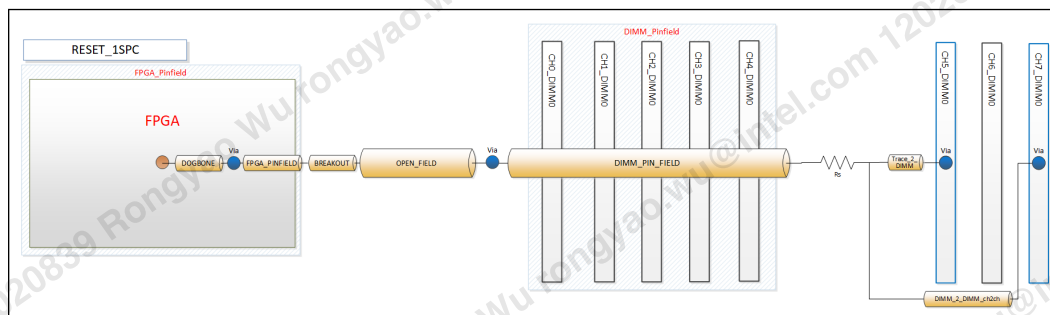


Table 72. Snippet of DDR5 Miscellaneous Memory signal Guidelines, RESET, FPGA to DIMM, 1SPC - 2S Monoplanar Length Routing Guidelines

Routing Section	Max(mils) Base Board Trace Length
DOGBONE	30
FPGA_PINFIELD	620
BREAKOUT	440
OPEN_FIELD	28500
DIMM_PIN_FIELD	4830
TRACE_TO_DIMM	500
DIMM_TO_DIMM_ch2ch	1500

Figure 39. DDR5 Miscellaneous Memory Signal Guidelines, RESET. FPGA to DIMM, 2SPC - 2S Monoplanar

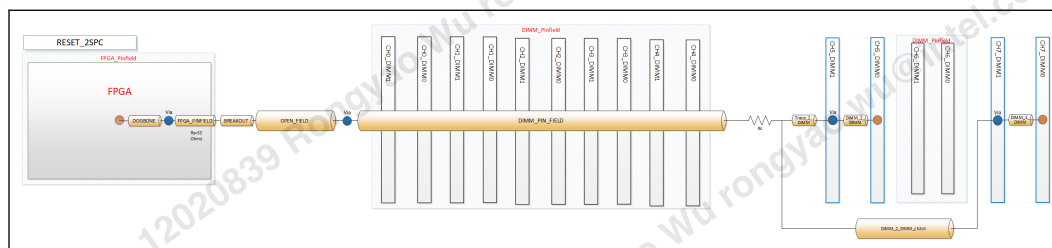


Table 73. Snippet of DDR5 Miscellaneous Memory signal Guidelines, RESET, FPGA to DIMM, 2SPC 2S Monoplanar Length Routing Guidelines

Routing Section	Max(mils) Base Board Trace Length
DOGBONE	30
FPGA_PINFIELD	620
BREAKOUT	440
OPEN_FIELD	28500
DIMM_PIN_FIELD	4830
TRACE_TO_DIMM	500
DIMM_TO_DIMM_ch2ch	1500
DIMM_TO_DIMM	520

2.16.3.2 Oak Stream Platform Debug Probe Interfaces for S3M Streaming Tracing

Background

Secured Startup Services (S3M) streaming traces provide critical runtime debug information for debugging issues related to early boot, reset flow, and other flows.

S3M 3.0 introduces the ability to use **I3C*** over **Type-C** ^[1] or **MIPI10** as debug probe interface for S3M streaming trace capability^[2]. These interfaces provide enhanced tracing capabilities compared to S3M UART logging.

Intel is planning to **drop S3M UART logging capability in Oak Stream generation**. As a reminder, the S3M UART logging has never been a Plan of Record (POR) feature.

Contact your Intel representative if you have feedback on S3M streaming tracing capabilities.

Hardware Requirements

The following table highlights the hardware requirements to enable S3M streaming traces debug capability. Intel strongly recommends supporting either I3C* over Type-C^[1] or MIPI10 debug probe interface and ensuring that Closed Chassis Adapter (CCA) debug device is available.

If hardware requirements are not met, debug capabilities will be limited.

Table 74. Oak Stream Hardware Requirements for S3M Debug Capabilities

Debug Probe Interface	Debug Device	Key Debug Function
I3C* over Type - C ^[1]	CCA ^[3] Lauterbach* (LTB)	Run control and Streaming
MIPI10	CCA ^[3] LTB	Run control and Streaming
Parallel Tracing Interface (PTI)	LTB	Run control and Streaming
eXtended Debug Port (XDP)	Intel® In-Target Probe (Intel® ITP)	Run control and Trace dump

NOTES

1. See PDG collaterals for I3C* Design Guidelines.
2. See the *S3M 3.0 Overview, Debug and Authentication*, document number [851297](#), for more details on S3M tracing.
3. Shift-level adapters are required for CCA.

2.16.4 Tools and Debug

2.16.4.1 FISH/FISHer Tool will be End of Life (EOL) in Oak Stream Platform

The OEM Fault Injection Software Hook (FISH/FISHer) tool, used for error injection and RAS validation, will no longer be supported starting with Oak Stream.

The *Oak Stream Platform RAS Technology Integration and Validation Guide*, document number 846674, will only support CScripts and ACPI EINJ error injection.

For any questions and assistance, contact your Intel support representative.

2.17 WW35-2025

Updated:

- [\(Updated WW35\) Oak Stream Key Collaterals Availability and Forecast](#) on page 209

New:

- [Oak Stream 16-Channel Mackinaw City Design Archive Release Notification](#) on page 210
- [Oak Stream 8-Channel Platform Pin List Release Notification](#) on page 211

2.17.1 Documentation

2.17.1.1 (Updated WW35) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 75. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	WW42'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW36'25
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces Oak Stream Platforms BIOS Writer's Guide, document number 820107.	839071	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.5	1.0	WW36'25
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW36'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	TBD	N/A	0.5	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	0.9	Q4'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	0.95	WW40'25
Oak Stream 8-Channel Platform Pin List	821023	0.5	0.7	Q4'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.31	0.5	WW38'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	WW39'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q3'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.17.1.2 Oak Stream 16-Channel Mackinaw City Design Archive Release Notification

The *Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive*, document number [821026](#), revision 0.3, is now available in the Resource and Documentation Center (RDC).

Overview of 0.3 Changes:

- Initial Release:
 - Design Guideline
 - Reference Guide Diagrams

Mackinaw City is an Oak Stream 16-Channel 2-Socket 32 DIMMs reference design aligned to the NVIDIA MGX* specification. This design fits in 19-inch racks. The next revision of this collateral, targeted for October 2025, will include the following updates:

- Board file
- BOM file
- Schematics file
- Stack-up file
- Updates to Reference Guide for HPM Header, Switch, and LED definitions.

Refer to *Oak Stream 16-Channel Platform 2-Socket in 19-Inch Rack NVIDIA MGX* Design Guidance*, document number [860677](#), for HW, SI, thermal, mechanical, and power design considerations for NVIDIA MGX* reference designs. Oak Stream 16-Channel Platform Design Guidelines (PDG) rules are applicable to this reference design.

NOTE

Customers are responsible for final signal and power integrity simulations and validation.

2.17.1.3 Oak Stream 8-Channel Platform Pin List Release Notification

The *Oak Stream 8-Channel Pin List*, document number [821023](#), revision 0.5, is now available in the Resource and Documentation Center (RDC).

Overview of 0.5 changes:

- Initial Release

2.18 WW34-2025

New:

- [Intel® Platform Firmware Resilience \(Intel® PFR\) Plan of Record \(POR\) Update](#) on page 216
- [Oak Stream 16-Channel Narrow Backplate Constraints for Pre-ES1 Narrow TFLM HW](#) on page 214
- [Oak Stream 16-Channel Platform Electrical Specification Release Notification](#) on page 212

Updated:

- [\(Updated WW34\) Oak Stream Key Collaterals Availability and Forecast](#) on page 211
- [\(Updated\) I3C_MNG Has Limited Functionality in Diamond Rapids IMH1-based SKUs - Clarification](#) on page 213
- [\(Updated\) Transition to Pay-For-Service Model for EDA Conversion Service - Clarification](#) on page 214

2.18.1 Documentation

2.18.1.1 (Updated WW34) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 76. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	WW42'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW36'25
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i>	839071	0.9	1.0	Q4'25
<i>continued...</i>				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Note:</i> This collateral replaces <i>Oak Stream Platforms BIOS Writer's Guide</i> , document number 820107.				
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.5	1.0	WW36'25
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW36'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	TBD	N/A	0.5	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	0.9	Q4'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	0.95	WW40'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	WW37'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.31	0.5	WW37'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	WW39'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q3'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.18.1.2 Oak Stream 16-Channel Platform Electrical Specification Release Notification

The *Oak Stream 16-Channel Platform Electrical Specification*, document number 787006, revision 0.9, is now available in the Resource and Documentation Center (RDC).

Overview of 0.9 changes:

- General Guidelines:
 - Added strap signal notes (example: **ESPI_EDAF_EN**) in Power-On Configuration Options
 - Updated Signal Group Summary with failsafe signals
 - Added termination details for unused IOs under the **Signals with On-Die Weak PU/PD** table
- Essential IO Signals:
 - Updated descriptions for various EIO signals
 - Revised the **Processor Essential IO Unused Signal Terminations** table

2.18.2 Enabling

2.18.2.1 (Updated) I3C_MNG Has Limited Functionality in Diamond Rapids IMH1-based SKUs - Clarification

NOTE

Updated information is highlighted in **bold text**.

- **Background**
 - In Diamond Rapids processors, **Partition ID (PID) is used to configure the platform in partition mode (2x1S configuration). Both PID and SOCKET_ID are used to define I3C_MNG target instance ID in a system.** Each I3C_MNG target instance requires a unique ID.
- **Problem Statement**
 - In IMH1-based SKUs **2x1S configurations**, the I3C_MNG target instance ID is duplicated **in both sockets**.
- **Root cause**
 - IMH1-based SKUs do not route PID strap pins correctly, resulting in duplicated I3C_MNG target addresses **for both sockets**.
- **Implication**
 - I3C_MNG will not be functional on **platforms configured in partition mode (2x1S configurations).**
 - I3C_MNG is functional in single-socket platforms and multi-socket systems **that do not configure partition mode.**
- **Status**
 - A microcode workaround for IMH1-based SKUs is under evaluation. Monitor the Message of the Week (MoW) for updates.

2.18.2.2 (Updated) Transition to Pay-For-Service Model for EDA Conversion Service - Clarification

NOTE

Updated information is highlighted in **bold text**.

Intel's reference designs schematic and layout files are provided in Cadence Allegro* versions 22.1 and 23.1. Customers that require files in other output formats must submit their request through the EDA Conversion Service.

The EDA Conversion Service will transition to a pay-for-service model. To request Intel's reference designs to be converted from original Cadence* formats to other EDA formats, customers must pay an annual access fee of 2,799 USD per Company Account ID. **This annual access grants unlimited conversions from different collaterals and formats.** Payments must be made through the [Intel® Design-In Tools Store website](#).

Once the EDA conversion service is purchased, customers can send their conversion requests to Intel EDA Conversion Service. Converted files will be available for download on the Resource and Documentation Center (RDC) website.

Table 77. EDA Conversion Service Output Formats

Layout Files Output Format	Schematic Files Output Format
Mentor (.pcb +. prj)	Mentor (.prj)
Zuken CR8000 (.dsgn)	Zuken (.sdm)
Zuken CR5000(.rul/ .pcb)	OrCAD (.dsn, .iscf)
ODB++ (.tgz)	Concept HDL (.cpm)
Altium(.PcbDoc)	Altium(.PrjPcb)
-	Main device Symbol (.olb)

NOTE

Conversions to older Cadence* Allegro versions (example: 17.1) are also covered by EDA Conversion Service.

For detail instructions on how to request and EDA Conversion Service, see the *Intel EDA Conversion Service Transition: Important Update & FAQ Guide*, document number 861035.

2.18.2.3 Oak Stream 16-Channel Narrow Backplate Constraints for Pre-ES1 Narrow TFLM HW

As communicated in WW23'25, the [Narrow Torsion Fastener Loading Mechanism \(TFLM\) Execution Plan](#) consists of five milestones. The third milestone is mapped to ES1-A samples availability.

The narrow backplates that will be available before the ES1-A samples may accommodate system configurations within the following conditions:

- Open chassis without board pan for any Host Processor Module (HPM) form factor
- 1 Socket HPM in open or closed chassis
- 2 Socket HPM in shadowed core configuration or M-FLW form-factor in open or closed chassis
- 2 Socket HPMs in M-SDNO form-factor or MGX form-factor in closed chassis with board pan if all the following conditions are met:
 - Sockets are centered across the CL-datum (Centerline) of the HPM (see figure *Socket Centering, Orientation, and Offset for NVIDIA MGX* - Top View*)
 - DIMMs between the 2 sockets have a minimum pitch of 6.66 mm (262 mil) (M-SDNO only)
 - The system design accounts for a 2.8 mm backplate thickness (See figure *Cross-Section of Board, Backplate, and Board Pan*)
 - The datum (0,0) and the north side of the package on both sockets face the same edge of the board (See figure *Socket Centering, Orientation, and Offset for NVIDIA MGX* - Top View*)
 - North side of the socket backplate Keep Out Zone (KOZ) is a minimum of 71 mm from the X-axis datum (NVIDIA MGX* specification only)

Figure 40. Socket Centering, Orientation, and Offset for NVIDIA MGX* - Top View

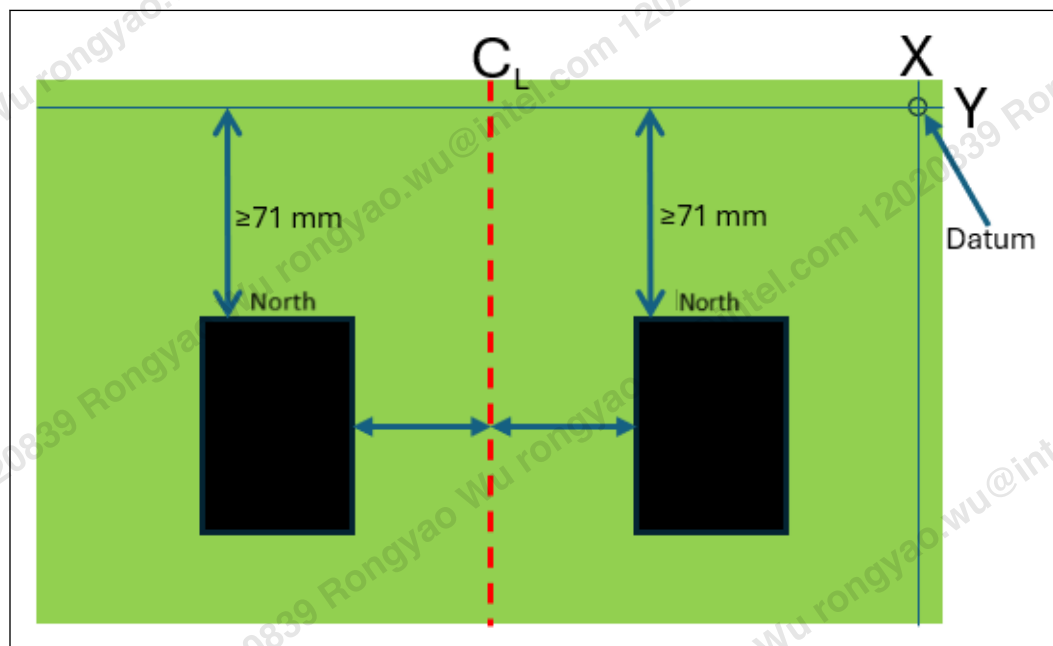
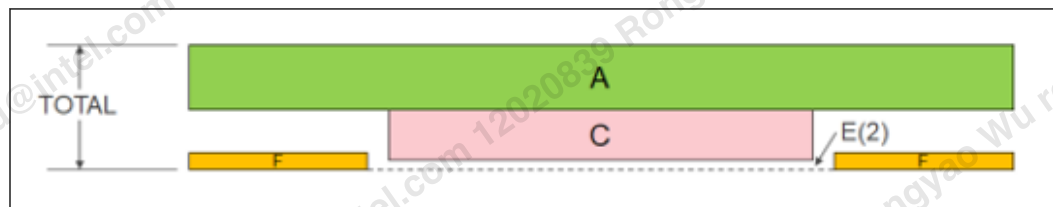


Figure 41. Cross-Section of Board, Backplate, and Board Pan



NOTE

A=Motherboard, C= Backplate, F= Board Pan. Figure courtesy of OCP* DC-MHS M-SDNO specification, Revision 1.0

Customers need to evaluate other form factors outside of enabled specifications (DC-MHS M-FLW and M-S DNO class C, and NVIDIA* MGX).

Intel is assessing refinements to the Oak Stream 16-Channel backplate design to better align with these specifications' requirements, and with DDR5 physical space requirements.

The narrow backplates that will be available at the ES1-A milestone are targeted to accommodate M-FLW, M-SDNO class C, and MGX* specifications. Monitor the Message of the Week for a narrow backplate update by early October 2025.

2.18.2.4 Intel® Platform Firmware Resilience (Intel® PFR) Plan of Record (POR) Update

Intel® PFR is no longer a POR feature starting with Oak Stream platforms.

Background

Intel® PFR was designed as a complete firmware resilience and recovery solution compliant with NIST SP 800-193. It was first offered on 3rd Gen Xeon Scalable Processors. Since Intel® PFR's introduction, the ecosystem has created a range of NIST800-193 compliant solutions tailored to the needs of enterprise and cloud data center needs.

In alignment to industry trends, Intel will no longer develop, validate, or support Intel® PFR solutions. Intel will continue to focus on industry-standard based CPU attestation capabilities and CPU functions needed to support customer's Platform Root of Trust (PRoT) solutions.

Contact your Intel representative if you have feedback on Xeon platform capabilities needed to support your PRoT or NIST SP 800-193 solutions.

CPU Attestation Support

CPU attestation will continue to be POR in Oak Stream.

- Johnson City Reference Platforms will validate CPU attestation and its related functions and interfaces in a Platform Root of Truste (ProT) solution.
- SPDM 1.2 CPU attestation will be validated an Altera® FPGA as our PRoT proxy. This is **validation configuration only** and will not be supported as a feature or customer reference architecture by Intel.
- Intel will release Intel PFR Altera FPGA source code for Johnson City reference platforms with limited support.

For platform attestation, firmware resilience, and recovery, customers are encouraged to leverage their BMC* PRoT component or CPLD solution from providers in the ecosystem.

Documentation

An Intel® PFR Descoping section will be added to the *Intel® Platform Firmware Resilience (Intel® PFR) 5.0 Architecture and Hardware Design Guide*, document number [814910](#), revision 0.9. This will be the final revision of this collateral.

Oak stream Platform Root of Trust (PROT) Design Guide collateral will be released to help customers design their own ProT solutions. This document will detail CPU supported SPD 1.2 protocols, AFM and measurement calculation and other ProT features for customer reference.

Monitor the Message of the Week (MoW) for updates on collaterals availability.

2.19 WW33-2025

New:

- [I3C_DBG Has Limited Functionality in Diamond Rapids IMH1-Based SKUs](#) on page 217
- [I3C_MNG Has Limited Functionality in Diamond Rapids IMH1-Based SKUs](#) on page 218
- [Oak Stream 16-Channel Customer Reference Board \(CRB\) Johnson City Ordering Window Now Open](#) on page 218
- [Transition to Pay-For-Service Model for EDA Conversion Service](#) on page 219

2.19.1 Enabling

2.19.1.1 I3C_DBG Has Limited Functionality in Diamond Rapids IMH1-Based SKUs

- **Background**
 - In Diamond Rapids processors, there are two I3C_DBG target instances per socket, one in each IMH die. Each I3C_DBG target instance requires a unique ID.
- **Problem Statement**
 - In IMH1-based SKUs, the I3C_DBG target instance ID in the secondary IMH die is incorrectly set.
- **Root cause**
 - IMH1-based SKUs do not route SOCKET_ID and PID strap pins to the secondary IMH correctly, resulting in duplicated I3C_DBG target addresses in multi-socket configurations.
- **Implication**
 - I3C_DBG is not functional in multi-socket platforms with IMH1-based SKUs.
 - I3C_DBG is functional in single-socket platforms.
- **Status**
 - No fix for IMH1-based SKUs. This issue will be fixed in IMH2-based SKUs.
- **Mitigation**
 - For IMH1-based SKUs multi-socket configurations, use XDP as debug probe.

2.19.1.2 I3C_MNG Has Limited Functionality in Diamond Rapids IMH1-Based SKUs

- **Background**
 - In Diamond Rapids processors, there are two I3C_MNG target instances per socket, one in each IMH die. Each I3C_MNG target instance requires a unique ID.
- **Problem Statement**
 - In IMH1-based SKUs, the I3C_MNG target instance ID in both primary and secondary IMH die is duplicated.
- **Root cause**
 - IMH1-based SKUs do not route PID strap pins to the primary and secondary IMH correctly, resulting in duplicated I3C_MNG target addresses.
- **Implication**
 - I3C_MNG will not be functional on multi-socket platforms that use socket-level partitioning (1x2S configurations).
 - I3C_MNG is functional in single-socket platforms and multi-socket platforms that do not use socket-level partitioning (2S configuration).
- **Status**
 - A microcode workaround for IMH1-based SKUs is under evaluation. Monitor the Message of the Week (MoW) for updates.

2.19.1.3 Oak Stream 16-Channel Customer Reference Board (CRB) Johnson City Ordering Window Now Open

As communicated in WW24'25, the ordering window for [Oak Stream 16-Channel Customer Reference Board \(CRB\) Program](#) is now open.

There are two CRB products for Oak Stream 16-Channel Platform:

- [JNC1 - 1 DIMM slot per channel, 16 slots](#)
- [JNC2 - 2 DIMM slot per channel, 32 slots](#)

To access these products, you must sign into the [Design-In Tools \(DIT\)](#) website using your Intel Premier Account.

Coupon Information

Selected coupons will be distributed through your Intel Representative starting WW34'2025. These coupon codes can be used on DIT website during checkout. **Ensure that you are using the correct coupon code for the correct product offering.**

Coupon codes have the following notation "JNC_ SS _AAA_XXXXXX ", where:

- "SS" represents board configuration: 1S for 1 board, 2S for 2 boards, and 4S for 4 boards).
- "AAA" represents company code name
- "XXXXXX" is the unique coupon number

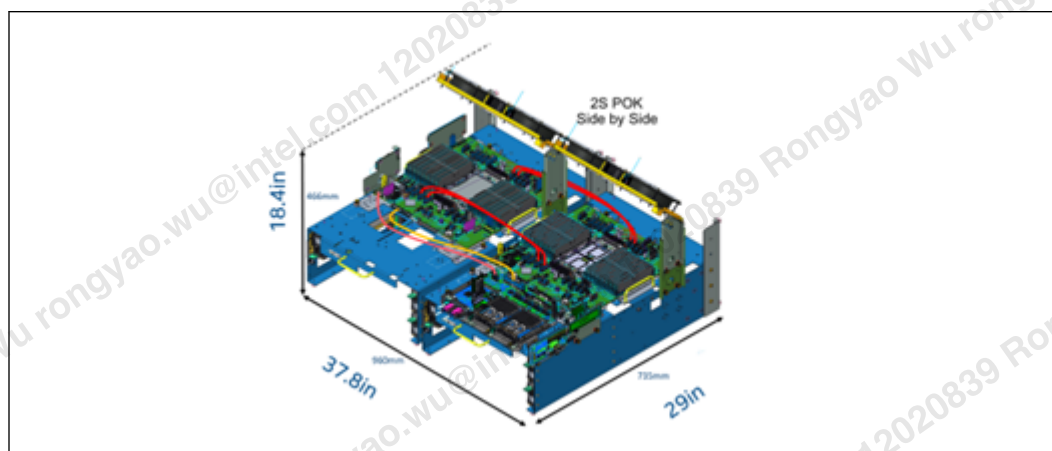
Each CRB includes:

- 1 Power On Kit (POK): 1 Chassis, 1 Power Distribution Board (PDB), 1 PSU, fans, brackets, Boot Module, EMCC module, and cables.
- 1 Baseboard (pre-assembled on POK, includes bolster plate and backplate)
- Data Center Secure Control Module (DC-SCM) (pre-installed on baseboard)
- 1 Diamond Rapids Processor (pre-installed on baseboard, includes heatsink)
- 2 DDR5 RDIMMs

For 2S and 4S board configurations, the following accessories are also included: Aggregator board, PDB, and cables.

The following figure showcases a 2-Socket board setup:

Figure 42. Johnson City 2-Socket Cabled Board Setup



Assembly instructions will be included in the shipment detailing 2-Socket cabling connectivity (red and orange cables from the figure).

Shipment Schedule

First CRB shipments will align with A-step samples availability.

Additional Information

Johnson City Theory of Operations (TOPs) will contain instructions for platform assembly, debug, and operation. Monitor the Message of Week for updates on collaterals availability.

2.19.1.4 Transition to Pay-For-Service Model for EDA Conversion Service

The EDA Conversion Service will transition to a pay-for-service model. To maintain access to Intel reference designs converted from original Cadence formats to other EDA formats, an annual access fee of \$2,799 USD will be applied per Company Account ID. Payments can be made through the [Intel® Design-In Tools Store website](#).

Converted files will be available for download on the Resource and Documentation Center (RDC) website.

Table 78. EDA Conversion Service Output Formats

Layout Files Output Format	Schematic Files Output Format
Mentor (.pcb +. prj)	Mentor (.prj)
Zuken CR8000 (.dsgn)	Zuken (.sdm)
Zuken CR5000(.rul/ .pcb)	OrCAD (.dsn, .iscf)
ODB++ (.tgz)	Concept HDL (.cpm)
Altium(.PcbDoc)	Altium(.PrjPcb)
	Main device Symbol (.olb)

For detail instructions on how to request and EDA Conversion Service, see the *Intel EDA Conversion Service Transition: Important Update & FAQ Guide*, document number 861035.

2.20 WW32-2025

Updated:

- (Updated WW32) Oak Stream Key Collaterals Availability and Forecast on page 220

2.20.1 (Updated WW32) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 79. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	Q4'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW36'25
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> <i>Note: This collateral replaces Oak Stream Platforms BIOS Writer's Guide, document number 820107.</i>	839071	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.5	1.0	Late Q3'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW36'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	WW31'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	0.9	Q4'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	0.95	WW40'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	WW37'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.31	0.5	WW37'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	WW39'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q3'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.21 WW31-2025

New:

- [Oak Stream 16-Channel Platform MDP2 Design Archive Release Notification](#) on page 223
- [Processor Sensors Available via DMTF Platform Level Data Model \(PLDM\) Commands](#) on page 223

Updated:

- [\(Updated WW31\) Oak Stream Key Collaterals Availability and Forecast](#) on page 222

2.21.1 Documentation

2.21.1.1 (Updated WW31) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 80. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	Q4'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW36'25
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces Oak Stream Platforms BIOS Writer's Guide, document number 820107.	839071	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.5	1.0	Late Q3'25
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW36'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	WW31'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.71	0.9	Q4'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	0.95	WW40'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q3'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.31	0.5	WW33'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	WW39'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q3'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.21.1.2 Oak Stream 16-Channel Platform MDP2 Design Archive Release Notification

The *Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive*, document number [828483](#), revision 0.71, is now available in the Resource and Documentation Center.

Per the [Oak Stream 16-Channel Platform Board Landscape](#) article, MDP2 is an Oak Stream 16-Channel 2-Socket (2S) monoplanar design in a M-SDNO C335 board form factor, that now employs the narrow Torsion Fastener Loading Mechanism (TFLM) and tighter DIMM-to-DIMM pitch.

An overview of revision 0.71 changes is available in the change list PDF included in the compressed file.

2.21.2 Enabling

2.21.2.1 Processor Sensors Available via DMTF Platform Level Data Model (PLDM) Commands

Starting on the Oak Stream platform, Intel will expose a subset of Intel® Xeon® processor sensors via industry standard DMTF PLDM Type 2 commands over DMTF Management Component Transport Protocol (MCTP) as an alternative to current Plan of Record (POR) Intel proprietary features: System Management API (previously known as PECI commands) and Intel® Platform Monitoring Technology.

PLDM Sensor List

- CPU Power Average
- CPU Power Cap
- DIMM Power Average
- DIMM Power Cap per CPU
- DTS (temperature)
- Die (temperature)
- DIMM Temperature
- DIMM TS0 Temperature

- DIMM TS1 Temperature
- CPU Energy
- DIMM Energy per CPU
- Resolved Cores
- CPU Model/Stepping
- CPU Microcode Update Version
- CPU PPIN

PLDM Commands Supported on Oak Stream

Type 0 - Discovery and Control Commands:

- Get/Set TID
- Get PLDM Version
- Get PLDM Types
- Get PLDM Commands

Type 2 - Platform Monitoring and Control Commands:

- Set Numeric Sensor Enable
- Get Sensor Reading
- Get PDR Repository Info
- Get PDR

Platform Descriptor Records:

- Terminus Locator
- Processor Entity
- Entity Association
- Compact Numeric Sensor

PLDM support is a step towards replacing proprietary interfaces with industry standards. More details will be provided at the October Intel Datacenter Design Conference (DCDC). DCDC Registration details will be communicated through the corresponding October Message of the Week (MoW).

Contact your Intel representative if you have feedback on PLDM support.

2.22

WW30-2025

New:

- [Oak Stream 16-Channel Platform Narrow KOZ, ICD, and 3D Models Collateral Clarification](#) on page 226
- [Diamond Rapids Server Platform BIOS Developer's Guide Release Notification](#) on page 228
- [Intel® Modular Flash Image Tool \(Intel® Modular FIT\) GUI Deprecation](#) on page 228

Updated:

- (Updated WW30) Oak Stream Key Collaterals Availability and Forecast on page 225

2.22.1 Documentation

2.22.1.1 (Updated WW30) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 81. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	Q4'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW36'25
<i>Diamond Rapids Server Platform BIOS Developer's Guide</i> Note: This collateral replaces Oak Stream Platforms BIOS Writer's Guide, document number 820107.	839071	0.9	1.0	Q4'25
<i>Oak Stream 16-Channel 25 in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.5	1.0	Late Q3'25
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW36'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	WW31'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	WW31'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	0.95	WW40'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q3'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.31	0.5	WW33'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q3'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.22.1.2 Oak Stream 16-Channel Platform Narrow KOZ, ICD, and 3D Models Collateral Clarification

New revisions of the following mechanical collaterals are now available in the Resource and Documentation Center (RDC):

- *Preliminary Narrow Keep Out Zone (KOZ) and Interface Control Drawings (ICD) for Oak Stream 16-Channel*, document number [851010](#), revision 0.35.
- *Oak Stream 16 Channel Narrow Mechanical Models*, document number [852564](#), revision 0.35.

The Thermal Solution Interface Control Drawing (ICD) and its 3D model are updated in these new revisions. ICD updates include:

- Updated the location of corner holes for Anti-Tilt Nuts in the thermal solution base. Holes position shifted by 0.35 mm.
- Added an additional dimension for Heatsink Hardware holes that was not included in the previous release.

On thermal solutions designed against the original ICD, it is likely that Anti-Tilt Nuts will still align properly with the mating Anti-Tilt Stud features on the Bolster Plate. In this case, a thermal solution may properly assemble with a MB; however, it comes with the risk of rubbing of the AT Nut against the inside of the thermal solution hole. If this risk is observed, the hole in the thermal solution base may be enlarged to encompass the new hole location.

All Heatsink Hardware properly aligns with the updated Anti-Tilt Nut Hole in the thermal solution base. The only risk is to the Anti-Tilt Nut rubbing against the inside of the thermal solution base hole.

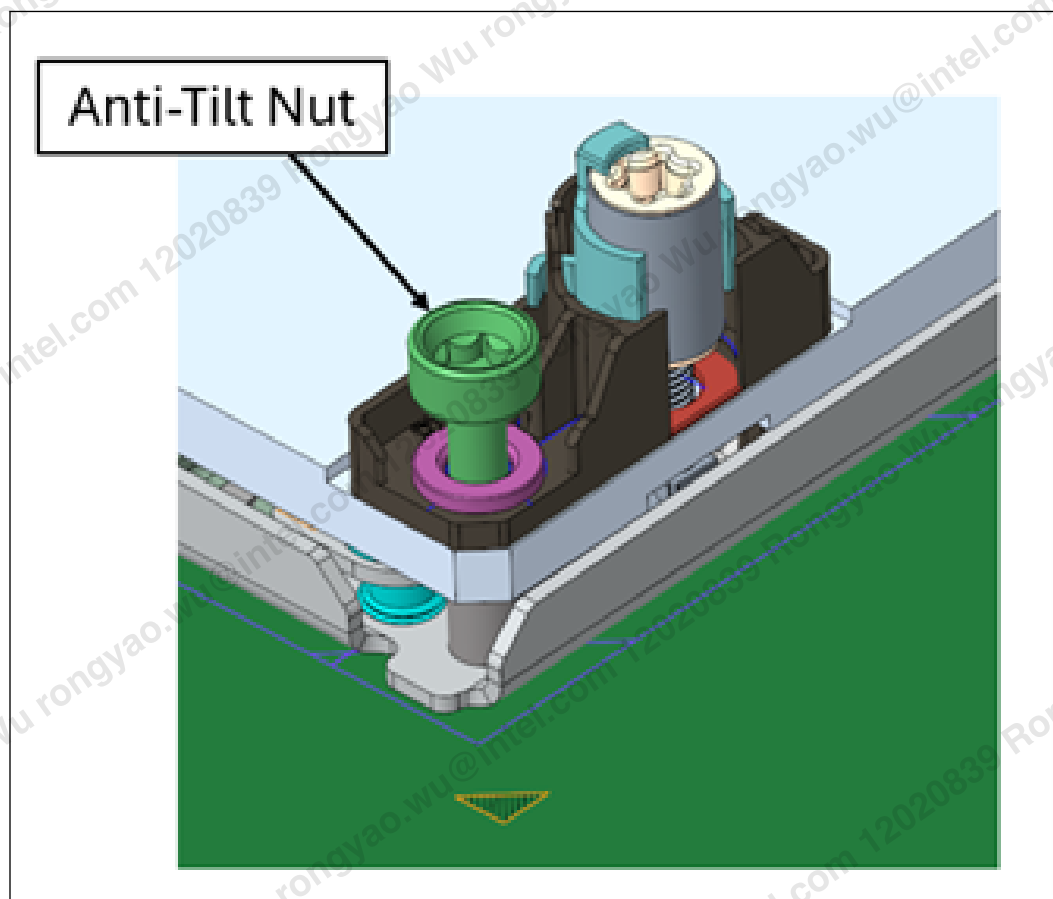
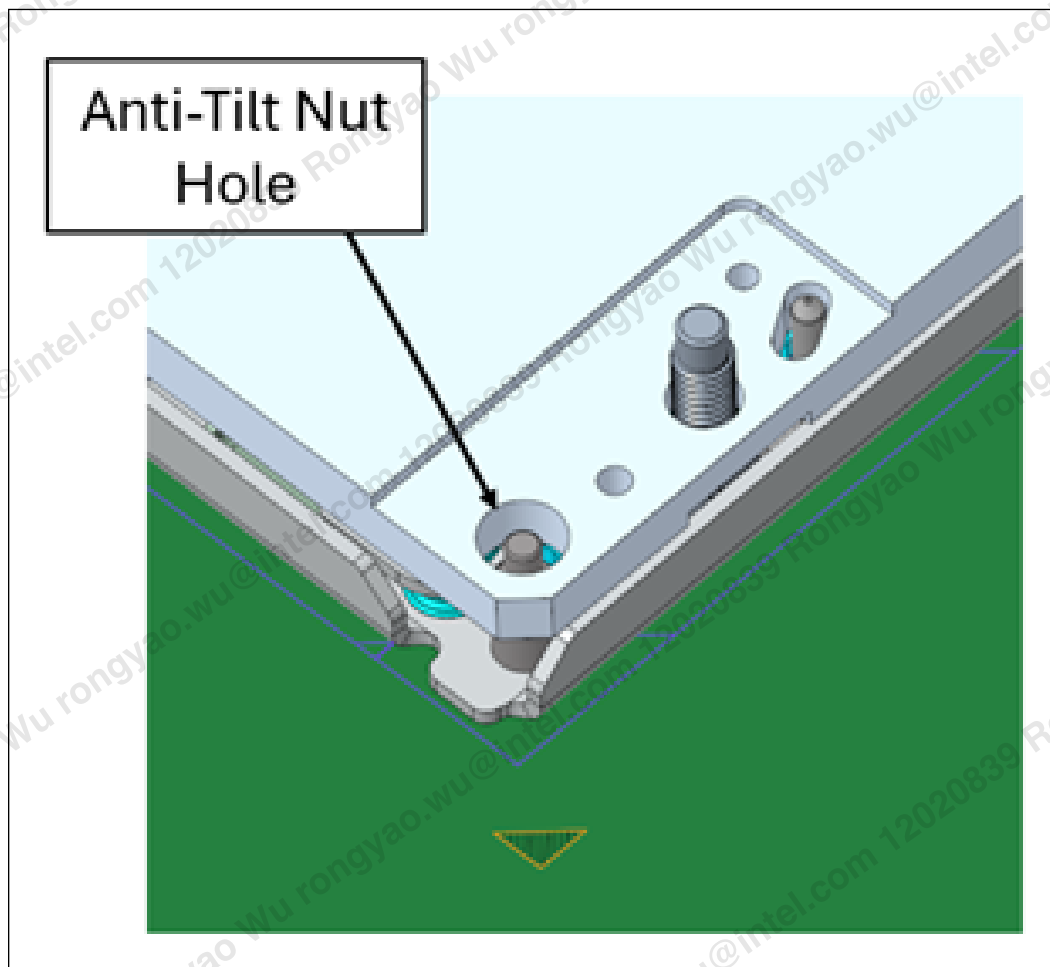
Figure 43. Anti-Tilt Nut

Figure 44. Anti-Tilt Nut Hole



2.22.1.3 Diamond Rapids Server Platform BIOS Developer's Guide Release Notification

The *Diamond Rapids Server Platform BIOS Developer's Guide*, document number [839071](#), Revision 0.9, is now available on the Resource and Documentation Center.

This BIOS Developer's Guide replaces the *Oak Stream Platform BIOS Writer's Guide*, document number 820107. Developer Guide is a processor-based document and will cover guidelines for both Xeon® and Xeon-D processor segments.

2.22.2 Enabling

2.22.2.1 Intel® Modular Flash Image Tool (Intel® Modular FIT) GUI Deprecation

Intel Modular FIT will only have command line interface in the Oak Stream Platform. Some of the key advantages of command line interface are:

- Allows configuring multiple settings in one command.

- Verifies all settings provided by the user are compatible with each other.

The *Intel® Modular Flash Image Tool (Intel® Modular FIT) User Guide for Oak Stream*, document number [834939](#), will be updated to reflect command line usage only.

2.23 WW29-2025

New:

- [Diamond Rapids Processor Architecture Specification Release Notification](#) on page 230
- [Cadence* SKILL Script for CuMax Delay Calculation Clarification](#) on page 231
- [Oak Stream 16-Channel Platform DDR_RESET \(FPGA to DIMM\) Topology Update](#) on page 231

Updated:

- [\(Updated WW29\) Oak Stream Key Collaterals Availability and Forecast](#) on page 229

2.23.1 Documentation

2.23.1.1 (Updated WW29) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 82. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.9	1.0	Q4'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW36'25
<i>Oak Stream 16-Channel 2S in 19-Inch Design Feasibility</i>	843581	1.6	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform CPLD Reference Code</i>	860022	0.5	1.0	Late Q3'25
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW36'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	WW31'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	WW30'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	1.0	Q4'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q3'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.31	0.5	WW33'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q3'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	WW30'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.23.1.2 Diamond Rapids Processor Architecture Specification Release Notification

The *Diamond Rapids Processor Architecture Specification*, document number 792359, revision 0.9, is now available on the Resource and Documentation Center.

Chapters updated in Revision 0.9:

- Overview, Features, and Topologies
- Flexible I/O (FIO)
- Intel® Out-of-Band Management Services Module (Intel® OOBM Services Module)
- Memory Subsystem
- Power Controller Unit (PCU)
- Reliability, Availability, and Serviceability (RAS)
- Processor Information ROM (PIROM)

See the revision history for the complete list of changes.

2.23.2 Enabling

2.23.2.1 Cadence* SKILL Script for CuMax Delay Calculation Clarification

Cadence SKILL Script for CuMax Delay Calculation, document number [857251](#), only functions if the **SNAKE_ROUTETAB** property is present in the layout .brd file.

To confirm if SNAKE_ROUTEAB property is present:

1. Open layout file.
2. Make a routing layer visible.
3. Zoom into an area containing a corner of the processor.
4. In the menu bar, select **Display** and then **Element**. This will enable the **Show Element** window.
5. In the **Options | Find | Visibility** menu, select **Find**. Within the "Design Object Find Filter" sub-window, Select **Shapes**.
6. Select aCuMax structure. If the **Show Element** window is not visible, right-click to raise the context menu and select **Show Element**.
7. If present in the database, *SNAKE_ROUTETAB* will be shown under **Properties attached to shape**.

Due to a known issue, in some cases the script's "Report" feature may generate an error and a 0-length output file when used under Cadence Allegro* 24.1. The "Report" function is only a convenience to help users avoid having to check the final mismatches in Allegro's Constraint Manager. The workaround is to use the "Add Delay" function of the script, then use Constraint Manager to check the delays, avoiding the "Report" function altogether. Users may make layout adjustments to match electrical delays and use Constraint Manager directly to check them, as the "Pin Delay" property added by the script will be part of the electrical delays calculated by Constraint Manager.

Reminder

Adding CopperMax (CuMax) tabs to a printed circuit board layout is only supported in Cadence* Allegro 23.1 and later, with the appropriate licenses. Contact cumax-info@intel.com for CuMax support.

2.23.3 Tools and Debug

2.23.3.1 Oak Stream 16-Channel Platform DDR_RESET (FPGA to DIMM) Topology Update

The Johnson City reference platforms implement a different DDR_RESET (FPGA to DIMM) topology for both 1SPC and 2SPC than what is currently supported in the *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#), revision 0.81. The current PDG topologies may be difficult to route within the given PDG revision 0.81 guidelines.

The DDR_RESET (FPGA to DIMM) topologies for both 1SPC and 2SPC will be updated to align with the Johnson City reference platform implementation. The updated topologies and a summary of the length routing guideline updated values are shown below. These updates will be intercepted in PDG revision 0.9.

NOTE

The recommended value for R_s is 100 Ohms, assuming the FPGA has a typical drive strength of 25 Ohms or weaker. If customer drivers are stronger (<25 Ohms typical), R_s may need to be modified to manage over/undershoot and slope reversal. Johnson City 2 uses $R_s = 33$ Ohms, which works because lengths are short enough; however, 100 Ohms is better at reducing slope reversal and overshoot.

Figure 45. PDG 0.81 Current DDR5 Miscellaneous Memory Signal Guidelines, RESET, FPGA to DIMM, 1SPC

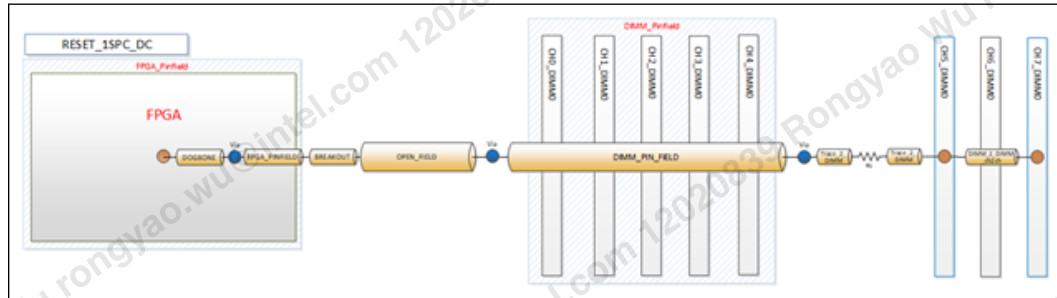


Figure 46. PDG 0.9 Updated DDR5 Miscellaneous Memory Signal Guidelines, RESET, FPGA to DIMM, 1SPC

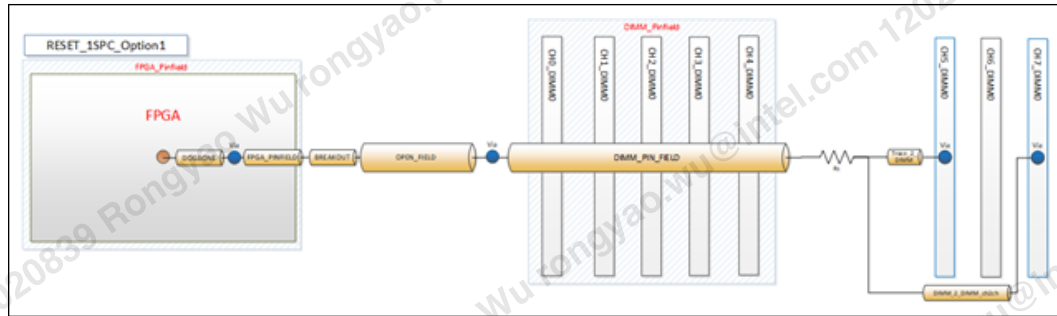


Table 83. PDG 0.9 Summary of Updated DDR5 RESET, FPGA to DIMM 4H5 (120 mils) and 3H4 (104 mils) Length Routing Guidelines, 1SPC

Routing Section	Layer Type	Layer Allowed	Via Count Base Board	Min(mils) Base Board Trace Length	Max(mils) Base Board Trace Length	4H5 (120 mil) Stack Up Trace Width (mils)	3H4 (104 mil) Stack Up Trace Width (mils)	Single Ended Impedance	Tol(+/-) % (Single) Impedance	Reference to Far Layer	Reference to Near Layer	Reference Distance To Plane Edge (mils)
DOGBONE	Top	Yes	1	0	25	4.8	4.7	-	-	-	-	-
FPGA_PINFIELD	Stripline	Yes	0	0	620	4.4	3.2	-	-	-	-	-
BREAKOUT	Stripline	Yes	0	0	165	4.4	3.2	50	10	-	-	-
OPEN_FIELD	Stripline	Yes	1	0	11235	4.4	3.2	50	10	-	-	-
DIMM_PINFIELD	Bottom	Yes	0	0	4050	4.8	4.7	50	15	-	-	-
TRACE_TO_DIMM	Bottom	Yes	1	0	75	4.8	4.7	50	15	-	-	-
DIMM_TO_DIMM_ch2ch	Bottom	Yes	1	0	850	4.8	4.7	50	15	-	-	-
ENTIRE_SIGNAL	All	-	4	-	-	-	-	-	-	GND	GND	-

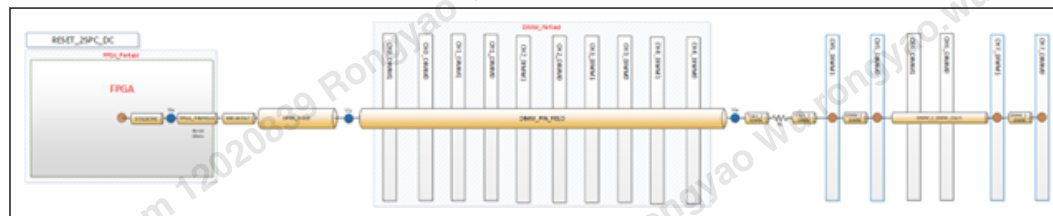
NOTEUpdated values are shown in **bold**.**Figure 47. PDG 0.81 Current DDR5 Miscellaneous Memory Signal Guidelines, RESET, FPGA to DIMM, 2SPC**

Figure 48. PDG 0.9 Updated DDR5 Miscellaneous Memory Signal Guidelines, RESET, FPGA to DIMM, 2SPC

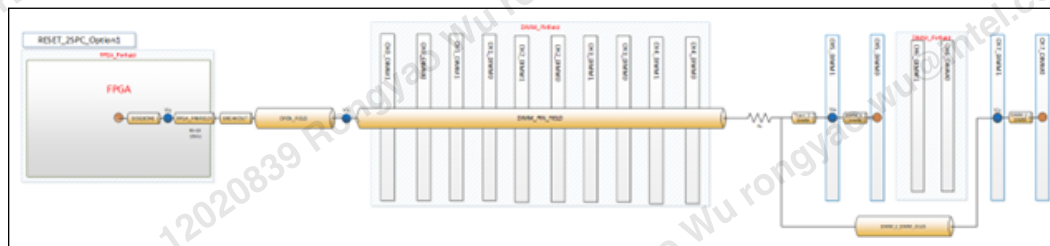


Table 84. PDG 0.9 Summary of Updated DDR5 RESET, FPGA to DIMM 4H5 (120 mils) and 3H4 (104 mils) Length Routing Guidelines, 2SPC

Routing Section	Layer Type	Layer Allowed	Via Count Based Board	Min (mils) Base Board Trace Length	Max (mils) Base Board Trace Length	4H5 (120 mil) Stack Up Trace Width (mils)	3H4 (104 mil) Stack Up Trace Width (mils)	Single Ended Impedance	Tol (+/-) % (Single) Impedance	Reference to Far Layer	Reference to Near Layer	Reference Distance To Plane Edge (mils)
DOGBONE	Top	Yes	1	0	25	4.8	4.7	-	-	-	-	-
FPGA_PINFIELD	Strip line	Yes	0	0	620	4.4	3.2	-	-	-	-	-
BREAKOUT	Strip line	Yes	0	0	170	4.4	3.2	50	10	-	-	-
OPEN_FIELD	Strip line	Yes	1	0	11235	4.4	3.2	50	10	-	-	-
DIMM_PIN_FIELD	Bottom	Yes	0	0	4830	4.8	4.7	50	15	-	-	-
TRACE_TO_DIMM	Bottom	Yes	1	0	165	4.8	4.7	50	15	-	-	-
DIMM_TO_DIMM_ch2	Bottom	Yes	1	0	1500	4.8	4.7	50	15	-	-	-
DIMM_TO_DIMM	Top	Yes	0	200	500	4.8	4.7	50	15	-	-	-
ENTIRE_SIGNAL	All	-	4	-	-	-	-	-	-	GND	GND	-

NOTE

Updated values are shown in **bold**.

Contact your Intel representative if you have any questions or concerns with the updated DDR_RESET (FPGA to DIMM) topologies.

2.24 WW28-2025

New:

- Oak Stream 16-Channel Platform DDR Command Address (CA, PAR, CS_N) Open Field Spacing Rules Update for 104 mil/3H4 Stack Ups on page 235

2.24.1 Enabling

2.24.1.1 Oak Stream 16-Channel Platform DDR Command Address (CA, PAR, CS_N) Open Field Spacing Rules Update for 104 mil/3H4 Stack Ups

There is an update to CA open field routing spacing rules for Oak Stream 16-Channel Platform 104 mil/3H4 stackup designs. 120 mil/4H5 stackup spacing rules remain unchanged.

The summary of the update for 1SPC and 2SPC is captured in the following table:

Table 85. Update to DDR5 Source Clocked Signal Spacing Guidelines (CA, PAR, CS_N)

		120 mils 4H5 Stackup	120 mils 4H5 Stackup	104 mils 3H4 Stackup	104 mils 3H4 Stackup
	Stackup	PDG 0.81	PDG 0.9	PDG 0.81	PDG 0.9
Routing Section	Layer Type	Trace to Trace (mils)	Trace to Trace (mils)	Trace to Trace (mils)	Trace to Trace (mils)
DOGBONE	Top	-	-	-	-
CPU_PINFIELD	Stripline	-	-	-	-
BREAKOUT	Upper Stripline (L3, L5, L7, L9)	6	6	7.2	7.2
BREAKOUT	Lower Stripline (L14, L16, L18, L20)	10	10	11.2	11.2
OPEN_FIELD	Upper Stripline (L3, L5, L7, L9)	3.0*h (12mils)	3.0*h (12mils)	4.4*h (13.2mils)	4.0*h (12mils)
OPEN_FIELD	Lower Stripline (L14, L16, L18, L20)	3.6*h (14.4mils)	3.6*h (14.4mils)	5.2*h (15.6mils)	4.8*h (14.4mils)
DIMM_PIN_FIELD	Upper Stripline (L3, L5, L7, L9)	3.0*h (12mils)	3.0*h (12mils)	4.4*h (13.2mils)	4.0*h (12mils)
DIMM_PIN_FIELD	Lower Stripline (L14, L16, L18, L20)	3.6*h (14.4mils)	3.6*h (14.4mils)	5.2*h (15.6mils)	4.8*h (14.4mils)
DIMM_2_DIMM	Top	6.4*h	6.4*h	6.4*h	6.4*h
(CPU_ \$CH_DIMM1)	Stripline	-	-	-	-
ENTIRE_SIGNAL	All	-	-	-	-

Notes:

- h stands for the dielectric height from the nearest ground reference plane.

- This update will be intercepted in Oak Stream 16-Channel Platform Design Guide, document number [788196](#), Revision 0.9.
- Design review services do not comprehend latest guidelines. Review process will be aligned after PDG release.

2.25 WW27-2025

New:

- [Clock Jitter Tool 6.0.3 Release Notification](#) on page 236
- [Oak Stream 16-Channel Platform Thermal Mechanical Specification \(TMS\) Release Notification](#) on page 238
- [Oak Stream 8-Channel Platform Thermal Mechanical Specification \(TMS\) Release Notification](#) on page 239
- [Oak Stream 16-Channel Platform 2-Socket Stacked System Topology Feedback Request](#) on page 241

Updated:

- [\(Updated WW27\) Diamond Rapids Tools Readiness](#) on page 236
- [\(Updated WW27\) Oak Stream Key Collaterals Availability and Forecast](#) on page 240

2.25.1 Tools and Debug

2.25.1.1 Clock Jitter Tool 6.0.3 Release Notification

The *Clock Jitter Tool*, document number [803838](#), version 6.0.3, is now available in the Resource and Documentation Center (RDC).

Version 6.0.3 is a bug-fix release to address a command line interface issue with 6.0.2 version that prevented the use of the tool through the command line due to a splash screen getting stuck.

The feature set of 6.0.3 version is unchanged from version 6.0.2.

NOTE

Free MathWorks MATLAB® Runtime version 9.10, available directly from the MathWorks website, is required to execute CJT 6.0.3 version.

2.25.1.2 (Updated WW27) Diamond Rapids Tools Readiness

Updated information highlighted in **bold text**

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to ITP Adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support
 - Two Streams Tracing

- LA-3907 preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports four trace ports)
- LA-3892 converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- ASSET* Arium - contact ASSET Arium for support
- Compact Debug Port (CDP) + MIPI10 Adapter for I3C*
 - Type-C connection: voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- Vtune™
- Intel® SoC Watch
- Intel® System Debugger
 - Token generation for Vendor Authorize Boot (VAB) / Vendor Authorize Debug (VAD)

Table 86. Diamond Rapids Tools Readiness

Tool	Trend
ASSET Intertech Arium *	TBD
ASSET* Close Chassis Controller (CCC)	TBD
Autonomous Crashdump (ACD)	Q3'25
CScripts	Q4'25
Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card <i>Note: Contact your Intel representative for availability.</i>	Available
Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C* (MM 99CJRJ)	Available
Intel® I3C* USB Type-C Level Shift Adapter (MM 99CN2C)	Available
Intel® At-Scale Debug (Intel® ASD) BMC-JTAG* Software Guide, document number 594948	Q3'25
W: Integrated Channel Analysis Tool (ICAT) Suite (document number 782966)	Available
W: Intel® Platform Clock Jitter Analysis Tool (document number 619648)	Available
16-Ch: PCIe* and Intel® UPI Load Board Solution Set	Available
16-Ch: PCIe* Load Board Solution Set	Available
8-Ch: Load Board Solution Sets	Q4'25
W: SelfTest 7 (document number 524152)	Available
Intel® SVT (Silicon View Technology) Closed Chassis Adapter (MM 921521)	Available
Intel® System Bring-up Toolkit NDA	Q4'25
Intel® In Field Scan	Q4'25
W: Intel® In-Target Probe (Intel® ITP) eXtended Debug Port (XDP) (MM 940990)	Available
W: Intel® IO Margin Tool and User Guide	Q4'25
L: Intel® IO Margin Tool and User Guide	Q4'25
Lauterbach*	TBD
Memory Assistant	Q4'25
<i>continued...</i>	

Tool	Trend
W: Memory Error Injection (MEI) DDR5 Card (MM 99CDV1)	Available
W: Memory Error Injection (MEI) DDR5 Lite Card (8000 MT/s)	WW32'25
PCIe* 5.0 Error Injection (PEI) Card (MM 99C3FA)	Available
PCI Express* (PCIe*) PEI 6.0 Card	TBD
Intel® System Stress and Memory Diagnostics <i>Note:</i> Replaces Intel® PTAT	Q3'25
Stitched Rank Margining Tool (RMT), Target Loader RMT (EFI)	Q4'25
Intel® Structural Test Plus (Intel® STP)	Available
16-ch: Almondsville Thermal Test Vehicle (TTV)	Q3'25
8-ch: Turkey Canyon TTV	Q4'25
<i>Server Security Toolkit, document number 819397</i>	Available
W:Gen5 VR Test Tool (VRTT)	Available
16-Ch: Red Interposer for Gen5 VRTT (MM 99D5W1)	Available
16-Ch: Blue Interposer for Gen5 VRTT (MM 99D5W2)	Available
16-Ch: VRTT Mechanical Interposer Retention Mechanism Kit (MM 99CN1R)	Available
8-Ch: VRTT Interposers and Retention Mechanism Kit	TBD

NOTE

W: Windows*. L: Linux*. If not specified, the tool is OS agnostic.

NOTE

Contact your Intel Representative for Intel® Simics support.

2.25.2 Documentation

2.25.2.1 Oak Stream 16-Channel Platform Thermal Mechanical Specification (TMS) Release Notification

The *Oak Stream 16-Channel Platform Thermal Mechanical Specification*, document number [786521](#), revision 0.9 is now available in the Resource and Documentation Center (RDC).

Overview of revision 0.9 changes:

- Updated Chapter 4: Package and Socket Stack
 - Added document number for LGA9324 Thermal Solution Stiffness Guideline Application Note in Mechanical Load Specification.
 - Added Processor and Socket Stack-Up Height
- Updated Chapter 5: Processor Thermal Specifications and Management
 - Added Processor Thermal Features.

- Added Thermal Control Circuit Activation Temperature.
- Updated Thermal Management.
- Revised Almondsville TMTV die number sequence to match with the actual TMTV, ensuring proper power application for each die type.
- Updated Chapter 7: Thermal Design Guideline
 - Updated thermal proxy for 2U 2S Cloud Front I/O system from 21" to 19", fan power consumption, system total airflow in the table "Equivalent System Thermal Proxy Boundary Condition Assumptions".
 - Updated DDR5 Power Estimate for Oak Stream 16-Channel Platform Thermal Sizing.
 - Updated 1U tall DIMM cooling capability.
 - Updated 2U tall DIMM cooling capability.
- Updated Chapter 8: Processor Heatsink Module and Loading Mechanism
 - Added discussion on platform transition to narrow hardware in Processor Heatsink Module and Loading Mechanism
- Added Chapter 11: Supplier Listing

NOTE

The enabled loading mechanism, carrier, and heatsink hardware are being updated for a narrower footprint on the MB and to enable Oak Stream 16-channel 2-socket in 19-inch designs. **All customer Oak Stream 16-Channel based designs must transition to the Narrow TFLM solution.** The current "wide" solution is still discussed in the latest TMS revision during the transition window. The narrow hardware stack will be included in a future revision of the TMS. See the *Oak Stream 16-Channel 2S in 19-Inch Design Feasibility Technical Article*, document number [843581](#), for related details.

2.25.2.2 Oak Stream 8-Channel Platform Thermal Mechanical Specification (TMS) Release Notification

The *Oak Stream 8-Channel Platform Thermal Mechanical Specification*, document number [786524](#), revision 0.5, is now available in the Resource and Documentation Center (RDC).

Overview of revision 0.5 changes:

- Updated Chapter 1: Introduction
 - Added *Turkey Canyon Thermal Mechanical Test Vehicle (TMTV) Package Thermal Model User Guide*, document number [852612](#) to Reference Documents table.
- Updated Chapter 5: Processor Thermal Specifications and Management
 - Added Turkey Canyon Thermal Mechanical Test Vehicle (TMTV) Overview and Diamond Rapids HCC/LCC Product Package Type.
 - Added Turkey Canyon TMTV TCASE Methodology.
 - Added Turkey Canyon TMTV to Diamond Rapids HCC/LCC Product TCASE Guidance and Thermal Boundary Conditions.
- Added Chapter 7: Thermal Design Guideline.

2.25.2.3 (Updated WW27) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 87. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	WW28'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW36'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW35'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	WW31'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	Mid Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	WW29'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.9	1.0	Q4'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q3'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.31	0.5	WW33'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Aggregator Board Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.5	0.7	Q4'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Mid Q3'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	WW29'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

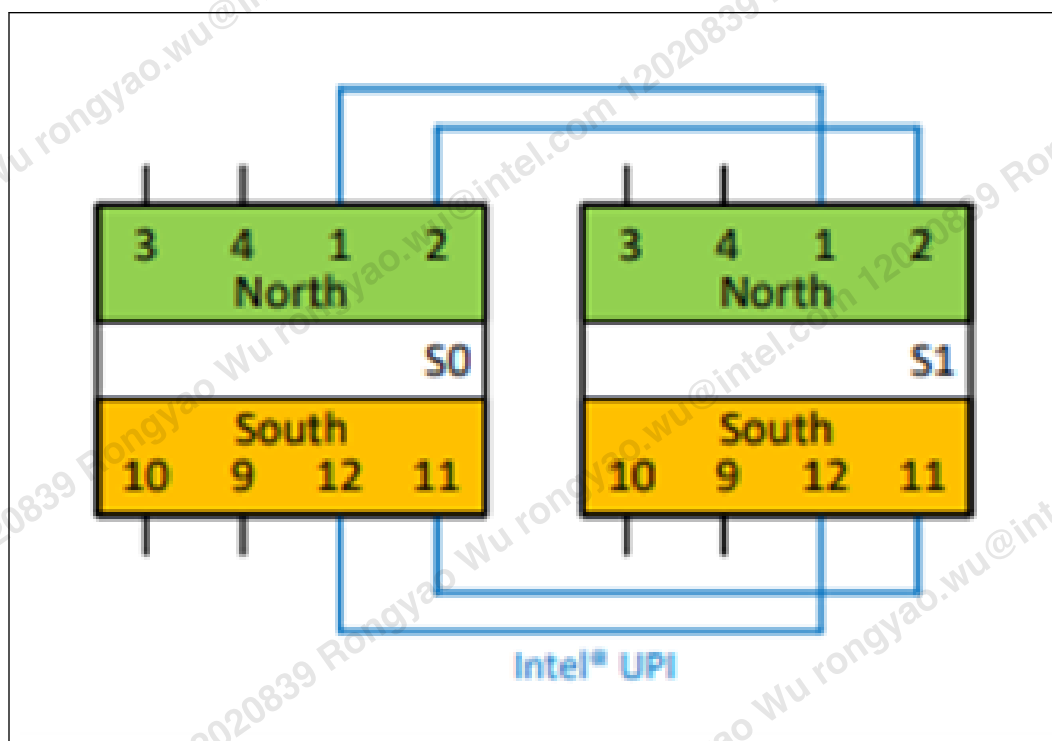
2.25.3 Customer Feedback

2.25.3.1 Oak Stream 16-Channel Platform 2-Socket Stacked System Topology Feedback Request

Intel is requesting customer feedback on the Oak Stream 16-Channel Platform 2 Socket (2S) topology in a stacked socket system.

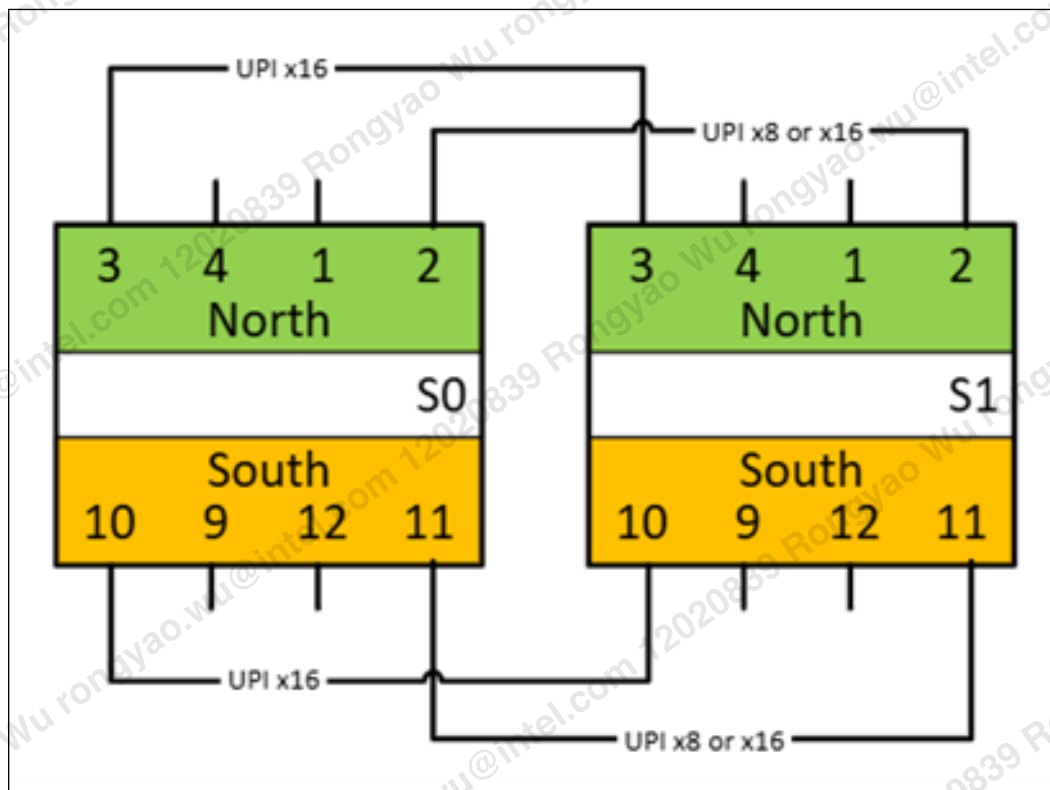
Currently, Intel's Plan of Record (POR) configuration for stacked systems is 2S4L Configuration #5, as documented in *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#). This topology uses FIO ports 1, 2, 11 and 12:

Figure 49. Current 2S4L Configuration #5



Intel is considering modifying the port mapping to use FIO ports 3, 2, 10 and 11 to accommodate more system configurations without increasing Intel's validation effort:

Figure 50. Proposed FIO Port Mapping Update for 2S4L Configuration #5



For customers implementing 2S4L Configuration #5, **assess if the proposed FIO port mapping update is suitable for your Oak Stream based designs.** Provide your input by **July 15th** to your Intel representative and to steven.e.shallenberg@intel.com.

2.26 WW26-2025

Updated:

- (Updated WW26) Diamond Rapids Tools Readiness on page 242

2.26.1 Tools and Debug

2.26.1.1 (Updated WW26) Diamond Rapids Tools Readiness

Updated information highlighted in **bold text**

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to ITP Adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support
 - Two Streams Tracing

- LA-3907 preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports four trace ports)
- LA-3892 converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- ASSET* Arium - contact ASSET Arium for support
- Compact Debug Port (CDP) + MIPI10 Adapter for I3C*
 - Type-C connection: voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- Vtune™
- Intel® SoC Watch
- Intel® System Debugger
 - Token generation for Vendor Authorize Boot (VAB) / Vendor Authorize Debug (VAD)

Table 88. Diamond Rapids Tools Readiness

Tool	Trend
ASSET *Intertech* Arium	TBD
ASSET* Close Chassis Controller (CCC)	TBD
Autonomous Crashdump (ACD)	Q3'25
CScripts	Q4'25
Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card <i>Note: Contact your Intel representative for availability.</i>	Available
Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C* (MM 99CJRJ)	Available
Intel® I3C* USB Type-C Level Shift Adapter (MM 99CN2C)	Available
Intel® At-Scale Debug (Intel® ASD) BMC-JTAG* Software Guide, document number 594948	Q3'25
W: Integrated Channel Analysis Tool (ICAT) Suite (document number 782966)	Available
W: Intel® Platform Clock Jitter Analysis Tool (document number 619648)	Available
16-ch: PCIe* and Intel® UPI Load Board Solution Set	Available
16-ch: PCIe* Load Board Solution Set	Available
8-Ch: Load Board Solution Sets	Q4'25
W: SelfTest 7 (document number 524152)	Available
Intel® SVT (Silicon View Technology) Closed Chassis Adapter (MM 921521)	Available
Intel® System Bring-up Toolkit NDA	Q4'25
W: Intel® In-Target Probe (Intel® ITP) eXtended Debug Port (XDP) (MM 940990)	Available
W: Intel® IO Margin Tool and User Guide	Q4'25
L: Intel® IO Margin Tool and User Guide	Q4'25
Lauterbach*	TBD
Memory Assistant	Q4'25
continued...	

Tool	Trend
W: Memory Error Injection (MEI) DDR5 Card (MM 99CDV1)	Available
W: Memory Error Injection (MEI) DDR5 Lite Card (8000 MT/s)	WW32'25
PCIe* 5.0 Error Injection (PEI) Card (MM 99C3FA)	Available
PCI Express* (PCIe*) PEI 6.0 Card	TBD
Intel® System Stress and Memory Diagnostics <i>Note:</i> Replaces Intel® PTAT	Q3'25
Stitched Rank Margining Tool (RMT), Target Loader RMT (EFI)	Q4'25
Intel® Structural Test Plus (Intel® STP)	Available
16-ch: Almondsville Thermal Test Vehicle (TTV)	Q3'25
8-ch: Turkey Canyon TTV	Q4'25
Server Security Toolkit, document number 819397	Available
W:Gen5 VR Test Tool (VRTT)	Available
16-Ch: Red Interposer for Gen5 VRTT (MM 99D5W1)	Available
16-Ch: Blue Interposer for Gen5 VRTT (MM 99D5W2)	Available
16-Ch: VRTT Mechanical Interposer Retention Mechanism Kit (MM 99CN1R)	Available
8-Ch: VRTT Interposers and Retention Mechanism Kit	TBD

NOTE

W: Windows*. L: Linux*. If not specified, tool is OS agnostic.

NOTE

Contact your Intel Representative for Intel® Simics support.

2.27 WW25-2025

Updated:

- [\(Updated WW25\) Oak Stream Key Collaterals Availability and Forecast](#) on page 245

New:

- [Cadence* SKILL Script for CuMax Delay Calculation Release Notification](#) on page 246
- [Intel® Platform Clock Jitter Analysis Tool 6.0.3 Release Notification](#) on page 246

2.27.1 Documentation

2.27.1.1 (Updated WW25) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 89. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
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<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	Late Q2'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Late Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	636257	0.31	0.5	WW33'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.4	0.5	WW28'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Q3'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	WW26'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.27.1.2 Cadence* SKILL Script for CuMax Delay Calculation Release Notification

The Cadence* *SKILL Script for CuMax Delay Calculation*, document number [857251](#), is now available in RDC.

The use of CopperMax* ("CuMax*") technology for PCB DDR routing complicates the calculation of electrical delays for length matching. This Cadence* SKILL script will automatically compute the delays resulting from CuMax tabs across a platform and will adjust the electrical delays reported in Cadence Allegro's Constraint Manager*. The script will also generate reports to confirm the delays and matching on the platform. A full set of instructions is included with this collateral.

NOTE

Cadence Allegro 23.1 or later, with appropriate CuMax feature enablement, is required to add CuMax tabs to a design. For CuMax information, contact cumax-info@intel.com.

2.27.1.3 Intel® Platform Clock Jitter Analysis Tool 6.0.3 Release Notification

The Intel® *Platform Clock Jitter Analysis Tool*, version 6.0.3, document number [619648](#), is now available in RDC.

The release contains a user guide and executable to enable design and analysis of phase jitter on PCI Express* (PCIe*) reference clock architectures and topologies through PCIe* 6.0. The tool can also analyze Intel® UPI clock structures. The tool assumes that a clock source waveform or jitter periods file is available from measurement. Tool output is the RMS jitter at the receiver.

This revision includes an improved user interface, new batch mode features, pass/fail evaluation, command-line support, and the ability to save reports. This tool contains more features and capabilities than the PCI-SIG* software, *Clock Jitter Tool*, document number [803838](#), though the core algorithms are identical.

2.28 WW24-2025

New:

- [Oak Stream 16-Channel VR Test Tool \(VRTT\) Update](#) on page 247
- [Oak Stream 16-Channel Customer Reference Board \(CRB\) Program](#) on page 247

Updated:

- [\(Updated WW24\) Oak Stream Key Collaterals Availability and Forecast](#) on page 248

2.28.1 Tools and Debug

2.28.1.1 Oak Stream 16-Channel VR Test Tool (VRTT) Update

To support a narrow, Keep Out Zone (KOZ) (103.5 x 158 mm), Intel is offering a new Oak Stream 16-Channel VRTT Retention Mechanism that supports both the 103.5 x 158 mm KOZ (narrow Torsion Fastener Loading Mechanism [TFLM]) and 109.5 x 158 mm KOZ (wide TFLM). Customers can pre-order the samples at **Intel® Design-In Tools** website:

- [Pre-Order Oak Stream 16-Channel VRTT Retention Mechanism](#)

Samples are targeted to be shipped by WW26'2025.

To support a narrow DIMM pitch (0.257 inch or 0.262 inch), Intel is offering a new header board for DDR5 VR Test Tool required for narrow DIMM pitch. The header board is available for shipping and customers can order the samples at **Intel Design-In Tools** website:

- [Order Header board for DDR5 VR Test Tool narrow DIMM pitch](#)

2.28.2 Enabling

2.28.2.1 Oak Stream 16-Channel Customer Reference Board (CRB) Program

The Customer Reference Board (CRB) Program provides reference platforms to enable early comparative hardware testing and validation against Intel platform-based designs. The CRB offers easy access to oscilloscope probing and other Intel recommended debug procedures.

Product Offering

Intel will be offering two CRB products for Oak Stream 16-Channel Platform:

- Johnson City 1: 120 mil - 1 DIMM slot per channel
- Johnson City 2: 104 mil - 2 DIMMs slot per channel

Each CRB includes:

- 1 Power On Kit (POK): 1 Chassis, 1 Power Distribution Board (PDB), 1 PSU, and system cabling.
- 1 1-Socket Baseboard (pre-assembled on POK)
- Data Center Secure Control Module (DC-SCM) (pre-installed on baseboard)
- 2 DDR5 RDIMMs
- 1 Diamond Rapids Processor (pre-installed on baseboard, heatsink included)

The CRBs are assembled and shipped from the United States.

Eligibility

CRBs are loaned through the Data Center Customer and Systems Engineering (DCSE) CRB Program to approved and selected customers only. Inventory will be limited.

Preliminary Schedule

Below are the current trends for Oak Stream 16-Channel CRB Program:

- CRB Coupon distribution will start by Late July'25. Contact your Intel representative for CRB coupon inquiries.
- CRB products will be available on the Intel® Design-In Tools website by Mid Aug'25.
- CRB shipment will align with Diamond Rapids UCC Engineering samples availability.

Intel will continue to provide CRB Program updates through the Message of the Week (MoW).

2.28.3 Documentation

2.28.3.1 (Updated WW24) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 90. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
Diamond Rapids Processor Architecture Specification	792359	0.8	0.9	WW28'25
Diamond Rapids Processor Registers Specification	793272	0.9	1.0	WW36'25
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	0.9	As needed	As needed
Oak Stream 16-Channel Platform Design Guide	788196	0.81	0.9	WW35'25
Oak Stream 16-Channel Platform Electrical Specification	787006	0.8	0.9	WW31'25
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	0.9	1.0	Q3'25
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258	0.9	1.0	Q3'25
Oak Stream 16-Channel Platform Pin List	636260	0.9	1.0	Q3'25
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	N/A	0.5	Late Q2'25
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	0.5	0.7	Late Q2'25
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	0.7	0.9	Mid Q2'25
Oak Stream 8-Channel Platform Basin City Pin List	TBD	N/A	0.3	Mid Q2'25
Oak Stream 8-Channel Platform Basin City Design Archive	636257	0.31	0.5	WW33'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.4	0.5	Mid Q2'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Q3'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	WW26'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.29 WW23-2025

New:

- [Oak Stream 16-Channel Platforms Design Archive Addendum Release Notification](#) on page 250
- [Oak Stream 8-Channel Platform Basin City Design Archive Release Notification](#) on page 251
- [Narrow Torsion Fastener Loading Mechanism \(TFLM\) Execution Plan](#) on page 251
- [Oak Stream 16-Channel Plan of Record \(POR\) Change to Narrow Torsion Fastener Loading Mechanism \(TFLM\)](#) on page 253
- [Oak Stream 16-Channel Platform Board Landscape](#) on page 253

Updated:

- [\(Updated WW23\) Oak Stream Key Collaterals Availability and Forecast](#) on page 249

2.29.1 Documentation

2.29.1.1 (Updated WW23) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 91. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
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Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	0.9	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW35'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	WW31'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	Late Q2'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Late Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q2'25
Oak Stream 8-Channel Platform Basin City Design Archive	636257	0.31	0.5	WW33'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Early Q4'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.4	0.5	Mid Q2'25
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<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	WW26'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.29.1.2 Oak Stream 16-Channel Platforms Design Archive Addendum Release Notification

The *Oak Stream 16-Channel Platforms Design Archive Addendum*, document number [856301](#), revision 1.0, is now available in the Resource and Documentation Center (RDC).

These files are intended to be paired with the *Oak Stream 16-Channel Platform Johnson City 1 Design Archive*, document number [823065](#), revision 0.9, and the *Oak Stream 16-Channel Platform Johnson City 2 Design Archive*, document number [636258](#), revision 0.9.

The addendum includes the board files that implement the [DDR Command Address \(CA\) Length Matching Update](#) communicated in WW20'25. Customers can replicate the changes into their designs by referencing the CA routing or the complete DDR routing on the addendum files and incorporating them into their designs.

NOTE

The DDR CA Length Matching Update will be intercepted in revision 1.0 of both Johnson City Design Archives. The addendum intent is to provide DDR CA update reference prior to the 1.0 release. Monitor the Message of the Week for collateral forecast availability.

2.29.1.3 Oak Stream 8-Channel Platform Basin City Design Archive Release Notification

The *Oak Stream 8-Channel Platform Basin City Design Archive*, document number [636257](#), revision 0.31, is now available in the Resource and Documentation Center (RDC).

Overview of 0.31 changes:

- Initial Release

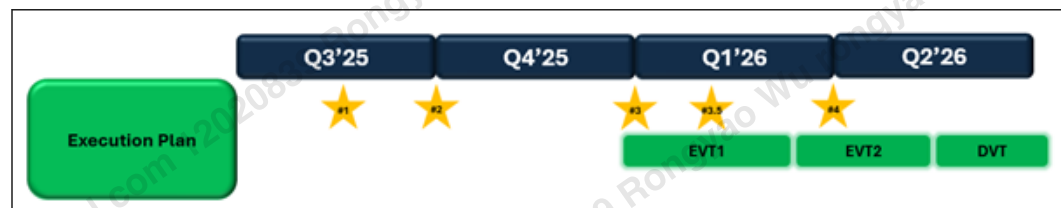
2.29.2 Enabling

2.29.2.1 Narrow Torsion Fastener Loading Mechanism (TFLM) Execution Plan

Following the Oak Stream 16-Channel Plan of Record (POR) Change to Narrow Torsion Fastener Loading Mechanism (TFLM), Intel is planning the following staggered execution plan for Oak Stream 16-Channel Platform based designs.

The following figure showcases the execution plan for the Narrow Torsion Fastener Loading Mechanism (NTFLM), each star represents a milestone of NTFLM Hardware (NTFLM HW) availability and maturity.

Figure 51. Narrow TFLM Execution Plan



First Milestone – Mid Q3'25

First milestone is focused on enabling early Power On preparations for customers with accelerated schedules:

- The quantity of NTFLM HW available at this point will be extremely limited as the prototype parts are still being tested:
- The NTFLM HW includes the carrier, socket, bolster plate, reference thermal solution, and anti-tilt solution.
- Availability will start around WW35'25. ***The NTFLM HW deployed during the first milestone will be explicitly approved and controlled by Intel.***
- Please contact your Intel representative if you have inquiries on hardware availability.
- There is a risk that the design of the hardware stack (bolster plate, carrier, socket cover, and heatsink hardware) can undergo significant changes during this milestone.
- Preliminary drawings and CAD files for the Narrow Reference Heatsink will be available in the mid Q3'25 timeframe.

Second Milestone - Early Q4'25

Second milestone is when the narrow TFLM design is frozen:

- From this milestone onward no changes that affect the form and fit are expected; There may be minor modifications in future milestones to meet the qualification criteria.
- Hard Tooling (HT) development starts at this milestone. Customers can start reaching out to the suppliers directly; No explicit approval from Intel will be needed. However, Intel expects limited availability of samples in this time frame. During this milestone, samples will still be considered prototypes. Therefore, there won't be Manufacturing Part Numbers (MPNs). Customers can use Intel internal Part Number (IPN) as reference for their communications with vendors.
- Narrow Reference Thermal solution hard tooling is expected to complete by mid Q4'25.
- Qualification of the HT heatsink parts will start, and limited samples of the heatsink will be available .

Third Milestone - Early and Mid Q1'26

Third milestone is when HT development completes:

- MPNs will be available at this point.
- Qualification of the HT parts starts; There will be a limited number of parts available to support EVT Builds.
- Only T0 load, Socket Durability and Heatsink Hardware Testing will be completed.
- Narrow Reference Thermal Solution HT is expected to be completed. Customers can contact suppliers directly.
- Shock and vibration testing is not expected to be completed until mid Q1'26.
- Availability of HT parts is scheduled to start early Q1'26.
- Customers can decide to wait until Early Q1'26 for the first viable HT parts or until Mid Q1'26 to source parts with shock and vibration testing completed. Alternatively, customers can source parts from the second milestone if they want to start EVT at the very beginning of Q1'26.
- The qualified HT Narrow Heatsink Thermal Reference solution is expected to be completed at this point.

Fourth Milestone - Early Q2'26

Fourth milestone is when the final qualification of the HT parts completes:

- Some minor changes for quality improvement after this milestone can be expected until PRQ, but these changes won't affect form, fit or function of the parts.
- Production parts are expected to be available in early Q2'26.

2.29.2.2 Oak Stream 16-Channel Plan of Record (POR) Change to Narrow Torsion Fastener Loading Mechanism (TFLM)

Intel is making a POR change on the Oak Stream 16-Channel Torsion Fastener Loading Mechanism (TFLM), changing from the 109.5mm x 158mm TFLM to a narrower **103.5mm** x 158mm TFLM (Narrow TFLM).

Narrow TFLM impacts the following mechanical components: Topside Keep out Zone (KOZ), Bolster plate, carrier, socket cover, thermal solutions, and anti-tilt system. The following components remain unchanged and common to both wide and narrow TFLM: bottom KOZ, socket, backplate, motherboard mounting holes patterns and thickness.

With this POR change, **all Oak Stream 16-Channel designs will need to transition to the Narrow TFLM** as pre-production and production parts will have carriers that are compatible only with Narrow TFLM. If customer's PO designs are not able to transition to Narrow TFLM, please contact your Intel representative for further guidance.

The *Oak Stream 16-Channel 2S in 19 Inch Design Feasibility Technical Article*, document number [843581](#), provides a system level overview of the Narrow TFLM and the potential design areas impacted. Additional Narrow TFLM mechanical and thermal enabling collaterals are also referenced in this technical article.

The *Narrow Torsion Fastener Loading Mechanism (TFLM) Execution Plan* article in the Oak Stream Platform Message of the Week (MoW) provides additional information on the Narrow TFLM key milestones and availability timeline. Intel will continue to provide communication on each key milestones along with any additional guidance on Narrow TFLM ordering. These updates will be communicated through the Message of the Week.

2.29.2.3 Oak Stream 16-Channel Platform Board Landscape

The following table provides an overview of the Oak Stream 16-Channel Platform Reference Designs:

Table 92. Oak Stream 16-Channel Platform Board Landscape

	Johnson City 1	Johnson City 2	MDP1	MDP2 (CPU1)	MDP2 (CPU0)
Size and PCB Stack-up	M-SDNO B335 11.842 in. x 14.318 in. 22L, ULL1+	14.002 in. x 18.598 in. 22L, ULL2	M-SNDO B335 9.8 in. x 13.2 in. 22L, ULL1+	M-SDNO C335 – 16.7", 22L ULL1+	M-SDNO C335 – 16.7", 22L ULL1+
TDP	500W	500W	500W	500W	500W
Memory Config	1 SPC. 16 × DDR5 8000 MT/s, MRDIMMs 12800 MT/s	2 SPC. 32 × DDR5 8000 MT/s, 2DPC 6000 MT/s	1 SPC. 16 × DDR5 8000 MT/s, MRDIMMs 12800 MT/s	1 SPC. 16 × DDR5 8000 MT/s, MRDIMMs 12800 MT/s	1 SPC. 16 × DDR5 8000 MT/s, MRDIMMs 12800 MT/s
<i>continued...</i>					

	Johnson City 1	Johnson City 2	MDP1	MDP2 (CPU1)	MDP2 (CPU0)
	0.325" DIMM-2-DIMM pitch	0.297" DIMM-2-DIMM pitch	0.325" DIMM-2-DIMM pitch	0.262" DIMM-2-DIMM pitch	0.262" DIMM-2-DIMM pitch
I/O lanes	Up to 128L PCIe 6.2: 4x16 N, 4x16 S Up to 8L PCIe 4.0: 1x4 N, 1x4 S	Up to 128L PCIe 6.2: 4x16 N, 4x16 S Up to 8L PCIe 4.0: 1x4 N, 1x4 S	Up to 128L PCIe 6.2: 4 x16 N, 4x16 S Up to 8L PCIe 4.0: 1x4 N, 1x4 S	Up to 64L PCIe 6.2: 2x16 N, 2x16 S Up to 64L Intel® UPI: 2x16 N, 2x16 S Up to 6L PCIe 4.0: 1x2 N, 1x4 S	Up to 64L PCIe 6.2: 2x16 N, 2x16 S Up to 64L Intel® UPI: 2x16 N, 2x16 S
I/O Near Side (Primary IMH)	FIO1-4: 2x8 SFF-TA-1016	FIO1-4: 2x8 SFF-TA-1016	FIO1,2,4: 2x8 SFF-TA-1016 FIO3: SFF-TA-1020 4C+	FIO3, 4: x16 Intel® UPI routing to CPU0 FIO1: SFF-TA-1020 4C+ FIO2: 2x8 SFF-TA-1016	FIO1, 2: x16 Intel® UPI routing to CPU1 FIO3, 4: 2x8 SFF-TA-1016
I/O Config (Primary IMH)	All ports support Intel® UPI cabling, CEM riser, CEM adapter, or EDSFF drive mid/back plane	All ports support Intel® UPI cabling, CEM riser, CEM adapter, or EDSFF drive mid/back plane	Storage - FIO1,2: x4 E1.S, FIO3: OCP 3.0, FIO4: NIC/Smart NIC. GPU - FIO1,2: x4 E1.S, FIO3: OCP 3.0, FIO4: GPU Computing - FIO1,4: NIC, FIO2: x4 E1.S, FIO3: OCP 3.0	Storage - FIO1: OCP 3.0, FIO2: x4 E1.S GPU - FIO1: OCP 3.0, FIO2: x4 U.2 Computing - FIO1: OCP 3.0, FIO2: x4 U.2	Storage - FIO2: x4 E1.S, FIO3: NIC GPU - FIO2: x4 U.2, FIO3: NIC Computing - FIO2: x4 U.2, FIO3: NIC
I/O Far Side (Secondary IMH)	FIO9,10,11,12: 2x8 SFF-TA-1016	FIO9,10,11,12: 2x8 SFF-TA-1016 FIO12[15:8]: 1x8 SFF-TA-1020 FIO12[7:0]: 1x8 SFF-TA-1016	FIO9,10,11,12: 2x8 SFF-TA-1016	FIO9,10: 2x8 SFF-TA-1016 FIO9,10: x16 Intel® UPI routing to CPU1	FIO9,10: x16 Intel® UPI routing to CPU0 FIO11,12: 2x8 SFF-TA-1016
I/O Config (Secondary IMH)	All ports support Intel® UPI cabling, CEM riser, CEM adapter, or EDSFF drive mid/back plane	FIO12[15:8] does not support Intel® UPI cabling, all others support Intel® UPI cabling, CEM riser, CEM adapter	Storage - FIO9,11,12: Reserved, FIO10: NIC/Smart NIC GPU - FIO9: Reserved, FIO10: GPU, FIO11,12: NIC Computing - FIO9: Reserved, FIO10,11,12: NIC	Storage - FIO8: M.2, FIO11: Reserved, FIO12: NIC GPU - FIO8: M.2, FIO11: Reserved, FIO12: GPU Computing - FIO11, 12: NIC	Storage - FIO9,10: NIC GPU - FIO9: GPU, FIO10: NIC Computing - FIO9,10: NIC
AC DC	All ports to be set in AC coupled mode	All ports to be set in AC coupled mode	All ports to be set in AC coupled mode	FIO1,2,11,12: AC FIO3,4,9,10: DC	FIO1,2,11,12: DC FIO3,4,9,10: AC

2.30 WW22-2025

New:

- Almondsville TMTV Package Thermal Model User Guide Collateral Release Notification on page 255
- Turkey Canyon TMTV Package Thermal Model User Guide Collateral Release Notification on page 255

2.30.1 Documentation

2.30.1.1 Almondsville TMTV Package Thermal Model User Guide Collateral Release Notification

The *Almondsville Thermal Mechanical Test Vehicle (TMTV) Package Thermal Models and User Guide*, document number 820835, revision 0.7, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 0.7 changes:

- Updated Almondsville TMTV die number sequence to match with the actual TMTV, ensuring proper power application for each die type.

2.30.1.2 Turkey Canyon TMTV Package Thermal Model User Guide Collateral Release Notification

The *Turkey Canyon Thermal Mechanical Test Vehicle (TMTV) Package Thermal Model User Guide*, document number 852612, revision 0.6, is now available in the Resource and Documentation Center (RDC).

Overview of Revision 0.6 changes:

- Added Applied Power Guidance Examples for 165W, 205W, and 250W SKUs.

2.31 WW21-2025

Updated:

- [\(Updated WW21\) Oak Stream Key Collaterals Availability and Forecast](#) on page 255

New:

- [Oak Stream Platforms System Boards Design Archive Release Notification](#) on page 256
- [Oak Stream 16-Channel Platform Johnson City 1 Design Archive Release Notification](#) on page 257
- [PCI Express* \(PCIe*\) 6.0 Hot Swap Backplane \(HSBP\) 8x4 Drive Bay Design Guide Release Notification](#) on page 257
- [Data Center Design Conference \(DCDC\) 2025 Asia Event Registration](#) on page 257

2.31.1 Documentation

2.31.1.1 (Updated WW21) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 93. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	WW27'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW36'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW35'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	WW31'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.9	1.0	Q3'25
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<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	Late Q2'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Late Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.4	0.5	Mid Q2'25
<i>Oak Stream Platforms System Boards Design Archive</i>	821025	0.5	0.7	Q3'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	WW24'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.31.1.2 Oak Stream Platforms System Boards Design Archive Release Notification

The *Oak Stream Platforms System Boards Design Archive*, document number [821025](#), revision 0.5, is now available in the Resource and Documentation Center (RDC).

Overview of 0.5 changes:

- Initial release.

NOTE

This revision only includes Power Distribution Board (PBD) 2 Power Supply Unit (PSU).

2.31.1.3 Oak Stream 16-Channel Platform Johnson City 1 Design Archive Release Notification

The *Oak Stream 16-Channel Platform Johnson City 1 Design Archive*, document number [823065](#), revision 0.9, is now available in the Resource and Documentation Center (RDC).

Overview of 0.9 changes:

- Updated HPM routing
- Included Bill of Materials (BOM)
- Added architecture block diagrams in schematic PDF

NOTE

This 0.9 revision does not intercept the DDR Command Address (CA) length matching updated from [WW20'25 MoW](#). The update will be intercepted in a future intermediate revision.

2.31.1.4 PCI Express* (PCIe*) 6.0 Hot Swap Backplane (HSBP) 8x4 Drive Bay Design Guide Release Notification

The *PCI Express* (PCIe*) 6.0 Hot Swap Backplane (HSBP) 8x4 Drive Bay Design Guide*, document number [852244](#), revision 1.0, is now available in the Resource and Documentation Center (RDC).

This Design Guide provides guidelines for implementing a three-connector topology for PCIe* 6.0 drives on a Hot-Swap Backplane (HSBP) designed to contain hard drives in a removable configuration.

Overview of 1.0 changes:

- Initial release

2.31.2 Enabling

2.31.2.1 Data Center Design Conference (DCDC) 2025 Asia Event Registration

Intel is excited to welcome you to Intel's 2025 Data Center Design Conference (DCDC) Asia Event. This year's conference highlights on Innovate on Compute, Accelerate AI Transformation, and System and Rack Design and Trends built on Intel® Xeon® solutions.

Registration for DCDC 2025 Asia Event is now open. Register for the virtual live events at [Tech Training](#).

Table 94. DCDC Asia Event Information

Geography	PRC	Taiwan	ASIA Virtual
Event Format	Face to Face	Face to Face	Virtual Live
Event Date/Schedule	June 24, 2025	June 26, 2025	June 30 – July 4, 2025
Language	Mandarin	Mandarin	Mandarin/English
Registration Period	Now to June 13, 2025	Now to June 11, 2025	Now to June 20, 2025
On-demand Videos	No recording at the event	No recording at the event	The event will be recorded and available later

NOTE

Seats for the face to face (F2F) event are limited. Contact your local Intel representative for more details.

2.32 WW20-2025

Updated:

- [\(Updated WW20\) Oak Stream Key Collaterals Availability and Forecast](#) on page 258

New:

- [Oak Stream 16-Channel Platform Mechanical Collaterals Release Notification](#) on page 259
- [6400-Rated RDIMM Support for 16-Channel Diamond Rapids-UCC Samples](#) on page 260
- [DDR Command Address \(CA\) Length Matching Update](#) on page 261

2.32.1 Documentation

2.32.1.1 (Updated WW20) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

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<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW35'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	WW31'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.8	0.9	Late Q2'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	Late Q2'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Late Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.4	0.5	Mid Q2'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	WW24'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.32.1.2 Oak Stream 16-Channel Platform Mechanical Collaterals Release Notification

Three Oak Stream 16-Channel Platform Mechanical Collaterals are now available in the Resource and Documentation Center (RDC):

- *Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models*, document number [786523](#), revision 0.7
- *Oak Stream 16 Channel Narrow Mechanical Models*, document number [852564](#), revision 0.3

- *Preliminary Narrow Keep Out Zone (KOZ) and Interface Control Drawings (ICD) for Oak Stream 16-Channel*, document number [851010](#), revision 0.3

Wide Torsion Fastener Loading Mechanism (TFLM) Collaterals

Overview of the *Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models* revision 0.7 changes:

- Added thermal reference solutions.
- Added Socket Cover.
- Updated all other components to their latest revisions. Detailed revision history that includes a list of components and part numbers is available in the PDF within the collateral.

Narrow TFLM Collaterals

Overview of the *Oak Stream 16-Channel Narrow Mechanical Models* revision 0.3 changes:

- Initial release. CAD files include:
 - Preliminary Topside Keep-Out Zone (KOZ) for Narrow TFLM designs.
 - First revision of the narrow bolster plate
 - 3D representation of the Interface Control Drawing (ICD)
- CAD files do not include the plastic carrier, thermal reference solution, nor heatsink hardware. These components are under development and will be included in a future revision of the collateral.

Overview of *Preliminary Narrow Keep Out Zone (KOZ) and Interface Control Drawings (ICD) for Oak Stream 16-Channel* revision 0.3 changes:

- Initial release includes:
 - First revision of the Thermal Interface Control Drawing (ICD)
 - First of the Narrow Topside Keep Out Zone (KOZ) Drawing.

NOTE

First revisions are subject to significant changes, these early revisions are being provided to give the initial direction the Narrow TFLM solution. First revisions should not be considered for production designs.

2.32.2 Enabling

2.32.2.1 6400-Rated RDIMM Support for 16-Channel Diamond Rapids-UCC Samples

Intel will provide support for 6400-rated RDIMM as non-POR memory configurations during early Oak Stream platform development phases to ensure customers validation activities are not limited by the number of DDR5-8000 RDIMM samples they have access to.

6400-rated RDIMM support will be provided under these conditions:

- Diamond Rapids-UCC samples (IMH1)

- Raw card types: 1Rx8 16 GB, 2Rx8 32 GB, 2Rx4 64 GB from all three RDIMM vendors
- Population configuration: 1DPC at 6400 MT/s

Intel's initial testing will cover 16-Channel 1-Socket 1DPC DDR5 6400 configurations. **Intel's support for 2DPC configurations with 6400-rated RDIMM will be limited.** If 6400-rated RDIMMs are used in 2DPC configurations, the RDIMMs should be configured at 5200 MT/s speed.

Intel strongly recommends that any issues found be reproduced with an 8000-rated RDIMM before contacting Intel for support.

The required BIOS knobs settings for 6400-rated RDIMM with UCC samples are as follows:

For 1DPC configuration:

- EnforcePopulationPor = Disabled
- EnforceDdrMemoryFreqPor = Enforce POR
- HostDdrFreqLimit = Auto

For 2DPC configuration:

- EnforcePopulationPor = Disabled
- EnforceDdrMemoryFreqPor = Disabled
- HostDdrFreqLimit = 5200 Non-POR

2.32.2.2 DDR Command Address (CA) Length Matching Update

Problem Statement

To improve DDR Command Address (CA) margins, Intel is tightening the length matching requirement between all CA Signals (CA[6:0] and PAR) within a sub-channel from 120 mils for each DIMM **to 50 mils**.

This change is expected to improve CA eye margins for Diamond Rapids processors and reduce the risk of future board changes for Oak Stream 16-Channel platform-based designs.

Table 96. DDR Signals Impacted by CA Length Matching Update

Channels	CA Matching groups	Signals
DDR[15:0]	CA Sub-Channel A	SA_CA[6:0], SA_PAR
DDR[15:0]	CA Sub-Channel B	SB_CA[6:0], SB_PAR

Length matching updated requirements:

- The parity (S[A/B]_PAR) signals for each sub-channel are part of the CA group and must be matched with the other CA signals in the sub-channel.
- The 50 mils requirement applies separately to each sub-channel.
- The 50 mils requirement must be applied to each DIMM in 2 SPC designs.

Implication

Intel will update Johnson City 1 and Johnson City 2 designs to align to this length matching requirement. Electrical validation will be performed with this change implemented.

IMPORTANT

Customers must update their designs to ensure adequate margins and future compatibility.

Customers who have taped-out Oak Stream 16-Channel based designs can still use 120 mils length matching for their Power On and most EVT activities (excluding DDR5 electrical validation); still, customers must update their designs for production.

IMPORTANT

Customer DDR5 electrical validation must be performed on boards with the new length matching target.

Status

The following collaterals will be updated to reflect Plan of Record (POR) change:

- *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#).
- *Oak Stream 16-Channel Platform Johnson City 1 Design Archive*, document number [823065](#).
- *Oak Stream 16-Channel Platform Johnson City 2 Design Archive*, document number [636258](#).

Figure 52. Preview of Upcoming PDG Changes for CA Length Matching Requirements

6.13 DDR5 Length Matching					
6.13.1 Length Matching					
Table 460. Length Matching Table					
Groups Involved	Type	Source (From)	Target (To)	Matching Type Tolerance (mils)	Notes
DDR[15:0]_S[A/B]_DQS[9:0]_D[P/N]	Length Matching(Pin-Pin)	CPU	EACH_DIMM	5	6
DDR[15:0]_CLK[1:0]_D[P/N]	Length Matching(Pin-Pin)	CPU	EACH_DIMM	5	6
DDR[15:0]_S[A/B]_CSA2CSB	Length Matching(Pin-Pin)	CPU	EACH_DIMM	100	7
DDR[15:0]_S[A/B]_CA2CA	Length Matching(Pin-Pin)	CPU	EACH_DIMM	50	8

NOTES					
1. Length match the DQ signals within each byte group.					
2. Signal from same DLL group need to be routed on same layer.					
3. Length match each bytegroup(DQ/ECC) to its associated DQS.					
4. To reduce DQ jitter, it is required DQs signal to be shorter than DQ within a byte, no upper limit on how much longer DQ than DQs is in same nibble if maximum DQ length met.					
5. Within Sub-channel, all CA signals (CA [6:0] and PAR) need to match CS signal within 300mils					
6. Differential signal P/N match to each within 5 mils.					
7. Between Sub-channels, all CS signals at a DIMM must match with 100 mils. At DIMM0, DDRn_SA_CS0, DDRn_SA_CS1, DDRn_SB_CS0, and DDRn_SB_CS1 must match. At DIMM1, DDRn_SA_CS2, DDRn_SA_CS3, DDRn_SB_CS2, and DDRn_SB_CS3 must match.					
8. With Sub-channel, all CA signals (include CA [6:0] and PAR) need to match within 50 mils					

Design review services will incorporate the 50 mils CA length matching requirement. Monitor the Oak Stream Platform Message of the Week (MoW) for updates on collaterals and updated design review services availability.

2.33 WW19-2025

Updated:

- [Oak Stream Collaterals](#) on page 373

New:

- [Clock Jitter Tool 6.0.2 Release Notification](#) on page 264
- [Diamond Rapids-UCC PCIe and Intel UPI Load Boards are Available at Intel® Design-in Tools Store](#) on page 264
- [Oak Stream 16-Channel 2-Socket \(2S\) in 19-inch Design Feasibility Technical Article Release Notification](#) on page 265
- [Oak Stream 16-Channel Platform MRDIMM Plan of Record \(POR\) Support](#) on page 265

2.33.1 Tools and Debug

2.33.1.1 Clock Jitter Tool 6.0.2 Release Notification

The *Clock Jitter Tool (CJT)*, document number 803838, version 6.0.2, is now available for download from the Resource and Documentation Center (RDC).

Version 6.0.2 is a bug-fix release to address a compilation issue in 6.0.1 version that prevented waveform evaluation on machines unless MathWorks* MATLAB and specific toolboxes were already present on the machine.

The feature set of 6.0.2 version is unchanged from versions 6.0.1 and 6.0; As a reminder, version 6.0.1 was released only to remove the MathWorks MATLAB Runtime installer from the distribution package.

NOTE

Free MathWorks MATLAB Runtime version 9.10, available directly from the MathWorks website, is required to execute CJT 6.0.2 version

2.33.1.2 Diamond Rapids-UCC PCIe and Intel UPI Load Boards are Available at Intel® Design-in Tools Store

The PCI Express* (PCIe*) and Intel® Ultra Path Interconnect (Intel® UPI) Load Board set for the Diamond Rapids-UCC processor socket is available for sale at the Intel Design-in Tools Store website.

The Diamond Rapids-UCC PCIe and Intel UPI Load Boards main features are:

- Retention block designed for 21-inch chassis Torsion Fastener Loading Mechanism (TFLM) solution.
- Full coverage of all the PCIe and Intel UPI UPI lanes.
- Coverage of specific Clock nets (REFCLK and PFT).
- Coverage of specific Essential IO lines (SPI, ESPI, I3C, SVID, SBLINK).
- Coverage of specific DDR5 lines (DDR_CH0_SB, DDR_CH7_SA).
- 2x-Through de-embedding structures.

The Load Boards can be purchased through the Intel Design-In Tools Store:

- [DMR-UCC PCIe* and Intel® UPI Load Board Solution Set](#)
- [DMR-UCC PCIe* Load Board Solution Set](#)

NOTE

If narrow TFLM becomes POR, retention blocks for 19-inch chassis will be distributed to customers that purchase Load Boards through DIT. See *Oak Stream 16-Channel 2S in 19-Inch Design Feasibility*, document number 843581, for details on narrow TFLM.

2.33.2 Documentation

2.33.2.1 Oak Stream 16-Channel 2-Socket (2S) in 19-inch Design Feasibility Technical Article Release Notification

The *Oak Stream 16-Channel 2S in 19-Inch Design Feasibility*, document number 843581, revision 1.4, is now available in the Resource and Documentation Center (RDC).

This Technical Article provides a snapshot of the on-going design feasibility studies to enable Oak Stream 16-Channel 2S in 19-inch chassis designs.

Overview of revision 1.4 changes:

- Updated Customer Enabling information.
- Added Diamond Rapids ES1 samples support considerations for different Host Processor Module (HPM) form factors.

2.33.3 Enabling

2.33.3.1 Oak Stream 16-Channel Platform MRDIMM Plan of Record (POR) Support

MRDIMMs are now supported in 1 DIMM per channel (1DPC) / 2 Slots per Channel (2SPC) configurations, with a maximum speed of 11200 MT/s. 2SPC support configurations details are:

- Supported in 1-Socket and 2-Socket configurations.
- Supported DIMM capacities are 64 GB (16 Gb), 96 GB (24 Gb), and 128 GB (32 Gb).
- 1DPC/2SPC will not be supported in 12-Channel SKUs.

Table 97. Oak Stream 16-Channel Platform MRDIMM 1DPC/2SPC Update

DIMM Type	DIMM Capacity (GB) and Density (Gb)	DIMM Capacity (GB) and Density (Gb)	DIMM Capacity (GB) and Density (Gb)	DIMM Capacity (GB) and Density (Gb)	DIMM Capacity (GB) and Density (Gb)	DIMM Capacity (GB) and Density (Gb)
MRDIMM	16 Gb	16 Gb	24 Gb	24 Gb	32 Gb	32 Gb
MRDIMM	1DPC	2DPC	1DPC	2DPC	1DPC	2DPC
MRDIMM	64 GB	N/A	96 GB	N/A	128 GB	N/A

Oak Stream collaterals, including Snapshot and Platform Design Guide, will be updated to reflect MRDIMM POR support.

NOTE

Intel will complete MRDIMM 1DPC/2SPC Validation Post-Launch. Customers can incorporate MRDIMM 1DPC/2SPC configurations into their Engineering Validation and Testing (EVT)/ Design Validation and Testing (DVT) planning, as Intel's BKC will include support for 1DPC/2SPC configurations.

2.34 WW18-2025

Updated:

- (Updated WW18) Oak Stream Key Collaterals Availability and Forecast on page 266

New:

- Turkey Canyon TMTV Package Thermal Model User Guide Release Notification on page 267
- Oak Stream 16-Channel Platform Johnson City 1 Design Archive Release Notification on page 267

2.34.1 Documentation

2.34.1.1 (Updated WW18) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 98. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	WW27'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW36'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW35'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	WW31'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.8	0.9	Late Q2'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	Late Q2'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Late Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.4	0.5	Mid Q2'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	WW21'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.34.1.2 Turkey Canyon TMTV Package Thermal Model User Guide Release Notification

The *Turkey Canyon TMTV Package Thermal Model User Guide*, document number [852612](#), revision 0.5 is now available in the Resource and Documentation Center (RDC).

Overview of Revision 0.5 changes:

- Initial release: This first revision allows customers to begin designing and evaluating thermal solutions, as well as conducting system-level thermal simulations for the Oak Stream 8-Channel Platform.

2.34.1.3 Oak Stream 16-Channel Platform Johnson City 1 Design Archive Release Notification

The *Oak Stream 16-Channel Platform Johnson City 1 Design Archive*, document number [823065](#), revision 0.8 is now available in the Resource and Documentation Center (RDC).

NOTE

Revision 0.8 does not include Johnson City 1 Bill of Materials (BOM) and Change list. These documents will be included in revision 0.9.

2.35 WW17-2025

Updated:

- (Updated WW17) Oak Stream Key Collaterals Availability and Forecast on page 268
- (Updated WW17) Q1'25 Oak Stream Quarterly Technical Review (QTR) Online Session - Material Available on page 269

New:

- [Oak Stream 16-Channel / 12-Channel Platform DDR5 VRTT Test Plan Release Notification](#) on page 269

2.35.1 Documentation

2.35.1.1 (Updated WW17) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 99. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	WW27'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	WW36'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	WW35'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	WW31'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.8	WW18'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	Late Q2'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Late Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.4	0.5	Mid Q2'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	WW21'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.35.1.2 (Updated WW17) Q1'25 Oak Stream Quarterly Technical Review (QTR) Online Session - Material Available

NOTE

Updated information is highlighted in **bold text**.

The Customer and Systems Engineering (CSE) team is excited to invite you to the Q1'25 Oak Stream Quarterly Technical Review (QTR) online session. This session will allow customers to find the latest enabling updates for the next-generation Oak Stream platforms.

The online session **was** hosted on April 22, at 5:00-7:00 PM Pacific Time (PDT).

The material presented is now available in the Resource and Documentation Center.

- **Recording: Oak Stream Platform Quarterly Technical Review (QTR) Q1'25 Recording**, document number **853183**
- **Presentation: Oak Stream Platform Quarterly Technical Review (QTR) Q1'25 - 848790**, document number **848790**

2.35.1.3 Oak Stream 16-Channel / 12-Channel Platform DDR5 VRTT Test Plan Release Notification

The *Oak Stream 16-Ch / Oak Stream 12-Ch Platform DDR5 Voltage Regulator Test Tool (VRTT) Test Plan*, document number **832854**, revision 0.71 is now available in RDC.

Overview of 0.71 changes:

- Updated memory configuration from MRDIMM 2U 24 Gb to 32 Gb
- Added DIMM pitch row and updated test methodology to support narrower DIMM pitch test
- Fixed some equations for power conversion

2.36 WW15-2025

New:

- [Q1'25 Oak Stream Quarterly Technical Review \(QTR\) Online Session](#) on page 272
- [Oak Stream 16-Channel Platform Design Guide \(PDG\) Clarification: Layout RSS Maxima](#) on page 271
- [Oak Stream 16-Channel Platform Electrical Specification DRAM_PWR_OK Update](#) on page 272
- [Diamond Rapids Processor Registers Specification Release Notification](#) on page 272

Updated:

- (Updated WW15) Oak Stream Key Collaterals Availability and Forecast on page 270

2.36.1 Documentation

2.36.1.1 (Updated WW15) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 100. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	Late Q2'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.81	0.9	Mid Q2'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	Q2'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	WW17'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.9	1.0	Q3'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	Late Q2'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Late Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Late Q2'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.4	0.5	Mid Q2'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	Q2'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.36.1.2 Oak Stream 16-Channel Platform Design Guide (PDG) Clarification: Layout RSS Maxima

The *Oak Stream 16-Channel Platform Design Guide*, document number 788196, Revision 0.81, Chapter 3, section 3.11.2 Minimizing the Effect of Fiber-weave for Data Rates at 32 Gbps and Above, will be updated in a future revision of the Platform Design Guide (PDG). These changes ensure consistency between the PDG recommendations and the Reference Platform implementations.

A preview of the changes is included below. Updated content is highlighted in **bold text**; minor typographical edits are not shown.

Minimizing the Effect of Fiber-weave for Data Rates at 32 Gbps and Above

Guideline

The fiber-weave effect becomes more prominent at higher data rates, and we recommend that only spread-glass materials are used.

The maximum allowed RSS lengths vs. data transfer rates are shown below, assuming spread-glass materials are used.

Table 101. Max RSS Length versus Transfer Speed

Transfer Speed (GT/s)	Glass Type	Max RSS Length – See Notes
32	E Glass	0.8" (See Notes 1, 3, 4)
32	Low-Dk Glass	1" (See Notes 2, 4)
48/56/64	Any Glass Type	< 0.1" for the routing outside of CPU pin field (this may be relaxed to 0.3" maximum if congestion prevents implementation of angled routing). See notes below for mitigation within the CPU pin field. Notes: - For routing outside of the CPU pin field, keep RSS length as small as possible (e.g., utilize angled routing). - CPU pin field routing can be a significant contributor to the fiber-weave induced skew due to space constraints (which make routing relatively straight/parallel to board edge). Intel has performed detailed analysis and optimization to mitigate the fiber-weave impact. It is highly recommended that Intel's reference design pin field routing is strictly followed to reduce the signal integrity risk in the design. Refer to Chapter 8 for more implementation details.

2.36.1.3 Diamond Rapids Processor Registers Specification Release Notification

The *Diamond Rapids Processor Registers Specification*, document number 793272, revision 0.9, is now available in the Resource and Documentation Center.

Overview of revision 0.9 changes:

- Added new sections:
 - Intel® QuickAssist Technology (Intel® QAT) Memory Space Registers
 - Host Input Output Processor (HIOP) Memory Space Registers
 - Scalable Coherent Agent (SCA) RDT Memory Space Registers
 - Converged Coherent Fabric (CCF) Registers
 - MCU_STAGING Interface Registers
 - TPMI In Band Access
- Updated register access information
 - Reformatted all chapters introduction to consolidate register access information in tables
 - Updated access information for all chapters
 - Added summary of PCI Configuration Space, Memory Space, and I/O space access information in Registers Overview Chapter
- Updated Power Control Unit (Punit) Registers chapter
 - Updated scope information
 - Removed unused registers
- Removed unused RASIP registers in Root and Root-1 sections

2.36.2 Enabling

2.36.2.1 Q1'25 Oak Stream Quarterly Technical Review (QTR) Online Session

Customer and Systems Engineering (CSE) team is excited to invite you to the Q1'25 Oak Stream Quarterly Technical Review (QTR) online session. This session will allow customers to find the latest enabling updates for the next-generation Oak Stream platforms.

The online session will be hosted on **April 22, at 5:00-7:00 PM Pacific Time (PDT)**.

The material presented will be recorded and made available afterward in the Resource and Documentation Center.

To join the online session:

- [Join Teams Live Event](#)

2.36.2.2 Oak Stream 16-Channel Platform Electrical Specification DRAM_PWR_OK Update

Problem Statement

The current implementation of the DRAM_PWR_OK receiver requires the use of fast drivers to meet the input rise and fall time specifications. This poses challenges to customers designs, as it often necessitates the addition of extra components in their designs.

Slew-Rate Specifications Update

The DRAM_PWR_OK slew rate was relaxed to allow for the use of slightly slower drivers. This slew rate (measured between $0.3 \times VDD$ to $0.7 \times VDD$) has been updated to 400 ns rise/fall time for the DRAM_PWR_OK receiver, leading to a new value of 1.1 mV/ns as opposed to 10 mV/ns.

Status

The DRAM_PWR_OK slew rate spec will be updated in the next revision of the *Oak Stream 16-Channel Platform Electrical Specification*, document number 787006.

2.37 WW14-2025

New:

- [Oak Stream Platform 16-Channel Platform 500W Cookbook Release Notification](#) on page 275

Updated:

- [\(Updated WW14\) Oak Stream Key Collaterals Availability and Forecast](#) on page 273
- [Boot Security Technology Enablement Feedback Request Friendly Reminder](#) on page 274

2.37.1 Documentation

2.37.1.1 (Updated WW14) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 102. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	Q2'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	WW15'25
<i>continued...</i>				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	0.7	As needed	As needed
Oak Stream 16-Channel Platform Design Guide	788196	0.81	0.9	Mid Q2'25
Oak Stream 16-Channel Platform Electrical Specification	787006	0.8	0.9	Q2'25
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	0.7	0.9	WW17'25
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258	0.9	0.1	Q2'25
Oak Stream 16-Channel Platform Pin List	636260	0.9	1.0	Q3'25
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	N/A	0.5	WW19'25
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	0.5	0.7	WW19'25
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	0.7	0.9	Q2'25
Oak Stream 8-Channel Platform Basin City Pin List	TBD	N/A	0.3	Mid Q2'25
Oak Stream 8-Channel Platform Basin City Design Archive	TBD	N/A	0.3	Mid Q2'25
Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive	TBD	N/A	0.5	Mid Q3'25
Oak Stream 8-Channel Platform Design Guide	813596	0.3	0.5	Mid Q3'25
Oak Stream 8-Channel Platform Electrical Specification	794719	0.3	0.5	Late Q2'25
Oak Stream 8-Channel Platform Thermal Mechanical Specification	786524	0.4	0.5	Mid Q2'25
Oak Stream Platforms BIOS Writer's Guide	820107	0.7	0.9	Q2'25
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.37.2 Customer Feedback

2.37.2.1 Boot Security Technology Enablement Feedback Request Friendly Reminder

Intel is requesting feedback from customers on boot security technologies enablement plans on Eagle Stream, Birch Stream, and Oak Stream based designs.

Provide your answers to the following questions to your Intel representative. The deadline for collecting feedback is now extended to WW16'25.

Your feedback will help plan support resources for these boot security technologies.

Do you enable Intel® Boot Guard on or plan to enable it on:

- Eagle Stream
- Birch Stream
- Oak Stream

Do you enable Intel® Trusted Execution Technology (Intel® TXT) on or plan to enable it on:

- Eagle Stream
- Birch Stream
- Oak Stream

Do you enable Intel® Platform Firmware Resilience (Intel® PFR) on or plan to enable it on:

- Eagle Stream
- Birch Stream
- Oak Stream

Do you enable CPU Attestation on or plan to enable it on:

- Birch Stream
- Oak Stream

Do you plan to enable an Intel vendor-authorized boot on Oak Stream?

- Yes
- No

2.37.3 Tools and Debug

2.37.3.1 Oak Stream Platform 16-Channel Platform 500W Cookbook Release Notification

The *Oak Stream 16-Channel Platform 500W Cookbook*, document number 851059, revision 1.09.02, is now available in the Resource and Documentation Center (RDC).

Overview of 1.09.02 revision:

- Initial release

This cookbook can be used for platform CPU VR rails validation using the Gen 5 Extended VRTT tool.

2.38 WW13-2025

New:

- [Oak Stream Platform 16-Channel Power Delivery Schematic Layout Checklist Release Notification](#) on page 276
- [ESPI Reset and CS0_N update for DC-SCM 2.1 Specification Compliance](#) on page 276
- [DC-SCM Design Update for SPI Flash Specification Compliance](#) on page 279

2.38.1 Documentation

2.38.1.1 Oak Stream Platform 16-Channel Power Delivery Schematic Layout Checklist Release Notification

The *Oak Stream Platform 16-Channel Power Delivery Schematic Layout Checklist*, document number 844697, revision 1.0, is now available in Resource and Documentation Center (RDC) .

This collateral provides a general power delivery schematic and layout checklist.

Overview of 1.0 revision:

- Initial release

2.38.2 Enabling

2.38.2.1 ESPI Reset and CS_{0_N} update for DC-SCM 2.1 Specification Compliance

Description

The BMC subsystem is required to be operational during the transition from STBY to Pre-S5 states, per Data Center Secure Control Module (DC-SCM) OCP* specification. As such, the BMC boots prior to the CPU reaching the S5 (powered on) state. This necessitates the isolation of BMC and CPU voltage IO domains.

Operational Scenarios

- **Legacy CPU Powered OFF:** When the Legacy CPU is off (AUX_PWRGOOD is low), the Quick Switch must remain disabled. This prevents leakage from the energized DC-SCM to the de-energized Legacy CPU.
- **Legacy CPU Powered ON (S5 State):** When the Legacy CPU is powered on (AUX_PWRGOOD is high) and the eSPI CS_{1_N} signal is low (indicating the selection of the Host Processor Module [HPM] Programmable Logic Device [PLD] eSPI device), the Quick Switch should be disabled. This action disconnects the BMC from the eSPI interface, eliminating signal integrity risk.

Implementation Details

Isolation can be achieved using a Quick Switch, which is controlled by the CPU AUX_PWRGOOD and eSPI CS_N signals. See the *Oak Stream 16-Channel Platform Design Guide*, document number 788196, for details on Quick Switch functionality.

Ensure proper termination resistors are installed on the BMC side when the eSPI interface is disconnected to the reference platform. See the following figures for implementation of device isolation.

Figure 53. Single Device eSPI Isolation

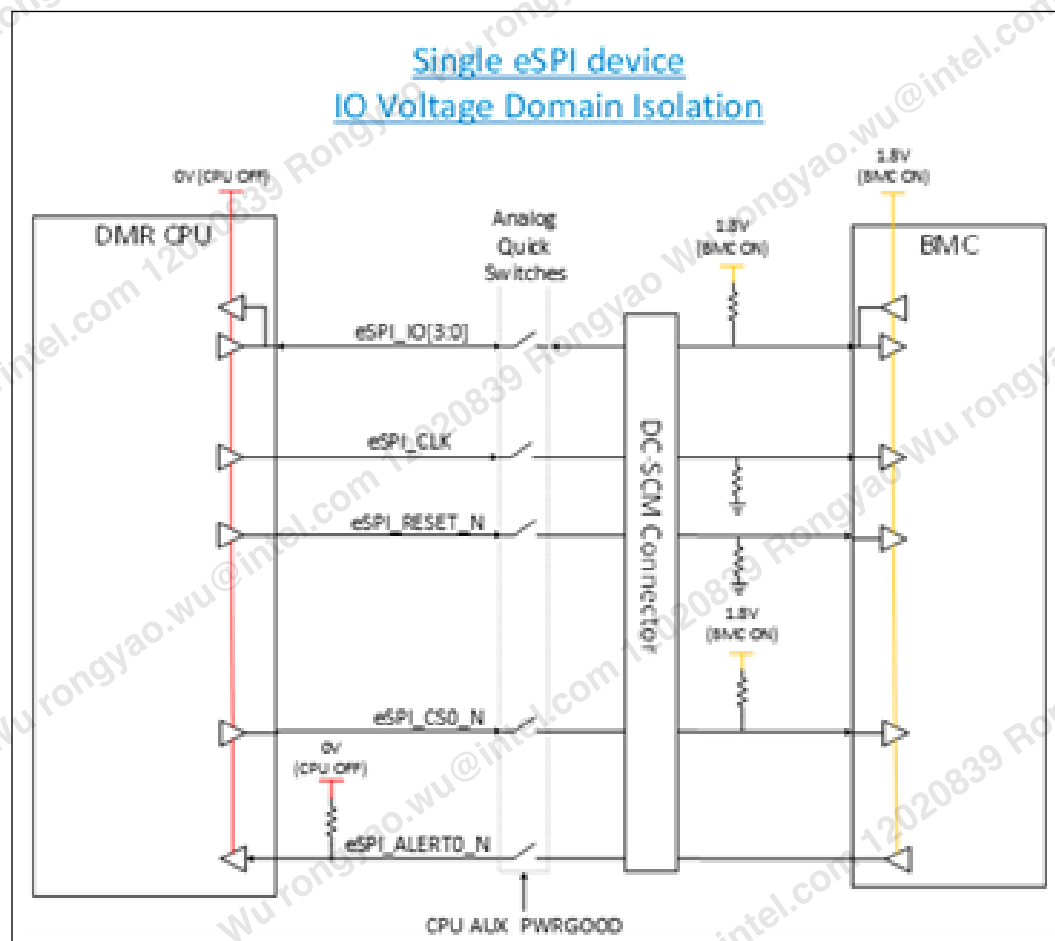
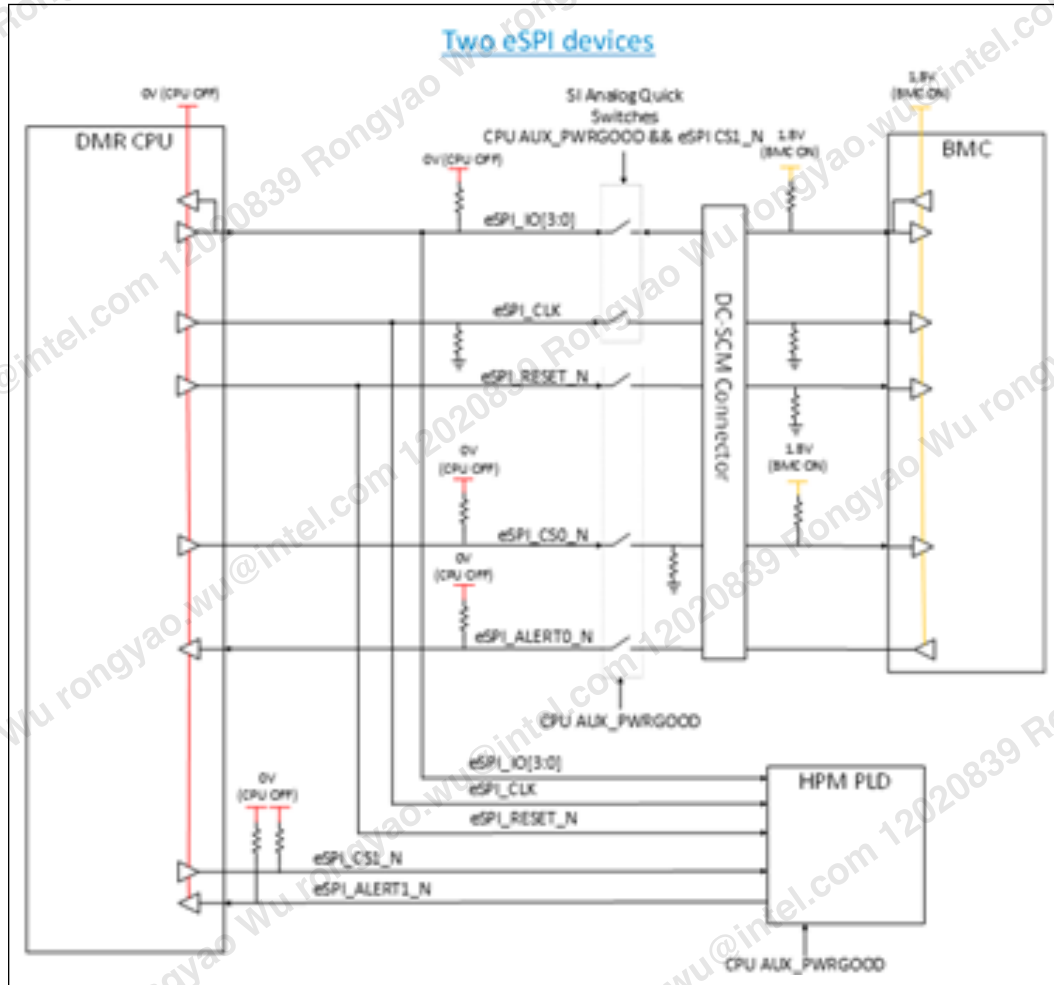


Figure 54. Two Device eSPI Isolation



NOTE

On the Oak Stream reference platform, eSPI device 1 is implemented on the HPM Programmable Logic Device (PLD). The HPM PLD uses AUX_PWRGOOD to set the eSPI device interface to High-Z, with isolation managed by the PLD logic. When AUX_PWRGOOD is LOW, the PLD sets the eSPI device IOs to High-Z and isolates the eSPI device internally. Once AUX_PWRGOOD is HIGH, the PLD enables the IOs and connects them to the eSPI device IP. If another device is used for eSPI device 1 without these isolation capabilities, external isolation switches are required, like those used for the BMC eSPI device.

Problem Statement

In current Oak Stream reference platforms, the eSPI Reset and CS0_N signals are not isolated, current Quick Switch does not include these signals.

Intel conducted a review with the reference platform BMC supplier. The supplier confirmed that the lack of eSPI Reset and CS0_N isolation poses no risk or reliability concerns as the eSPI IOs and CLK are in stable/idle logic state.

Mitigation/Fix

The *Oak Stream 16-Channel Platform Design Guide*, document number 788196, will integrate the CS0_N and Reset signals into the Quick Switch to align with OCP* DC-SCM specification, as shown in the previous figures.

Additionally, the DC-SCM design features an unpopulated pull-up resistor on the CS0_N line. This pull-up should remain unpopulated to prevent any potential leakage on Oak Stream reference platform-based designs. However, customers with designs that include Quick Switch isolation can populate the BMC side CS0_N pull up.

Status

The following collaterals will be updated to reflect the new pull-up resistor on CS_N:

- *Oak Stream 16-Channel Platform Design Guide*, document number 788196
- *Oak Stream Platform DC-SCM 2.1 Design Archive*, document number 636262

For questions or assistance, contact your Intel representative.

2.38.2.2 DC-SCM Design Update for SPI Flash Specification Compliance

Description

According to common industry SPI Flash Device specifications, the CS_N signal is required to de-assert (low to high rise) simultaneously with the flash device Voltage Common Collector (VCC). However, in the current Oak Stream reference platform design, there is a 100 ms delay in the de-assertion of CS_N following VCC ramp up.

Intel conducted a review with the two primary SPI Flash suppliers for the Oak Stream reference platforms. Both suppliers confirmed the 100 ms delay poses no risk or reliability concerns, but it is still recommended to meet the device ramp up specification.

Implication

While no functional failures have been identified so far, the validation is still ongoing. To ensure the design meets the device specifications, the Oak Stream DC-SCM design will be revised to incorporate a 10K ohm pull-up resistor on the CS_N signal for BIOS SPI Flash, BMC SPI Flash and IFWI Recovery Flash (Intel® Platform Firmware Resilience [Intel® PFR]).

Figure 55. Before CS₀_N Pull-Up Resistor

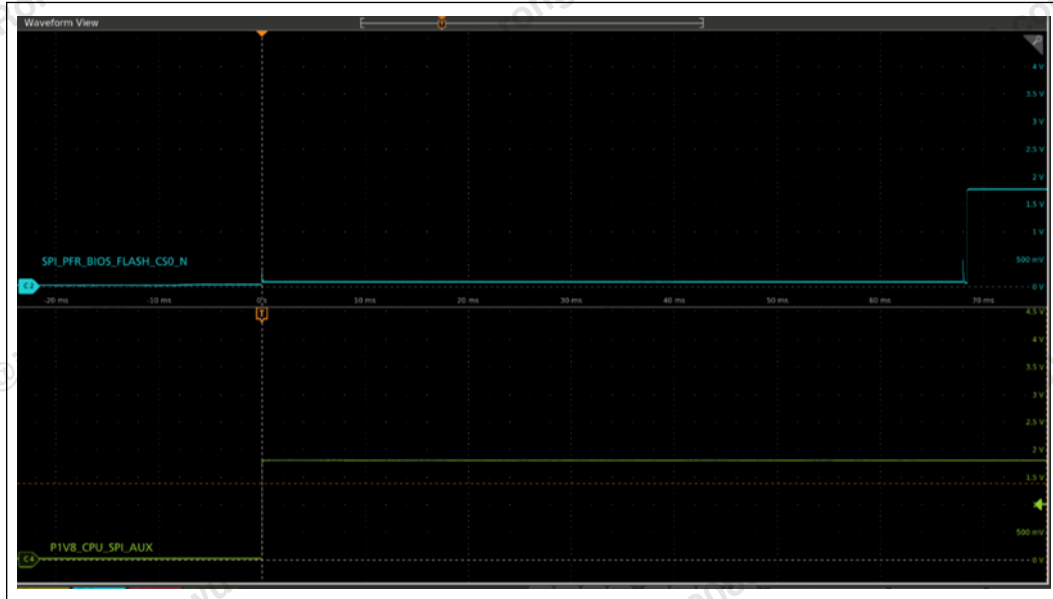
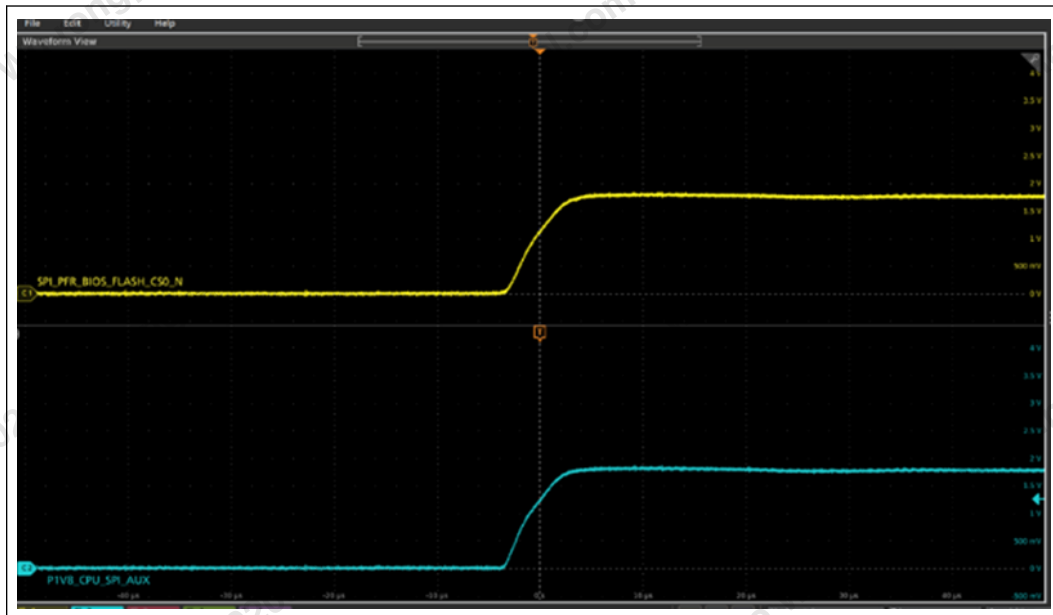


Figure 56. After CS₀_N Pull-Up Resistor



Root Cause

As the SPI is energized first during the boot flow, the CS_N signal is not actively driven when the DC-SCM is not yet energized. There is a missing pull-up resistor on the connection between the CS_N pin of the SPI flash and the DC-SCM. The pull-up resistor purpose is to keep the CS_N inactive until the DC-SCM is energized.

Mitigation/Fix

The Oak Stream DC-SCM design will be revised to incorporate a 10K ohm pull-up resistor on the CS_N signal, ensuring compliance with the device specification.

Status

The following collaterals will be updated to reflect the new pull-up resistor on CS_N:

- *Oak Stream 16-Channel Platform Design Guide*, document number 788196
- *Oak Stream Platform DC-SCM 2.1 Design Archive*, document number 636262

For questions or assistance, contact your Intel representative.

2.39 WW12-2025

New:

- [Oak Stream 16-Channel Platform Design Guide Release Notification](#) on page 281

Updated:

- [\(Updated WW12\) Oak Stream Key Collaterals Availability and Forecast](#) on page 282

2.39.1 Documentation

2.39.1.1 Oak Stream 16-Channel Platform Design Guide Release Notification

The *Oak Stream 16-Channel Platform Design Guide (PDG)*, document number [788196](#), revision 0.8, is now available in the Resource and Documentation Center (RDC).

Overview of key revision 0.8 changes:

- Chapter 1: Introduction and Overview
 - The Diamond Rapids UCC processor will be a pre-production only SKU; UCC SKUs will be used for Engineering Samples 1 (ES1) only. Diamond Rapids UCC will have support for UPI on Ports 1 and 9. UCC SKUs will not be productized.
- Chapter 4: Platform Clocking
 - Updated all topologies.
 - Updated 25-MHz Crystal Requirements table and notes
- Chapter 5: Processor Essential IO Signals Guidelines
 - Updated Terminating Unused SPI Signals.
 - Updated General Design Consideration in table Critical Signals Routing Summary
 - Updated Width (W) / Spacing (S) in Critical Signals Routing Summary table
- Chapter 6: DDR5 Memory Routing Guidelines
 - Updated the raw cards config list
 - Updated CuMax and SMT sections with newer details, images and guidelines.
 - Added DQ Bit Swizzling section under Memory Design Considerations as sub-chapter.
- Chapter 7: I3C* Design Guidelines

- Updated termination resistor from 2K to 10K ohms in the aggregator due to internal pull-up identified by recent board testing.
- Chapter 8: FIO Guidelines
 - Updated unused ports section
 - Added Routing FIO Nets Through a DIMM Pinfield Section
- Chapter 10: Intel® UPI 3.0 Guidelines
 - Added 2S 2 x16 port topologies as POR with intended AI head node use model
 - Updated 2-Connector Cabled Topology 1 Trade Off Table showing new extended lengths
 - Added 2-Connector Cabled Topology 1 Trade Off Table rows for 48 GT/s data rate
- Chapter 14: Power Distribution Guidelines
 - Updated Table 729: Processor Absolute Minimum and Maximum Ratings
 - Added new Section 14.6.10: CPU Power Rail Inrush current and Inductor Peak current (I_{Peak}) during a load transition.
 - Added new Section 14.6.13: Voltage Connections for CPU I/O Signal Design Considerations.

Refer to the document revision history for full list of changes.

Revision 0.8 Clarifications intercepted in 0.81 Revision

Customers need to be aware of the following clarifications for 0.8 revision:

- Chapter 2: Printed Circuit Board (PCB) Design Guidelines
 - Figure 15 0.104 Inch, 22-Layer Stackup is reflecting 120 stack-up.
- Chapter 4: Platform Clocking
 - Note "Ensure S0_PWROK asserted before enabling PFT_IN signal" is not applicable. Correct Note is "Ensure PFT_IN is enabled after CPU VCC_VNN is enabled, but before asserting S0_PWROK"
- Chapter 14: Power Distribution Guidelines
 - Table 739 Other Processor VR Power Parameters Table 2 has a typo. PVCDDR_HV1 for 16-CH and 12-CH The Load Line Used in RP Design (mOhm) values is 0.4, not 0.5.

These clarification updates are intercepted in the PDG revision 0.81.

2.39.1.2 (Updated WW12) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 103. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	Q2'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	WW15'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
Oak Stream 16-Channel Platform Design Guide	788196	0.81	0.9	Mid Q2'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	Q2'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	WW14'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	0.1	Q2'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	WW14'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW19'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	WW19'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Late Q2'25
Oak Stream 8-Channel Platform Thermal Mechanical Specification	786524	0.4	0.5	Mid Q2'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	Q2'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.40 WW11-2025

New:

- [Boot Security Technology Enablement Feedback Request](#) on page 284
- [Oak Stream 8-Channel Thermal Mechanical Collaterals Release Notification](#) on page 284

2.40.1 Customer Feedback

2.40.1.1 Boot Security Technology Enablement Feedback Request

Intel is requesting feedback from customers on boot security technologies enablement plans on Eagle Stream, Birch Stream, and Oak Stream based designs.

Provide your answers to the following questions to your Intel representative. The deadline for collecting feedback is WW14'25.

Do you enable Intel® Boot Guard on or plan to enable it on:

- Eagle Stream
- Birch Stream
- Oak Stream

Do you enable Intel® Trusted Execution Technology (Intel® TXT) on or plan to enable it on:

- Eagle Stream
- Birch Stream
- Oak Stream

Do you enable Intel® Platform Firmware Resilience (Intel® PFR) on or plan to enable it on:

- Eagle Stream
- Birch Stream
- Oak Stream

Do you enable CPU Attestation on or plan to enable it on:

- Birch Stream
- Oak Stream

Do you plan to enable an Intel vendor-authorized boot on Oak Stream?

- Yes
- No

2.40.2 Documentation

2.40.2.1 Oak Stream 8-Channel Thermal Mechanical Collaterals Release Notification

Initial releases of the *Oak Stream 8-Channel Platform Socket Enabling Components Mechanical Drawings*, document number [786525](#), revision 0.5, and the *Oak Stream 8-Channel Platform Socket Enabling Components Mechanical Models*, document number [786526](#), revision 0.5, are now available in the Resource and Documentation Center (RDC).

The *Oak Stream 8-Channel Platform Thermal Mechanical Specification (TMS)*, document number [786524](#), revision 0.4, is now available in RDC.

NOTE

Revision 0.4 covers mechanical updates. Revision 0.5 will cover thermal updates. Monitor the Oak Stream Key Collaterals Availability and Forecast Message of the Week (MoW) article for collateral release ETA.

Overview of revision 0.4 changes:

- Chapter 3: Socket Specifications - Added socket details, including Socket Loading and Deflection Specifications.
- Chapter 4: Package and Socket Stack Mechanical Design - Initial documentation of chapter details, including details on supported main board thicknesses and PHM Load Specification and Enabled Thermal Solution Mass (ETSM).
- Chapter 8: Processor Heatsink Module and Loading Mechanism - Added detail on bolster plate, backplate, package carrier, heatsink hardware, and socket cover.
- Chapter 10: Thermal Solution Designs- Initial documentation of chapter details. Updated socket contact count throughout.

2.41 WW10-2025

New:

- [Signal Integrity Collateral List for the Oak Stream Platform Release Notification](#) on page 286

Updated:

- [\(Updated WW10\) Oak Stream Key Collaterals Availability and Forecast](#) on page 285

2.41.1 Documentation

2.41.1.1 (Updated WW10) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 104. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	Q2'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	WW13'25
<i>continued...</i>				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.8	WW11'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.8	0.9	Q2'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	WW12'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	0.1	Q2'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW12'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	WW19'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Mid Q2'25
<i>Oak Stream 8-Channel Platform Marketing Development Platform 3 (MDP3) Design Archive</i>	TBD	N/A	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Mid Q3'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Late Q2'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.4	WW11'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	Q2'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.95	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.41.1.2 Signal Integrity Collateral List for the Oak Stream Platform Release Notification

The *Signal Integrity Collateral List for the Oak Stream Platform*, document number 822391, Revision 2025WW09, is now available in the Resource and Documentation Center (RDC).

The Signal Integrity Collateral List document provides a simple, central reference for key signal integrity (SI) simulation models, tools, support documents, and other information.

Overview of Revision 2025WW09 Changes:

- Updated revision numbers for several signal integrity simulation model sets and modeling tools
- Added new collaterals for cable/connector and re-timer design guidance.

Updated collaterals are indicated individually, for convenience.

2.42 WW09-2025

New:

- [Oak Stream 16-Channel Platform Electrical Specification Release Notification](#) on page 288
- [Diamond Rapids Processor Post Internal Error \(IERR\) Harvesting](#) on page 288
- [Power and Thermal Software Solution Update](#) on page 291

Updated:

- [\(Update WW9\) Diamond Rapids Tools Readiness](#) on page 289
- [\(Updated WW9\) Oak Stream Key Collaterals Availability and Forecast](#) on page 287

2.42.1 Documentation

2.42.1.1 (Updated WW9) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 105. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	Q2'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	WW13'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.8	WW11'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.9	1.0	Q2'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	WW12'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	0.1	Q2'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW12'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Q2'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	Q2'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.9	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.42.1.2 Oak Stream 16-Channel Platform Electrical Specification Release Notification

The *Oak Stream 16-Channel Platform Electrical Specification*, document number 787006, Revision 0.8, is now available in the Resource and Documentation Center (RDC).

Overview of 0.8 Changes:

- Updated signal naming convention to match schematic and pin list nomenclature
- Updated overshoot/undershoot waveforms
- Revised AC specifications numbers for multiple signals: Debug, I3C*, Clocking

2.42.2 Enabling

2.42.2.1 Diamond Rapids Processor Post Internal Error (IERR) Harvesting

Diamond Rapids Primary In-Memory Hierarchy (IMH) Machine Check Architecture (MCA) banks Model-Specific Registers (MSRs) are accessed through a core within even numbered module, while secondary IMH MCA bank MSRs are accessed through a core within an odd numbered module.

When an IERR fault occurs (CATERR# hold), and Diamond Rapids ondie Crashlog completes collecting the error logs (CLIP# de-asserted), platform may choose to trigger a warm reset to enter "BIOS based error harvesting" boot cycle. Since post IERR "BIOS based error harvesting" occurs pre-MRC, the monarch core executes without the ability to call other cores. This limits BIOS' ability to access MCA bank MSRs in the other IMH.

OEMs can retrieve MCA bank information from crashlog and can make use of BMC to harvest MCA MSRs Out-Of-Band (OOB) prior or post warm reset.

See *Oak Stream Platform RAS Deep Dive Technical Training*, document number 824347, and *Diamond Rapids Processor Architecture Specification*, document number 792359, for details on Diamond Rapids machine check architecture.

2.42.3 Tools and Debug

2.42.3.1 (Update WW9) Diamond Rapids Tools Readiness

Updated information highlighted in **bold text**

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to ITP Adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support
 - Two Streams Tracing
 - LA-3907 preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports four trace ports)
 - LA-3892 converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- ASSET* Arium - contact ASSET Arium for support
- Compact Debug Port (CDP) + MIPI10 Adapter for I3C*
 - Type-C connection: voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- Vtune™
- Intel® SoC Watch
- Intel® System Debugger
 - Token generation for Vendor Authorize Boot (VAB) / Vendor Authorize Debug (VAD)

Table 106. Diamond Rapids Tools Readiness

Tool	Trend
ASSET *Intertech* Arium	TBD
ASSET* Close Chassis Controller (CCC)	TBD
Autonomous Crashdump (ACD)	Q3'25
CScripts	Q3'25
Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card	Q3'25
Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C for CCA	Available
Intel® I3C USB Type-C Level Shift Adapter	Available
Intel® At-Scale Debug (Intel® ASD)	Q3'25
W: Intel Channel Analyzer Tool (ICAT)	Available
<i>continued...</i>	

Tool	Trend
W: Intelligent Debug Validation (IDV MM#99A8C)	Not Supported in Oak Stream
16-Ch Intel's PLB/ULB Test Cards	WW25'25
8-Ch: Intel's PLB/ULB Test Cards	Q4'25
W: UEFI/JTAG-Intel SelfTest	Q3'25
Intel® SVT (Silicon View Technology) Closed Chassis Adapter (CCA)	Available
Intel® System Bring-up Toolkit NDA	Q4'25
W: Intel XDP3 (MM#940990)	Available
W: Intel® IO Margin Tool and User Guide	Q4'25
L: Intel® IO Margin Tool and User Guide	Q4'25
Lauterbach*	TBD
Memory Assistant	Q3'25
W: Memory Error Injection (MEI) DDR5 Card (MM#99CDV1)	Available
W: Memory Error Injection (MEI) DDR5 Lite Card (8000 MT/s)	WW27'25
PCIe* PEI 5.0 Card	Available
PCI Express* (PCIe*) PEI 6.0 Card	TBD
Intel® System Stress and Memory Diagnostics	Q3'25
Intel® Power and Thermal Analysis Tool (Intel® PTAT)	Not Supported in Oak Stream. Replaced by Intel® System Stress and Memory Diagnostics
Stitched Rank Margining Tool (RMT), Target Loader RMT (EFI)	Q4'25
Intel® Structural Test Plus (Intel® STP)	Available
16-Ch Thermal Test Vehicle (TTV)	Q2'25
8-ch TTV	Q3'25
Server Security Toolkit	TBD
W: VRTT Gen5	Available
16-Ch VRTT Interposer	WW20'25
8-Ch VRTT Interposer	TBD

NOTE

W: Windows*. L: Linux*. If not specified, tool is OS agnostic.

NOTE

Contact your Intel Representative for Intel® Simics support.

2.42.3.2 Power and Thermal Software Solution Update

The power and thermal software solutions have continuously evolved to meet the growing needs of Intel customers. From the introduction of the Intel® Power Thermal Utility (Intel® PTU) to the advancement of the Intel® Power and Thermal Analysis Tool (Intel® PTAT), Intel has consistently strived to enhance tools and capabilities.

Intel is introducing the next generation of power and thermal software solutions: Intel® System Stress and Memory Diagnostics, the new software solution for comprehensive power and thermal analysis. **Intel System Stress and Memory Diagnostics will replace Intel PTAT in Oak Stream.**

Intel System Stress and Memory Diagnostics will become the primary software solution for customers to collect power, thermal, and frequency system data. The tool will offer the same capabilities as Intel PTAT, with a modern software architecture that will provide additional flexibility to the users.

Comprehensive details regarding the Intel System Stress and Memory Diagnostics software architecture, feature set, available tests, and command line usage will be documented in the tool's user guide. The Intel System Stress and Memory Diagnostics User Guide is expected to be released in Q3'2025. See the Oak Stream Message of The Week for release notification updates.

Intel System Stress and Memory Diagnostics transition starts with Clear Water Forest and onwards. Intel PTAT will continue to be used in earlier generations.

2.43 WW08-2025

New:

- Documentation:
 - [Oak Stream Platforms DC-SCM 2.1 Design Archive Release Notification](#) on page 291
 - [Oak Stream RDC Collection Layout Updates](#) on page 292
- Enabling:
 - [GLOBAL_RESET_N and eSPI Global Reset Signal Clarification](#) on page 293
 - [Max PCB Length Change for PCI Express* 6.0 and Intel® Ultra Path Interconnect Version \(Intel® UPI\) 3.0 Topologies](#) on page 295

Updated:

- [\(Updated WW8\) Oak Stream Key Collaterals Availability and Forecast](#) on page 292

2.43.1 Documentation

2.43.1.1 Oak Stream Platforms DC-SCM 2.1 Design Archive Release Notification

The *Oak Stream Platforms DC-SCM 2.1 Design Archive*, document number [636258](#), revision 0.9, is now available in RDC.

Overview of 0.9 changes:

- A spreadsheet with the change list is included as part of the DA compressed file.

2.43.1.2 Oak Stream RDC Collection Layout Updates

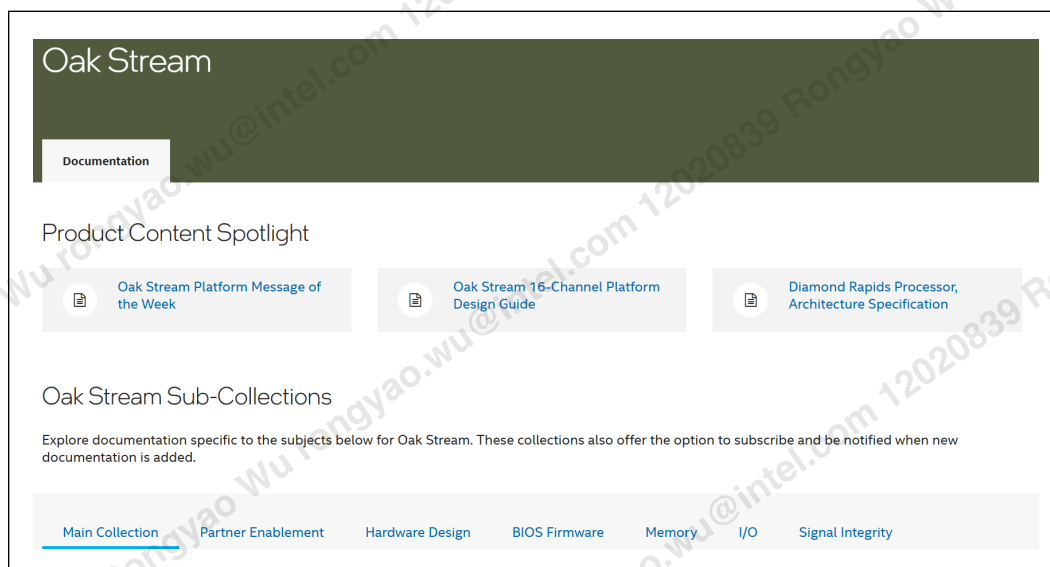
The [Intel Resource and Documentation Center \(RDC\) Oak Stream Collection](#) website has been updated to highlight key content.

The **Product Content Spotlight** section is positioned at the top of the page and includes links to the three most frequently accessed content.

The **Oak Stream Sub-Collections** tabs organize collaterals into six categories based on relevance to key design-in areas: Partner Enablement, Hardware Design, BIOS Firmware, Memory, I/O, and Signal Integrity.

The **Main Collection** tab includes all Oak Stream Collaterals.

Figure 57. Oak Stream Collection Layout



The **Search this Collection** search box will search on the current active sub-collection.

Customers can receive email notifications on the collaterals of each sub-collections using the **Follow this collection ->** button.

Contact your Intel Representative if you have questions on the new layout.

2.43.1.3 (Updated WW8) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 107. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	Q2'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	WW13'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.8	WW11'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	WW9'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	WW12'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	0.1	Q2'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW12'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	Q2'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.9	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.43.2 Enabling

2.43.2.1 GLOBAL_RESET_N and eSPI Global Reset Signal Clarification

Description

The CPU AUX_PWRGD signal's behavior has changed from Birch Stream to Oak Stream.

Oak Stream introduces the GLOBAL_RESET_N signal, which now handles the functionality previously managed by AUX_PWRGOOD in Birch Stream. As a result, AUX_PWRGOOD does not toggle during global resets in Oak Stream; instead, GLOBAL_RESET_N does.

See the *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#) for more details on GLOBAL_RESET_N functionality.

In Birch Stream, the eSPI_Reset# signal is asserted during an asynchronous global reset, coinciding with AUX_PWRGOOD de-assertion. The eSPI SLP_Sx virtual wire reset control is also governed by AUX_PWRGOOD.

See the *Enhanced Serial Peripheral Interface (eSPI) Compatibility Specification Self-Boot Server Platforms*, document number [627433](#), for more details on eSPI_Reset# signal.

Requirement

For proper synchronization during a global reset, the BMC eSPI virtual wires (VW), including SLP_Sx signals, must be de-asserted when the GLOBAL_RESET_N signal de-asserts.

Implication

AUX_PWRGOOD is connected to the BMC eSPI external VW reset (GPIOB0 pin in Reference Platform BMC). As a result of this connectivity and the change to AUX_PWRGOOD behavior in Oak Stream, **during a global reset, the SLP_Sx virtual wires do not reset**, causing the BMC eSPI device virtual wires to become out of sync.

Root Cause

The Oak Stream platform uses AUX_PWRGOOD instead of GLOBAL_RESET_N, affecting global reset handling and BMC eSPI virtual wire synchronization.

Mitigation/Fix

The Oak Stream platform SCM and HPM PLD reference code will accurately reflect the connection of GLOBAL_RESET_N to the BMC GPIOB0 pin. The release of the PLD reference code will be communicated through the Oak Stream Platform Message of the Week (MOW).

Status

The following collaterals will be updated to reflect GLOBAL_RESET_N connection update:

- *Enhanced Serial Peripheral Interface (eSPI) Compatibility Specification Self-Boot Server Platforms*, document number 627433
- *Oak Stream Platforms DC-SCM 2.1 Design Archive*, document number 636262
- *Oak Stream Platforms DC-SCM 2.1 Integration Guide*, document number 812906

For questions or concerns, contact your Intel representative.

2.43.2.2 Max PCB Length Change for PCI Express* 6.0 and Intel® Ultra Path Interconnect Version (Intel® UPI) 3.0 Topologies

To assist customers designing and evaluating Oak Stream 16-channel platform based designs, a summary of upcoming PCIe* 6.0 and Intel® UPI 3.0 PCB channel length updates is provided in the following tables.

These updates will be reflected in the upcoming revision of the *Oak Stream 16-Channel Platform Design Guide*, document number 788196.

Table 108. Updates to Max PCB Trace Length for PCI Express* 6.0 Topologies

Chapter	Topology	Current Max PCB Length	Updated Max PCB Length	Note
9.7.1.1	1 conn., EDSFF	13.5 inch	16.5 inch	
9.7.1.2	1 conn., OCP	13.5 inch	14.5 inch	
9.7.1.3	1 conn., AIC	10.0 inch	13.0 inch	
9.7.2.1	2 conn., Cable, EDSFF	9.0 inch	11.0 inch	20 inch Cable
9.7.2.2	2 conn., Cable, AIC	7.0 inch	7.5 inch	20 inch Cable
9.7.2.3	2 conn., Riser, AIC ¹	12.0 inch	12.5 inch	5 inch Riser
9.7.3.1	3 conn., Cable, HSBP, EDSFF	7.0 inch	9.0 inch	20 inch Cable

NOTE

¹ AIC and Mainboard: -0.85 dB/in and Riser board -0.53 dB/in at 16 GHz.

Table 109. Updates to Max PCB Trace Length for Intel® UPI 3.0 Topologies

Chapter	Topology	Current Max PCB Length	Updated Max PCB Length	Note
10.4.1	2 conn., Topology 1, 56 GT/s	12.0 inch	14.0 inch	20 inch Cable
10.4.1	2 conn., Topology 1, 56 GT/s	13.0 inch	15.0 inch	12 inch Cable
10.4.1	2 conn., Topology 1, 56 GT/s	N/A	12.5 inch	28 inch Cable

continued...

Chapter	Topology	Current Max PCB Length	Updated Max PCB Length	Note
10.4.1	2 conn., Topology 1, 56 GT/s	N/A	11.0 inch	36 inch Cable
10.4.2	0 conn., Topology 2, 56 GT/s	19.0 inch	21.0 inch	

NOTE

N/A represents not documented in the current revision.

2.44 WW07-2025

New:

- [Oak Stream 16-Channel Platform Johnson City 2 Design Archive Release Notification](#) on page 296
- [Oak Stream 16-Channel Platform Shadow Core Technical Article Release Notification](#) on page 297

Updated:

- (Updated WW7) [Oak Stream Key Collaterals Availability and Forecast](#) on page 297
- Appendix: [Oak Stream Collaterals](#) on page 373

2.44.1 Documentation

2.44.1.1 Oak Stream 16-Channel Platform Johnson City 2 Design Archive Release Notification

The *Oak Stream 16-Channel Platform Johnson City 2 Design Archive*, document number 636258, Revision 0.9, is now available in RDC.

Overview of 0.9 changes:

- Aligned design to 104 mil, 22-layer stackup for thinner PCB.

Intel supports Cadence Allegro* versions 22.1 and 23.1. The Design Archive (DA) compressed file includes .brd files for both Cadence Allegro version 23.1 and 22.1 (Compatible with 17.x).

For additional details on DA deliverables, refer to [WW31-24 Design Archive MoW Article](#)

Intel does not endorse or recommend specific software vendors or products.

2.44.1.2 Oak Stream 16-Channel Platform Shadow Core Technical Article Release Notification

The *Oak Stream 16-Channel Platform 22-Layer Shadow Core Platform Design Considerations*, document number 845670, Revision 0.5, is now available in RDC.

Overview of 0.5 changes:

- First release

2.44.1.3 (Updated WW7) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 110. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.8	0.9	Q2'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	WW13'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.8	WW11'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	WW9'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	WW9'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.9	0.1	Q2'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW10'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	Q2'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	WW11'25

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.45 WW05-2025

New:

- [Diamond Rapids Processor, Architecture Specification Release Notification](#) on page 298

Updated:

- [\(Updated WW5\) Oak Stream Key Collaterals Availability and Forecast](#) on page 298

2.45.1 Documentation

2.45.1.1 Diamond Rapids Processor, Architecture Specification Release Notification

The *Diamond Rapids Processor, Architecture Specification*, document number 792359, Revision 0.8 is now available in RDC.

Overview of Revision 0.8 changes:

- Added new Processor Information ROM (PIROM) chapter
- Updated Overview, Features and Topologies Chapter
 - Added I/O Hub (IOH) Functional Description and Block Diagrams
 - Updated IMH1 and IMH2 Functional Description and Block Diagrams
 - Updated SoC Architecture Diagrams for all Diamond Rapids SKUs
- Updated Intel® Out-of-Band Management Services Module Chapter
- Updated Reliability, Availability, and Serviceability (RAS) Chapter
- Updated Intel® In Field Scan Section and moved section under RAS Chapter
- Removed Signal Definitions Chapter

2.45.1.2 (Updated WW5) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 111. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
Diamond Rapids Processor Architecture Specification	792359	0.8	0.9	Q2'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	WW13'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.8	WW11'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	WW8'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	WW9'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.7	0.9	WW7'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW10'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	Q2'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	WW8'25

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.46 WW03-2025

New:

- [Intelligent Debug and Validation \(IDV\) Tool Support Drop](#) on page 299

2.46.1 Tools and Debug

2.46.1.1 Intelligent Debug and Validation (IDV) Tool Support Drop

The Intelligent Debug and Validation (IDV) tool will no longer be supported starting with Oak Stream.

2.47 WW02-2025

New:

- [Clock Jitter Tool \(CJT\) Release Notification](#) on page 301
- [Tabbed Routing Design and Characterization Tool \(TRDCT\) Release Notification](#) on page 302

Updated:

- [\(Updated WW2\) Oak Stream Key Collaterals Availability and Forecast](#) on page 300

Follow up:

- Make sure to follow up on the Copper Max (CuMax) notes in the following section: [\[Follow Up\] Copper Max \(CuMax\) Support on the Oak Stream Platform](#) on page 300

2.47.1 Enabling

2.47.1.1 [Follow Up] Copper Max (CuMax) Support on the Oak Stream Platform

As noted in the Data Center Design Conference (DCDC) presentation, *Oak Stream PCB Design Challenges and Industry Readiness*, document number [818071](#), Intel is enabling a new impedance-control technology called Copper Max (CuMax) for DDR routing on the Oak Stream platform.

To ensure use of the technology exclusively on Intel processor-based platforms, CuMax support in the Cadence* Allegro family of layout software requires installation of an additional free license. Customers interested in obtaining a license must send a request to the following e-mail address: cumax-info@intel.com

Contingent on Intel approval, Cadence* will be notified and will contact the requestor(s) directly with additional instructions.

Enabling of other layout software is under discussion. Customers should contact their Intel Field representatives if CuMax support is needed in other PCB layout software products.

Intel does not endorse or recommend specific software vendors or products.

2.47.2 Documentation

2.47.2.1 (Updated WW2) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 112. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.7	0.8	WW6'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	WW11'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.8	WW11'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	WW8'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	WW9'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.7	0.9	WW7'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW10'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Q2'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.7	0.9	Q2'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	Q2'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	WW8'25

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

2.47.3 Tools and Debug

2.47.3.1 Clock Jitter Tool (CJT) Release Notification

The *Clock Jitter Tool (CJT)*, document number 803838, version 6.0.1, is now available on RDC. This software supports the calculation of reference clock jitter for PCI Express* 1.0 through 6.0 using measured waveforms.

CJT requires the installation of MathWorks* MATLAB Runtime 9.10 libraries, or the full MATLAB 2021a product is needed to run CJT.

This release removes the automated installer, in accordance with Intel's software security policies. Compared to the previous version, no new features were added, and no changes were made to existing functions or analysis method.

2.47.3.2 Tabbed Routing Design and Characterization Tool (TRDCT) Release Notification

The *Tabbed Routing Design and Characterization Tool (TRDCT)*, document number 560170, version 1.6.1, is now available on RDC.

TRDCT is intended for use with Ansys* HFSS and Electronic Desktop* for creating models and calculating parameters for tabbed transmission lines and related PCB structures. The installation of MathWorks* MATLAB Runtime 9.10 libraries or the full MATLAB 2021a product is needed to run TRDCT.

Summary of 1.6.1 changes:

- Supports both tabbed routing and CuMax calculations.
- Addresses a microstrip modeling issue reported against version 1.5.
- Corrects a bug related to image display reported against version 1.6.

3.0 2024

3.1 WW51-2024

New:

- [Intel® Confidential Computing BIOS Writer's Guide Release Notification](#) on page 303
- [Oak Stream 16-Channel Platform Essential I/O: Models Release Notification](#) on page 303
- [Oak Stream 16-Channel Platform Marketing Development Platform 2 \(MDP2\) Design Archive Release Notification](#) on page 304
- [ICAT 6.200.50 Execution Errors](#) on page 305

Updated:

- [\(Updated WW51\) Oak Stream Key Collaterals Availability and Forecast](#) on page 304

NOTE

The Oak Stream Platform Message of the Week (MoW) will not be released in WW52-2024 and WW01-2025.

3.1.1 Documentation

3.1.1.1 Intel® Confidential Computing BIOS Writer's Guide Release Notification

The *Intel® Confidential Computing BIOS Writer's Guide*, document number [823668](#), revision 0.5 is now available in RDC.

This document is a resource for BIOS developers and engineers who are working with Intel's confidential computing technologies, Intel® Software Guard Extensions (Intel® SGX) and Intel® Trust Domain Extensions (Intel® TDX) on the Oak Stream platform.

3.1.1.2 Oak Stream 16-Channel Platform Essential I/O: Models Release Notification

The *Oak Stream 16-Channel Platform Essential I/O: Models*, document number 822178, revision 1.0, is now available in RDC.

This release contains updated signal integrity models for components used on the Essential I/O signal group, including SPI, I2C*, and other interfaces. The documentation has been updated to include specific processor pin numbers and signal directions.

3.1.1.3 Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive Release Notification

The *Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive*, document number [828483](#), revision 0.5, is now available in RDC.

Overview of Revision 0.5 changes:

- Initial release

MDP2 is a reference design for 16-Channel 2-Socket (2S) monoplanar designs. Some changes have been identified as required for this configuration. These changes are detailed in the README file included in the compressed file. Customers using MDP2 DA as reference should note the changes and implement them on their designs.

Revision 0.7 will implement the 2S monoplanar design required changes and will also include the Bill of Materials (BOM) information.

3.1.1.4 (Updated WW51) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 113. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.7	0.8	WW5'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.8	WW9'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	Jan'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	0.5	0.7	Q1'25
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.7	0.9	Q1'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	Q1'25

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

3.1.2 Tools and Debug

3.1.2.1 ICAT 6.200.50 Execution Errors

1. Some users have reported issues with the latest release of the Integrated Channel Analysis Tool (ICAT) Suite, 6.200.50; errors are observed when opening and attempting to simulate topology files generated using previous releases.
2. This issue can be resolved by moving the desired file to the default environment:
 - a. Open the file in ICAT
 - b. Click "Yes" in the ICAT dialog box: "The environment paths set in the project do not match the default ones. Would you like to switch to the default environment?"

3.2 WW49-2024

New:

- [Oak Stream 12-Channel Platform AI Head Node Design Considerations Release Notification](#) on page 307
- [Engineering Validation and Test \(EVT\) Enabling for Oak Stream 16-Channel 1 Socket \(1S\)/ 2 Sockets \(2S\) Designs](#) on page 307
- [VCCDDR_HV0 and VCCDDR_HV1 Voltage Rails \(VR\) Updates](#) on page 309

Updated:

- [\(Updated WW49\) Oak Stream Key Collaterals Availability and Forecast](#) on page 306
- [\(Updated WW49\) Q4 Oak Stream Quarterly Technical Review \(QTR\) Online Session](#) on page 307

3.2.1 Documentation

3.2.1.1 (Updated WW49) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are tracked in the [Appendix](#) section.

Table 114. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.7	0.8	WW5'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	Q1'25
<i>Oak Stream 12-Channel AI Head Node Design Considerations Technical Article</i>	823064	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	Jan'25
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW50'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.7	0.9	Q1'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	Q4'24

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

3.2.1.2 Oak Stream 12-Channel Platform AI Head Node Design Considerations Release Notification

The *Oak Stream 12-Channel Platform AI Head Node Design Considerations*, document number 823064, Revision 1.0, is now available in RDC.

Overview of Revision 1.0 changes:

- Added System Construct Chapter
- Added Printed Circuit Board (PCB) Design Chapter
- Added Power Delivery Chapter
- Added Thermal and Mechanical Design Considerations Chapter
- Added Memory and HSIO Design Considerations Chapter

3.2.2 Enabling

3.2.2.1 (Updated WW49) Q4 Oak Stream Quarterly Technical Review (QTR) Online Session

NOTE

Updated information is highlighted in **bold text**.

Customer and Systems Engineering team is excited to invite you to the Oak Stream Quarterly Technical Review (QTR) online session. This session will allow customers to find the latest enabling updates for the next-generation Oak Stream platforms.

The online session **was** hosted on December 3rd, at 5:00-7:00PM Pacific Time (PDT).

The material presented is now available in the Resource and Documentation Center.

- **Recording: Oak Stream Platform Quarterly Technical Review (QTR) Q4'24 Recording**, document number **842289**
- **Presentation: Oak Stream Platform Quarterly Technical Review (QTR) Q4'24**, document number **837295**

3.2.2.2 Engineering Validation and Test (EVT) Enabling for Oak Stream 16-Channel 1 Socket (1S)/ 2 Sockets (2S) Designs

To enable customer 1 Socket (1S)/ 2 Sockets (2S) Time to Market (TTM) launch readiness aligning to Oak Stream 16-Channel platform schedule, Intel will enable 1S/2S customers' **EVT** activities by separating Engineering Validation and Test (EVT) into **EVT 1** and **EVT 2** phases.

EVT 1 activities will use UCC IMH1-based ES1 A-step samples and **EVT 2** activities will use UCC-M IMH2-based ES0 B-step samples.

See the latest *CNDA Snapshot for Intel Xeon Oak Stream Platform* for the detailed Oak Stream schedule and key sampling milestones.

A high-level mapping of expected customers EVT 1 and EVT 2 activities is detailed in Figure 58 on page 308:

Figure 58. Customer Validation Phase Mapping

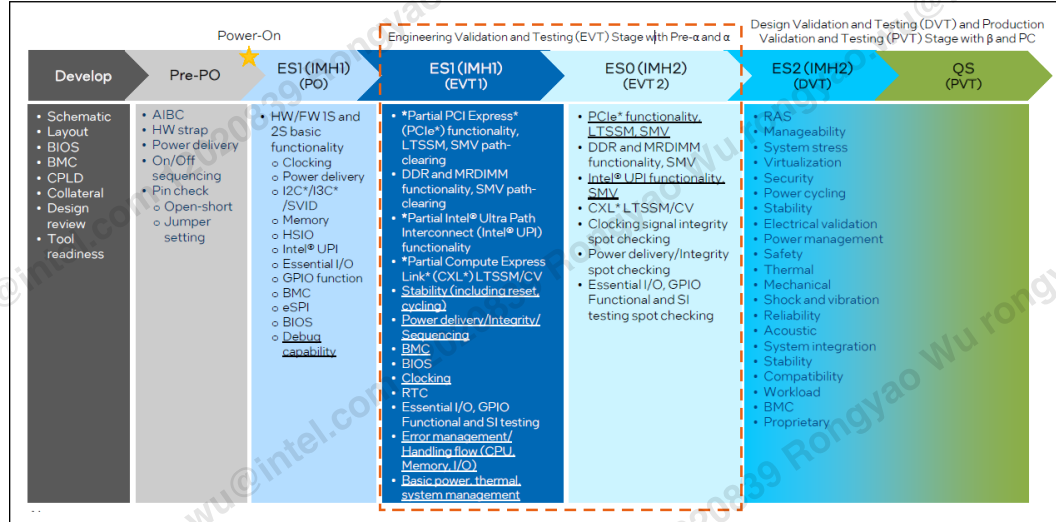
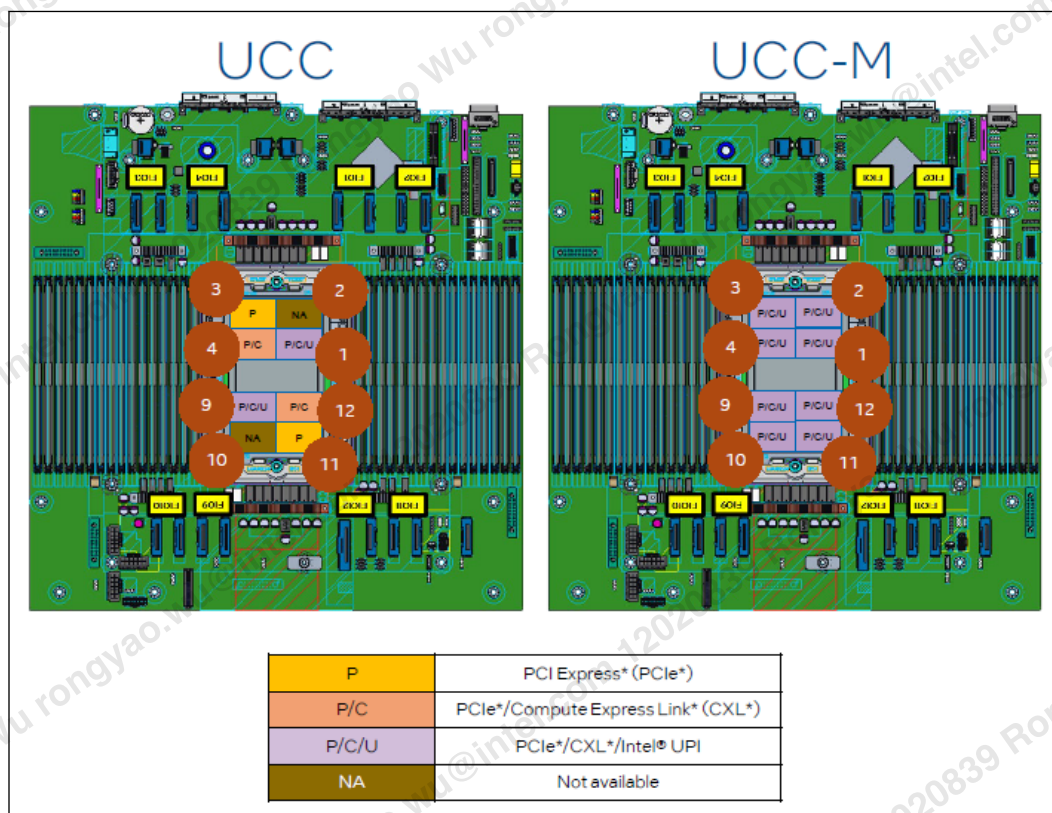


Figure 59 on page 309 contains the high-level comparison of the UCC and UCC-M HSIO capabilities. Differences in the HSIO capabilities are comprehended in the mapping of customers' **EVT 1** and **EVT 2** activities.

Figure 59. UCC and UCC-M Port Mapping



IMH1-based ES1 samples have Intel® Ultra Path Interconnect (Intel® UPI) capabilities on ports 1 and 9. 2S designs must use ports 1 and 9 for Intel® UPI connectivity to be able to take advantage of ES1 samples for earlier 2S EVT start.

Intel® UPI connectivity for UCC is:

- Socket 0 port 1 to Socket 1 port 1
- Socket 0 port 9 to Socket 1 port 9

3.2.2.3 VCCDDR_HV0 and VCCDDR_HV1 Voltage Rails (VR) Updates

On Oak Stream 16-Channel Johnson City Reference Platform, VCCDDR_HV0 and VCCDDR_HV1 VR phase counts were reduced from 4-phase TLVR to 3-phase TLVR due to an update in current specifications.

See the *Oak Stream 12-Channel Platform AI Head Node Design Considerations*, document number [823064](#), for VCCDDR_HV0 and VCCDDR_HV1 specifications for the 12-Channel configuration.

Decoupling caps near these VRs were also reduced as indicated in the following tables:

Table 115. VCCDDR_HV0 and VCCDDR_HV1 for Oak Stream 16-Channel Platform: 1 of 2

Power Rails	TDC Current (A)	Peak Current (A)	Max Current Step Size (A)	Max Current Step Slew Rate di/dt (A/us)	Min Load (A)
PVCCDDR_HV0/HV1 (for 16-Channel)	71	Old: 85 New: 71	Old: 64 New: 54	Old: 40 New: 30	Old: 21 New: 17

Table 116. VCCDDR_HV0 and VCCDDR_HV1 for Oak Stream 16-Channel Platform: 2 of 2

Capacitance (uF)	Voltage (V)	Tolerance (+/-%)	Type	Size	Quantity per RB	Location
47	4	20	X6S	0805	Old: 12 New: 9	VR
10	4	20	X6S	0402	12	VR
1	16	10	X6S	0402	Old: 4 New: 3	VR

These updates will be included in the next revision of the *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#).

3.3 WW48-2024

New:

- Appendix: [Oak Stream Collaterals](#) on page 373

Updated:

- [\(Updated WW48\) Oak Stream Key Collaterals Availability and Forecast](#) on page 310
- [\(Updated WW48\) Q4 Oak Stream Quarterly Technical Review \(QTR\) Online Session - Reminder](#) on page 311

3.3.1 Documentation

3.3.1.1 (Updated WW48) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Additional collaterals are now tracked in the Appendix section.

Table 117. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.7	0.9	Q1'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	Q1'25
<i>Oak Stream 12-Channel AI Head Node Design Considerations Technical Article</i>	823064	N/A	1.0	As needed
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW48'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW50'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.7	0.9	Q1'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	Q4'24

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

3.3.2 Enabling**3.3.2.1 (Updated WW48) Q4 Oak Stream Quarterly Technical Review (QTR) Online Session - Reminder****NOTE**

Updated information is highlighted in **bold text**.

Customer and Systems Engineering team are excited to invite you to the Oak Stream Quarterly Technical Review (QTR) online session. This session will allow customers to find the latest enabling updates for the next-generation Oak Stream platforms.

The online session will be hosted on **December 3rd, at 5:00-7:00PM Pacific Time (PDT)**.

The material presented will be recorded and made available afterward in the Resource and Documentation Center.

To join the online session:

- Join [Teams Live Event](#)

Save the date and Teams' link for the next week's online session. Topics include:

- **Oak Stream 8-Channel Design Enabling**
- **Oak Stream 16-channel Design Enabling**
- **HSIO Technology Enabling**

The material presented will be recorded and made available afterward in the Resource and Documentation Center (RDC).

3.4 WW47-2024

New:

- [Oak Stream Design Discovery Form \(DDF\) Release Notification](#) on page 312
- [Q4 Oak Stream Quarterly Technical Review \(QTR\) Online Session](#) on page 313

Updated:

- [\(Updated WW47\) Diamond Rapids Tools Readiness](#) on page 313

3.4.1 Documentation

3.4.1.1 Oak Stream Design Discovery Form (DDF) Release Notification

The *Oak Stream Design Discovery Form (DDF)*, document number 838537, revision 0.5, is now available in the Resource and Documentation Center.

DDF datasheet is intended to gauge customer physical design and analysis interests and capture product usage models, implementation details, and timelines.

Customers must submit their filled-out DDF datasheet as part of their Design Review IPS. Completing the DDF datasheet enables Intel to provide tailored support when needed.

Key changes in 0.5 revision include:

- Initial release of the document. Added Instructions Project, Si and Layout, Topologies, and Manageability tabs.

3.4.2 Enabling

3.4.2.1 Q4 Oak Stream Quarterly Technical Review (QTR) Online Session

Customer and Systems Engineering team are excited to invite you to the Oak Stream Quarterly Technical Review (QTR) online session. This session will allow customers to find the latest enabling updates for the next-generation Oak Stream platforms.

The online session will be hosted on **December 3rd, at 5:00-7:00PM Pacific Time (PDT)**.

The material presented will be recorded and made available afterward in the Resource and Documentation Center.

To join the online session:

- Join [Teams Live Event](#)

3.4.3 Tools and Debug

3.4.3.1 (Updated WW47) Diamond Rapids Tools Readiness

Updated information highlighted in **bold text**

Intel® JTAG Debugger / USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to ITP Adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach for support
 - Two Streams Tracing
 - LA-3907 preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports four trace ports)
 - LA-3892 converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- ASSET* Arium - contact ASSET Arium for support
- Compact Debug Port (CDP) + MIPI10 Adapter for I3C*
 - Type-C connection: voltage level shifter and existing Closed Chassis Adapter (CCA) are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- Vtune™
- Intel® SoC Watch
- Intel® System Debugger
 - Token generation for Vendor Authorize Boot (VAB) / Vendor Authorize Debug (VAD)
- Intelligent Debug and Validation Tool (IDV Tool) Application

Table 118. Oak Stream and Diamond Rapids Tools Readiness

Tool	Trend
ASSET Intertech* Arium	TBD
ASSET Close Chassis Controller (CCC)	TBD
<i>continued...</i>	

Tool	Trend
Autonomous Crashdump (ACD)	Alpha
CScripts	Q3'25
Field Programmable Gate Arrays (FPGA) Image for Compute Express Link* (CXL*) Test Card	Q3'25
Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C for CCA	Available
Intel® I3C USB Type-C Level Shift Adapter	Available
Intel® At-Scale Debug (Intel® ASD)	Alpha
W: Intel Channel Analyzer Tool (ICAT)	Q3'25
W: Intelligent Debug Validation (IDV MM#99A8C)	Q3'25
16-Ch Intel's PLB/ULB Test Cards	WW18'25
8-Ch: Intel's PLB/ULB Test Cards	Q2'25
W: UEFI/JTAG-Intel SelfTest	Q3'25
Intel® SVT (Silicon View Technology) Closed Chassis Adapter (CCA)	Available
Intel® System Bring-up Toolkit NDA	Q3'25
W: Intel XDP3 (MM#940990)	Available
W: Intel® IO Margin Tool and User Guide	Q3'25
L: Intel® IO Margin Tool and User Guide	Q3'25
Lauterbach*	Q3'25
Memory Assistant	Q3'25
W: Memory Error Injection (MEI) DDR5 Card (MM#99CDV1)	Available
W: Memory Error Injection (MEI) DDR5 Lite Card (8000 MT/s)	Q1'25
PCI Express* (PCIe*) Interface Hardware (MM#STLGRN99)	Available
W: PCIe* Interface Hardware Error Injection Tool Kit, document number 541780	Available
PCIe* Error Injection (PEI) 4.0 Card	Available
PCIe* PEI Collaterals, document number 610667	Available
PCIe* PEI 5.0 Card	Available
PCI Express* (PCIe*) PEI 6.0 Card	WW27'25
Intel® Power and Thermal Analysis Tool (Intel® PTAT) Note: Limited version will be available in ES1.	ES2
Stitched Rank Margining Tool (RMT) Available, Target Loader RMT (EFI)	Q3'25
Intel® Simics®	Q4'24
Intel® Structural Test Plus (Intel® STP)	WW32'24
Thermal Test Vehicle (TTV)	WW32'24
Server Security Toolkit	Q3'25
W: VRTT	WW35'24
16-Ch VRTT Interposer	WW30'24
8-Ch VRTT Interposer	WW09'25

NOTE

W: Windows*. L: Linux*. If not specified, tool is OS diagnostic.

NOTE

The Intel® Simics® release previously communicated "as available in Q3" is an early engineering stage. The general release is targeted for early Q4. Contact your Intel representative if you have further inquiries on Intel® Simics® availability.

3.5 WW46-2024

New:

- [Integrated Channel Analysis Tool \(ICAT\) Suite 6.200.50 Release Notification](#) on page 315
- [Oak Stream Platform BIOS Writer's Guide Release Notification](#) on page 316
- [Oak Stream RAS Deep Dive Technical Training \(DDTT\) Release Notification](#) on page 316
- [Oak Stream 16-Channel Platform Johnson City 1 Design Archive: Standalone Schematic and Layout Files Release Notification](#) on page 317

Updated:

- [\(Updated WW46\) Oak Stream Key Collaterals Availability and Forecast](#) on page 317

3.5.1 Tools and Debug

3.5.1.1 Integrated Channel Analysis Tool (ICAT) Suite 6.200.50 Release Notification

Integrated Channel Analysis Tool (ICAT) Suite, document number 782966, version 6.200.50 is now available in the Resource and Documentation Center.

Memory mode changes include:

- Added support for configuration files to include controller jitter in DOE analyses

SerDes mode changes include:

- Improved empirical jitter correlation between built-in device and IBIS AMI-based simulation modes.
- Improved transmitter equalization optimization in IBIS AMI, including crosstalk superposition in analytical-only IBIS AMI contexts.

Other changes include:

- IMADC now supports reporting of fitted insertion loss data at multiple Nyquist frequencies or channel modeling and analysis.

See the release notes in the ICAT distribution package for additional details.

3.5.2 Documentation

3.5.2.1 Oak Stream Platform BIOS Writer's Guide Release Notification

The *Oak Stream Platform BIOS Writer's Guide*, document number 820107, Revision 0.7 is now available in the Resource and Documentation Center.

Key changes in this revision include:

- Added new RAS chapter
- Merged PCI express, Compute express Link and Intel® Ultra Path Interconnect Version 3.0 chapters into a new I/O chapter
- Migrated Integrity and Data Encryption chapter content to security enabling collateral. Monitor the Message of the Week for future release notifications.
- Removed Intel® Total Memory Encryption – Multi-key (Intel® TME-MK) information (feature not supported)
- Updated S3M registers details in the Security Considerations for BIOS section.

See the document's revision history for complete list of changes.

3.5.2.2 Oak Stream RAS Deep Dive Technical Training (DDTT) Release Notification

The *Oak Stream Platform RAS Deep Dive Technical Training*, document number 824347, Revision 0.7 is now available in the Resource and Documentation Center.

Key changes in this revision include:

- Added MCA error signaling flows
- Added hint registers populated per die per error source
- Added CBB RAS content
- Added RREL error examples for x8 DRAM DIMMs
- Added memory error logs in HAMVF
- Added DDR5 transaction correction flow
- Added ECC error classifier content
- Added ADDDC content
- Added high-level PPR content
- Added IMH IO content
- Added RASIP content
- Added internal D2D content
- Added PCIe* and CXL* content
- Added CSMI threshold content
- Added high level patrol scrub content
- Added Per Rank Corrected ECC Error Counter and Memory Leaky Bucket
- Added Poison and Viral Enabling

3.5.2.3 Oak Stream 16-Channel Platform Johnson City 1 Design Archive: Standalone Schematic and Layout Files Release Notification

Different schematic and layout file formats for the Johnson City 1 reference platform are now available in the Resource and Documentation Center. Files are available as stand-alone documents:

- *Oak Stream 16 Ch (Johnson City 1) - Mentor Schematics*, Revision 0.7, document number 838787
- *Oak Stream 16 Ch (Johnson City 1) - Zuken Schematics*, Revision 0.7, document number 838789
- *Oak Stream 16 Ch (Johnson City 1) - OrCAD Schematics*, Revision 0.7, document number 838790
- *Oak Stream 16 Ch (Johnson City 1) - Zuken Layout*, Revision 0.7, document number 838791
- *Oak Stream 16 Ch (Johnson City 1) - ODB Layout*, Revision 0.7, document number 838792
- *Oak Stream 16 Ch (Johnson City 1) - Mentor Layout*, Revision 0.7, document number 838793

Stand-alone schematic and layout files for the Johnson City 2 reference platform are planned to be released in WW47'24.

3.5.2.4 (Updated WW46) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Table 119. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.7	0.9	Q1'25
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<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.7	0.9	Q1'25
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.5	As needed	As needed
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.5	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW48'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW50'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
Oak Stream Platform BIOS Writer's Guide	820107	0.7	0.9	Q1'25
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	Q4'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q1'25
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

3.6 WW45-2024

New:

- [Oak Stream Support for 2S 2X16 Topology](#) on page 318

Updated:

- [\(Updated WW45\) Oak Stream Key Collaterals Availability and Forecast](#) on page 320

3.6.1 Enabling

3.6.1.1 Oak Stream Support for 2S 2X16 Topology

Plan of Record (POR) Change

Accelerator server head nodes implementing 2 Socket (2S) Oak Stream topologies require more x16 Flexible I/O (FIO) ports than current plan of record due to Intel® Ultra Path Interconnect (Intel® UPI) 3.0 usage on four FIO ports between sockets.

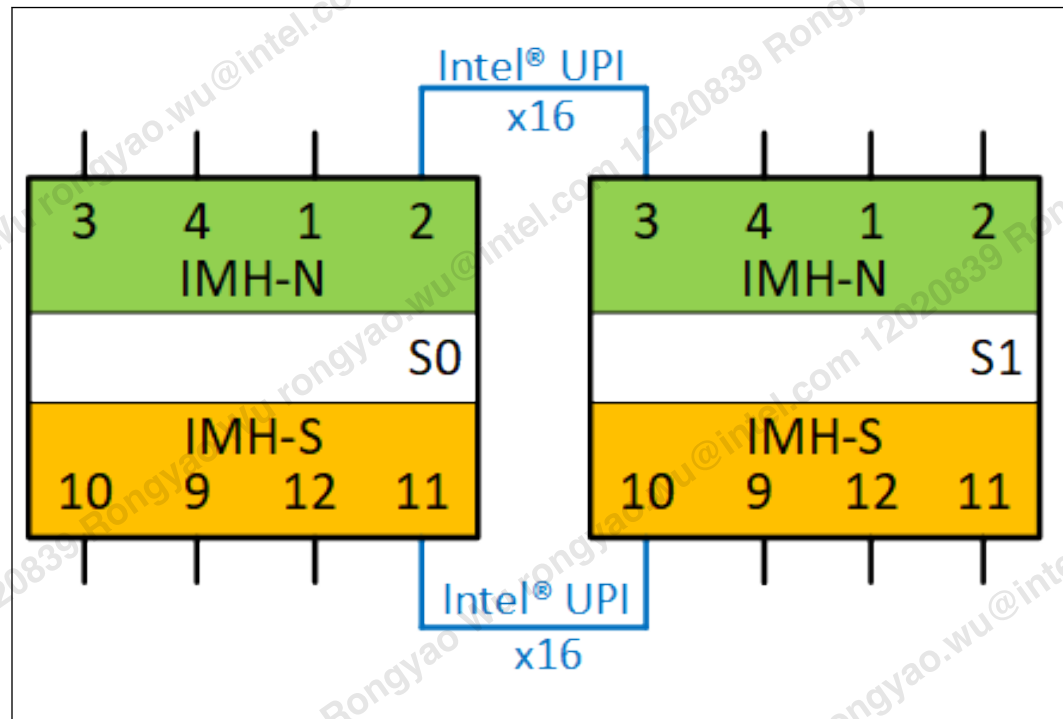
Intel is updating plan of record, adding a 2S 2X16 Intel UPI 3.0 (2S2L) topology as a solution. This topology increases available FIO ports for PCIe* 6.2 or CXL* 3.1 usage to 6 x16 for a total of 96 lanes; remaining 32 lanes are used for Intel UPI 3.0.

Customers interested in implementing this topology should consider targeted system workloads as lower socket-to-socket bandwidth may impact performance.

Connectivity requirements remain the same with one x16 FIO as Intel UPI 3.0 connecting IMH-N dies between sockets and one x16 FIO as Intel UPI 3.0 connecting IMH-S. FIO ports used as Intel UPI 3.0 may be customized to design.

Next is an example 2S system with FIO ports 2 and 11 on Socket 0 connected to FIO ports 3 and 10 respectively.

Figure 60. 2S2L Configuration



Intel will validate the 2S 2X16 (2S2L) topologies shown in the following table. For illustration purposes, sockets are detailed as socket 0 and socket 1. Designs may reverse the socket IDs of these CPUs; the port mappings will be considered as validated by Intel.

Table 120. 2S2L Configurations Validated by Intel

Configuration	Link Details	Source	Destination
#1	1 x16 North	S0 FIO Port 2	S1 FIO Port 3
#1	1 x16 South	S0 FIO Port 11	S1 FIO Port 10
continued...			

Configuration	Link Details	Source	Destination
#2	1 x16 North	S0 FIO Port 1	S1 FIO Port 1
#2	1 x16 South	S0 FIO Port 9	S1 FIO Port 9
#3	1 x16 North	S0 FIO Port 2	S1 FIO Port 3
#3	1 x16 South	S0 FIO Port 10	S1 FIO Port 11
#4	1 x16 North	S0 FIO Port 1	S1 FIO Port 3
#4	1 x16 South	S0 FIO Port 11	S1 FIO Port 9
#5	1 x16 North	S0 FIO Port 1	S1 FIO Port 1
#5	1 x16 South	S0 FIO Port 11	S1 FIO Port 11

The next revision of the *Oak Stream 16-Channel Platform Design Guide*, document number **788196**, will be updated to indicate 2S2L support and validation.

3.6.2 Documentation

3.6.2.1 (Updated WW45) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Table 121. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.7	0.9	Q1'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	Q1'25
<i>Oak Stream 12-Channel AI Head Node Design Considerations</i>	823064	N/A	1.0	WW48'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.5	As needed	As needed
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.5	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW48'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW50'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.5	0.7	WW46'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	Q4'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q1'25
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

3.7 WW43-2024

New:

- [Johnson City Design Archive \(DA\) Release Notification](#) on page 321

Updated:

- [\(Updated WW43\) Oak Stream Key Collaterals Availability and Forecast](#) on page 322

3.7.1 Documentation

3.7.1.1 Johnson City Design Archive (DA) Release Notification

The *Oak Stream 16-Channel Platform Johnson City 1 Design Archive*, document number [823065](#), and the *Oak Stream 16-Channel Platform Johnson City 2 Design Archive*, document number [636258](#), revision 0.7, are now available in Resource and Documentation Center.

All Design Archives (DA) include:

- Bill of Materials (BOM) spreadsheet

- Schematic (SCH) PDF file
- Schematic Database folder
- Worklib and archive libraries
- Board files folder

Overview of Revision 0.7 changes:

- Aligned content to the latest reference platform (RP) definition
- Cleaned up comments on designer names, and undefined rules and values
- Removed legacy platform and interposer references
- Updated FIO P12 references

NOTE

In *CPU Terminations – Debug I3C/JTAG* diagram, resistor RB2L22 is a 1K PU for PMODE. The Platform Design Guide (PDG) correctly documents this resistor as unstuffed. Stuffing is a don't care as PMODE is driven push-pull.

3.7.1.2 (Updated WW43) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream Resource and Documentation Center Collection](#).

Table 122. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.7	0.9	Q1'25
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	Q1'25
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW47'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.7	0.9	Q1'25
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.5	As needed	As needed
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.5	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW48'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW50'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q1'25
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q1'25
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q1'25
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	Q4'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q1'25
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

3.8 WW42-2024

New:

- [Oak Stream 16-Channel Platform Design Guide Release Notification](#) on page 323

Updated:

- [\(Updated WW42\) Oak Stream Key Collaterals Availability and Forecast](#) on page 324

3.8.1 Documentation

3.8.1.1 Oak Stream 16-Channel Platform Design Guide Release Notification

The *Oak Stream 16-Channel Platform Design Guide*, document number 788196, revision 0.7, is now available in Resource and Documentation Center.

Overview of revision 0.7 changes:

- Updated chapters:
 - Platform Clocking

- Processor Essential IO Signals Guidelines
- DDR5 Memory Routing Guidelines
- I3C* Design Guidelines
- FIO Guidelines
- Intel UPI 3.0 Guidelines
- System Management
- Power Distribution Guidelines

See the document revision history for a complete list of changes.

3.8.1.2 (Updated WW42) Oak Stream Key Collaterals Availability and Forecast

NOTE

Updated information is highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream RDC Collection](#).

Table 123. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
Diamond Rapids Processor Architecture Specification	792359	0.7	0.9	As needed
Diamond Rapids Processor Registers Specification	793272	0.7	0.9	As needed
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW43'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
Oak Stream 16-Channel Platform Design Guide	788196	0.7	0.9	As needed
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
Oak Stream 16-Channel Platform Electrical Specification	787006	0.7	0.9	As needed
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.5	0.7	WW46'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW44'24
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.5	As needed	As needed
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.5	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW48'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW50'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	Q4'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q1'25
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

3.9 WW41-2024

New:

- [Diamond Rapids Processor Register Specification Release Notification](#) on page 325
- [Oak Stream 16-Channel Platform Electrical Specification Release Notification](#) on page 326

Updated:

- [\(Updated WW41\) Oak Stream Key Collaterals Availability and Forecast](#) on page 326

3.9.1 Documentation**3.9.1.1 Diamond Rapids Processor Register Specification Release Notification**

The *Diamond Rapids Processor Register Specification*, document number 793272, revision 0.7, is now available in Resource and Documentation Center.

Overview of revision 0.7 changes:

- Added new chapters:
 - Compute Express Link* Cache Memory Controller (CXL.CM) Registers
 - I/O Memory Management Unit (IOMMU) Registers
 - I3C* Registers
 - Intel® Trace Hub Registers

- Intel® Out-of-Band Management Services Module (Intel® OOBM Service Module) Registers
- Reliability Availability, and Serviceability IP (RASIP) Registers
- Memory Security Engine (MSE) Registers
- Non-CoherentUnit (NCU) Registers
- Power Control Unit (Punit) Registers
- Topology Aware and Power Management Interface (TPMI) Registers
- Added new subchapters:
 - PCIe 6.0 Integrity and Common Encryption Engine (CEE) Memory Space Registers
 - Universal Bridge (UBR) Memory Space Registers
 - S3M ACPI Memory Space Registers
 - PCIe* EXPPTMBARRoot Port Memory Space Registers
 - PCIe*RCRBBARRoot Port Memory Space Registers
- Updated register access information for all chapters
- Added register scope and Machine Check to Mode Specific Registers (MSRs) chapter
- Updated register definitions in PCIe* 6 Module, Secure Startup Module (S3M), Memory Controller (MC), Configuration Agent (UBOX), and MSRs Chapters

3.9.1.2 Oak Stream 16-Channel Platform Electrical Specification Release Notification

The *Oak Stream 16-Channel Platform Electrical Specification*, document number 787006, revision 0.7, is now available in Resource and Documentation Center.

Overview of revision 0.7 changes:

- Updated POC Straps information
- Updated the chapters:
 - Clocking
 - Essential IO Signals
 - DDR5
 - I3C* Registers
 - Debug
 - Voltage Regulation

See the revision history for a complete list of changes.

3.9.1.3 (Updated WW41) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream RDC Collection](#).

Table 124. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.7	0.9	Q4'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.7	0.9	Q4'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW43'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW42'24
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.5	0.7	WW46'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW44'24
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.5	As needed	As needed
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.5	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW48'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW50'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	Q4'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q1'25
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

3.10 WW40-2024

New:

- [Diamond Rapids Processor, Architecture Specification Release Notification](#) on page 328
- [Multiple Oak Stream VR Enabling Documents Release Notification](#) on page 328

Updated:

- [\(Updated WW40\) Oak Stream Key Collaterals Availability and Forecast](#) on page 329

3.10.1 Documentation

3.10.1.1 Diamond Rapids Processor, Architecture Specification Release Notification

The *Diamond Rapids Processor, Architecture Specification*, document number [792359](#), Revision 0.7 is now available in Resource and Documentation Center.

Overview of Revision 0.7 changes:

New Chapters:

- Intel® Out-of-Band Management Services Module (Intel® OOBM Services Module)
- Intel® System Debugger Functional Description
 - Added Intel® Trace Hub (Intel® TH) Subsection
- Power Control Unit Functional Description
 - Added Intel® Speed Select Technology - Base Frequency (Intel® SST-BF) subsection
 - Added Intel® Speed Select Technology - Turbo Frequency (Intel® SST-TF) subsection
- Utility Box (Ubox) Functional Description

Updated Chapters:

- Memory Subsystem
- Reliability, Availability, and Serviceability (RAS)
- Security

Refer to the revision history for the complete list of changes.

3.10.1.2 Multiple Oak Stream VR Enabling Documents Release Notification

The *Oak Stream 16-Ch / Oak Stream 12-Ch Platform DDR5 Voltage Regulator Test Tool (VRTT) Test Plan*, document number [832854](#), Revision 0.5 is now available in Resource and Documentation Center. This test plan is a guidance to customers to perform 12V power delivery test, which feeds into the DDR5 DIMM area.

The *Oak Stream 16-ch Platform Power Budget Calculator*, document number [833790](#), Revision 0.5 is now available in Resource and Documentation Center. This collateral is the system power budget calculator based on the Johnson City Reference Platform (RP), which customers can refer to for sizing their system power supplies.

The *Oak Stream 16-Ch Rpath Analysis Best-Known Method (BKM)*, document number [833706](#), Revision 0.5 is now available in Resource and Documentation Center. This BKM to show how to do Rpath analysis using the Cadence* PowerDC™ tool based on Johnson City RP design.

3.10.1.3 (Updated WW40) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream RDC Collection](#).

Table 125. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
Diamond Rapids Processor Architecture Specification	792359	0.7	0.9	Q4'24
Diamond Rapids Processor Registers Specification	793272	0.5	0.7	WW41'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW43'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW42'24
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.5	0.7	WW41'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.5	0.7	WW46'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW44'24
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.5	As needed	As needed
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.5	As needed	As needed
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW48'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW50'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	Q4'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q1'25
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

3.11 WW39-2024

Updated:

- (Updated WW39) Oak Stream Key Collaterals Availability and Forecast on page 330

3.11.1 (Updated WW39) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream RDC Collection](#).

Table 126. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	WW40'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.5	0.7	WW40'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW43'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW42'24
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.5	0.7	WW41'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.5	0.7	WW46'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW44'24
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings	786522	0.5	As needed	As needed
Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models	786523	0.5	As needed	As needed
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	N/A	0.5	WW48'24
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	N/A	0.5	WW50'24
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	0.5	0.7	Q4'24
Oak Stream 8-Channel Platform Basin City Pin List	TBD	N/A	0.3	Q4'24
Oak Stream 8-Channel Platform Basin City Design Archive	TBD	N/A	0.3	Q4'24
Oak Stream 8-Channel Platform Design Guide	813596	0.3	0.5	Q4'24
Oak Stream 8-Channel Platform Electrical Specification	794719	0.3	0.5	Q4'24
Oak Stream 8-Channel Platform Thermal Mechanical Specification	786524	0.3	0.5	Q4'24
Oak Stream Platform BIOS Writer's Guide	820107	0.5	0.7	WW44'24
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	0.7	0.9	Q4'24
Oak Stream Platforms DC-SCM 2.1 Integration Guide	812906	0.5	0.7	Q1'25
Oak Stream Platform Design In	816342	1.05	As needed	As needed

NOTE

The document has not been published yet if the latest revision column is marked as N/A.

3.12 WW37-2024

New:

- [BMC_INIT Support Drop on the Oak Stream Platform](#) on page 331

Updated:

- [\(Updated WW37\) Oak Stream Key Collaterals Availability and Forecast](#) on page 332

3.12.1 Enabling

3.12.1.1 BMC_INIT Support Drop on the Oak Stream Platform

Intel is phasing out BMC_INIT support on the Oak Stream and future platforms.

For details on BMC_INIT support drop, see [Customer Feedback Request on BMC_INIT](#) in the WW30-2024 article.

If this drop impacts your platform design, contact your Intel representative as soon as possible for assistance.

3.12.2 Documentation

3.12.2.1 (Updated WW37) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

All Oak Stream collaterals are available in the [Oak Stream RDC Collection](#).

NOTE

The 0.7 revisions of Johnson City 16-Channel Platforms and Marketing Development Platforms Design Archives are delayed to align with the latest revision of the Johnson City 16-Channel Platform Pin List.

Table 127. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	WW39'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.5	0.7	WW39'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW43'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.5	0.7	WW46'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW44'24
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	WW39'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	WW39'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW48'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW50'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.7	0.9	Q4'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q1'25
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

When Latest revision is marked as N/A it means that document has not been published yet.

3.13 WW36-2024

New:

- [Field Diagnostics Implementation Guide Release Notification](#) on page 333
- [Platform Design Guide Supplementary Articles Release Notification](#) on page 333

Updated:

- [\(Updated WW36\) Oak Stream Key Collaterals Availability and Forecast](#) on page 334

3.13.1 Documentation

3.13.1.1 Field Diagnostics Implementation Guide Release Notification

The *Field Diagnostics Implementation Guide With Guidelines for FRU Isolation on IERR*, document number 543790, Revision 5.0, is now available in RDC.

Overview of Revision 5.0 changes:

- Added chapter on Oak Stream Error Harvesting Strategy
- Added sections on MCERR strategy and impact of enabling viral mode
- Added and amended sections pertaining to previous generation platform debug including but not limited to S3M error handling

3.13.1.2 Platform Design Guide Supplementary Articles Release Notification

The *DFM Rules to Minimize Layer-to-Layer Misregistration for PCI Express* 6.0 Technical Article*, document number 831075, is now available in RDC. This document provides guidance on updated PCB spacing rules to maximize success for PCIe* 6.0 layout.

The technical article *Johnson City Copper Balance*, document number 830941, is now available in RDC. This document explains the purpose behind intentionally added copper voids and copper shapes in the Johnson City layout.

3.13.1.3 (Updated WW36) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 128. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	WW37'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.5	0.7	WW39'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW39'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.5	0.7	WW37'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.5	0.9	WW01'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW38'24
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	WW37'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	WW37'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW42'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW47'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	0.7	0.9	Q4'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q4'24
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

Latest revision read as N/A means that document has not been published yet.

3.14 WW35-2024

New:

- [Clarification on Targeted Degraded Topologies due to an Intel® Ultra Path Interconnect \(Intel® UPI\) Failure](#) on page 335
- [Signal Integrity Collateral List for the Oak Stream Platform Release Notification](#) on page 338

Updated:

- [\(Updated WW35\) Oak Stream Key Collaterals Availability and Forecast](#) on page 338

3.14.1 Enabling

3.14.1.1 Clarification on Targeted Degraded Topologies due to an Intel® Ultra Path Interconnect (Intel® UPI) Failure

This article describes the targeted Oak Stream platform degraded topologies due to an Intel® Ultra Path Interconnect (Intel® UPI) link failure. These topologies are driven from the baseline Intel® UPI connectivity requirements for Oak Stream:

- Same number of Intel® UPI links connected in each Integrated Memory and I/O (IMH) die for 16-channel platforms.
- Core Building Block (CBB) coherent traffic must not cross IMH dies.

These degraded topologies are under evaluation and are not currently Plan of Record (POR), meaning they are not current planned to be validated by Intel. Contact your Intel representative if you have opens regarding these targeted degraded topologies.

16-Channel (16-Ch) 2-Socket (2S) Topologies

All topologies require a minimum of 1x16 Intel® UPI links per IMH die to connect between sockets. A degraded topology will result independently of the link failure coming from IMH-N or IMH-S.

Figure 61. 16-Ch 2S 4x16 - Degraded Topology

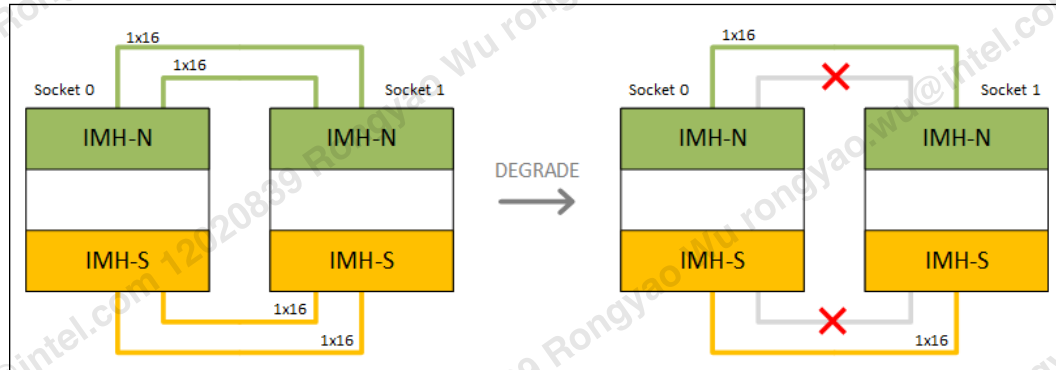
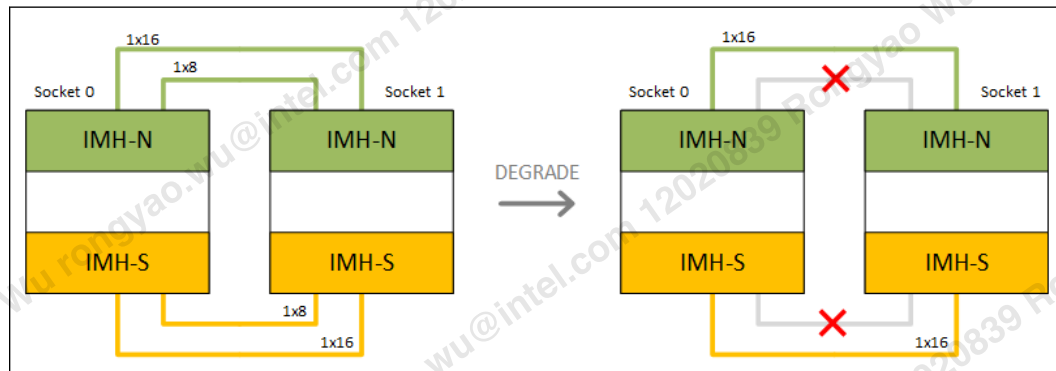
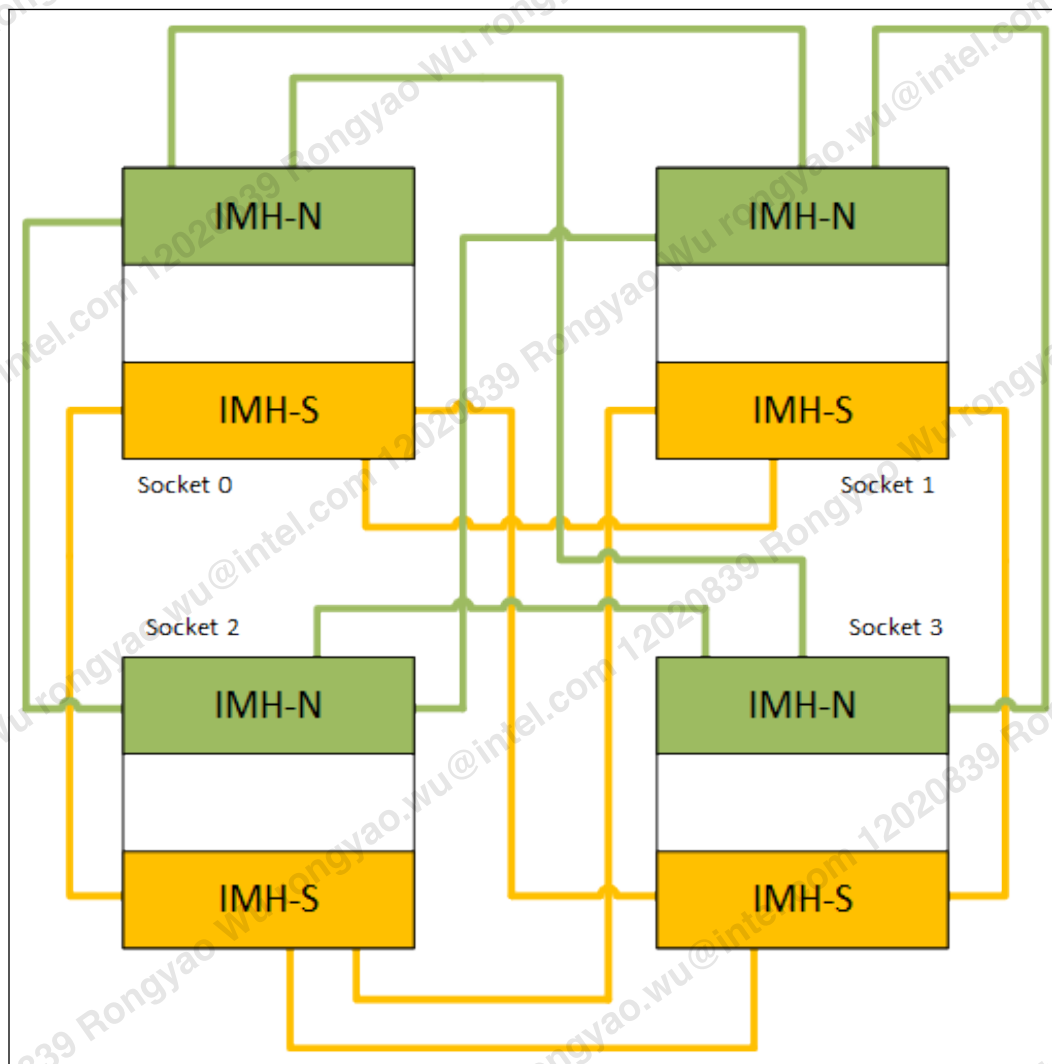


Figure 62. 16-Ch 2S 1x16+1x8 with x8 Link Failure - Degraded Topology



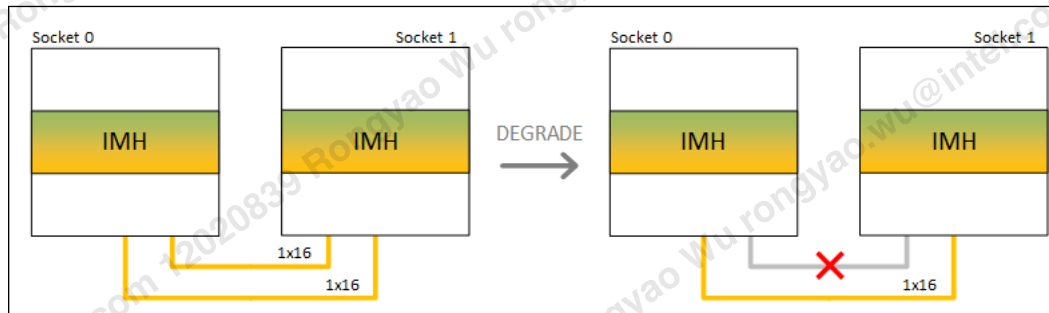
16-Channel (16-Ch) 4-Socket (4S) Topology

The targeted 4S topology is fully connected. This topology cannot be degraded as each IMH die has a single connection to another IMH die in another socket. Any link failure will create an isolated socket.

Figure 63. 16CH 4S Fully Connected - Not Degradable (No Boot)**8-Channel (8-Ch) 2-Socket (2S) Topology**

On 8-Ch, there is a single IMH die. Intel® UPI links are routed from the south ports. Degraded topology requires a minimum of 1x16 Intel® UPI links to connect between sockets.

Figure 64. 8-Ch 2S 2x16 - Degraded Topology



3.14.2 Documentation

3.14.2.1 Signal Integrity Collateral List for the Oak Stream Platform Release Notification

The *Signal Integrity Collateral List for the Oak Stream Platform*, document number 822391, revision 2024WW34 is now available in RDC.

Overview of 2024WW34 changes:

- Initial revision

This document provides a quick reference to Signal Integrity (SI) models, simulation, modeling software tools, and methodology training available on RDC for the Oak Stream platform. The signal integrity collateral list will be updated as more collaterals are released to RDC.

3.14.2.2 (Updated WW35) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 129. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	WW37'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.5	0.7	WW39'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW39'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.5	0.7	WW37'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.5	0.9	WW01'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW38'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	WW37'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	WW37'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW42'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW47'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.5	0.7	WW37'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q4'24
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

Latest revision read as N/A means that document has not been published yet.

3.15 WW34-2024

New:

- [Clarification on Cross RAS Full Memory Mirroring and Security Features](#) on page 339

Updated:

- [\(Updated WW34\) Oak Stream Key Collaterals Availability and Forecast](#) on page 340

3.15.1 Enabling

3.15.1.1 Clarification on Cross RAS Full Memory Mirroring and Security Features

Oak Stream Platform does not support cross functionality of Full Memory mirroring and security features:

- Intel® Software Guard Extensions (Intel® SGX)
- Intel® Total Memory Encryption – Multi-Key (Intel® TME-MK)
- Intel® Trust Domain Extensions (Intel® TDX)

Above security features can be enabled in address range mirroring mode of operation, but functionality will not be applicable to the mirrored regions.

OEMs may have enabled security features over mirrored regions in their lab environment and observed no functional issues, however, this cross functionality is not Plan of Record (POR).

Oak Stream collaterals will be updated on revision 0.7 to reflect no support of Cross RAS Memory Mirroring and these security features.

3.15.2 Documentation

3.15.2.1 (Updated WW34) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 130. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	WW36'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.5	0.7	WW39'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW39'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.5	0.7	WW37'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.5	0.9	WW01'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW38'24
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	WW35'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	WW35'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.5	0.7	Q4'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW42'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW47'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platform BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.5	0.7	WW35'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q4'24
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

Latest revision read as N/A means that document has not been published yet.

3.16 WW33-2024

New:

- [Oak Stream 16-Channel SPI, eSPI, and SVID Topologies Update](#) on page 341

Updated:

- [\(Updated WW33\) Oak Stream Key Collaterals Availability and Forecast](#) on page 342

3.16.1 Enabling**3.16.1.1 Oak Stream 16-Channel SPI, eSPI, and SVID Topologies Update**

Latest signal integrity assessments on the Oak Stream reference platform have resulted in the following bus operating limitations:

- Serial Peripheral Interface (SPI) Guidelines for 2-device daisy chain and star topologies will now operate at up to 40 MHz (previously 50 MHz)
- Enhanced Serial Peripheral Interface (eSPI) routing guidelines for DC-SCM 2-device topology will now operate at up to 33 MHz (previously 50 MHz)

Intel continues to investigate these assessments. There may be updates to these operating frequencies in a future revision of the *Oak Stream 16-Channel Platform Design Guide*, document number 788196.

Serial Voltage Identification Code (SVID) Topologies

Each socket has two SVID buses and Johnson City 2 (JC2) reference platform currently has up to three electrical Voltage Regulator (VR) loads on each SVID bus:

- Simulations have shown that up to four VR controller loads can be supported on each SVID bus, provided that the total bus length is 18" or less. The four VR load topology and guidelines will be updated in PDG revision 0.7.
- The five VR loads topology currently documented in the PDG revision 0.5 will be removed in revision 0.7.

3.16.2 Documentation

3.16.2.1 (Updated WW33) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 131. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	WW36'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.5	0.7	WW39'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW39'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)</i>	828193	1.0	As needed	As needed
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.5	0.7	WW37'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	0.5	0.9	WW01'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW38'24
<i>Oak Stream 16-Channel Platform Johnson City 2 120mil, 22-layer to 104mil, 22-layer Guidance Technical Paper</i> <i>Note: This collateral is deprecated. Information relevant to this technical paper is now covered in Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4), document number 828193.</i>	N/A	N/A	N/A	N/A
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.7	0.9	Q4'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	WW34'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	WW34'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.3	0.5	WW34'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW42'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW47'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.5	0.7	WW35'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	Q4'24
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

Latest revision read as N/A means that document has not been published yet.

3.17 WW31-2024

New:

- [Design Archives \(DA\) Deliverables and Cadence* Allegro Support Clarification](#) on page 343

Updated:

- [\(Updated WW31\) Oak Stream Key Collaterals Availability and Forecast](#) on page 344

3.17.1 Documentation

3.17.1.1 Design Archives (DA) Deliverables and Cadence* Allegro Support Clarification

Deliverables included in Design Archive (DA)

All Oak Stream DA available in RDC contain the following deliverables:

- Bill of Materials (BOM) spreadsheet
- Schematic (SCH) PDF file
- Schematic Database Folder
- — .lib file
- — .dat file
- — .cpm file
- — ptf files
- Worklib libraries
- Archive libraries
- Board files Folder

- — .brd files

NOTE

All DA collaterals must be paired with the corresponding Platform Pin List.

Cadence* Allegro Version Support Clarification

Intel supports Cadence* Allegro versions 22.1 and 23.1. Intel intends to provide .brd files for both versions in all their DA.

NOTE

22.1 .brd files are not directly compatible with a 23.1 license.

For customers interested in [Copper Max \(CuMax\) support](#), CuMax is supported only in Cadence* Allegro version 23.1 or later.

Table 132. Oak Stream Design Archives (DA)

Design Archive (DA)	Document Number	Current Version	Cadence* Version in DA
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	0.7	23.1 <i>Note:</i> 22.1 files will be included in a future revision.
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	0.5	23.1 and 22.1
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258	0.5	23.1 and 22.1
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	N/A	23.1 and 22.1
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	N/A	23.1 and 22.1
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	0.5	23.1 and 22.1

Intel does not endorse or recommend specific software vendors or products.

3.17.1.2 (Updated WW31) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 133. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	WW34'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.5	0.7	WW39'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW35'24
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	0.7	As needed	As needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel Platform Design Guide 104mil, 22-layer Guidance Technical Paper</i>	823062	N/A	1.0	WW39'24
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.5	0.7	WW37'24
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	0.5	0.9	WW01'25
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW38'24
<i>Oak Stream 16-Channel Platform Johnson City 2 120mil, 22-layer to 104mil, 22-layer Guidance Technical Paper</i>	823063	N/A	1.0	WW39'24
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.5	0.6	WW32'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	WW33'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	WW33'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.3	0.5	WW33'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive</i>	827452	N/A	0.5	WW39'24
<i>Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive</i>	828483	N/A	0.5	WW47'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.5	0.7	WW32'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.5	0.7	WW34'24
<i>Oak Stream Platform Design In</i>	816342	1.05	As needed	As needed

NOTE

Latest revision as N/A means document is not published yet.

3.18 WW30-2024

New:

- [Copper Max \(CuMax\) Support on the Oak Stream Platform](#) on page 346
- [PCI Express* \(PCIe*\) 6.0 Re-drivers Outside of Support Plan-of-Record \(POR\)](#) on page 346
- [BMC_INIT Feedback Request](#) on page 350

Updated:

- [\(Updated WW30\) Oak Stream Key Collaterals Availability and Forecast](#) on page 347
- [\(Updated WW30\) Diamond Rapids Tools Readiness](#) on page 348

3.18.1 Enabling

3.18.1.1 Copper Max (CuMax) Support on the Oak Stream Platform

As noted in the Data Center Design Conference (DCDC) presentation, *Oak Stream PCB Design Challenges and Industry Readiness*, document number [818071](#), Intel is enabling a new impedance-control technology called Copper Max (CuMax) for DDR routing on the Oak Stream platform.

To ensure use of the technology exclusively on Intel processor-based platforms, CuMax support in the Cadence* Allegro family of layout software requires installation of an additional free license. Customers interested in obtaining a license must send a request to the following e-mail address: cumax-info@intel.com

Contingent on Intel approval, Cadence* will be notified and will contact the requestor(s) directly with additional instructions.

Enabling of other layout software is under discussion. Customers should contact their Intel Field representatives if CuMax support is needed in other PCB layout software products.

Intel does not endorse or recommend specific software vendors or products.

3.18.1.2 PCI Express* (PCIe*) 6.0 Re-drivers Outside of Support Plan-of-Record (POR)

Intel will not provide PCI Express* (PCIe*) 6.0 topologies containing re-drivers as part of the Oak Stream Platform Design Guide (PDG) or reference platforms. In addition, Intel will not be providing enabling guidance to the industry for the development of PCIe* 6.0 re-driver components or solutions (note that Platform Development Kits will be available, though these will not feature re-driver-based topologies). Customers who design Oak Stream-based platforms using re-driver components on PCIe* 6.0 topologies must do so with their own resources. This includes simulating, validating, margining, and debugging.

Topologies that only support PCIe* 5.0 and earlier generations may continue to use the following design guidance related to re-drivers:

PCI Express 5.0/Compute Express Link 1.1 Linear Re-driver Characterization Methodology*, document number 627286

PCI Express* (PCIe*) 5.0 Re-driver Vendor Status Update, document number 627287

Be aware that this documentation is not being actively updated. Technical information from third parties may be outdated, and is subject to change without notice.

3.18.2 Documentation

3.18.2.1 (Updated WW30) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 134. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
Diamond Rapids Processor Architecture Specification	792359	0.5	0.7	WW34'24
Diamond Rapids Processor Registers Specification	793272	0.5	0.7	WW39'24
Oak Stream 12-Channel Platform Technical Paper	823064	N/A	1.0	WW35'24
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	0.5	0.7	WW31'24
Oak Stream 16-Channel Platform Design Guide	788196	0.5	0.7	WW39'24
Oak Stream 16-Channel Platform Design Guide 104mil, 22-layer Guidance Technical Paper	823062	N/A	1.0	WW39'24
Oak Stream 16-Channel Platform Electrical Specification	787006	0.5	0.7	WW37'24
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	N/A	0.5	WW30'24
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258	0.5	0.7	WW38'24
Oak Stream 16-Channel Platform Johnson City 2 120mil, 22-layer to 104mil, 22-layer Guidance Technical Paper	823063	N/A	1.0	WW39'24
Oak Stream 16-Channel Platform Pin List	636260	0.5	0.6	WW32'24
Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings	786522	0.3	0.5	WW31'24
Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models	786523	0.3	0.5	WW31'24
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	0.3	0.5	WW31'24
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452	N/A	0.5	WW39'24
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483	N/A	0.5	WW47'24
Oak Stream 8-Channel Platform Basin City Pin List	TBD	N/A	0.3	Q4'24
Oak Stream 8-Channel Platform Basin City Design Archive	TBD	N/A	0.3	Q4'24
Oak Stream 8-Channel Platform Design Guide	813596	0.3	0.5	Q4'24
Oak Stream 8-Channel Platform Electrical Specification	794719	0.3	0.5	Q4'24
Oak Stream 8-Channel Platform Thermal Mechanical Specification	786524	0.3	0.5	Q4'24
Oak Stream Platforms BIOS Writer's Guide	820107	0.5	0.7	WW44'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	0.5	0.7	WW32'24
Oak Stream Platforms DC-SCM 2.1 Integration Guide	812906	0.5	0.7	WW34'24
Oak Stream Platform Design In	816342	1.05	As needed	As needed

NOTE

Latest revision as N/A means document is not published yet.

3.18.3 Tools and Debug

3.18.3.1 (Updated WW30) Diamond Rapids Tools Readiness

Updated information highlighted in **bold text**

Intel® JTAG Debugger/USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to ITP Adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support
 - 2 Streams Tracing
 - LA-3907 Preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports 4 trace ports)
 - LA-3892 Converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- Asset Arium* - Contact Asset Arium* for support
- Compact Debug Port (CDP) + MIPI10 Adapter for I3C*
 - Type-C connection: Voltage Level Shifter and existing CCA are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- Vtune™
- Intel® SoC Watch
- Intel® System Debugger
 - Token Generation for Vendor Authorize Boot (VAB)/Vendor Authorize Debug (VAD)
- Intelligent Debug and Validation Tool (IDV Tool) Application

Table 135. Oak Stream and Diamond Rapids Tools Readiness

Tool	Trend
ASSET Intertech Arium*	TBD
ASSET* Close Chassis Controller (CCC)	TBD
continued...	

Tool	Trend
Autonomous Crashdump (ACD)	Alpha
CScripts	Q3'25
FPGA Image for Compute Express Link* (CXL*) Test Card	Q3'25
Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C for CCA	Available
Intel® I3C USB Type-C Level Shift Adapter	Available
Intel® At-Scale Debug (Intel® ASD)	Alpha
W: Intel Channel Analyzer Tool (ICAT)	Q3'25
W: Intelligent Debug Validation (IDV MM#99A8C)	Q3'25
16-Ch Intel's PLB/ULB Test Cards	WW06'25
8-Ch: Intel's PLB/ULB Test Cards	Q2'25
W: UEFI/JTAG-Intel SelfTest	Q3'25
Intel® SVT (Silicon View Technology) Closed Chassis Adapter (CCA)	Available
Intel® System Bring-up Toolkit NDA	Q3'25
W: Intel XDP3 (MM#940990)	Available
W: Intel® IO Margin Tool and User Guide	Q3'25
L: Intel® IO Margin Tool and User Guide	Q3'25
Lauterbach*	Q3'25
Memory Assistant	Q3'25
W: Memory Error Injection (MEI) DDR5 Card (MM#99CDV1)	Available
W: Memory Error Injection (MEI) DDR5 Lite Card (8000 MT/s)	Q1'25
PCI Express* (PCIe*) Interface Hardware (MM#STLGRN99)	Available
W: PCIe* Interface Hardware Error Injection Tool Kit , document number 541780	Available
PCIe* Error Injection (PEI) 4.0 Card	Available
PCIe* PEI Collaterals , document number 610667	Available
PCIe* PEI 5.0 Card	Available
PCIe* PEI 6.0 Card	WW27'25
Intel® Power and Thermal Analysis Tool (Intel® PTAT) <i>Note:</i> Limited version will be available in ES1.	ES2
Stitched RMT Available, Target Loader RMT (EFI)	Q3'25
Intel® Simics®	Q4'24
Intel® Structural Test Plus (Intel® STP)	WW32'24
Thermal Test Vehicle (TTV)	WW32'24
Server Security Toolkit	Q3'25
W:VRTT	WW35'24
16-Ch VRTT Interposer	WW30'24
8-Ch VRTT Interposer	WW09'25

NOTE

W: Windows*. L: Linux*. If not specified, tool is OS agnostic.

NOTE

The Intel® Simics® release previously communicated "as available in Q3" is an early engineering stage. The general release is targeted for early Q4. Contact your Intel representative if you have further inquiries on Intel® Simics® availability.

3.18.4 Customer Feedback

3.18.4.1 BMC_INIT Feedback Request

As part of ongoing efforts to enhance products and services, Intel is seeking valuable feedback from customers regarding the potential implementation of the BMC_INIT soft strap on the Oak Stream Platform.

The BMC_INIT soft strap enables the System Service Processor (SSP) boot mode. This mode involves the SSP (BMC) configuring boot parameters such as the path to flash, UPI link initialization, and others via I3C_MNG.

Prior legacy usages are deprecated, and new capabilities are addressed without using SSP boot mode.

Intel is interested in understanding your plans and considerations regarding this feature. Your insights are invaluable and will help shape future offerings. Respond to the following questions:

1. Do you plan to implement the BMC_INIT soft strap on Oak Stream based designs?
 - Yes/No
2. If the answer is yes, briefly describe the implementation plan.

Your feedback will be kept confidential and will only be used to improve our products and services. **Provide your responses by the end of WW 31 (August 2)** to your designated CEA representative.

3.19 WW28-2024

New:

- [Oak Stream Platform BIOS Writer's Guide Release Notification](#) on page 351

Updated:

- [\(Updated WW28\) Oak Stream Key Collaterals Availability and Forecast](#) on page 351

3.19.1 Documentation

3.19.1.1 Oak Stream Platform BIOS Writer's Guide Release Notification

The *Oak Stream Platform BIOS Writer's Guide*, document number 820107, Revision 0.5, is now available in RDC.

Overview of Revision 0.5 changes:

- Initial Revision
- New chapters:
 - Platform Overview
 - Integrated Boot
 - Processor Initialization and Configuration
 - PCI Express* (PCIe*)
 - Compute Express Link* (CXL*) 3.0
 - Intel® Ultra Path Interconnect Version 3.0 (Intel® UPI 3.0)
 - Memory Subsystem
 - Security Considerations for BIOS
 - Integrity and Data Encryption

3.19.1.2 (Updated WW28) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 136. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	WW33'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.5	0.7	WW36'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW35'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.5	0.7	WW31'24
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel Platform Design Guide 104mil, 22-layer Guidance Technical Paper</i>	823062	N/A	1.0	WW32'24
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.4	0.5	WW31'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	N/A	0.5	WW30'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	WW30'24
<i>Oak Stream 16-Channel Platform Johnson City 2 120mil, 22-layer to 104mil, 22-layer Guidance Technical Paper</i>	823063	N/A	1.0	WW30'24
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.5	0.6	WW30'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	WW29'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	WW29'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.3	0.5	WW29'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	0.5	0.7	WW44'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.5	0.7	WW32'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.3	0.5	Q4'24
<i>Oak Stream Platform Design In</i>	816342	1.04	1.05	WW29'24

3.20 WW26-2024

Updated:

- (Updated WW26) Oak Stream Key Collaterals Availability and Forecast on page 352

3.20.1 Documentation

3.20.1.1 (Updated WW26) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 137. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	WW33'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.5	0.7	WW36'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW35'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.5	0.7	WW31'24
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel Platform Design Guide ULL2 Guidance Technical Paper</i>	823062	N/A	1.0	WW30'24
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.4	0.5	WW31'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	N/A	1.0	WW28'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	Q3'24
<i>Oak Stream 16-Channel Platform Johnson City 2 ULL1+ to ULL2 Guidance Technical Paper</i>	823063	N/A	1.0	WW30'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.5	0.6	WW30'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	WW29'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	WW29'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.3	0.5	WW29'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	N/A	0.0	WW27'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.5	0.7	Q3'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.3	0.5	Q4'24
<i>Oak Stream Platform Design In</i>	816342	1.04	As Needed	As Needed

3.21 WW24-2024

New:

- [Diamond Rapids Processor Registers Specification Release Notification](#) on page 353
- [Oak Stream RAS Deep Dive Technical Training Release Notification](#) on page 354

Updated:

- [\(Updated WW24\) Oak Stream Key Collaterals Availability and Forecast](#) on page 354
- [\(Updated WW24\) Oak Stream and Diamond Rapids Tools Readiness](#) on page 355

3.21.1 Documentation

3.21.1.1 Diamond Rapids Processor Registers Specification Release Notification

The *Diamond Rapids Processor Registers Specification*, document number [793272](#), Revision 0.5 is now available in RDC.

Overview of Revision 0.5 changes:

- Added new Chapters:
 - Universal Bridge (UBR)
 - Host I/O Processor (HIOP)
 - PCI Express* (PCIe*) and Compute Express Link (CXL.io)

- DDR Physical Layer (DDRIO)
- Model Specific Registers (MSR)
- Secure Startup Services Module (S3M)
- Updated Accelerator Chapter with Host Accelerator Port (HAP), IOSF2SFI, and PCIe* 4.0 Registers Sections.
- Updated Register Overview Chapter with PCI Configuration Space Device ID (DID) Information
- Added initial Register Access Information in each chapter introduction.

3.21.1.2 Oak Stream RAS Deep Dive Technical Training Release Notification

The *Oak Stream Platform, RAS Deep Dive Technical Training*, document number [824347](#), Revision 0.5 is now available in RDC.

Overview of Revision 0.5 changes:

- Initial release. Added content for:
 - Oak Stream RAS Features
 - RAS Functional and Signaling Block Diagrams.
 - Machine Check Architecture
 - Memory Subsystem
 - I/O Subsystem
 - RAS Offload

3.21.1.3 (Updated WW24) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 138. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	WW33'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.5	0.7	WW36'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW35'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.5	As Needed	As Needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel Platform Design Guide ULL2 Guidance Technical Paper</i>	823062	N/A	1.0	WW30'24
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.4	0.5	WW31'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	N/A	1.0	WW28'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	Q3'24
<i>Oak Stream 16-Channel Platform Johnson City 2 ULL1+ to ULL2 Guidance Technical Paper</i>	823063	N/A	1.0	WW30'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.5	0.6	WW29'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	Q3'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	Q3'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.3	0.5	WW25'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	N/A	0.0	WW27'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.5	0.7	WW27'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.3	0.5	Q4'24
<i>Oak Stream Platform Design In</i>	816342	1.04	As Needed	As Needed

3.21.2 Tools and Debug

3.21.2.1 (Updated WW24) Oak Stream and Diamond Rapids Tools Readiness

Updated information highlighted in **bold text**

JTAG/USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to ITP Adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support
 - 2 Streams Tracing
 - LA-3907 Preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports 4 trace ports)
 - LA-3892 Converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- Asset Arium* - Contact Asset Arium* for support
- Compact Debug Port (CDP) + MIPI10 Adapter for I3C*
 - Type-C connection: Voltage Level Shifter and existing CCA are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- Vtune™
- Intel® SoC Watch
- Intel® System Debugger

- Token Generation for Vendor Authorize Boot (VAB)/Vendor Authorize Debug (VAD)
- Intelligent Debug and Validation Tool (IDV Tool) Application

Table 139. Oak Stream and Diamond Rapids Tools Readiness

Tool	Trend
ASSET Intertech Arium*	TBD
ASSET* Close Chassis Controller (CCC)	TBD
Autonomous Crashdump (ACD)	Alpha
CScripts	Q3'25
FPGA Image for Compute Express Link* (CXL*) Test Card	Q3'25
Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C for CCA	Available
Intel® I3C USB Type-C Level Shift Adapter	Available
Intel® At-Scale Debug (Intel® ASD)	Alpha
W: Intel Channel Analyzer Tool (ICAT)	Q3'25
W: Intelligent Debug Validation (IDV MM#99A8C)	Q3'25
16-Ch Intel's PLB/ULB Test Cards	WW06'25
8-Ch: Intel's PLB/ULB Test Cards	Q2'25
W: UEFI/JTAG-Intel SelfTest	Q3'25
Intel® SVT (Silicon View Technology) Closed Chassis Adapter (CCA)	Available
Intel® System Bring-up Toolkit NDA	Q3'25
W: Intel XDP3 (MM#940990)	Available
W: Intel® IO Margin Tool and User Guide	Q3'25
L: Intel® IO Margin Tool and User Guide	Q3'25
Lauterbach*	Q3'25
Memory Assistant	Q3'25
W: Memory Error Injection (MEI) DDR5 Card (MM#99CDV1)	Available
W: Memory Error Injection (MEI) DDR5 Lite Card (8000 MT/s)	Q1'25
PCI Express* (PCIe*) Interface Hardware (MM#STLGRN99)	Available
W: PCIe* Interface Hardware Error Injection Tool Kit, document number 541780	Available
PCIe* Error Injection (PEI) 4.0 Card	Available
PCIe* PEI Collaterals, document number 610667	Available
PCIe* PEI 5.0 Card	Available
PCIe* PEI 6.0 Card	WW27'25
Intel® Power and Thermal Analysis Tool (Intel® PTAT) Note: Limited version will be available in ES1.	ES2
Stitched RMT Available, Target Loader RMT (EFI)	Q3'25
Intel® Simics®	WW28'24
continued...	

Tool	Trend
Intel® Structural Test Plus (Intel® STP)	WW32'24
Thermal Test Vehicle (TTV)	WW27'24
Server Security Toolkit	Q3'25
W:VRTT	WW35'24
16-Ch VRTT Interposer	WW30'24
8-Ch VRTT Interposer	WW09'25

NOTE

W: Windows*. L: Linux*. If not specified, tool is OS agnostic.

3.22 WW23-2024

- New:
 - [Oak Stream 16-Channel Platform Design Guide Release Notification](#)
- Updated:
 - [\(Updated WW23\) Oak Stream Key Collaterals Availability and Forecast](#)

3.22.1 Documentation

3.22.1.1 Oak Stream 16-Channel Platform Design Guide Release Notification

The *Oak Stream 16-Channel Platform Design Guide*, document number [788196](#), Revision 0.5 is now available in RDC.

Overview of Revision 0.5 changes:

- Updated diagrams in Introduction Chapter.
- Updated chapters:
 - Introduction Updated Platform Clocking
 - Processor Essential IO Signals
 - I3C* Design
 - PCIe and CXL
 - Platform Debug
 - Power Distribution
- Added new Flexible I/O chapter.

Refer to revision history for complete list of changes.

3.22.1.2 (Updated WW23) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 140. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release ETA
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	Q3'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.3	0.5	WW24'24
<i>Oak Stream 12-Channel Platform Technical Paper</i>	823064	N/A	1.0	WW30'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	0.5	As Needed	As Needed
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.5	0.7	WW39'24
<i>Oak Stream 16-Channel Platform Design Guide ULL2 Guidance Technical Paper</i>	823062	N/A	1.0	WW30'24
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.4	0.5	WW31'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	N/A	1.0	WW28'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	Q3'24
<i>Oak Stream 16-Channel Platform Johnson City 2 ULL1+ to ULL2 Guidance Technical Paper</i>	823063	N/A	1.0	WW30'24
<i>Oak Stream 16-Channel Platform Pin List</i>	636260	0.5	0.6	Q3'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	Q3'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	Q3'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.3	0.5	WW25'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	N/A	0.0	WW27'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.5	0.7	WW27'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.3	0.5	Q4'24
<i>Oak Stream Platform Design In</i>	816342	1.04	As Needed	As Needed

3.23 WW21-2024

- Updated:
 - (Updated WW21) Data Center Design Conference 2024 on page 359
 - (Updated WW21) Oak Stream Key Collaterals Availability and Forecast on page 359

3.23.1 Enabling

3.23.1.1 (Updated WW21) Data Center Design Conference 2024

We are excited to welcome you to Intel's 2024 Data Center Design Conference (DCDC). This year's conference focuses on Sierra Forest, Granite Rapids, Birch Stream, and Oak Stream.

[View the complete course list and register here](#)

The live virtual event will allow customers and Intel employees to view all sessions as they are happening on a secure streaming platform. The virtual event begins Monday, June 3rd and concludes Friday, June 14th, 2024. All sessions will be recorded and available on-demand after the event. One to two sessions per track will be presented each day (all times are listed in Pacific time zones).

NOTE

The following section has been updated; please review it for the latest information.

DCDC Asia Event

Registration is now open for the Data Center Design Conference (DCDC) Asia Event (Mandarin only).

[Virtual-live event registration link and course list.](#)

Table 141. DCDC Asia Event Information

Geography	Taiwan	PRC	ASIA Virtual
Format	Face to Face	Face to Face	
Date	June 20, 2024	June 25, 2024	July 1 - 9, 2024
Language	Mandarin	Mandarin	Mandarin
Registration Period	Now to June 5, 2024	Now to June 7, 2024	Now to June 23, 2024
On-demand Videos	No recording at the event	No recording at the event	The event will be recorded and available later

3.23.2 Documentation

3.23.2.1 (Updated WW21) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 142. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release
Diamond Rapids Processor Architecture Specification	792359	0.5	0.7	Q3'24
Diamond Rapids Processor Registers Specification	793272	0.3	0.5	WW23'24
Oak Stream 12-Channel Platform Technical Paper	823064	N/A	1.0	WW30'24
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	0.5	As Needed	As Needed
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.3	0.5	WW23'24
<i>Oak Stream 16-Channel Platform Design Guide ULL2 Guidance Technical Paper</i>	823062	N/A	1.0	WW30'24
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.4	0.5	WW24'24
<i>Oak Stream 16-Channel Platform Johnson City 1 Design Archive</i>	823065	N/A	1.0	WW28'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.5	0.7	Q3'24
<i>Oak Stream 16-Channel Platform Johnson City 2 ULL1+ to ULL2 Guidance Technical Paper</i>	823063	N/A	1.0	WW30'24
<i>Oak Stream 16-Channel Platform Johnson City Pin List</i>	636260	0.5	0.6	Q3'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	Q3'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	Q3'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.3	0.5	WW25'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	N/A	0.0	WW27'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.5	0.7	WW27'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.3	0.5	Q4'24
<i>Oak Stream Platform Design In</i>	816342	1.03	As Needed	As Needed

3.24 WW20-2024

- New:
 - [Data Center Design Conference \(DCDC\) 2024](#) on page 361
- Updated:
 - [\(Updated WW20\) Oak Stream Key Collaterals Availability and Forecast](#) on page 361
 - [\(Updated WW20\) Diamond Rapids Tools Readiness](#) on page 362
 - [\(Updated WW20\) Reminder: Diamond Rapids Third-Party Device Feedback Request](#) on page 364

3.24.1 Enabling

3.24.1.1 Data Center Design Conference (DCDC) 2024

We are excited to welcome you to Intel's 2024 Data Center Design Conference (DCDC). This year's conference focuses on Sierra Forest, Granite Rapids, Birch Stream, and Oak Stream.

[View the complete course list and register here](#)

The live virtual event will allow customers and Intel employees to view all sessions as they are happening on a secure streaming platform. The virtual event begins Monday, June 3rd and concludes Friday, June 14th, 2024. All sessions will be recorded and available on-demand after the event. One to two sessions per track will be presented each day (all times are listed in Pacific time zones).

3.24.2 Documentation

3.24.2.1 (Updated WW20) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 143. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release
Diamond Rapids Processor Architecture Specification	792359	0.5	0.7	Q3'24
Diamond Rapids Processor Registers Specification	793272	0.3	0.5	WW21'24
Oak Stream 12-Channel Platform Technical Paper	823064	N/A	1.0	WW30'24
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599	0.5	As Needed	As Needed
Oak Stream 16-Channel Platform Design Guide	788196	0.3	0.5	WW21'24
Oak Stream 16-Channel Platform Design Guide ULL2 Guidance Technical Paper	823062	N/A	1.0	WW30'24
Oak Stream 16-Channel Platform Electrical Specification	787006	0.4	0.5	WW24'24
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065	N/A	1.0	WW28'24
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258	0.5	0.7	Q3'24
Oak Stream 16-Channel Platform Johnson City 2 ULL1+ to ULL2 Guidance Technical Paper	823063	N/A	1.0	WW30'24
Oak Stream 16-Channel Platform Johnson City Pin List	636260	0.5	0.6	Q3'24
Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings	786522	0.3	0.5	Q3'24
Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models	786523	0.3	0.5	Q3'24
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521	0.3	0.4	WW21'24
Oak Stream 8-Channel Platform Basin City Pin List	TBD	N/A	0.3	Q4'24
Oak Stream 8-Channel Platform Basin City Design Archive	TBD	N/A	0.3	Q4'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release
Oak Stream 8-Channel Platform Design Guide	813596	0.3	0.5	Q4'24
Oak Stream 8-Channel Platform Electrical Specification	794719	0.3	0.5	Q4'24
Oak Stream 8-Channel Platform Thermal Mechanical Specification	786524	0.3	0.5	Q4'24
Oak Stream Platforms BIOS Writer's Guide	820107	N/A	0.0	WW27'24
Oak Stream Platforms DC-SCM 2.1 Design Archive	636262	0.5	0.7	WW27'24
Oak Stream Platforms DC-SCM 2.1 Integration Guide	812906	0.3	0.5	Q4'24
Oak Stream Platform Design In	816342	1.03	As Needed	As Needed

3.24.3 Tools and Debug

3.24.3.1 (Updated WW20) Diamond Rapids Tools Readiness

Updated information highlighted in **bold text**

JTAG/USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to ITP Adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach* for support
 - 2 Streams Tracing
 - LA-3907 Preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports 4 trace ports)
 - LA-3892 Converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- Asset Arium* - **Contact Asset Arium* for support**
- Compact Debug Port (CDP) + MIPI10 Adapter for I3C*
 - Type-C connection: Voltage Level Shifter and existing CCA are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

- Vtune™
- Intel® SoC Watch
- Intel® System Debugger
 - Token Generation for Vendor Authorize Boot (VAB)/Vendor Authorize Debug (VAD)
- Intelligent Debug and Validation Tool (IDV Tool) Application

Table 144. Tools Readiness

Tool	Trend
ASSET Intertech Arium*	TBD
ASSET* Close Chassis Controller (CCC)	TBD
continued...	

Tool	Trend
Autonomous Crashdump (ACD)	Alpha
CScripts	Q3'25
FPGA Image for Compute Express Link* (CXL*) Test Card	Q3'25
Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C for CCA	Available
Intel® I3C USB Type-C Level Shift Adapter	Available
Intel® At-Scale Debug (Intel® ASD)	Alpha
W: Intel Channel Analyzer Tool (ICAT)	Q3'25
W: Intelligent Debug Validation (IDV MM#99A8C)	Q3'25
16-Ch Intel's PLB/ULB Test Cards	WW44'24
W: UEFI/JTAG-Intel SelfTest	Q3'25
Intel® SVT (Silicon View Technology) Closed Chassis Adapter	Available
Intel® System Bring-up Toolkit NDA	Q3'25
W: Intel XDP3 (MM#940990)	Available
W: Intel® IO Margin Tool and User Guide	Q3'25
L: Intel® IO Margin Tool and User Guide	Q3'25
Lauterbach*	Q3'25
Memory Assistant	Q3'25
W-Memory Error Injection (MEI) DDR5 Lite Card (8000 MT/s)	WW25'25
PCI Express* (PCIe*) Interface Hardware (MM#STLGRN99)	Available
W: PCIe* Interface Hardware Error Injection Tool Kit , document number 541780	Available
PCIe* Error Injection (PEI) 4.0 Card	Available
PCIe* PEI Collaterals , document number 610667	Available
PCIe* PEI 5.0 Card	Available
PCIe* PEI 6.0 Card	WW25'25
Intel® Power and Thermal Analysis Tool (Intel® PTAT) - Limited*	ES1
Stitched RMT Available, Target Loader RMT (EFI)	Q3'25
Intel® Simics®	WW20'24
Intel® Structural Test Plus (Intel® STP)	WW32'24
Thermal Test Vehicle (TTV)	WW27'24
Server Security Toolkit	Q3'25
W:VRTT	WW35'24
16-Ch VRTT Interposer	WW30'24
8-Ch VRTT Interposer	WW09'25

NOTE

W: Windows*. L: Linux*. If not specified, tool is OS agnostic.

3.24.4 Software and Firmware

3.24.5 Customer Feedback

3.24.5.1 (Updated WW20) Reminder: Diamond Rapids Third-Party Device Feedback Request

Updated information highlighted in **bold text**

Intel is requesting customer feedback on their planned third party device usage for Oak Stream.

This information will enable Intel to tune our internal validation coverage. Your feedback will remain confidential.

ATTENTION

Make sure to provide your feedback by **WW23** to your DCPAE representative.

1. List the vendor devices that are of primary interest to you from the following categories:
 - Accelerator – Compute (example: Intel® Flex 1100)
 - Accelerator – Network (example: Dayton Peak "Mt. Evans")
 - Fabric – InfiniBand*
 - Network Adapter
 - SAS-PCI Express* (PCIe*)
 - SSD: PCIe-EDSFF, PCIe-M.2
 - Fiber Channel
 - Other

Table 145. Third Party Device Feedback Example

Device Type	Device Name	PCIe	SKU
Accelerator - Compute	Intel® Flex 1100	4.0	-
Accelerator - Network	Dayton Peak	4.0	-
Other	-	-	-

2. Will you have any custom devices?
 - Yes
 - No
3. If so, describe the customizations (form factor, chipset, firmware, or driver):

Table 146. Custom Device Feedback Example

Device Description	PCIe Gen	Form Factor	Chipset	Firmware	Driver
Other	-	-	-	-	-

4. Are you planning any custom configurations?

- Yes
- No

5. If so, describe your custom configurations:

3.25 WW19-2024

- New:
 - [Seamless Firmware Update Architecture Overview Release Notification](#) on page 365
- Updated:
 - [\(Updated WW19\) Oak Stream Key Collaterals Availability and Forecast](#) on page 365

3.25.1 Enabling

3.25.2 Documentation

3.25.2.1 Seamless Firmware Update Architecture Overview Release Notification

Intel's *Seamless Firmware Updates for Oak Stream Architecture Overview*, document number 819852, revision 0.5, is now available in Resource & Documentation Center (RDC).

Overview of revision 0.5 changes:

- Initial release of this document.

3.25.2.2 (Updated WW19) Oak Stream Key Collaterals Availability and Forecast

Updated information highlighted in **bold text**.

Table 147. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	Q3'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.3	0.5	WW21'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.3	0.5	WW20'24
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.3	0.5	WW21'24
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.4	0.5	WW24'24
<i>Oak Stream 16-Channel Platform Johnson City Pin List</i>	636260	0.5	0.6	Q3'24
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522	0.3	0.5	Q3'24
continued...				

Document Title	Document Number	Latest Revision	Next Revision	Next Release
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523	0.3	0.5	Q3'24
<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.3	0.4	WW21'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Design Guide</i>	813596	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
<i>Oak Stream Platforms BIOS Writer's Guide</i>	820107	N/A	0.0	WW26'24
<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	0.5	0.7	WW27'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.3	0.5	Q4'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	N/A	0.3	WW20'24
<i>Oak Stream Platform Design In</i>	816342	1.03	As Needed	TBD

3.25.3 Software and Firmware

3.25.4 Customer Feedback

3.26 WW18-2024

- New:
 - Oak Stream Platform Snapshot Memory Section Clarification on page 366
 - Oak Stream Key Collaterals Availability and Forecast on page 367
 - Diamond Rapids Tools Readiness on page 368
 - Third Party Device Feedback Request on page 369

3.26.1 Enabling

3.26.2 Documentation

3.26.2.1 Oak Stream Platform Snapshot Memory Section Clarification

The Oak Stream Platform Snapshot April revision was released in WW16.

Contact your Intel representative to request a copy of the Oak Stream Platform Snapshot.

1-DIMM Support Clarification

The “Proposed Limitations on 1 DIMM Per Socket Config” section of the April Oak Stream Platform Snapshot is not up to date. See the *Oak Stream Platform Design In*, document number 816342, for information on 1-DIMM per Socket configuration support.

2SPC/1DPC MRDIMM Support Clarification

2SPC/1DPC MRDIMM configuration for 16-channel and 12-channel platforms is not supported.

See the Oak Stream Platform Snapshot and the latest *Oak Stream Platform Design In*, document number 816342, for the latest information on MRDIMM support.

3.26.2.2 Oak Stream Key Collaterals Availability and Forecast

Table 148. Oak Stream Collaterals

Document Title	Document Number	Latest Revision	Next Revision	Next Release
<i>Diamond Rapids Processor Architecture Specification</i>	792359	0.5	0.7	Q3'24
<i>Diamond Rapids Processor Registers Specification</i>	793272	0.3	0.5	WW18'24
<i>Oak Stream 16-Channel Platform Johnson City 2 Design Archive</i>	636258	0.3	0.5	WW19'24
<i>Oak Stream 16-Channel Platform Design Guide</i>	788196	0.3	0.5	WW20'24
<i>Oak Stream 16-Channel Platform Electrical Specification</i>	787006	0.4	0.5	WW24'24
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<i>Oak Stream 16-Channel Platform Thermal Mechanical Specification</i>	786521	0.3	0.4	WW18'24
<i>Oak Stream 8-Channel Platform Basin City Pin List</i>	TBD	N/A	0.3	Q4'24
<i>Oak Stream 8-Channel Platform Basin City Design Archive</i>	TBD	N/A	0.3	Q4'24
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<i>Oak Stream 8-Channel Platform Electrical Specification</i>	794719	0.3	0.5	Q4'24
<i>Oak Stream 8-Channel Platform Thermal Mechanical Specification</i>	786524	0.3	0.5	Q4'24
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<i>Oak Stream Platforms DC-SCM 2.1 Design Archive</i>	636262	N/A	0.3	WW18'24
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906	0.3	0.5	Q4'24
<i>Oak Stream 16-Channel Platform Aggregator Board Design Archive</i>	820599	N/A	0.3	WW19'24
<i>Oak Stream Platform Design In</i>	816342	1.03	As Needed	TBD

3.26.3 Tools and Debug

3.26.3.1 Diamond Rapids Tools Readiness

JTAG/USB Hardware Debugger Support:

- Intel® In-Target Probe (Intel® ITP) XDP-3BR includes MIPI60 to ITP Adapter
- Close Chassis Adapter (CCA)
- Lauterbach* - Contact Lauterbach for support
 - 2 Streams Tracing
 - LA-3907 Preprocessor for Intel® Processor Trace (Intel® PT) Autofocus MIPI60-P (supports 4 trace ports)
 - LA-3892 Converged MIPI60-Q/-P or MIPI60-Cv2/-P to MIPI60
- Asset Arium*
- Compact Debug Port (CDP) + MIPI10 Adapter for I3C*
 - Type-C connection: Voltage Level Shifter and existing CCA are available for order, compatible with existing and future products.

Intel® System Bring-up Toolkit NDA includes the following tools:

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- Intel® SoC Watch
- Intel® System Debugger
 - Token Generation for Vendor Authorize Boot (VAB)/Vendor Authorize Debug (VAD)
- Intelligent Debug and Validation Tool (IDV Tool) Application

Table 149. Tools Readiness

Tool	Trend
ASSET Intertech Arium*	TBD
ASSET* Close Chassis Controller (CCC)	TBD
N/A-Autonomous Crashdump (ACD)	Alpha
CScripts	Q3'25
FPGA Image for Compute Express Link* (CXL*) Test Card	Q3'25
Intel® 10-Pin Debug Level Shift Adapter for MIPI I3C for CCA	Available
Intel® I3C USB Type-C Level Shift Adapter	Available
Intel® At-Scale Debug (Intel® ASD)	Alpha
W-Intel Channel Analyzer Tool (ICAT)	Q3'25
W-Intelligent Debug Validation (IDV MM#99A8C)	Q3'25
16CH: Intel's PLB/ULB Test Cards	WW44'24
W- UEFI/JTAG-Intel SelfTest	Q3'25
Intel® SVT (Silicon View Technology) Closed Chassis Adapter	Available
continued...	

Tool	Trend
Intel® System Bring-up Toolkit NDA	Q3'25
W-Intel XDP3 (MM#940990)	Available
W-Intel® IO Margin Tool and User Guide	Q3'25
L-Intel® IO Margin Tool and User Guide	Q3'25
W/L-Lauterbach*	Q3'25
Memory Assistant	Q3'25
W-Memory Error Injection(MEI) DDR5 Card (MM#99CDV1)	Available
PCI Express* (PCIe*) Interface Hardware (MM#STLGRN99)	Available
W- PCIe* Interface Hardware Error Injection Tool Kit, document number 541780	Available
PCIe* Error Injection (PEI) 4.0 Card	Available
PCIe* PEI Collaterals, document number 610667	Available
PCIe* PEI 5.0 Card	Available
PCIe* PEI 6.0 Card	WW42'25
Intel® Power and Thermal Analysis Tool (Intel® PTAT)	ES2
Stitched RMT Available, Target Loader RMT (EFI)	Q3'25
Intel® Simics®	WW20'24
Intel® Structural Test Plus (Intel® STP)	WW32'24
Thermal Test Vehicle (TTV)	WW27'24
Server Security Toolkit	Q3'25
W-VRTT	WW35'24
16-Ch VRTT Interposer	WW30'24
8-Ch VRTT Interposer	WW09'25

3.26.4 Software and Firmware

3.26.5 Customer Feedback

3.26.5.1 Third Party Device Feedback Request

Intel is requesting customer feedback on their planned third party device usage for Oak Stream.

This information will enable Intel to tune our internal validation coverage. Your feedback will remain confidential.

- List the vendor devices that are of primary interest to you from the following categories:
 - Accelerator – Compute (example: Intel® Flex 1100)
 - Accelerator – Network (example: Dayton Peak "Mt. Evans")
 - Fabric – InfiniBand*

- Network Adapter
- SAS-PCI Express* (PCIe*)
- SSD: PCIe-EDSFF, PCIe-M.2
- Fiber Channel
- Other

Table 150. Third Party Device Feedback Example

Device Type	Device Name	PCIe	SKU
Accelerator - Compute	Intel® Flex 1100	4.0	-
Accelerator - Network	Dayton Peak	4.0	-
Other	-	-	-

2. Will you have any custom devices?

- Yes
- No

3. If so, describe the customizations (form factor, chipset, firmware, or driver):

Table 151. Custom Device Feedback Example

Device Description	PCIe Gen	Form Factor	Chipset	Firmware	Driver
Other	-	-	-	-	-

4. Are you planning any custom configurations?

- Yes
- No

5. If so, describe your custom configurations:

Provide your feedback by WW21 to your DCPAE representative.

Appendix A Document Revisions Standard

Standard Revision Releases

Intel DCPAE team is standardizing document revision definitions. These standard revisions connect the document revision to the Intel phase, hardware, and software components in which its information was sourced.

Intermediate Revision Releases

Intermediate revisions can occur anytime critical information needs to be shared before the next standard revision release. Intermediate revisions do not have a standard definition and specific details are outlined in the revision history.

First Release

Based on content scope, the first release of a document may intercept a different standard revision from other documents of the same collection.

Releases without Update

Documents may now release a standard revision numbering change without updating technical information. This communicates that the document has been reviewed for and meets the criteria required for a given Intel execution phase, but no updates were needed.

Table 152. Pre-Silicon and Pre-Power-On Standard Revision Definitions

Revision	Document Source Data	Customer Focus
0.3	Intel® architecture and design documents	Development and Planning
0.5	Intel® architecture and design documents	Design Start
0.7	Intel® architecture and design documents	Stack-Up, Layout, Modeling
0.9	Intel® architecture and design documents	Design Update, Board Simulation
1.0	Intel® architecture and design documents	Pre Power-On Design Complete

Table 153. Post-Silicon and Intel Power-On Standard Revision Definitions

Revision	Document Source Data	Customer Focus
1.1	Intel completion and exit of ES0 silicon and Power-On software phase. (PO-Exit)	Product development and validation with ES1 silicon and Alpha software (EVT-Entry)
1.2	Intel completion and exit of ES1 silicon and Alpha software phase. (EVT-Exit)	Product development and validation with ES2 silicon and Beta software (DVT-Entry)
continued...		



Oak Stream Platform—Document Revisions Standard

Revision	Document Source Data	Customer Focus
1.5	Intel completion and exit of ES2 silicon and Beta software phase. (DVT-Exit)	Product development and validation with Quality Sample (QS) silicon and Production Candidate (PC) software (PVT-Entry)
2.0	Intel completion and exit of QS silicon and Production Candidate (PC) software phase. (PVT-Exit)	Product launch readiness with Production Release Qualification (PRQ) silicon and Production Version (PV) software. (First Launch)
2.1	Intel completion and exit of QS silicon and PC software phase. (PVT-Exit)	Subsequent product launch readiness with PRQ silicon and PV software (Second Launch)

Appendix B Oak Stream Collaterals

NOTE

Updated information is highlighted in bold text.

All Oak Stream collaterals are browsable in the [Oak Stream Resource and Documentation Center Collection](#).

Refer to Resource and Documentation Center (RDC) for latest revision of each collateral.

Table 154. Oak Stream Collaterals

Document Title	Document Number
2024 Intel Security Technology Workshop: Building Secure AI Systems on Intel® Xeon®	842293
2024 Interconnect Industry Association Seminar (TWN)	819687
2024 Interconnect Industry Association (IIA) (PRC)	819402
2025 Intel PCIe* Ecosystem Symposium Technical Training Materials	863516
2025 Interconnect Industry Association (IIA) Seminar	864929
48V Common Footprint (CFP) Electronic Fuse (eFuse) Enabling Specification	839076
48V to Core Voltage Regulator Common Module (VRCM) Enabling Technical Article	866972
Almondsville Thermal Mechanical Test Vehicle (TMTV) Application Note	852568
Almondsville Thermal Mechanical Test Vehicle (TMTV) Package Thermal Models and User Guide	820835
CK440Q Clock Generator Design Specification	610707
CPU Attestation and Trusted Telemetry with Reference Platform Root of Trust in Oak Stream	873234
CScripts for Oak Stream Platforms	821387
Cadence SKILL Script for CuMax Delay Calculation	857251
Clock Architecture, Devices, and Tools for Next Generation Servers	816339
Clock Sub-System Components for Intel® Servers PRC Amendment	848688
Common Footprint eFuse Enabling Specification	786600
Compute Express Link* (CXL*) Supplemental Validation Guide – Oak Stream	869657
continued...	

Document Title	Document Number
Customer Visible Server Intel® Firmware Support Package (Intel® FSP) Changes	847361
DB2000QL and DB1206 Differential Buffer Design Specification	602475
DDR Analysis Script Collection for RSM Fitting and UPM Calculation	638972
DDR5 12V Transient Test Setup for One Slot Per Channel Narrow DIMM Pitch	872598
DDR5 VRTT Load Card High Current Modification	792605
DFM Rules to Minimize Layer-to-Layer Misregistration for PCI Express* 6.0 Technical Article	831075
Diamond Rapids Double Data Rate Memory Validation Guide	867684
Diamond Rapids High Density (DMR HD) Update	870906
Diamond Rapids PCI Express* (PCIe*) 6.0 IBIS-AMI Models	868826
Diamond Rapids PCI Express* (PCIe*) and Intel® Ultra Path Interconnect (Intel® UPI) Estimated Package Losses	861617
Diamond Rapids Processor PCI Express* (PCIe*) 6.0 Signal Integrity (SI) Analysis on Package	854155
Diamond Rapids Processor Architecture Specification	792359
Diamond Rapids Processor DDR5 Signal Integrity (SI) Example for Oak Stream 16-Channel Platform	854485
Diamond Rapids Processor Essential IO: Sideband Link (SB) 1.0 Signal Integrity (SI) Model User Guide for Oak Stream Platform	849025
Diamond Rapids Processor Out-Of-Band (OOB) Accessible Registers	864991
Diamond Rapids Processor PCI Express* (PCIe*) 6.0 Signal Integrity (SI) Retimer Simulation Decks for Oak Stream 16-Channel Platform	855289
Diamond Rapids Processor PCI Express* (PCIe*) 6.0 Signal Integrity (SI) Simulation Decks for Oak Stream 16-Channel Platform	822170
Diamond Rapids Processor Registers Specification	793272
Diamond Rapids Roadmap Update November 2024	837847
Diamond Rapids Server Platform BIOS Developer's Guide	839071
Diamond Rapids Update External Broad Customers Nov 2025	868471
Diamond Rapids-UCC Processor DDR5 Signal Integrity Models	830146
Diamond Rapids-UCC/UCC-M Processor DDR5 Signal Integrity Model User Guide	830147
Early Boot Error and CATERR Triage Methods	864433
Effective Intra-Pair Skew Evaluation Tool	602348
continued...	

Document Title	Document Number
Enabling Signal Integrity (SI) Solutions for Intel® Xeon® Platforms - Frequently Asked Questions	786827
Enhanced Serial Peripheral Interface (eSPI) Compatibility Specification Self-Boot Server Platforms	627433
Integrated Channel Analysis Tool (ICAT)	782966
Intel Data Center Platforms CNDA Roadmap	846883
Intel EDA Conversion Service Transition: Important Update & FAQ Guide	861035
Intel Features in OpenBMC* Deep Dive	851920
Intel SGX BIOS Info Tool for Windows	556025
Intel Server Customer Scripts (CScripts)	821204
Intel Sustainable Server Design Guide	788905
Intel's Seamless Firmware Updates for Oak Stream Architecture Overview	819852
Intel(R) PMT XML Files for Diamond Rapids on Oak Stream	865036
Intel® Automatic In-board Characterization (Intel® AIBC) Certification	820098
Intel® Automatic In-board Characterization (Intel® AIBC) Update for Oak Stream	843525
Intel® Confidential Computing BIOS Developer's Guide	823668
Intel® Modular Flash Image Tool (Intel® Modular FIT) User Guide for Oak Stream	834939
Intel® Platform Firmware Resilience (Intel® PFR) 5.0 Architecture and Hardware Design Guide	814910
Intel® Platform Firmware Resilience (Intel® PFR) Tool Guide	611974
Intel® SGX Functional Validation Tool for Windows	567350
Intel® System Debugger Overview	784535
Intel® System Stress & Memory Diagnostics - Linux	818881
Intel® Tab Modeler and Calculator (Intel® TMC)	560500
Intel® Tab Modeler and Calculator (Intel® TMC) User Guide	560501
Intel® Xeon® Processors for AI	870638
Intel® Xeon® Reference Platforms Baseboard Management Controller (BMC) ASPEED® AST2700	861949
Intel's Seamless Firmware Updates for Oak Stream – SMM Runtime and Driver Updates Software Architecture Specification	834824
Intel's Architectural Implementation of MIPI Debug for I3C*	813296
Intel's Seamless Firmware Update for Oak Stream - SMM Runtime Update – Driver Update	856484
continued...	

Document Title	Document Number
Intel's Seamless Firmware Updates for Oak Stream - Out-Of-Band (OOB) BIOS Update Architecture Specification	834621
Intel's Seamless Firmware Updates for Oak Stream - System Management Mode (SMM) Runtime Update - Telemetry	835455
Intel's Seamless Firmware Updates for Oak Stream - BIOS Fast Boot Optimization Guideline	834260
Intel's Seamless Firmware Updates for Oak Stream - BMC-BIOS Coordination for Host SPI Flash Access Architecture Specification	832804
Intel's Seamless Firmware Updates for Oak Stream - OOB System Management Mode (SMM) Runtime Update	835478
Intel's Seamless Firmware Updates for Oak Stream - Out-of-Band (OOB) Microcode Update Architecture Specification	837563
Intel's Seamless Firmware Updates for Oak Stream - System Management Mode (SMM) Runtime Update - Code Injection	835645
Johnson City Copper Balance	830941
Johnson City Theory of Operations (TOPs)	864699
LGA9324 Thermal Solution Stiffness Guideline Application Note	795863
Layout Violations Report - Instructions Tab	837861
Narrow DIMM Connector Rework Recommendations	871481
Narrow KOZ OKS-16CH VRTT Retention Kit Installation Instructions	860569
OKS-16ch VR Test Tool Retention Mechanism Assembly Instructions	835411
Oak Stream (Diamond Rapids) Intel® Ultra Path Interconnect (Intel® UPI) 3.0 Simulation Model User Guide and Models	829339
Oak Stream 12-Channel Platform AI Head Node Design Considerations Technical Article	823064
Oak Stream 16 Ch (Johnson City 2) - Mentor Schematics	838877
Oak Stream 16 Channel Platform CPLD Reference Code	860022
Oak Stream 16-Ch Rpath Analysis Best-Known Method (BKM)	833706
Oak Stream 16-Channel 2-Socket in 19-Inch Racks Design Guidance	843581
Oak Stream 16-Channel Alberta City Design Archive	870365
Oak Stream 16-Channel PCB Design Considerations for Thinner Board Alternative (104 Mil, 22 Layer, 3H4)	828193
Oak Stream 16-Channel Platform 2-Socket in 19-Inch Rack NVIDIA MGX* Design Guidance	860677
Oak Stream 16-Channel Platform 500W Cookbook	851059
Oak Stream 16-Channel Platform Aggregator Board Design Archive	820599
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Document Title	Document Number
<i>Oak Stream 16-Channel Platform Clocks SIMUG</i>	822175
<i>Oak Stream 16-Channel Platform Debug I/O (JTAG): Decks and Models</i>	846457
<i>Oak Stream 16-Channel Platform Debug I/O: Models</i>	822176
Oak Stream 16-Channel Platform Design Archive (Johnson City) - Mentor* Layout	821071
Oak Stream 16-Channel Platform Design Archive (Johnson City) - ODB* Layout	821070
Oak Stream 16-Channel Platform Design Archive (Johnson City) - Zuken* Layout	821072
<i>Oak Stream 16-Channel Platform Design Discovery Form (DDF)</i>	838537
Oak Stream 16-Channel Platform Design Guide	788196
Oak Stream 16-Channel Platform Electrical Specification	787006
<i>Oak Stream 16-Channel Platform Essential I/O: Models</i>	822178
Oak Stream 16-Channel Platform Johnson City 1 Design Archive	823065
Oak Stream 16-Channel Platform Johnson City 2 Design Archive	636258
Oak Stream 16-Channel Platform Marketing Development Platform 1 (MDP1) Design Archive	827452
Oak Stream 16-Channel Platform Marketing Development Platform 2 (MDP2) Design Archive	828483
Oak Stream 16-Channel Platform Modular On-Board BMC Design Considerations	855287
Oak Stream 16-Channel Platform PCI Express* (PCIe*) and Intel® UPI Models; Part 1	822164
Oak Stream 16-Channel Platform PCI Express* (PCIe*) and Intel® UPI Models; Part 2	822165
Oak Stream 16-Channel Platform Pin List	636260
Oak Stream 16-Channel Platform Power Budget Calculator	833790
<i>Oak Stream 16-Channel Platform Shadow Core Design Considerations Technical Article</i>	845670
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Drawings</i>	786522
<i>Oak Stream 16-Channel Platform Socket Enabling Components Mechanical Models</i>	786523
Oak Stream 16-Channel Platform Thermal Mechanical Specification	786521
Oak Stream 16-Channel Platforms Design Archive Addendum	856301
Oak Stream 16-Channel/Oak Stream 12-Channel Platform DDR5 Voltage Regulator Test Tool (VRTT) Test Plan	832854
continued...	

Document Title	Document Number
Oak Stream 16CH VR Test Tool Interposer Installation Steps (Video)	851767
Oak Stream 8-Channel Pin List	821023
Oak Stream 8-Channel Platform Aggregator Board Design Archive	865794
Oak Stream 8-Channel Platform Basin City Design Archive	636257
Oak Stream 8-Channel Platform Design Guide	813596
Oak Stream 8-Channel Platform Electrical Specification	794719
Oak Stream 8-Channel Platform Socket Enabling Components Mechanical Drawings	786525
Oak Stream 8-Channel Platform Socket Enabling Components Mechanical Models List of Components	786526
Oak Stream 8-Channel Platform Thermal Mechanical Specification	786524
Oak Stream Diamond Rapids 16-Channel Mackinaw City Design Archive	821026
Oak Stream Dual Partitioning Boot Hardware Design Checklist	844091
Oak Stream Firmware Deep Dive - Oak Stream Architecture and RAS Overview	851626
Oak Stream Firmware Deep Dive Compute Express Link* (CXL*)	851589
Oak Stream Firmware Deep Dive – Diamond Rapids Memory MicroController (MMC)/ zCode Overview	848466
Oak Stream Firmware Deep Dive – Diamond Rapids System Memory Address Map	851254
Oak Stream Firmware Deep Dive – Intel® Out-of-Band Management Services Module (Intel® OOBM Services Module) and Telemetry	851007
Oak Stream Firmware Deep Dive – Intel® Ultra Path Interconnect (Intel® UPI) 3.0	852034
Oak Stream Firmware Deep Dive – Manageability/ Debugging	851752
Oak Stream Firmware Deep Dive – PCI Express* (PCIe*)	851755
Oak Stream Firmware Deep Dive – PCIe/CXL*	852184
Oak Stream Firmware Deep Dive – S3M 3.0	851753
Oak Stream Intel® OpenBMC Extensions Development Guide	834088
Oak Stream Platform 16-Channel Interposer Port Models and Socket RL Values	842679
Oak Stream Platform 16-Channel Power Delivery Schematic Layout Checklist	844697
Oak Stream Platform 64 GT/s PAM4 Limit Proposal SFF-TA-1002 Connector, Straight, Right Angle, Straddle Mount	786076
continued...	

Document Title	Document Number
<i>Oak Stream Platform 64 GT/s PAM4 Limit Proposal, SFF-TA-1002 Connector, Straight</i>	786236
Oak Stream Platform BIOS Writer's Guide	820107
Oak Stream Platform DC-SCM 2.1 Design Archive	636262
<i>Oak Stream Platform Design In</i>	816342
Oak Stream Platform Design Options and Considerations Overview Technical Article	858912
Oak Stream Platform Diamond Rapids 12-/16-Channel Dashboard	863599
Oak Stream Platform Essential I/O: I2C/I3C* Timing Analysis	869554
Oak Stream Platform Essential I/O: eSPI /SPI Timing Analysis	869552
<i>Oak Stream Platform High-Speed Differential Signal Crossing Voltage Regulator Design Guide</i>	823169
Oak Stream Platform Memory Subsystem CLTT BIOS MRC Implementation Technical Article	862446
Oak Stream Platform Message of the Week	820238
Oak Stream Platform Power-On Debug Handbook Tool Package Scripts	854296
Oak Stream Platform Power-On and Reset Issue Debug Handbook	835693
<i>Oak Stream Platform Quarterly Technical Review (QTR) Q1'25 Recording</i>	853183
<i>Oak Stream Platform Quarterly Technical Review (QTR) Q1'25 - 848790</i>	848790
<i>Oak Stream Platform Quarterly Technical Review (QTR) Q4'24</i>	837295
<i>Oak Stream Platform Quarterly Technical Review (QTR) Q4'24 Recording</i>	842289
Oak Stream Platform RAS Technology Integration and Validation Guide	846674
Oak Stream Platform Root of Trust (PROM) Design Guide	862993
<i>Oak Stream Platform SPI Programming Guide</i>	820596
Oak Stream Platform Security Update	869452
Oak Stream Platform Tokens, Vendor Authorized Debug (VAD), and Intel® Vendor Authorized Boot Integration and Validation Guide	863386
Oak Stream Platform Update and Design Considerations	867308
<i>Oak Stream Platform, RAS Deep Dive Technical Training</i>	824347
<i>Oak Stream Platforms DC-SCM 2.1 Integration Guide</i>	812906
Oak Stream Platforms System Boards Design Archive	821025
Oak Stream Power-On Checklist	859423
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Document Title	Document Number
Oak Stream-UCC/UCC-M Platform PCI Express* (PCIe*) Load Board (PLB) and Intel® UPI Load Board (ULB) User Guide and Socket Pin Map	855442
Oak Stream/Diamond Rapids Update: Sample Communications	847016
PCI Express* (PCIe*) 6.0 1x16 Slot Cabled Riser Platform Architecture Specification	805690
PCI Express* (PCIe*) 6.0 Compliance Test Best Known Method (BKM)	872609
PCI Express* (PCIe*) 6.0 Hot Swap Backplane (HSBP) 8x4 Drive Bay Design Guide	852244
PCI Express* (PCIe*) 6.0 Supplemental Validation Guide – Oak Stream	867608
PCI Express* (PCIe*) Lane Margin Tool (LMT) for PCIe* 4.0/5.0/6.0	639433
PCI Express* 6.0 Cable Assembly Design Specification	764126
PCI Express* 6.0 Connector Status	780797
PCI Express* 6.0 Internal Cable Measurement Procedure Specification	766311
PCI Express* 6.0 Mated Cable Assembly Status	780796
PCI Express* 6.0 SFF-TA-1002 Orthogonal and Straight Connector Measurement Procedure Specification	814603
PCI Express* 6.0 SFF-TA-1033 Straight Connector Measurement Procedure Specification	781300
PCIe Ecosystem Symposium: Enabling Data Center AI Server Solutions	831542
Partitioned Boot Architecture with Bootable Intel® Xeon® Processors	789728
Printed Circuit Board Plated Over Filled Vias (POFV) Technical Article	844506
Signal Integrity Collateral List for the Oak Stream Platform	822391
Signal Integrity Jitter Simulation for PCI Express* (PCIe*) Reference Clocks	872782
System Management Specification	648590
Tabbed Routing Design and Characterization Tool (TRDCT)	560170
Tabbed Routing Design and Characterization Tool [TRDCT] User Guide	574704
Test Seed Loader (TSL) and Dragon Seed Content	620322
Turkey Canyon Thermal Mechanical Test Vehicle (TMTV) Package Thermal Model User Guide	852612
Unified Power Delivery Tester (UPDT) - VR Test Tool Host Software	788921